Pharmacology

Credits: The course is made up of 3 credit hours

Course description The course aims to provide the students of the essentials pharmacology, with the emphasis on, drug actions, uses, routs of administration, dosages, adverse reactions, contra-indications, nursing considerations etc.

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Teaching Method : Lectures, Discussion and Group work

Evaluation

- Midterm exam 30%
- Assignment 30%
- Final exam 40%





- **Pharmacology** is a Latin word meaning drug knowledge. **Pharmaco-** means drug, **-ology** means science or knowledge.

- **Pharmacology** is the study of chemicals (drugs) effects on living tissues and how those chemicals help diagnose, treat, cure, and prevent disease or correct the pathophysiology of living tissues.





- **Pharmacopeia:** is a book containing formulas, which are standard for preparation and administration of drug. If a drug is included in this book it has met the standards of quality, purity, and strength.
- Most countries publish an authoritative lists of drugs called a pharmacopoeia such as
  - British Pharmacopoeia (BP)
  - United States Pharmacopeia (USP)

Drug is defined by US FDA as:



- **A drug:** is a substance or chemical capable of altering a biochemical or physiological processes in the body; these responses may be desirable (therapeutic) or undesirable (adverse).

- **A drug:** is a substance intended for use in the diagnosis, cure, mitigation, treatment, or prevention of disease.

Nursing Responsibilities:

- Registered nurses, midwives are accountable for their own acts.

The nurse must:

- **know the therapeutic uses** of the medicine to be administered, its normal dosage, side-effects, precautions and contraindications
- be certain of the **identity of the patient** to whom the medicine is to be administered
- **check that the prescription**, or the label on the medicine dispensed by a pharmacist, is clearly written and unambiguous

- Traditionally, drugs were derived from natural sources, such as:
    - plants
    - animals
    - minerals.



- The earliest drug concoctions from plants used everything: the leaves, roots, bulb, stem, seeds, buds, and blossoms. As a result, harmful substances often found their way into the mixture.

These active constitutes are

a. **Alkaloids:** atropine, morphine, codeine and heroine etc. they are toxic, has bitter taste, alkali-like, highly potent, insoluble in water

b. **Glycosides:** Such as Digoxin.

c. **Gums:** Such as agar

d. **Resins:** They are fatty like substances that are greasy and irritant such as cathartics

e. **Oils:**

1. Volatile oil such as peppermint oil and clove oil.
2. Fixed such as olive oil and castor oil.



Aid from animals

The body fluids or glands of animals are also natural drug sources.

The drugs obtained from animal sources include:

- *hormones,* such as insulin
- *oils* and *fats* (usually fixed), such as cod-liver oil
- *enzymes,* which are produced by living cells and act as catalysts, such as pancreatin and pepsin
- *vaccines,* which are suspensions of killed, modified, or attenuated microorganisms.

Many minerals

- Metallic and nonmetallic minerals provide various inorganic materials not available from plants or animals.
- Mineral sources are used as they occur in nature or they’re combined with other ingredients. Examples of drugs that contain minerals are iron, iodine, and Epsom salts.
- Today, most drugs are produced in laboratories. Examples of such drugs include thyroid hormone (from natural sources) and cimetidine (from synthetic sources).

- Drugs that share similar characteristics are grouped together as **a *pharmacologic class*** (or family). Beta-adrenergic blockers are an example of a pharmacologic class.
- A second type of drug grouping is the ***therapeutic class***, which categorizes drugs by therapeutic use. Antihypertensive are an example of a therapeutic class.

Classification of drugs according to prescription

a) Controlled drug:

These are medicines that have the potential for addiction and subsequent abuse (supplied by a registered pharmacy on receipt of a written prescription from a registered medical practitioner)(codeine).

b) General sale list (OTC):

Medicines sold directly to a consumer without a prescription from a healthcare professional (analgesics).

Dosage forms of drugs



Oral preparations

Liquids

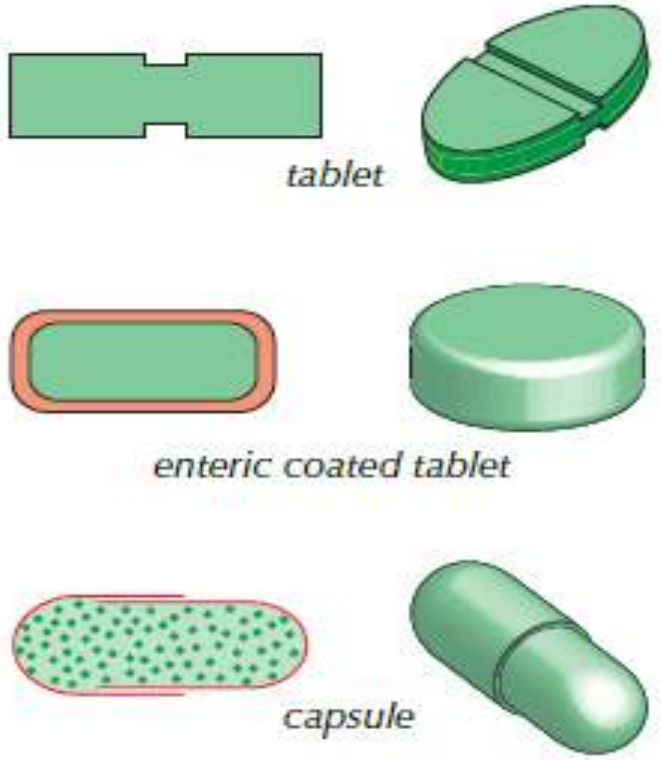
- ✓ **Emulsions** Mixture of two immiscible liquids (oil and water) by means of emulsifying agent (e.g. Liquid paraffin).
- ✓ **Oral suspensions** Dispersion through a liquid of a solid in finely divided particles of a size.
- ✓ **Elixir** Sweetened, flavored hydro alcoholic solution containing drug
- ✓ **Syrup** Concentrated solution of sugar containing flavoring, coloring and therapeutically active substances.

Dosage forms of drugs

Oral preparations

Solids

- Tablets: Compressed powdered dosage form into hard disks or cylinders.
- Capsules: Shells of gelatin containing drug for oral use medication in powder or oil form



Dosage forms of drugs

Oral preparations

- **Capsules:**
  - ➢ **Used for drugs that have unpleasant taste**
  - ➢ **Resist the action of gastric juice**
  - ➢ **Sustained release**





Dosage forms of drugs

Rectal preparations

- **Suppository:** solid preparations for insertion into rectum e.g. Aminophylline supp.
- **Enema:** liquid preparations for insertion into rectum.





Dosage forms of drugs

Parenteral preparations

Sterile solution or suspension containing drug that may be dispensed in:

a. Ampules: containing single dose

b. Vials: rubber capped bottles containing a number of doses.





Dosage forms of drugs



Inhalational preparations

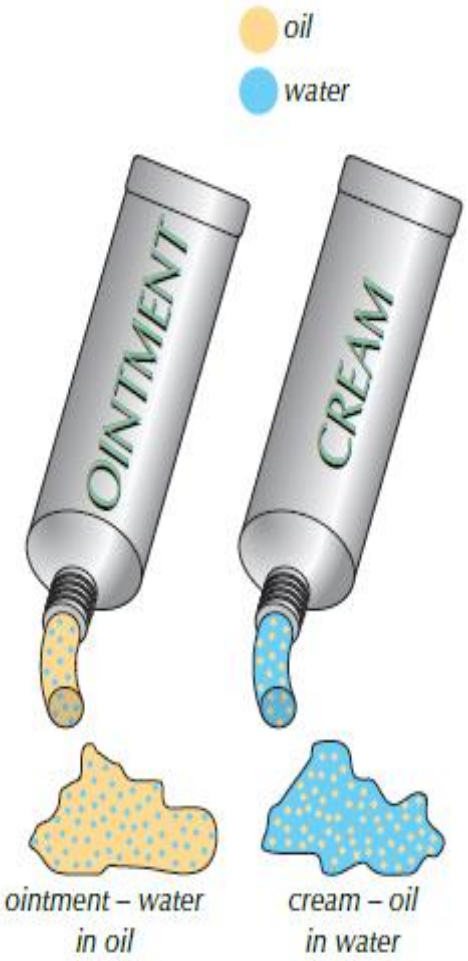
a. Gases: administration thru special devices.
b. Volatile liquid
c. Steam inhalation
d. Aerosols: released into respiratory passages in the form of a fine mist of liquid droplets (bronchodilation).

Dosage forms of drugs

Topical preparations

- Ointments
- Cream
- Pastes
- Ophthalmic ointment
- Lotion
- Eye, Ear and Nose Drops
- Powders
- Vaginal douches

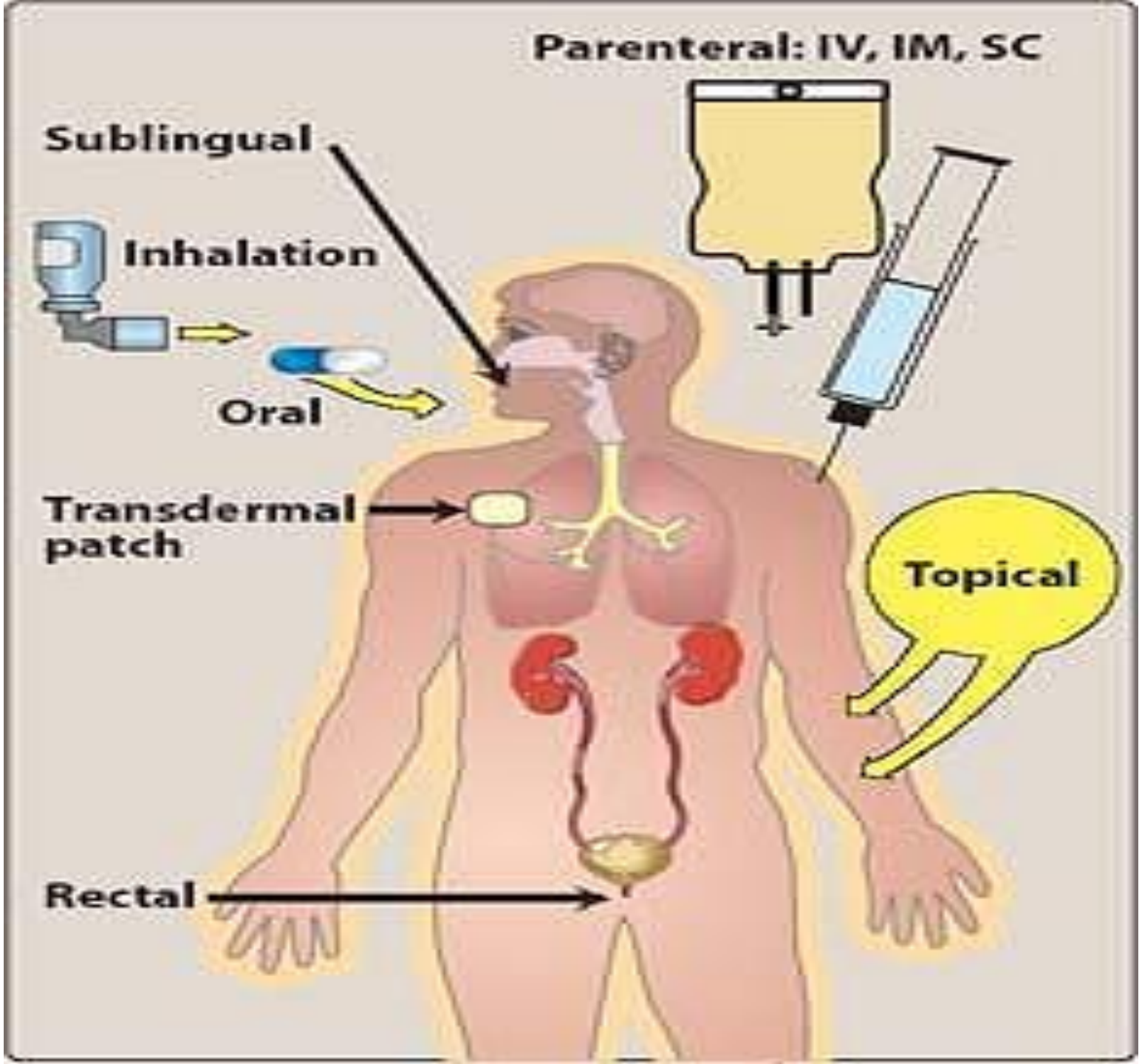




Dosage forms of drugs

- Disposable syringes
- Large volume intravenous solutions
- Intermittent IV solution




Parenteral: IV, IM, SC
Sublingual
Inhalation
Oral
Transdermal patch
Topical
Rectal

❑ Dosage:

It refer to the determination of size, frequency and number of doses (quantity to be administered in one time)

❑ Loading / priming dose:

A dose that quickly raises the concentration of drug in plasma to target concentration (that will produce the desired therapeutic effect

- Amoxi 500 mg Cap.

  1X3X5

- Decort .5mg Tab

  1X3X3

  1X2X3

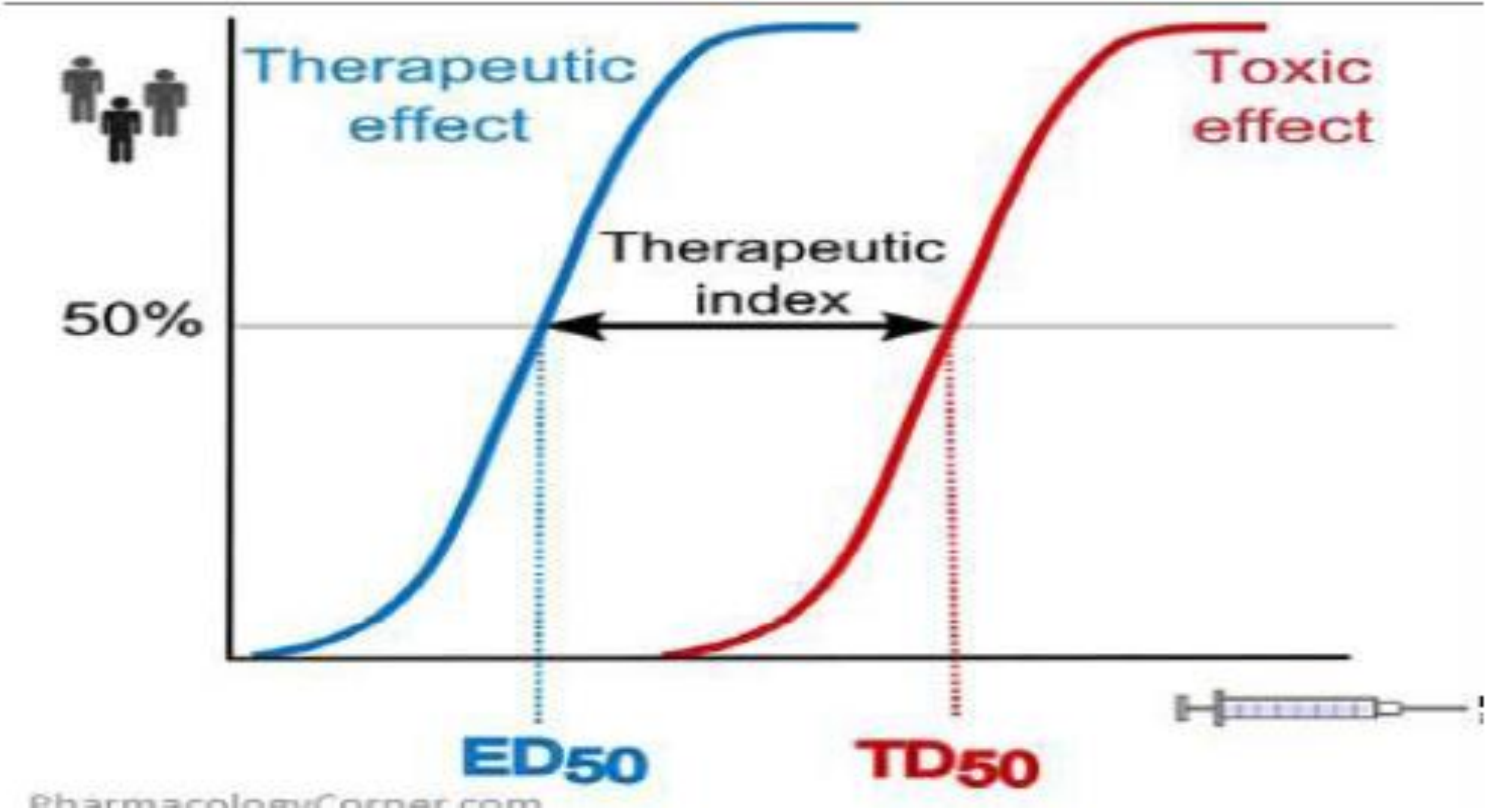
  1X1X3

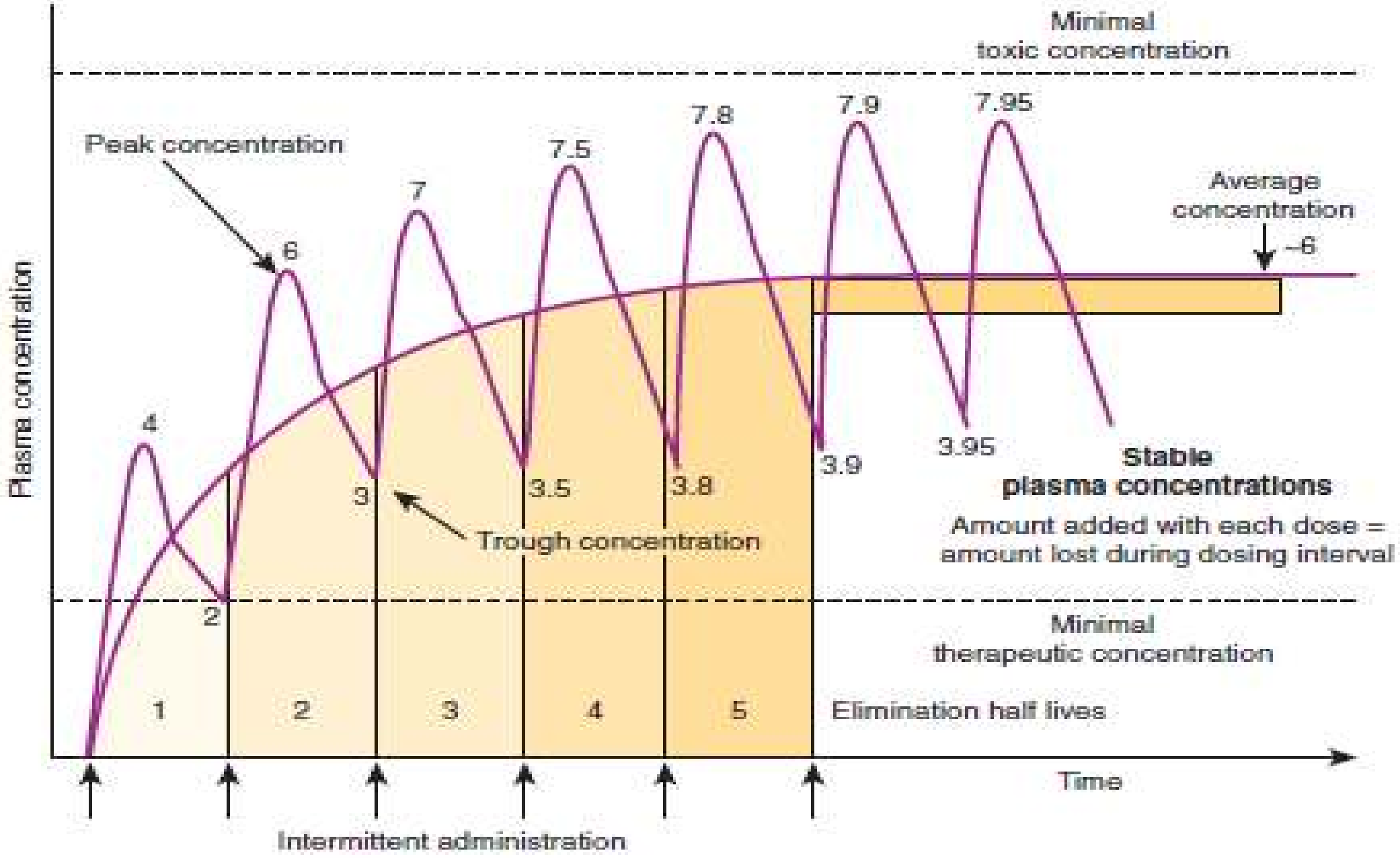
- Doxi 100mg Tab.

  1X2X1

  1x1x5

- Maintenance dose or dose rate: a dose that maintain a steady state of drug in body (dose of drug that replace the drug eliminated since proceeding dose).
- Maximal tolerated dose: largest dose of the a drug that can be taken safely.
- Fatal dose: a dose that produces death.


Therapeutic effect
Toxic effect
Therapeutic index
50%
ED50
TD50
PharmacologyCorner.com

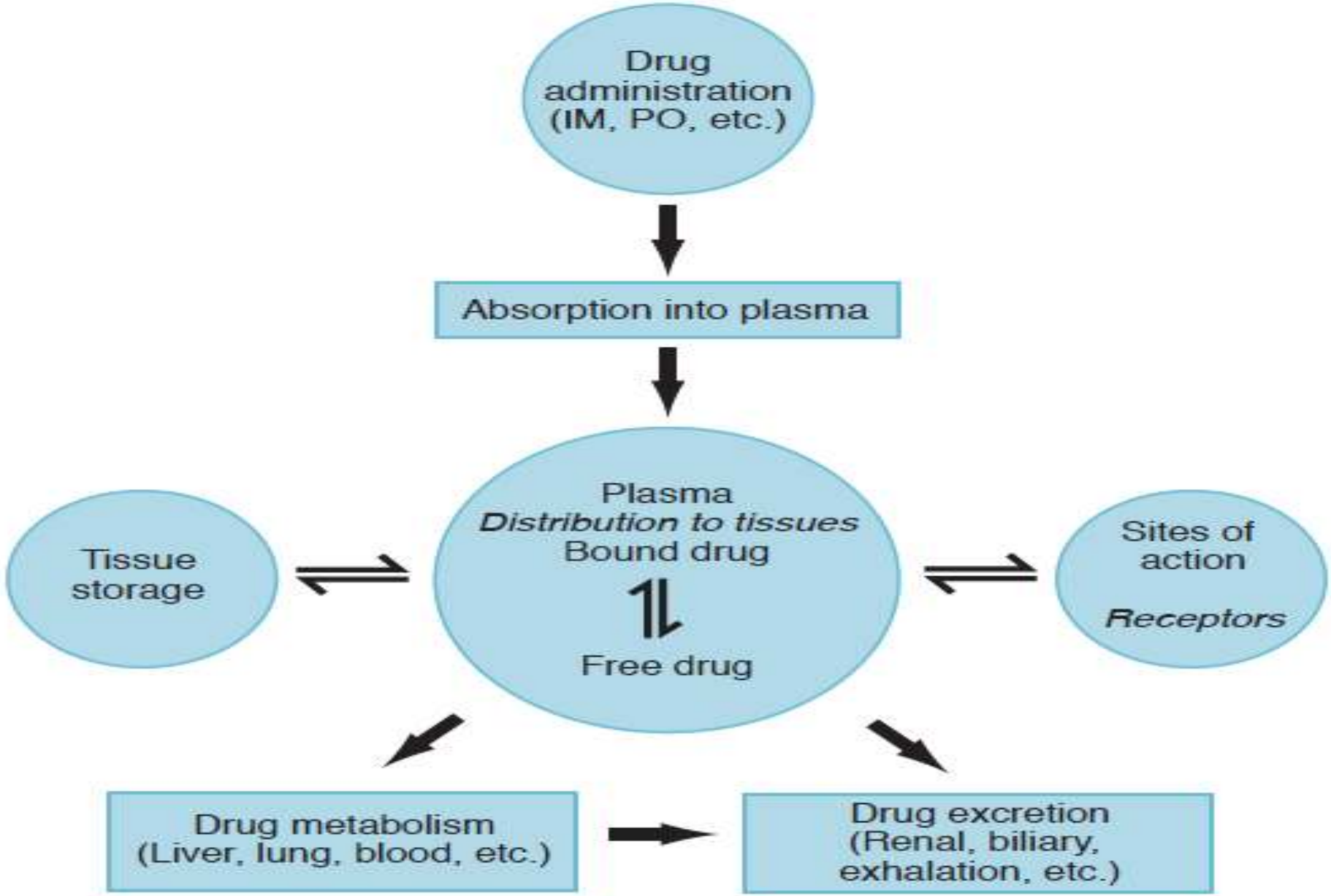

Minimal
toxic concentration
Peak concentration
Average
concentration
~6
Plasma concentration
4
6
7
7.5
7.8
7.9
7.95
2
3
3.5
3.8
3.9
3.95
Trough concentration
Stable
plasma concentrations
Amount added with each dose =
amount lost during dosing interval
Minimal
therapeutic concentration
1
2
3
4
5
Elimination half lives
Time
Intermittent administration

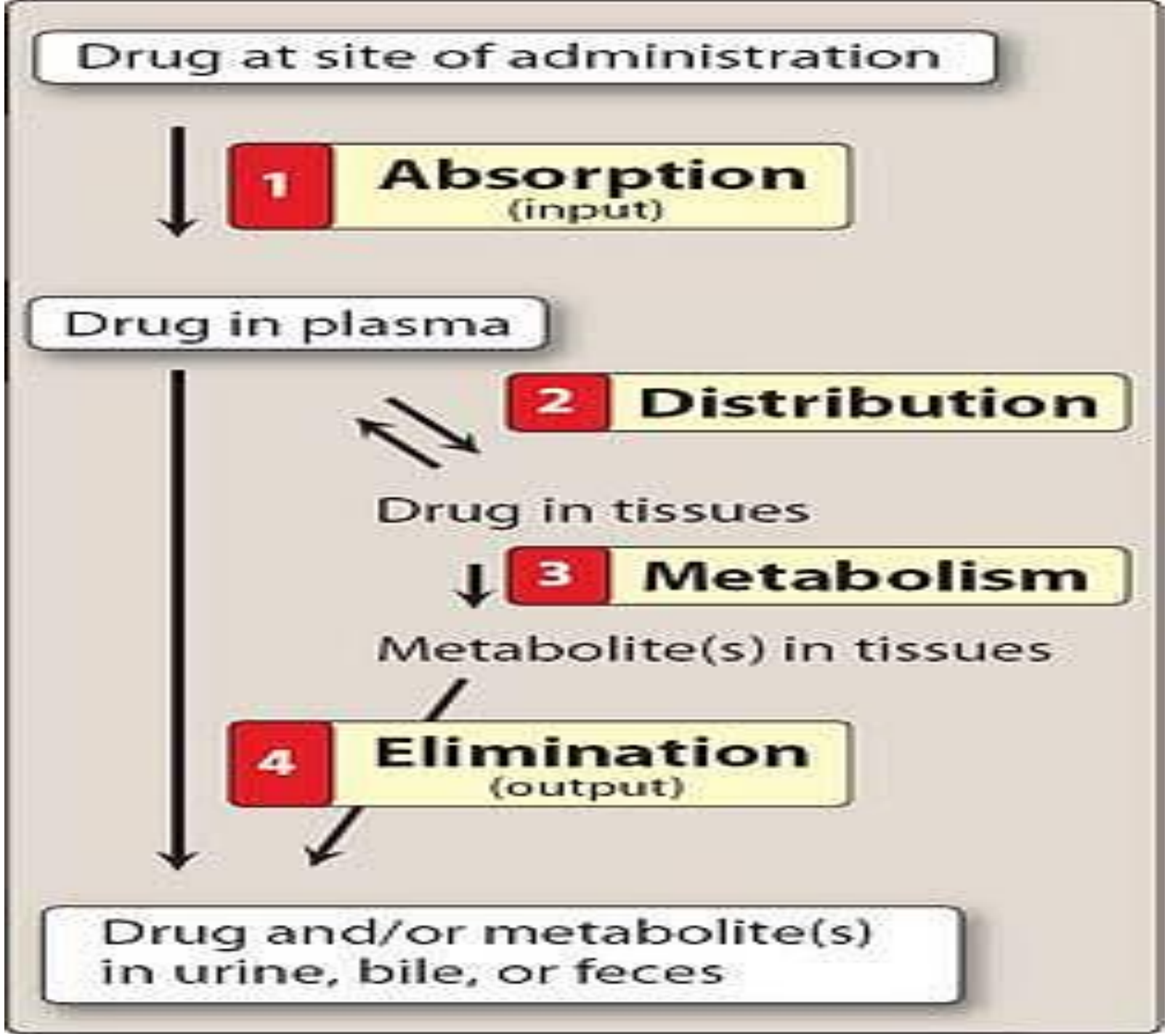
Pharmacokinetics

The term *kinetics* refers to movement. Pharmacokinetics deals with a drug's actions as it moves through the body. Therefore, pharmacokinetics discusses how a drug is:

- absorbed (taken into the body)
- distributed (moved into various tissues)
- metabolized (changed into a form that can be excreted)
- excreted (removed from the body).

This branch of pharmacology is also concerned with a drug's onset of action, peak concentration level, and duration of action.


Drug administration (IM, PO, etc.)
Absorption into plasma
Plasma
Distribution to tissues
Bound drug
Free drug
Tissue storage
Sites of action
Receptors
Drug metabolism (Liver, lung, blood, etc.)
Drug excretion (Renal, biliary, exhalation, etc.)


Drug at site of administration
1 Absorption
(input)
Drug in plasma
2 Distribution
Drug in tissues
3 Metabolism
Metabolite(s) in tissues
4 Elimination
(output)
Drug and/or metabolite(s)
in urine, bile, or feces

I. Drug Absorption

- It refers to passage of a drug from its site of administration into blood stream and distributed in the body, it comes in contact with numerous membranes.
- Drugs pass some membranes but not others.
- On a cellular level, drugs are absorbed by several means—primarily through active or passive transport.
- Absorption process varies according to the route of administration, dosage form, and dose of the drug.

Factor affecting drug absorption:

a. Drug factors:

- Degree of ionization
- Solubility (lipophilicity),
- Formulation (solution vs. tablet).

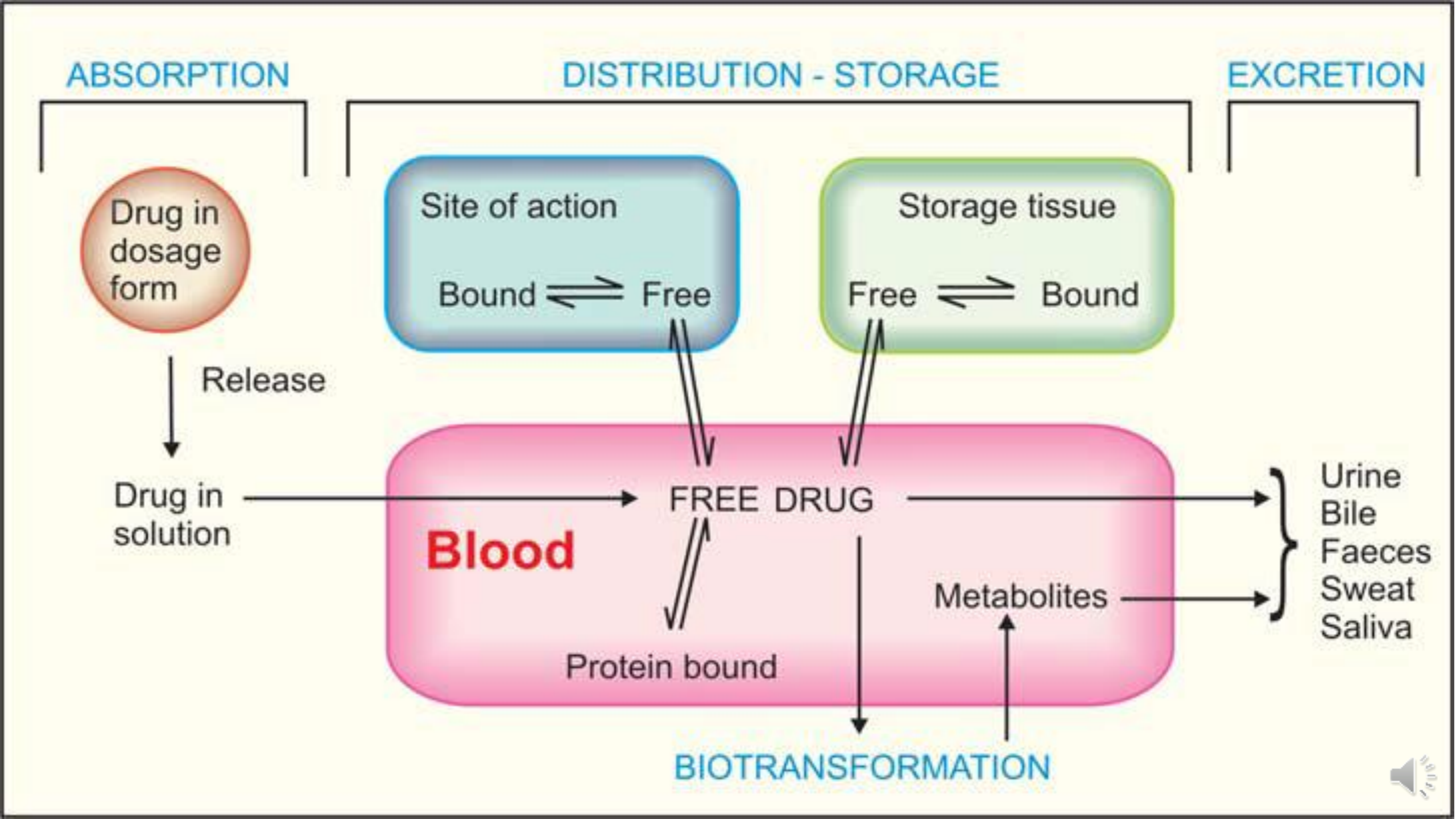
Factor affecting drug absorption:

b. Patient factors :

- The route of administration. (oral, sublingual, rectal, inhalation)
- Blood flow
- Total surface area available for absorption
- Contact time

II. Drug Distribution

- It is the process by which the drug reversibly leaves the bloodstream and enters tissues and ultimately to its site of action.
- The distribution phase can be extremely slow for drugs that bind strongly to serum protein complex is unable to pass out of plasma.
- hypoalbuminemia, as in liver disease, burn and malnutrition may affect absorption and distribution of drugs.


ABSORPTION
DISTRIBUTION - STORAGE
EXCRETION
Drug in dosage form
Release
Drug in solution
Site of action
Bound ⇌ Free
Storage tissue
Free ⇌ Bound
Blood
FREE DRUG
Protein bound
Metabolites
BIOTRANSFORMATION
Urine
Bile
Faeces
Sweat
Saliva

Factor affecting drug distribution:

Membrane permeability:

- lipid soluble drugs (unionized)
- BBB
- placenta barrier

Factor affecting drug distribution:

1. Plasma protein binding: :

- Some drugs can bind non specifically and reversibly to various plasma protein e.g. albumin and globulin.
- Bound and free drug reach an equilibrium.
- Only the free drug exerts a biologic effect
- Bound drug stays in vascular apace and is not metabolized or eliminating

Factor affecting drug distribution:

2. **Depot storage: :**

- Lipophilic drugs, such as the sedative thiopental, accumulate in fat.
- An obese person might be sedated for a greater period of time than a lean person to whom the same dose of thiopental had been administered.
- Calcium binding drugs, such as the antibiotic tetracycline, accumulate in bone and teeth.

Factor affecting drug distribution:

4. Blood flow :

- Most of drugs distributed initially to organs that have rich blood supply as the heart, liver and kidney. Delivery of the drug to the viscera, skin, bone and a depose tissue is slower.

5. Tissue affinity:

- Some tissue are localized in specific tissues which possess specific drug receptors e.g. iodine in thyroid gland and chloroquine in liver

III. Biotransformation (Drug Metabolism)

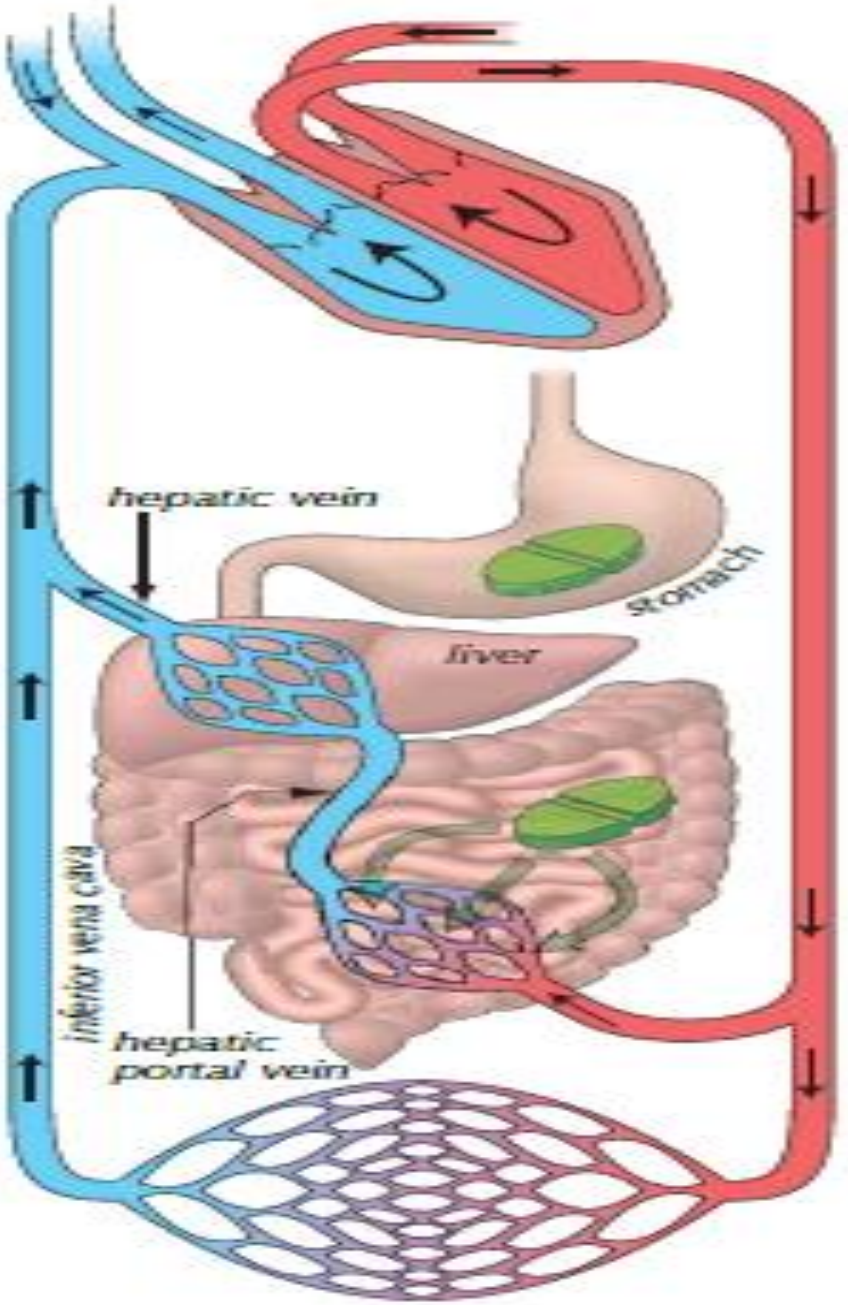
- Drugs are most often eliminated by biotransformation and/or excretion into the urine or bile.
- The process of metabolism transforms lipophilic drugs into more polar readily excretable products.

III. Biotransformation (Drug Metabolism)

- The liver is the major site for drug metabolism, but specific drugs may undergo biotransformation in other tissues

- Following an oral administration, many drugs are absorbed intact from small intestine and transported via portal system to liver where they undergo extensive metabolism referred as **first pass effect.**

Some agents are initially administered as inactive compounds (pro-drugs) and must be metabolized to their active forms.



III. Biotransformation (Drug Metabolism)

A. Phase I reactions:

- Alter chemical reactivity and increases aqueous solubility of drugs.
- It includes oxidation (cytochrome P450), reduction and hydrologists

III. Biotransformation (Drug Metabolism)

Phase I reactions:

Metabolic reactions can transform

- An active drug into less-active or inactive forms.
- A PRODRUG (inactive or less-active drug) into a more active drug.

III. Biotransformation (Drug Metabolism)

Phase II:

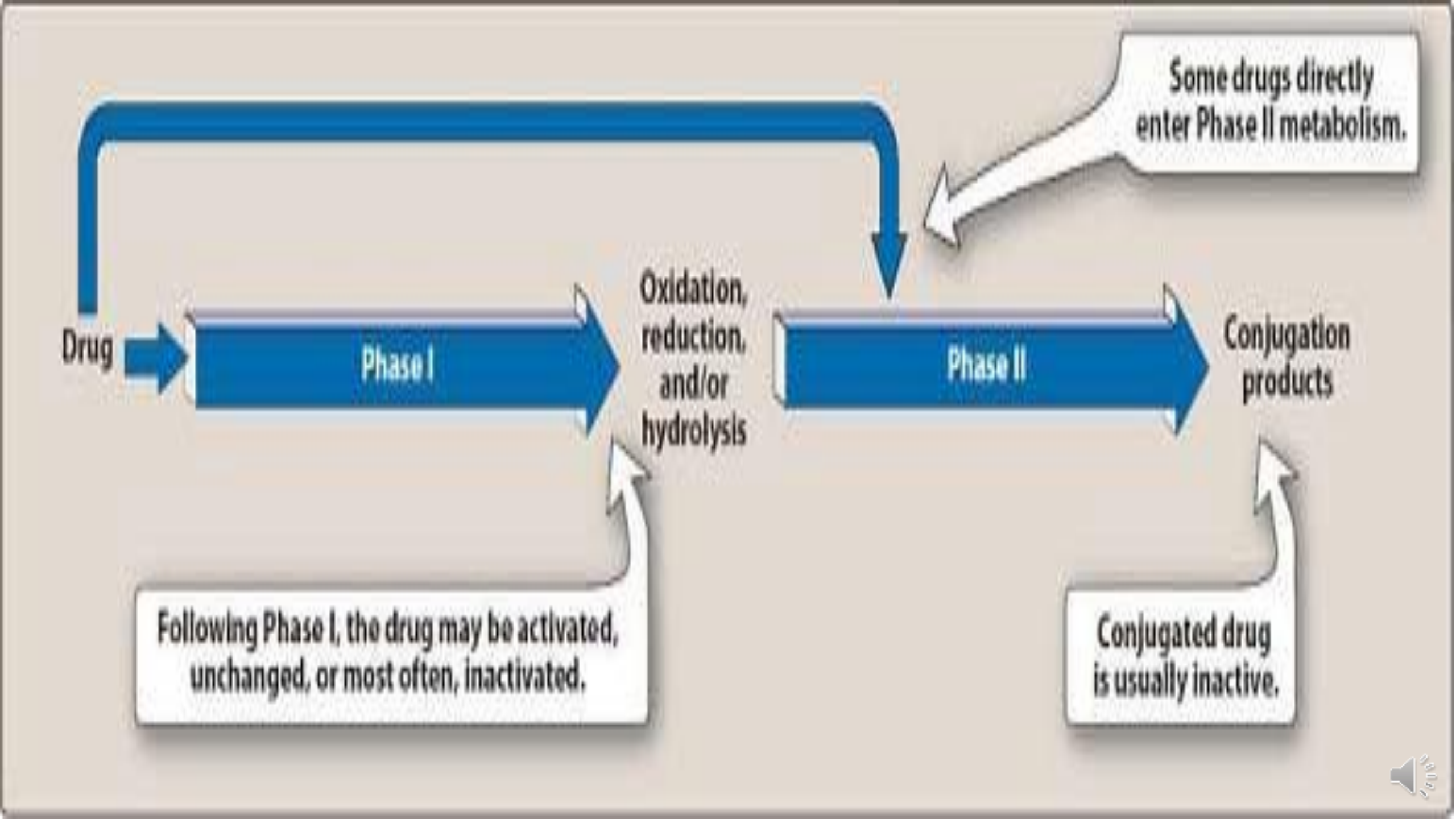
This phase consists of conjugation reactions.

- If the metabolite from Phase I metabolism is sufficiently polar, it can be excreted by the kidneys. However, many Phase I metabolites are too lipophilic to be retained in the kidney tubules.

III. Biotransformation (Drug Metabolism)

Phase II:

- A subsequent conjugation reaction with an endogenous substrate, such as glucuronic acid, results in polar, usually more water-soluble compounds that are most often therapeutically inactive.
- The highly polar drug conjugates may then be excreted by the kidney or bile.

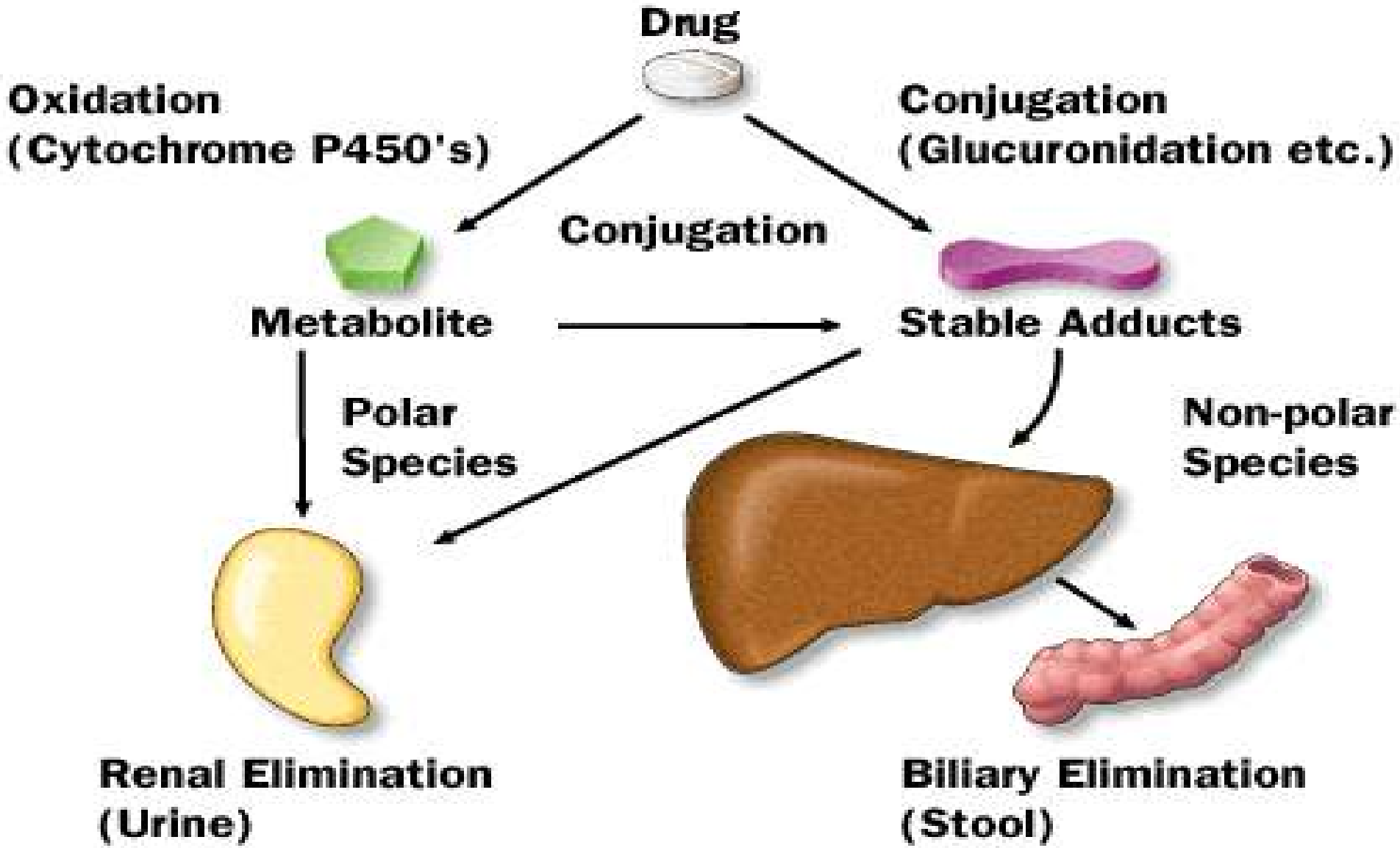

Some drugs directly enter Phase II metabolism.
Drug
Phase I
Oxidation, reduction, and/or hydrolysis
Phase II
Conjugation products
Following Phase I, the drug may be activated, unchanged, or most often, inactivated.
Conjugated drug is usually inactive.

Factor affecting metabolism

a. Genetics : it may alter enzyme level
b. Chemical properties of drug: certain drugs may stimulate or inhibit the metabolism of other drug
c. Rout of administration: oral route can result in extensive first pass effect.
d. Diet: starvation depletes glycine and alter glycine conjugation.
e. Dosage: toxic dose can deplete enzyme.
f. Age: liver cannot detoxify drugs as well in neonates than adults.
g. Disease: liver and kidney disease decrease drug metabolism.

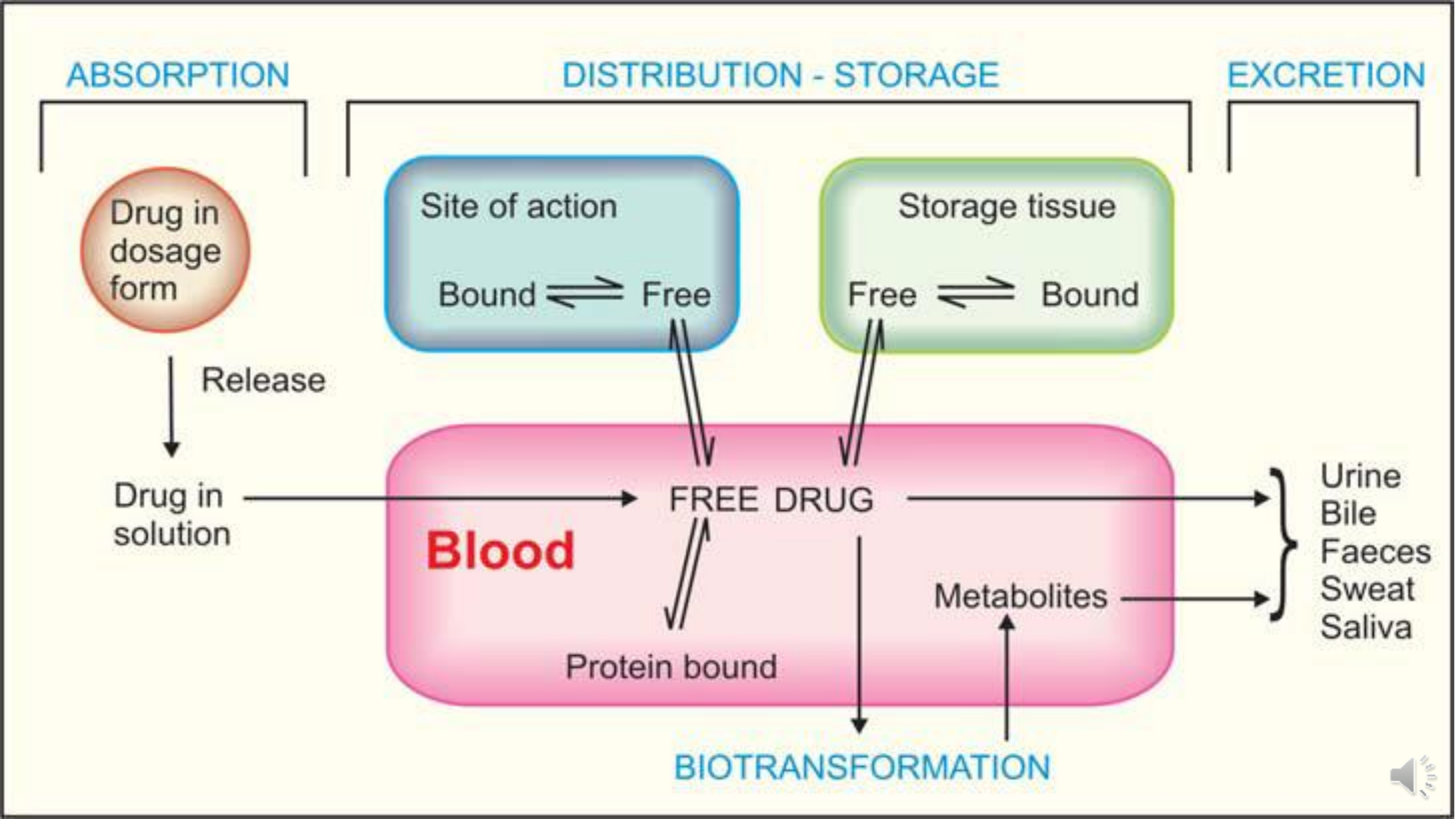
Drug Elimination (Excretion)

- Drug excretion refers to the elimination of drugs from the body.
- Most drugs are excreted by the kidneys and leave the body through urine.
- Drugs can also be excreted through the lungs, exocrine glands (sweat, salivary, or mammary glands), skin, and intestinal tract.


Drug
Oxidation
(Cytochrome P450's)
Conjugation
(Glucuronidation etc.)
Conjugation
Metabolite
Stable Adducts
Polar
Species
Non-polar
Species
Renal Elimination
(Urine)
Biliary Elimination
(Stool)

Routes of elimination:

- Kidney: excretion of drugs and their metabolites into urine.
- Liver: it can secrete drugs or their metabolites into bile (e.g. opioids).
- GIT: iodides, mercury in saliva
- Lung: gaseous and volatile general anesthetic are excreted in expired air.
- Other routes: sweat, tears and breast milk.


ABSORPTION
DISTRIBUTION - STORAGE
EXCRETION
Drug in dosage form
Release
Drug in solution
Site of action
Bound ⇌ Free
Storage tissue
Free ⇌ Bound
Blood
FREE DRUG
Protein bound
Metabolites
BIOTRANSFORMATION
Urine
Bile
Faeces
Sweat
Saliva

Onset, peak, and duration

In addition to absorption, distribution, metabolism, and excretion, three other factors play important roles in a drug's pharmacokinetics:

- onset of action
- peak concentration
- duration of action.

Onset of action

- Onset of action refers to the time interval that starts when the drug is administered and ends when the therapeutic effect actually begins.

- Rate of onset varies depending on the route of administration and other pharmacokinetic properties.

- As the body absorbs more drug, blood concentration levels rise.
- The peak concentration level is reached when the absorption rate equals the elimination rate.
- The duration of action is the length of time the drug produces its therapeutic effect.

Half-life: it is the time required to change the amount of drug in body by one-half during elimination.

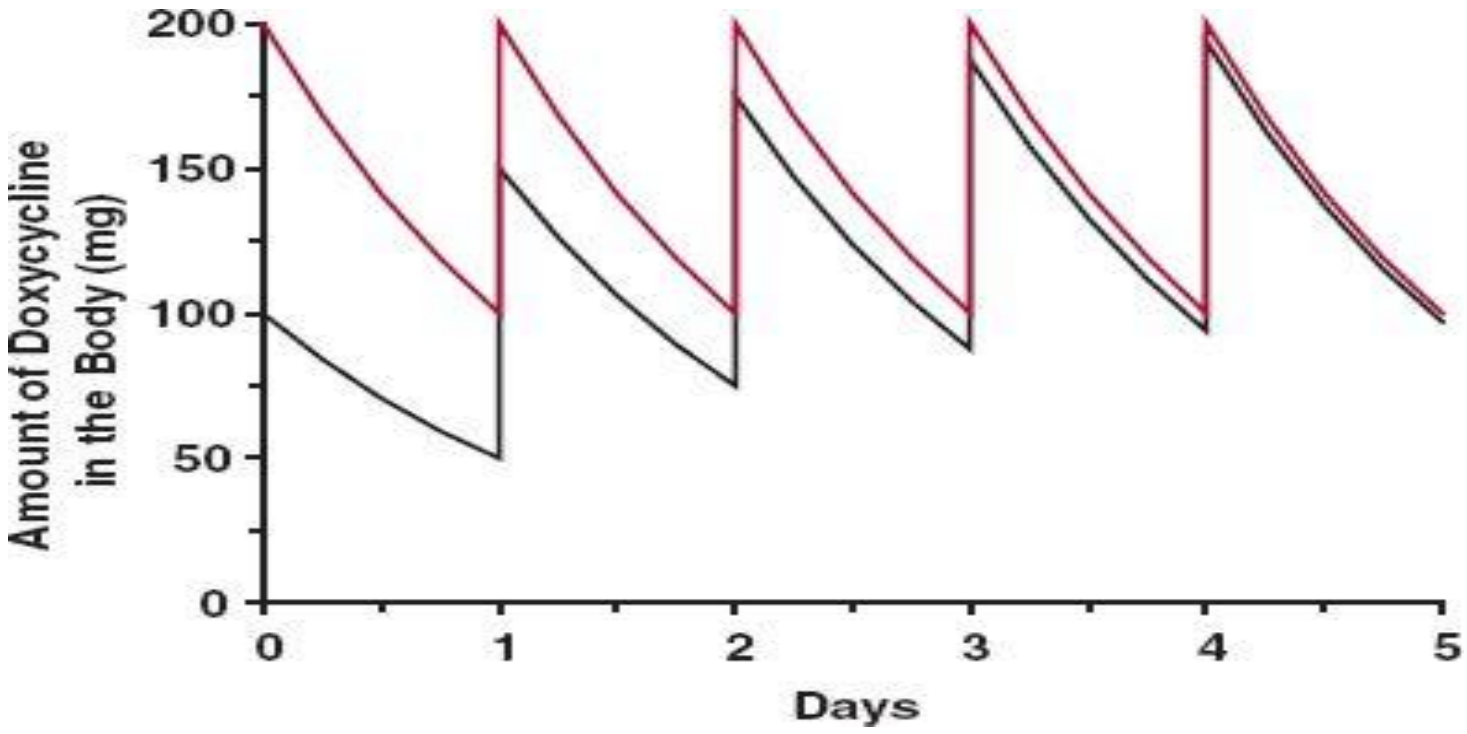
Note: if there is renal disease the risk of drugs accumulation and toxicity is increased.

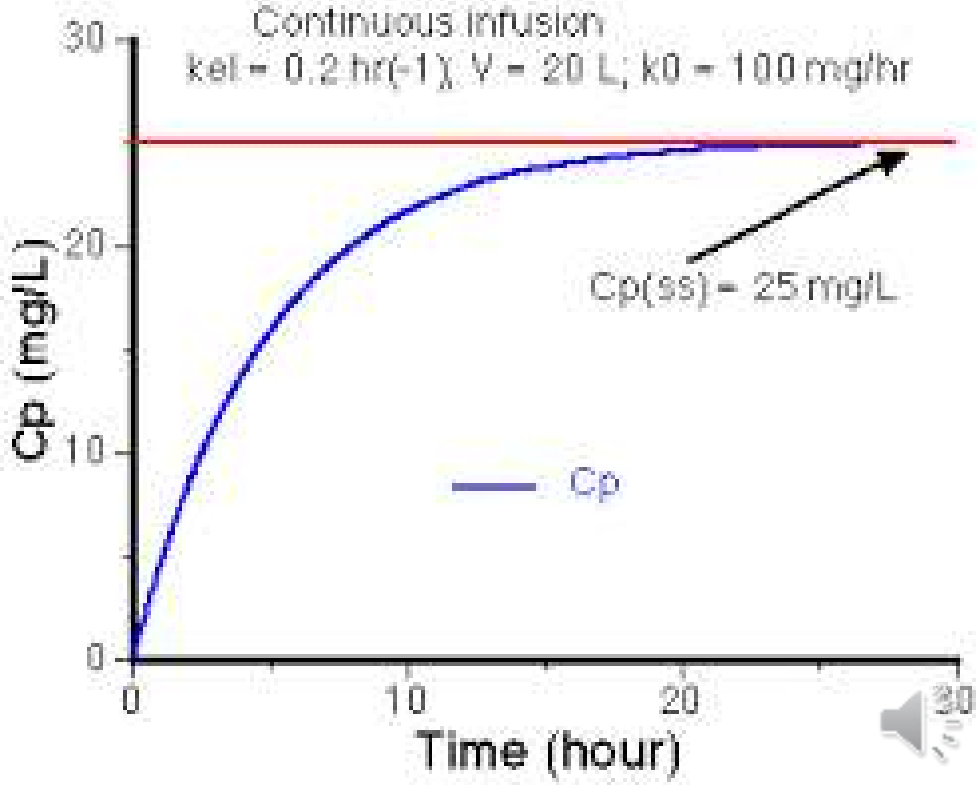
Factor increasing drug half-life:

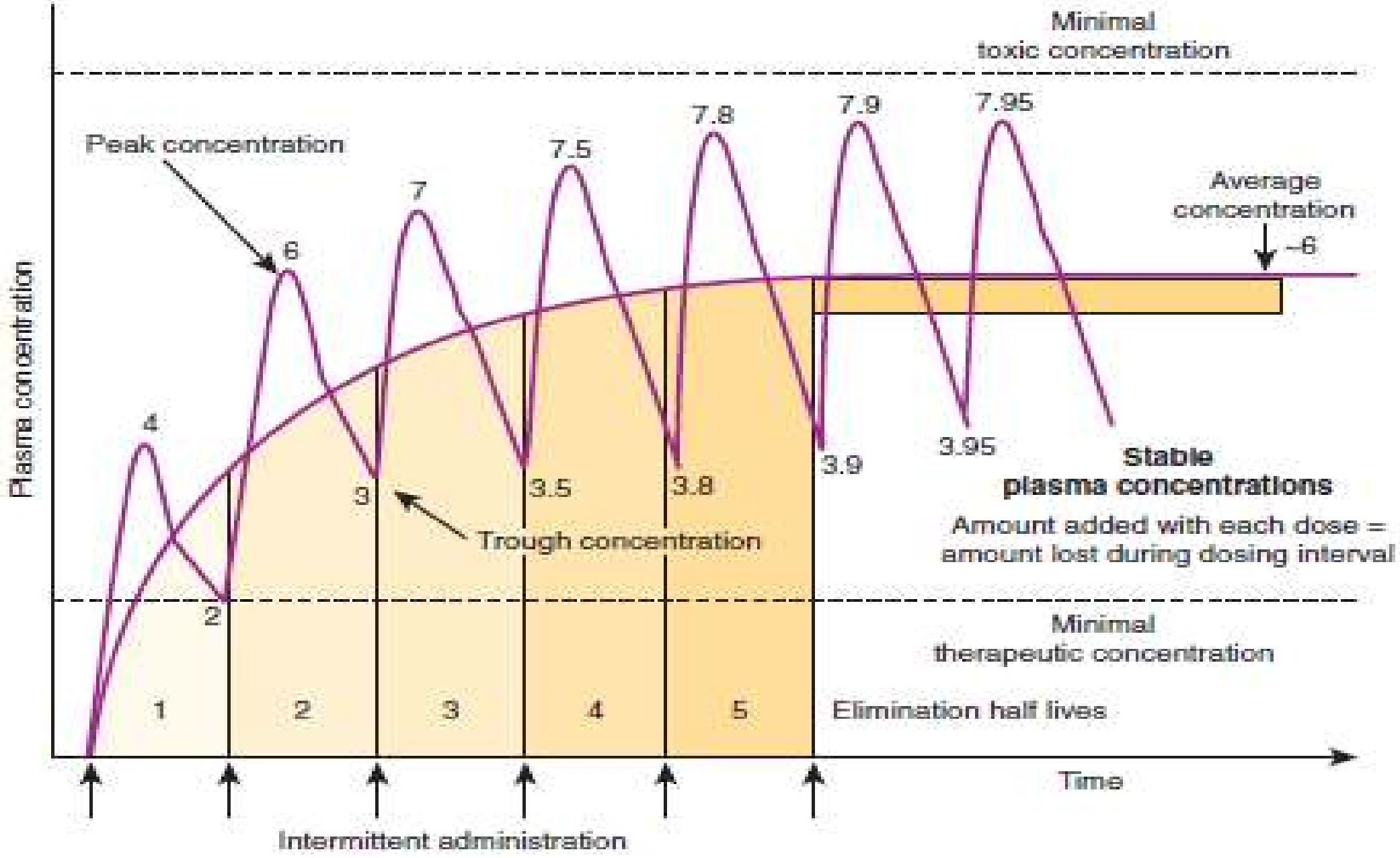
1. Decrease renal plasma flow e.g. cardiogenic shock, cardiac failure, hemorrhage.
2. Addition of second drug that displaces first from albumin, increasing Vd.
3. Decrease excretion ratio e.g. renal disease.
4. Decrease metabolism e.g. when another drug inhibits its biotransformation.

Steady state concentration:

A steady state plasma concentration of drug occurs when the rate of drug elimination is equal to the rate of administration.






Minimal
toxic concentration
Peak concentration
Average
concentration
~6
Plasma concentration
4
6
7
7.5
7.8
7.9
7.95
2
3
3.5
3.8
3.9
3.95
Trough concentration
Stable
plasma concentrations
Amount added with each dose =
amount lost during dosing interval
Minimal
therapeutic concentration
1
2
3
4
5
Elimination half lives
Time
Intermittent administration

Bioavailability

Bioavailability is expressed as the fraction of administered drug that gains access to the systemic circulation in a chemically unchanged form.

For example, if 100 mg of a drug are administered orally and 70 mg of this drug are absorbed unchanged, the bioavailability is 0.7 or seventy percent.

Factors affecting bioavailability

1. The drug itself.
2. The metabolism of the patient.
3. liberated rate (released) from its dosage form or from storage in the body.

Many drugs bind to serum protein (albumin) from which they released gradually, others stored at specific site as bone (tetracycline).

Drug isn't to be bioavailable if

1. Bound to protein or to any other substance that makes the drug permanently or temporarily inactive.
2. Not released from its dosage form or site of administration.
3. Partially or totally degraded.

Pharmacodynamics

- Pharmacodynamics is the study of the drug mechanisms that produce biochemical or physiologic changes in the body.
- The interaction at the cellular level between a drug and cellular components, such as the complex proteins that make up the cell membrane, enzymes, or target receptors, represents *drug action*.
- The response resulting from this drug action is called the *drug effect.*

How do drugs work?

1) Receptors:

It refers to specific drug binding sites in a cell or on its surface, which mediate the action of drug.

Nature of drug receptors

1. Regulatory proteins: e.g. neurotransmitters, hormone.
2. Enzyme e.g. dihydrofolate reductase.
3. Transport protein e.g. Na+ / k+ ATPase
4. Structural proteins, e.g. tubulin.

How do drugs work?

1) Agonist:

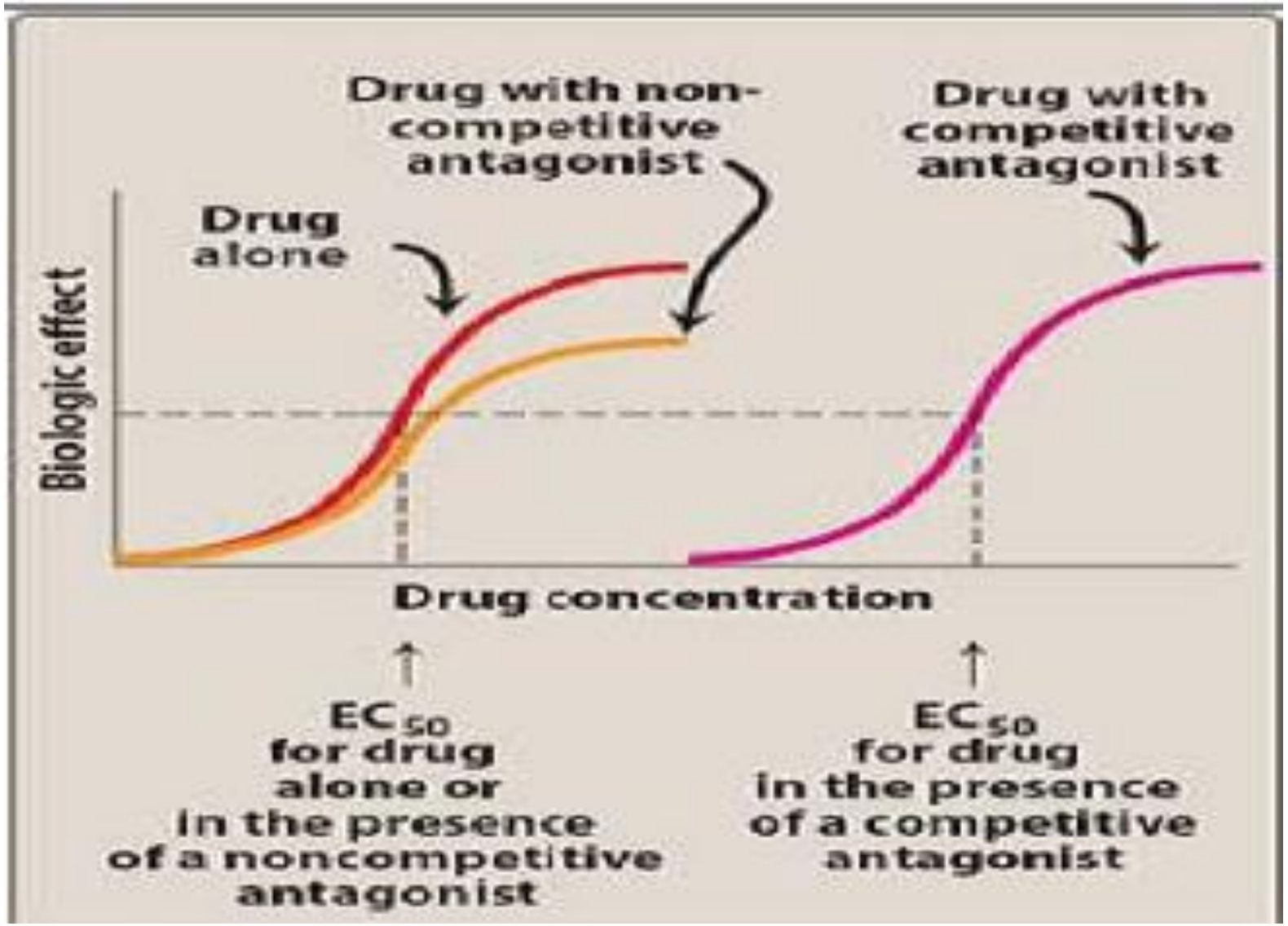
a drug that produce some of the effects of endogenous compounds when it interacts with receptor, e.g. isoproterenol at β adrenoceptors. Drug must have affinity and intrinsic activity to be agonist.

How do drugs work?

2) Antagonist:

- a drug that binds to receptor without activating it, thereby blocking endogenous agonist from exerting its effect
- e.g. propranolol at β adrenoceptors. Drug must have affinity but no intrinsic activity to be antagonist.

How do drugs work?



How do drugs work?

3) Partial agonist:

a drug that binds to receptor blocking access of natural agonist and also capable of a low degree of activation (ie, affinity + some intrinsic activity) e.g pindolol and oxprenolol at β adrenoceptors

How do drugs work?

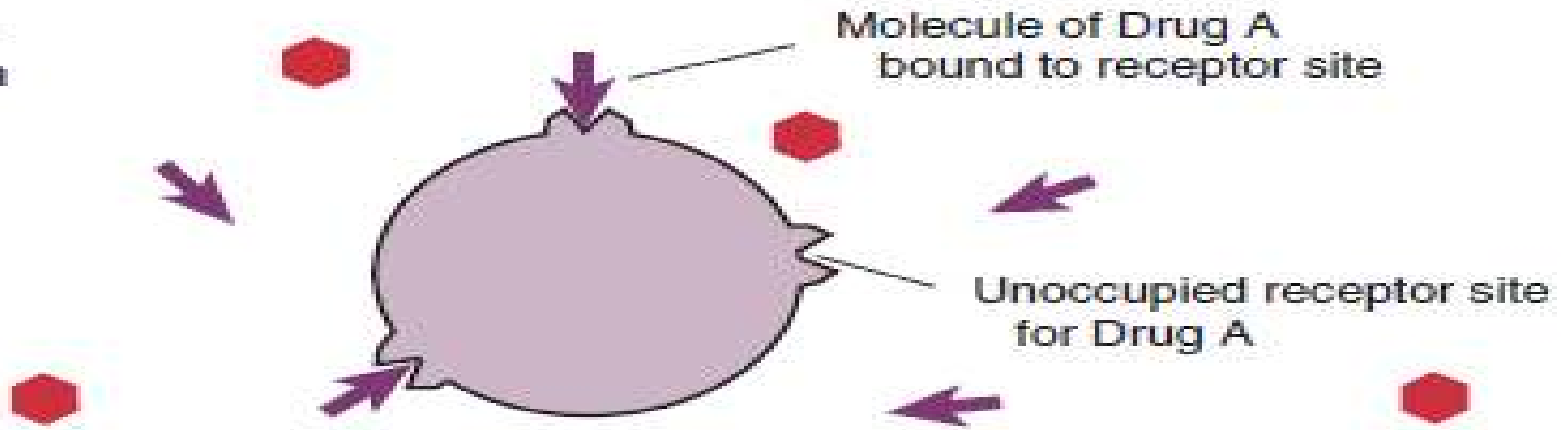
4. Inverse agonist:

is an agent that binds to the same receptor as an agonist but induces a pharmacological response opposite to that agonist, e.g. β – carboline at benzodiazepine receptors.

A. Agonist interaction with receptor site on cell

Drug A: An agonist that binds to receptor sites and produces an effect

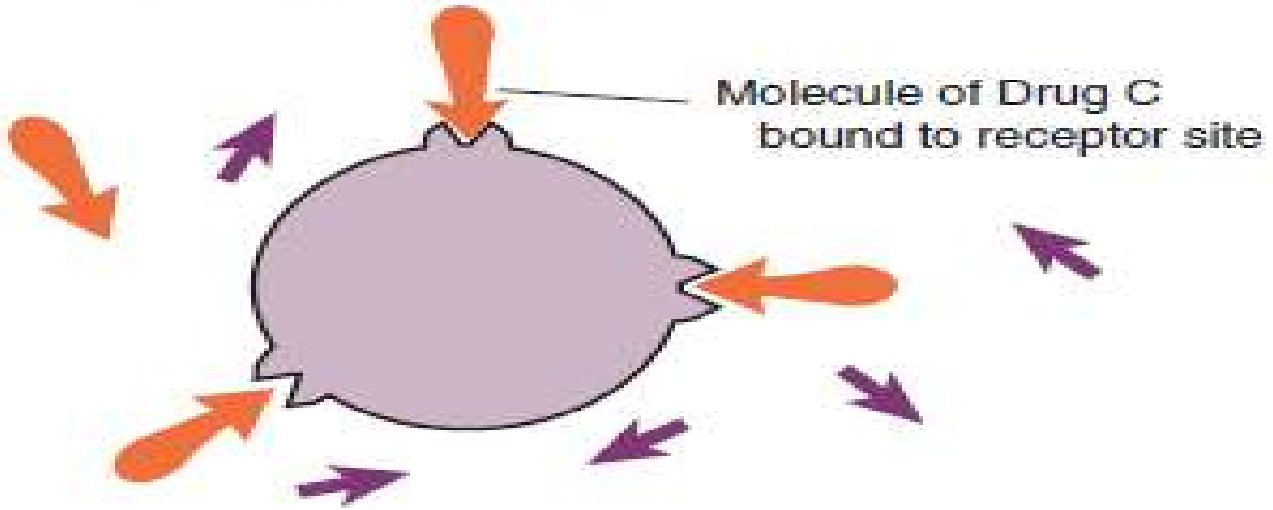
Drug B: Cannot bind to receptor sites on this cell and produces no effect



B. Competitive antagonism

Drug A: An agonist

Drug C: An antagonist of Drug A that binds to the same receptor sites as Drug A and prevents Drug A from binding



NB: If a drug acts on a variety of receptors, it's said to be *nonselective* and can cause multiple and widespread effects.

How do drugs work?

2) Antimetabolites:

- These drugs closely resemble substances which are used by the cells for nutrition and when absorbed, the cells cannot use them and so fail to multiply e.g. Sulfonamides are used to stop the multiplication of bacteria.

How do drugs work?

3) Enzyme Inhibitors:

- Enzymes are substances that speed up many chemical processes within the body.
- Some of these enzyme-activated processes are concerned with the transport of chemicals in and out of the cells. Certain drugs have the property of inhibiting their action and thus interfere with some of these processes, e.g. An angiotensin-converting-enzyme inhibitor (ACEI).

How do drugs work?

4) Action on cell membranes:

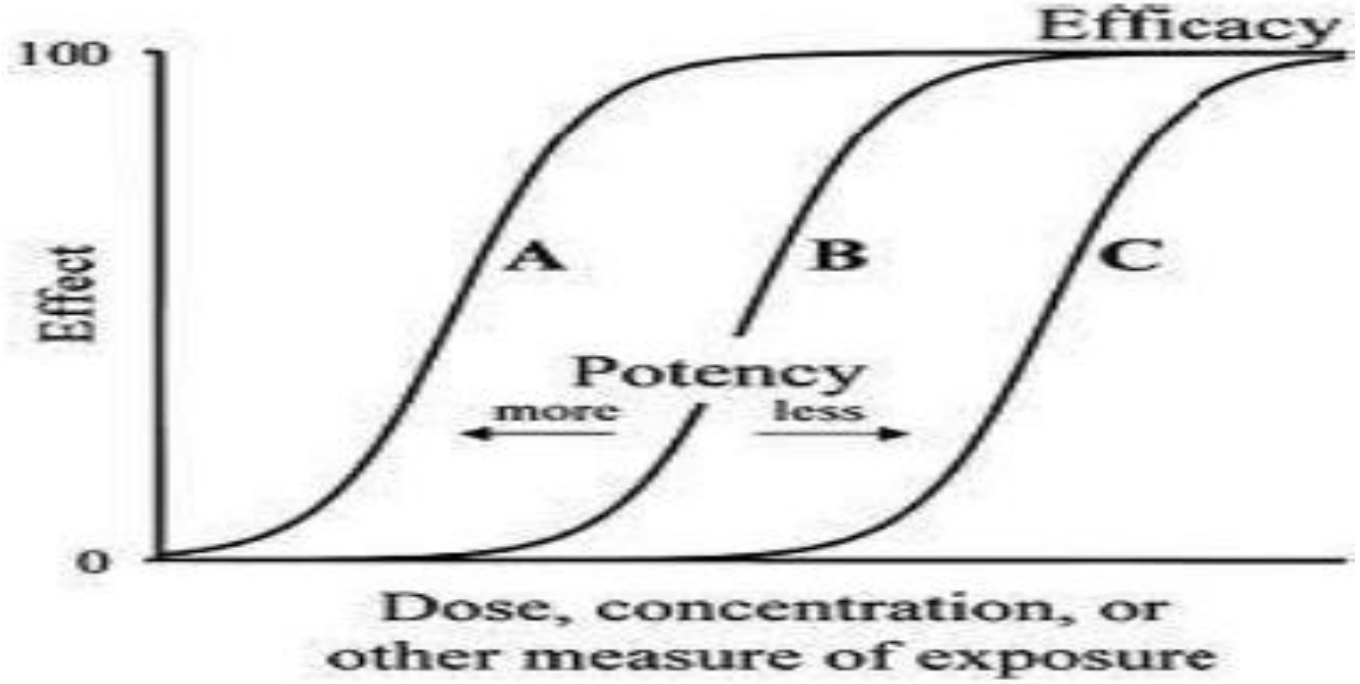
- The function of nerves and muscles depends on ions passing across the membranes surrounding these cells.
- Certain drugs interfere with movement of these ions and thus prevent nerve or muscle function as demonstrated by local anesthetics which block impulses passing up a sensory nerve.

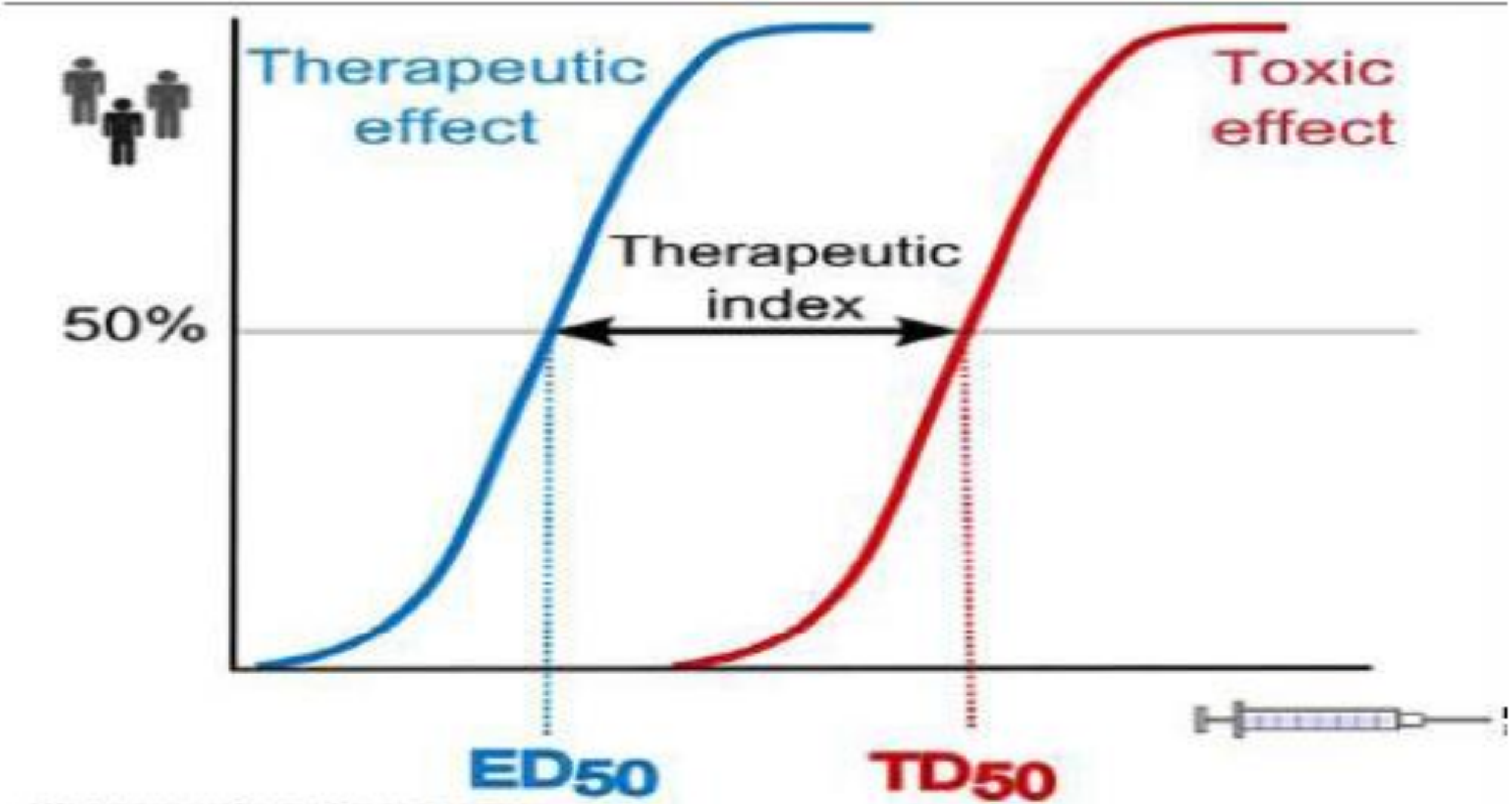
How do drugs work?

5) Cytotoxic Effect:

- Drugs may be used to kill bacteria or malignant cells without undue damage to the patient's cells. The way this is brought about varies between drugs.

Drug potency refers to the relative amount of a drug required to produce a desired response. Drug potency is also used to compare two drugs. If drug X produces the same response as drug Y but at a lower dose, then drug X is more potent than drug Y.




Therapeutic effect
Toxic effect
Therapeutic index
50%
ED50
TD50

a dose-response curve is used to graphically represent the relationship between the dose of a drug and the response it produces.

On the dose-response curve, a low dose usually corresponds with a low response. At a low dose, an increase in dose produces only a slight increase in response. With further increases in dose, there's a marked rise in drug response. After a certain point, an increase in dose yields little or no increase in response. At this point, the drug is said to have reached *maximum effectiveness.*

- As the dose of drug administered is increased, pharmacological effect will also increase and at certain dose, the resulting effect will reach a maximum level.
- Most drugs produce multiple effects. The relationship between a drug's desired therapeutic effects and its adverse effects is called the drug's *therapeutic index.* It's also referred to as its *margin of safety.*
- The therapeutic index usually measures the difference between:
    - an effective dose for 50% of the patients treated
    - the minimal dose at which adverse reactions occur.

- Pharmacotherapeutics is the use of drugs to treat disease.
- When choosing a drug to treat a particular condition, health care providers consider the drug's effectiveness as well as such factors as the type of therapy the patient will receive.

- The type of therapy prescribed depends on the severity, urgency, and prognosis of the patient's condition.

Therapy types include:

- acute therapy, if the patient is critically ill and requires acute intensive therapy
- empiric therapy, based on practical experience rather than on pure scientific data
- maintenance therapy, for patients with chronic conditions that don't resolve

Therapy types include:

- supplemental or replacement therapy, to replenish or substitute for missing substances in the body
- supportive therapy, which doesn't treat the cause of the disease but maintains other threatened body systems until the patient's condition resolves
- palliative therapy, used for end-stage or terminal diseases to make the patient as comfortable as possible.

- A patient’s overall health as well as other individual factors can alter his response to a drug.
- Coinciding medical conditions and personal lifestyle characteristics must also be considered when selecting drug therapy.

- Certain drugs have a tendency to create drug tolerance and drug dependence in patients.
- Drug tolerance occurs when a patient has a decreased response to a drug over time.
- The patient then requires larger doses to produce the same response.

Drug interactions

- Drug interactions can occur between drugs or between drugs and foods.
- They can interfere with the results of a laboratory test or produce physical or chemical incompatibilities.
- The more drugs a patient receives, the greater the chances are that a drug interaction will occur.

Potential drug interactions include:

- additive effects
- potentiation
- antagonistic effects
- decreased or increased absorption
- decreased or increased metabolism and excretion.

Types Drug Interactions:

TYPE OF INTERACTIONS	MATHEMATICAL MODEL
Addition: The response elicited by combined is EQUAL TO the combined responses of the individual drugs.	1+1=2
Synergism: The response elicited by combined drugs is GREATER than the combined of the individual drugs.	1+1=3
Potentiation: A drug which has no effect enhances the effect of a second drug.	0+1=2
Antagonism: Drug inhibits the effect of another drug. Usually, the antagonist has no inherent activity.	1+1=0

Drug Interactions:

- Addition: two drug with the same effect, when given together, produce an effect that is equal in magnitude to the sum of effects when the drugs are given individually (1+1=2) e.g. Giving two drugs together, such as two analgesics (pain relievers)

- Synergism: two drugs with same effect, when given together, produce an effect that is greater in magnitude than sum of effects when the drugs are given individually (1+1›2) e.g. Carbon tetrachloride and ethanol (ethyl alcohol) are individually toxic to the liver.

Drug Interactions:

- Potentiation: A drug lacking an effect of its own increases the effect of a second active drug (0+1› 1) e.g. promethazine would potentiate the effect of an opioid (Codeine).

- Antagonism: Drug inhibits the effect of another drug. Usually, the 1 + 1 = 0 antagonist has no inherent activity e.g. protamine sulfate is a heparin antagonist. This means that the administration of protamine sulfate completely neutralizes the effects of heparin in the body.

Drug Interactions:

- Altering absorption: Two drugs given together can change the absorption of one or both of the drugs. For example, drugs that change the acidity of the stomach can affect the ability of another drug to dissolve in the stomach. Other drugs can interact and form an insoluble compound that can't be absorbed. Sometimes, an absorption interaction can be avoided by separating drug administration by at least 2 hours.

Factors Influencing Drug Effects

a. Age

- children and older adults and recommended pediatric doses, bodies may respond very differently in all aspects of pharmacokinetics, less effective absorption, distribution, biotransformation or metabolism of drugs because of age-related liver changes, and less effective excretion owing to less efficient kidneys and by concurrent disease process such as C.V.D. (cardiovascular disease).
- decreased or adjustment for dose and monitor the patient closely for the desired effects

Factors Influencing Drug Effects

b. Weight:

- The greater the weight, the greater must be the dose.
- For very lean and very obese individuals, drug dosage is frequently determined on the basis of drug/kg of body weight or body surface area.

Factors Influencing Drug Effects

c. Sex:

- Females don’t always respond to the action of drug in the same manner as do men when giving IM injections.

- Women have more fat cells than men do, so drugs that deposit in fat may be slowly released and cause effects for a prolonged period.

Factors Influencing Drug Effects

During pregnancy, lactation, and menstruation, many drugs are stopped:

1) Aspirin: not used during menstruation
2) Drugs excreted in milk aren't given during lactation as penicillin.
3) Uterine stimulant should be avoided during pregnancy as they may produce abortion such as prostaglandin. Cytotec oral tablets contain either 100 mcg or 200 mcg of misoprostol, a synthetic prostaglandin E1 analog.
4) Drugs that might affect fetus and placenta should not be given during pregnancy.

Factors Influencing Drug Effects

d. Time of administration:

- Drug is more rapidly absorbed when the GIT is free of food, while irritating drugs are more tolerated if there is food in stomach.

- Body resistant to drug is generally greater in the early morning when the body is at its lowest point of physiologic functioning and conversely, the body is more sensitive to drugs effect during time of maximal activity.

Factors Influencing Drug Effects

e. Pathologic state:

- **The disease can change the functioning of the chemical reactions within the body and thus change the response to the drug.**

- **Pathological conditions can change the basic pharmacokinetics of a drug. For example, GI disorders can affect the absorption of many oral drugs.**

Factors Influencing Drug Effects

f. Genetic Factors

- Genetic differences can sometimes explain patients' varied responses to a given drug.

- Some people lack certain enzyme systems necessary for metabolizing a drug, whereas others have overactive enzyme systems that cause drugs to be broken down more quickly.

Factors Influencing Drug Effects

g. Immunological Factors

- People can develop an allergy to a drug. After exposure to its proteins, a person can develop antibodies to a drug. With future exposure to the same drug, that person may experience a full-blown allergic reaction.

- Sensitivity to a drug can range from mild (e.g., dermatological reactions such as a rash) to more severe (e.g., anaphylaxis, shock, and death).

Factors Influencing Drug Effects

h. Psychological Factors

- **The patient's attitude about a drug has been shown to have an effect on how that drug works.**
- **A drug is more likely to be effective if the patient thinks it will work than if the patient believes it will not work. This is called the placebo effect.**

Factors Influencing Drug Effects

i. Environmental Factors

- Some drug effects are enhanced by a quiet, cool, non-stimulating environment. For example, sedating drugs are given to help a patient relax or to decrease tension.
- Reducing external stimuli to decrease tension and stimulation help the drug be more effective.
- Other drug effects may be influenced by temperature. For example, antihypertensive that work well during cold, winter months may become too effective in warmer environments

Factors Influencing Drug Effects

j. Tolerance

- The body may develop a tolerance to some drugs over time.
- Tolerance may arise because of increased biotransformation of the drug, increased resistance to its effects, or other pharmacokinetic factors. When tolerance occurs, the drug no long causes the same reaction. Therefore, increasingly larger doses are needed to achieve a therapeutic effect. An example is morphine, an opiate used for pain relief.

Factors Influencing Drug Effects

j. Tolerance

- Clinically, this situation can be avoided by giving the drug in smaller doses or in combination with other drugs that may also relieve pain.
- Cross-tolerance—or resistance to drugs within the same class—may also occur in some situations.

Factors Influencing Drug Effects

k. Accumulation

- If a drug is taken in successive doses at intervals that are shorter than recommended, or if the body is unable to eliminate a drug properly, the drug can accumulate in the body, leading to toxic levels and adverse effects. This can be avoided by following the drug regimen precisely.

Factors Influencing Drug Effects

I. Interactions

- When two or more drugs or substances are taken together, there is a possibility that an interaction can occur, causing unanticipated effects in the body.
- Alternative therapies, such as herbal products, act as drugs in the body and can cause these same interactions.
- Certain foods can interact with drugs in much the same way. Usually this is an increase or decrease in the desired therapeutic effect of one or all of the drugs or an increase in adverse effects.

Side effects and drug toxicity

Adverse effects are undesired effects that may be unpleasant or even dangerous.

They can occur for many reasons, including the following:

1. **The drug may have other effects on the body besides the therapeutic effect.**
2. **The patient may be sensitive to the drug being given.**
3. **The drug's action on the body may cause other responses that are undesirable or unpleasant.**
4. **The patient may be taking too much or too little of the drug, leading to adverse effects.**

Adverse effects

Examples:

- **Dermatologic reactions "pruritis, urticaria, alopecia, rashes, photosensitivity.**

- **Blood dyscrasias: in certain patients, bone marrow is sensitive to certain drugs; this may result in the insufficient production of platelets, red blood cells & white blood cells.**
- **Patients who receive a drug that may cause bone-marrow depression are monitored closely by frequent blood counts & for early signs & symptoms of infection.**

Adverse effects

Some forms of blood dyscrasias:

a) **Agranulocytosis:** (reduction in the number of granulocytes).
b) **Aplastic anemia:** Pancytopenia (reduction of all formed elements of blood).
c) **Hemolytic anemia:** (reduced hemoglobin level due to lyses of RBCs).
d) **Thrombocytopenia:** (platelets deficiency)

Adverse effects

- **Hepatotoxicity:** (Liver damage), abdominal pain & jaundice.
- **Nephrotoxicity:** (Kidney damages), hematuria, anuria, proteinuria, edema.
- **Ototoxicity:** (ear damage), damage of vestibular or/and auditory portion of the 8th cranial nerve.



Adverse effects

- **CNS toxicity:** loss of judgment, loss of movement coordination, loss of consciousness, convulsions, tardive dyskinesia, reduce knee-jerk reflex, drowsiness, dizziness, …… .
- **GI disturbance:** nausea, vomiting, abdominal pain, diarrhea….. .
- **Sexual dysfunction:** change in libido.

Hypersensitivity:

It refers to allergic or other immunologic responses to drugs. It occurs when the patient shows extreme sensitivity to an effect of the drug. The response is the usual pharmacological effect; however, the effect is intense and exaggerated. A simple decrease in the dose may be sufficient to eliminate this type of reactions.

Drug Allergies:

- A drug allergy occurs when the body forms antibodies to a particular drug, causing an immune response when the person is re-exposed to the drug.
- Allergic responses to drugs occur in some patients and not in others.
- It is an unwanted response to a drug resulting from previous exposure to that drug or one closely related to it. So drug allergy is happens only after a second or subsequent exposure to the drug.

Allergic reactions to drugs differ from drug toxicity in the following ways.

1. The allergic reaction occurs in only a fraction of the population where as toxicity will occur in all individuals if the dose is high enough.
2. The allergic response is unusual in that a small amount of unsafe drug causes a severe reaction.
3. With allergy, the reaction is different from the usual pharmacological effect of the drug.

Types of allergic responses:

1. **Immediate reaction:** involving antigen and antibody resulting in the release of histamine.
   - In mild cases, the reaction is limited to urticaria (wheals and itching of the skin).
   - In severe cases (anaphylactic reaction), characterized by circulatory collapse or asphyxia due to swelling of the larynx & occlusion of the air way passages. Example: many patients are allergic to penicillin.

Types of allergic responses:

2. Delayed reaction:

- Occurring several days or even weeks after the drug has been administered. It Characterized by fever (drug fever), swelling of the joints, reaction may involves blood-forming organs and kidneys.
- Treatment of anaphylactic reaction may include administration of Epinephrine, Oxygen, Antihistamines, Corticosteroids.

Drug idiosyncrasies:

Idiosyncratic reactions are defined as those reactions that occur in patients who have abnormal genes which cause this abnormal response to that drug. Reaction may be excessive and unusual that causes by:

a) Genetic variation in drugs metabolism.

b) Immunologic mechanism, including allergic reactions

Example: Succinylcholine, a muscle relaxant drug, usually is broken down rapidly by enzymes in the plasma and liver so that the effects of the drug last for only a few minutes.

Drug toxicity:

- Excess dosage may be either accidental or intentional, results in an exaggerated response to this drug.

- It may be sever & may lead to respiratory depression. Cardio-vascular collapse and death may occur if the drug is not withdrawn and adequate treatment is started.

ROUTS OF ADMINISTRATION

Enteral	Parenteral	Others
- oral	- intravascular	-inhalation
-sublingual	-intramascular	-intranasal
-rectal	-subcutaneous	-intrathecal
		-intraventricular
		-topical
		-transdermal

Routs of Administration

❑The ORAL route: (P.O.)

Absorption Pattern:

- Variable; affected by many factors.

Advantages:

- Safest and most common, convenient, and economical route of administration.

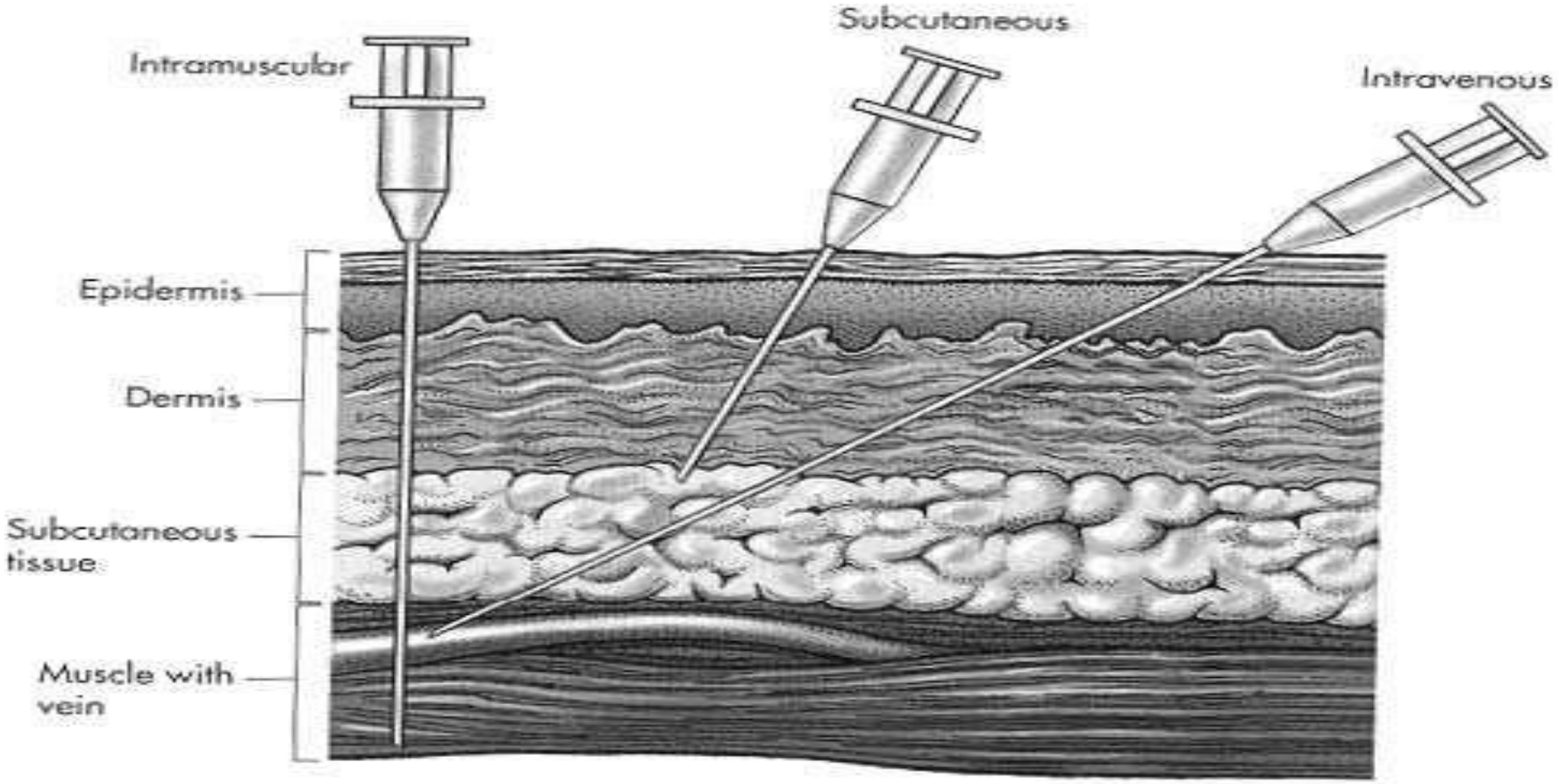
Routs of Administration

❑The ORAL route: (P.O.)

- **Inconsistent, slow absorption**
- **Food may affect absorption**
- **Patent compliance is necessary**
- **Drugs may be metabolized before systemic absorption and degraded by stomach acid and enzymes.**
- **Drug irritation may causes nausea & vomiting.**
- **Unsuitable in patients who are uncooperative, unconscious & vomiting.**

Routs of Administration

❑Parenteral Route:



Routs of Administration

- **Parenteral Route:**

Intravenous Injection : I V

Two methods of intravenous injection are used:

1) **Administering the drug as a bolus.**
2) **Continuous infusion of a drug, called IV drip**

Routs of Administration

❑ Parenteral Route:

Intravenous Injection : I V

IV solutions must be completely homogenous. No solution containing particulate matter, regardless of how fine the particles should ever be injected IV because of the danger of creating EMBLOI.

Routs of Administration

❑Parenteral Route: Intravenous

Advantages:

- Drugs that cannot be given through GIT can be given parenterally.
- Eliminate the unpredictability of absorption from GIT, the dose can be accurately titrated against response.
- Medication can be given to an unconscious patient (Valuable in emergency situations).
- Suitable for irritating substances and complex mixtures e.g. chemotherapy.
- Less painful during drug administration.

Routs of Administration

Parenteral Route: Intravenous

Disadvantages:

- Need sterile technique.
- More expensive than oral preparations.
- Patient usually not able to medicate himself
- Accidental penetration of blood vessel, especially if medication is not suitable for I.V. administration (complication).
- Unsuitable for oily substances.
- IV injection of drugs may cause local reactions

Routs of Administration

Parenteral Route:

C. Intravenous Injection : I V

Disadvantages:

- **Accidental overdose, and there is no way to withdraw the drug from circulation once it is injected.**
- **Thrombophlebitis from repeated injections.**

To minimize this phenomena (but doesn't exclude it), drug should be given by I.V. drip.



Parenteral Route: Subcutaneous Injection (SC)

- **Depends on drug diluents: Aqueous solutions: prompt**
  Depot preparations: slow and sustained

Advantages:

- **Suitable for slow-release drugs**
- **Ideal for some poorly soluble suspensions**

Routs of Administration

Parenteral Route: Subcutaneous Injection (SC)

Disadvantages:

- The most painful route.
- Slow rate of absorption.
- Not effective in emergency (poor blood supply).
- Unsuitable for drugs administered in large volumes.

❑ Parenteral Route:

Intramuscular Route: IM

Absorption Pattern:

- Depends on drug diluents: Aqueous solution: prompt Depot preparations: slow and sustained

Routs of Administration

❑ Parenteral Route:

- second most widely used route for injectable drugs.
- suitable for administration of both solutions and suspensions.
- Depending on formulation can have very long duration of action, e.g. depot antipsychotics and contraceptives (oily base).
- drug absorption is more quickly than SC route.
- Less painful (less nerve supply).
- Good absorption, especially for drugs with a low oral bioavailability
- Onset is more rapid than sc.

Routs of Administration

❑ Parenteral Route:

Disadvantages:

- Affects certain lab tests (creatine kinase)
- Can be painful
- Can cause intramuscular hemorrhage (precluded during anticoagulation therapy)

Routs of Administration

Parenteral Route:

Intradermal (ID):

- ID is a route of administration in which the drug is injected just below the epidermis.
- Injection is usually used for allergy test.
- Used for Tuberculosis (TB) tests and for penicillin sensitivity test.

Routs of Administration

Parenteral Route:

Intrathecal administration

- Intrathecal route of administration involves inserting a needle between 2 vertebrae and injecting the drug into the CSF.
- Used frequently in operating room (OR) when anesthesiologist desires to produce a spinal block and inject local anesthetic directly into the spinal cord.
- Used to administer Chemotherapy to prevent CNS metastases of some types of cancers.

Routs of Administration

Rectal, urethral and vaginal suppositories:

Absorption Pattern:

- Erratic and variable

Advantages:

- Partially bypasses first-pass effect
- Bypasses destruction by stomach acid
- Ideal if drug causes vomiting
- Ideal in patients who are vomiting, or comatose

Routs of Administration

Rectal, urethral and vaginal suppositories:

Disadvantages:

- Drugs may irritate the rectal mucosa
- Not a well-accepted route.
- Rectal route is ineffective when the patient suffers from diarrhea.

N.B.: rectal route can produce local effect or systemic effect.

Routs of Administration

Topical route:

- Are intended for use at the site of administration.
- There are exceptions e.g. Nitroglycerin ointment that is rubbed into the skin and systemic effect is achieved.

- Two topical preparations are frequently confused with each other (ointment and creams).

Routs of Administration

Topical route:

a) Ointments:

- Are usually made up of a petrolatum type of base.
- They are quite greasy, as they intended for prolonged contact with the skin.
- They don’t disappear or penetrate the skin, but remain on the surface.

Routs of Administration

Topical route:

b) Creams:

- Made of viscous water-soluble chemicals, which usually disappear after being placed to the skin.
- Patient prefers cream to ointment, as creams don't discolor clothing.

Routs of Administration: Inhalation

Absorption Pattern

- Systemic absorption occur, this is not always desirable.

Advantages:

- Absorption is rapid; can have immediate effects
- Ideal for gases
- Effective for patients with respiratory problems
- Dose can be titrated
- Localized effect to target lungs: lower doses used compared to that with oral or parental administration
- Fewer systemic side effects

Routs of Administration: Inhalation

Disadvantages:

- Most addictive route (drug can enter the brain quickly)
- Patient may have difficulty regulating dose.
- Some patients may have difficulty using inhalers.

Routs of administration: Sublingual

Absorption Pattern:

Depends on the drugs : (for example, nitroglycerin) have rapid, direct systemic absorption Most drugs erratically or incompletely absorbed

Routs of Administration: Sublingual

Advantages:

- Bypasses first-pass effect
- Bypasses destruction by stomach acid
- Drug stability maintained because the pH saliva relatively neutral
- May cause immediate pharmacological effects

Disadvantages:

- Limited to certain types of drugs
- Limited to drugs that can be taken in small doses
- May lose part of the drug dose if swallowed.

Prescription

- A medication order is written directions provided by a prescribing practitioner for a specific medication to be administered to an individual.

- Medicating a patient begins when the medication is suggested and authorized by a legal prescriber, usually a licensed physician, dentist.

Drugs orders and prescriptions

Prescription

- As an authorized provider, you must have a medication order in order to administer medications to individuals.

It consists of 4 parts:

1. The superscription:

a) It includes the patient's name, address, date and the symbol Rx.
b) The age of patient (infant or a child) should be written to permit the pharmacist to check the correctness of the dose.

2. The inscription:-

It states the name of the drug, dosage form, and the amount.

It consists of 4 parts:

3. The subscription:

It contains the directions of the pharmacist, now usually limited to the number of the doses to be dispensed.

4. The signature:

- **It is abbreviated by “S” or “Sig”**
- **It includes the prescriber’s signature.**

Elements of prescriber's order

1. Patient's name and other identifying data.
2. Date that the order was written.
3. Medication name.
4. Dosage to be administered each time.
5. Route of administration.
6. Frequency of administration and special instructions.
7. Prescriber's signature.

Example :

Name : Mohammed Ismail **Age: 38 years**

Address: Gaza, Khan younis **Date: 15/9/2015**

Rx

Amoxi 500 mg, 1 capsule every 8 hours after meal for 5 days.

Signature

mg	milligram		**mcg**	microgram
alt. hor.	every other hour		**A.M.or a.m.**	morning
mEq	millieguivalent		**ml**	milliliter
NG	Nasogastric		**NPO**	nothing by mouth
b.i.d.	two times a day		**BUN**	blood urea nitrogen
C	with		**s**	without
CHF	congestive heart failure		**os**	mouth
cm	centimeter		**Caps**	capsules
b	before		**p**	after

d.	day		dc	discontinue
p.o.	by mouth		per	by, through
dl	deciliter		P.R.	by rectum
elix.	elixir		prn	when necessary
g(gm)	gram		q	every
q.d.	every day		q2h	every 2 hours
gtt	a drop, drops		q.h.	every hour
GU	genitourinary		qhs	every night
H,hr	hour		h.s.	at bed time
qid	4 times daily		qod	every other day
® or Rt	right		Lt	left
I.V.	intravenous		IM	intramuscular
Kg	kilogram		Rx	symbol for a prescription

L	liter		**Min**	minute
Sc or SQ	subcutaneous		**Syr**	syrup
SL	sublingnal		**stat**	immediately
SOS	if necessary once only		**tab.**	tablet
Sol	solution		**t.i.d.**	3 times daily
Mcg	microgram		**U**	unit
V.O.	verbal order		**oz.**	once

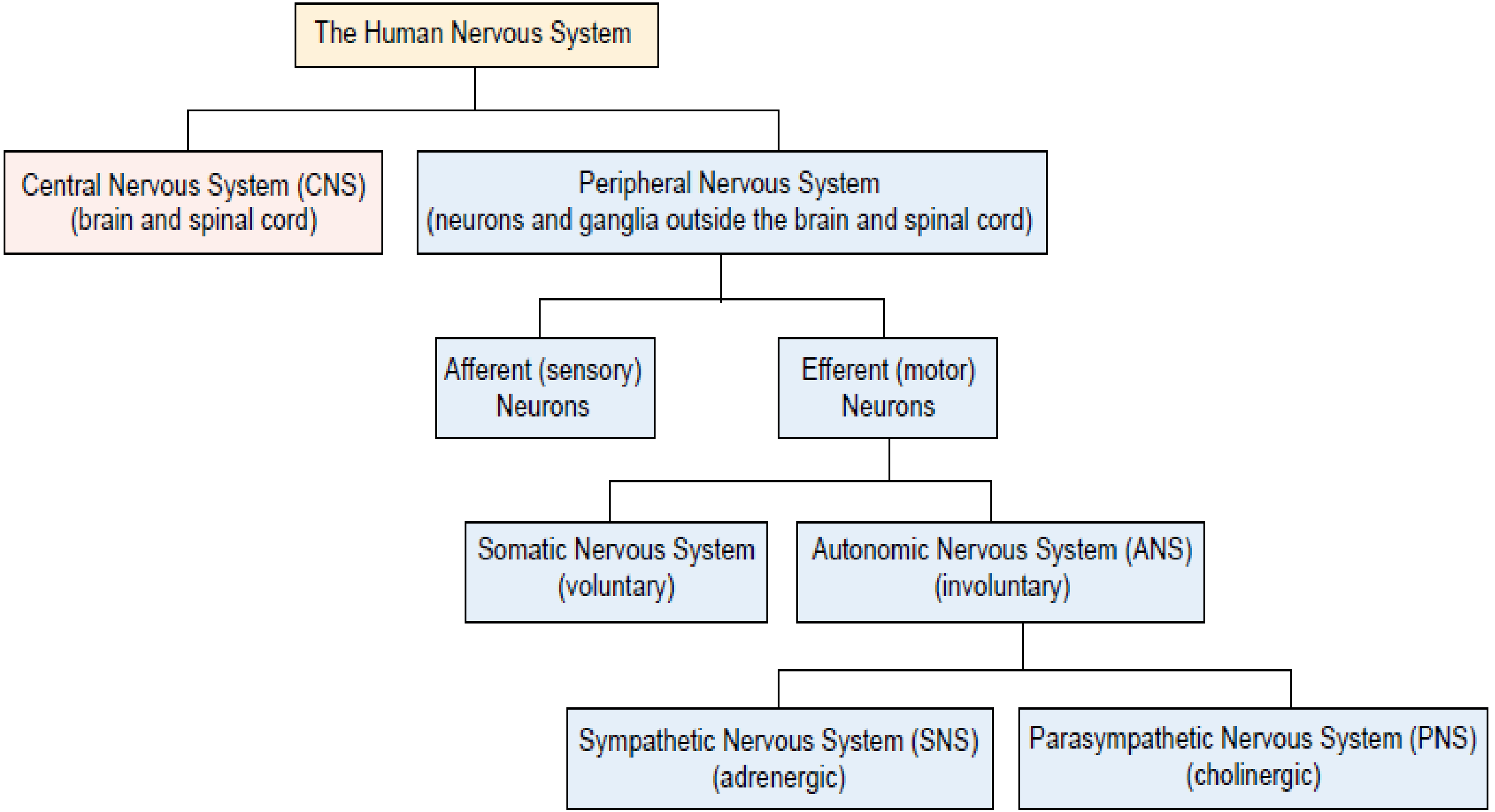
Weights & Measures

- 1 kg = 1000 gm
- 1 gm = 1000 mg
- 1 mg = 1000 Mcg
- 1 L = 1000 ML
- 1 teaspoonful = about 5 ml
- 1 tablespoonful = about 15 ml
- 1 ml = 15 gtt.
- 1 pound (lb.) = 454 gram
- 1 kg = 2.2 Libra (lb.)
- 1 gm = 15 grains

Risk category of drugs during pregnancy

Category		*Examples*
A	Adequate studies in pregnant women have failed to demonstrate a risk to the foetus	Inj. Mag. sulfate, thyroxine
B	Adequate human studies are lacking, but animal studies have failed to demonstrate a risk to the foetus or Adequate studies in pregnant women have failed to demonstrate a risk to the foetus, but animal studies have shown an adverse effect on the foetus	Penicillin V, amoxicillin, cefactor, erythromycin, paracetamol, lignocaine
C	No adequate studies in pregnant women and animal studies are lacking or have shown and adverse effect on foetus, but potential benefit may warrant use of the drug in pregnant women despite potential risk	Morphine, codeine, atropine, corticosteroids, adrenaline, thiopentone, bupivacaine
D	There is evidence of human foetal risk, but the potential benefits from use of the drug may be acceptable despite the potential risk	Aspirin, phenytoin, carbamazepine, valproate, lorazepam
X	Studies in animals or humans have demonstrated foetal abnormalities, and potential risk clearly outweighs possible benefit	Estrogens, isotretinoin, ergometrine

Nervous system: it is a system that along with endocrine system coordinates the regulation and integration of body functions.


The Human Nervous System
Central Nervous System (CNS)
(brain and spinal cord)
Peripheral Nervous System
(neurons and ganglia outside the brain and spinal cord)
Afferent (sensory)
Neurons
Efferent (motor)
Neurons
Somatic Nervous System
(voluntary)
Autonomic Nervous System (ANS)
(involuntary)
Sympathetic Nervous System (SNS)
(adrenergic)
Parasympathetic Nervous System (PNS)
(cholinergic)

The autonomic nervous system regulates the body's involuntary functions, including heart rate, respiratory rate, and digestion.

It works through a balance of its two main components, the sympathetic and parasympathetic nervous systems.

Types of drugs used to treat disorders of the autonomic nervous system include:

- cholinergic drugs
- anticholinergic drugs
- adrenergic drugs
- adrenergic blocking drugs.

Classification of drugs affecting the autonomic nervous system

1. Drugs acting on the sympathetic nervous system

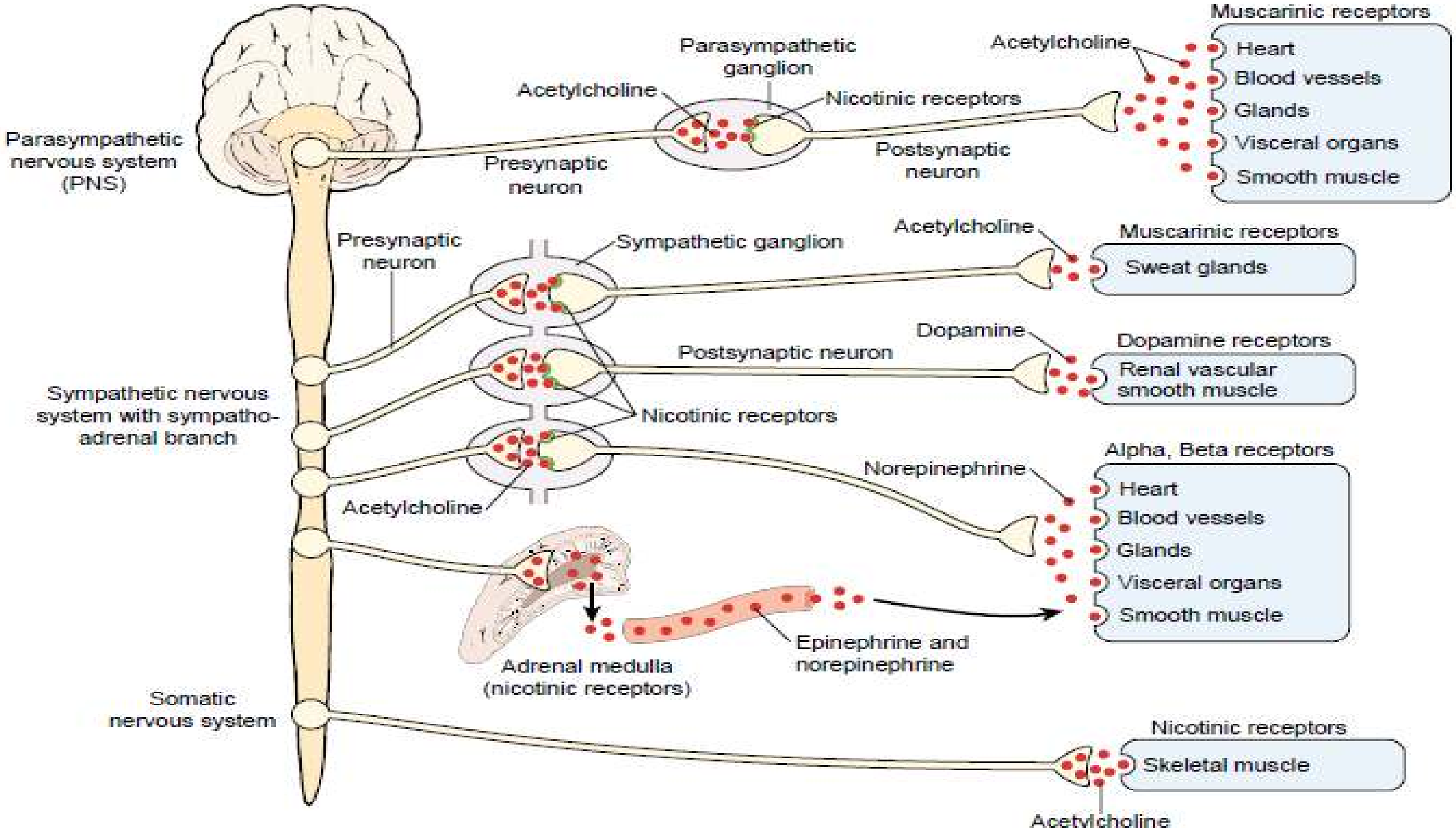
a) **Sympathomimetic or adrenergic drugs: are drugs that mimic the effects of sympathetic nerve stimulation.**

b) **Sympatholytic: are drugs that inhibit the activity of sympathetic nerve or that of sympathomimetic.**

Classification of drugs affecting the autonomic nervous system

2. Drugs acting on the parasympathetic nervous system

a. **Para-sympathomimetic or cholinergic drugs: are drugs which mimic acetylcholine or the effects of parasympathetic nerve stimulation.**
b. **Para-sympatholytic: are drugs that inhibit parasympathetic nervous system activity or that of cholinergic drugs.**


Parasympathetic nervous system (PNS)
Presynaptic neuron
Acetylcholine
Parasympathetic ganglion
Nicotinic receptors
Postsynaptic neuron
Acetylcholine
Muscarinic receptors
Heart
Blood vessels
Glands
Visceral organs
Smooth muscle
Sympathetic nervous system with sympatho-adrenal branch
Presynaptic neuron
Sympathetic ganglion
Acetylcholine
Muscarinic receptors
Sweat glands
Dopamine
Dopamine receptors
Renal vascular smooth muscle
Postsynaptic neuron
Nicotinic receptors
Acetylcholine
Norepinephrine
Alpha, Beta receptors
Heart
Blood vessels
Glands
Visceral organs
Smooth muscle
Adrenal medulla (nicotinic receptors)
Epinephrine and norepinephrine
Somatic nervous system
Nicotinic receptors
Skeletal muscle
Acetylcholine

A. Sympathomimetic (Adrenergic) Drugs

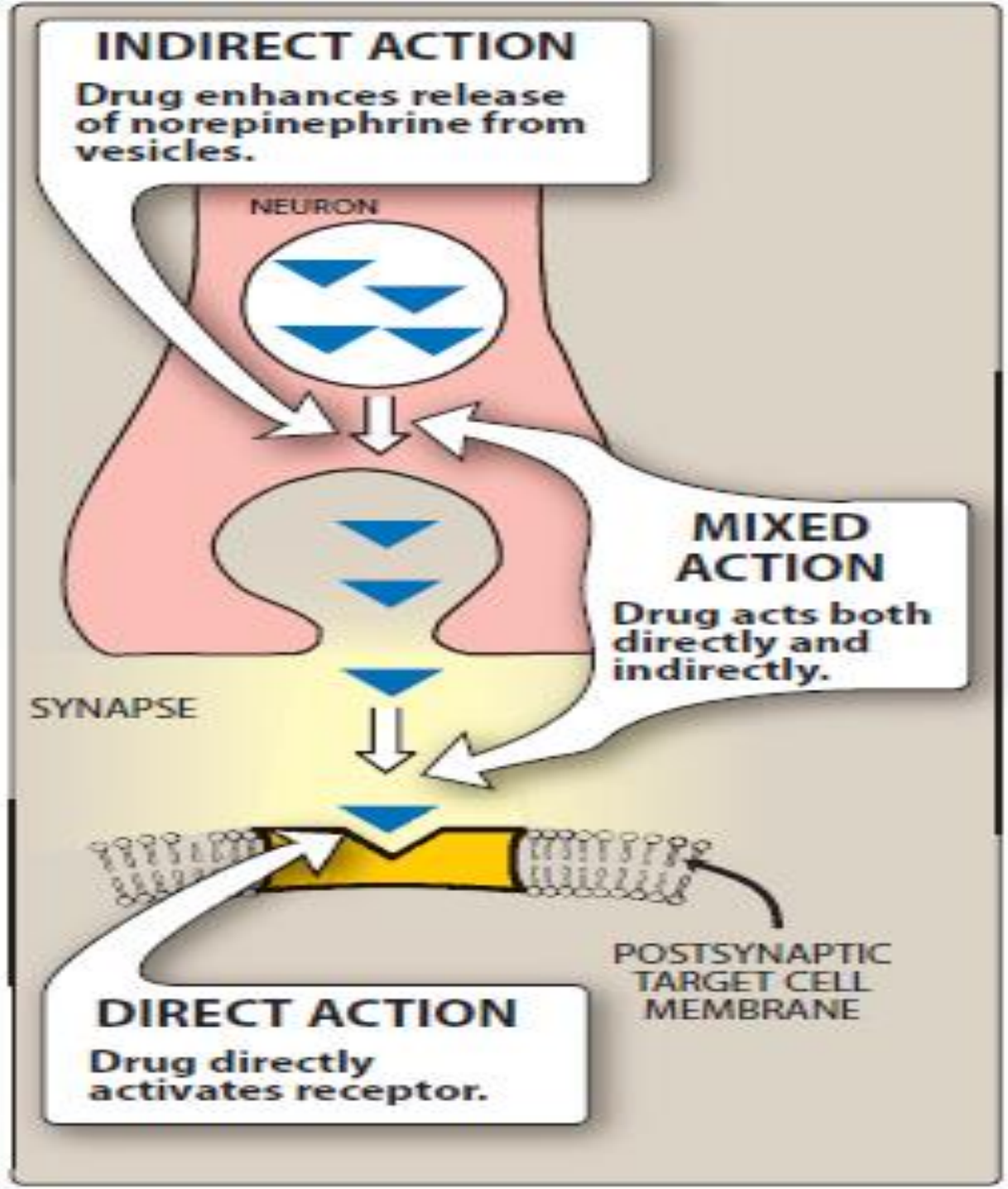
- **The adrenergic drugs supplement, mimic & reinforce the message transmitted by the natural neurohormones norepinephrine & epinephrine.**

- **These hormones are responsible for transmitting nerve impulses of the sympathetic nervous system.**

A. Actions of Sympathomimetic (Adrenergic) Drugs

A. According to mechanism of action:

1. **Direct acting agonist:** by bind to adrenergic receptors without interacting with the presynaptic neuron. mimic the action of nor epinephrine and epinephrine .
2. **Indirect acting agonist:** By regulating the release of the natural neurohormones from their storage sites at the never terminals.
3. **Mixed acting agonist:** By induce the release of norepinephrine from presynaptic terminals, and they activate adrenergic.


INDIRECT ACTION
Drug enhances release of norepinephrine from vesicles.
NEURON
MIXED ACTION
Drug acts both directly and indirectly.
SYNAPSE
POSTSYNAPTIC TARGET CELL MEMBRANE
DIRECT ACTION
Drug directly activates receptor.

B. According to receptor selectivity

These receptors are classified to receptor selectivity into:

1. **Alpha selective**
   - **Alpha 1 selective, e.g. phenylephrine**
   - **Alpha 2 selective, e.g. methylnorepinephrine.**
2. **Beta selective**
   - **Beta 1 selective, e.g. dobutamine**
   - **Beta 2 selective, e.g. terbutaline**
   - **Beta1 and 2 non selective e.g. isopreterenol**
3. **Alpha and beta non selective e.g. epinephrine, norepinephrine.**
4. **Dopamine selective e.g. dopamine.**

Sympathetic Nervous System:

The SNS is stimulated by physical or emotional stress, (fight-or-flight reaction).

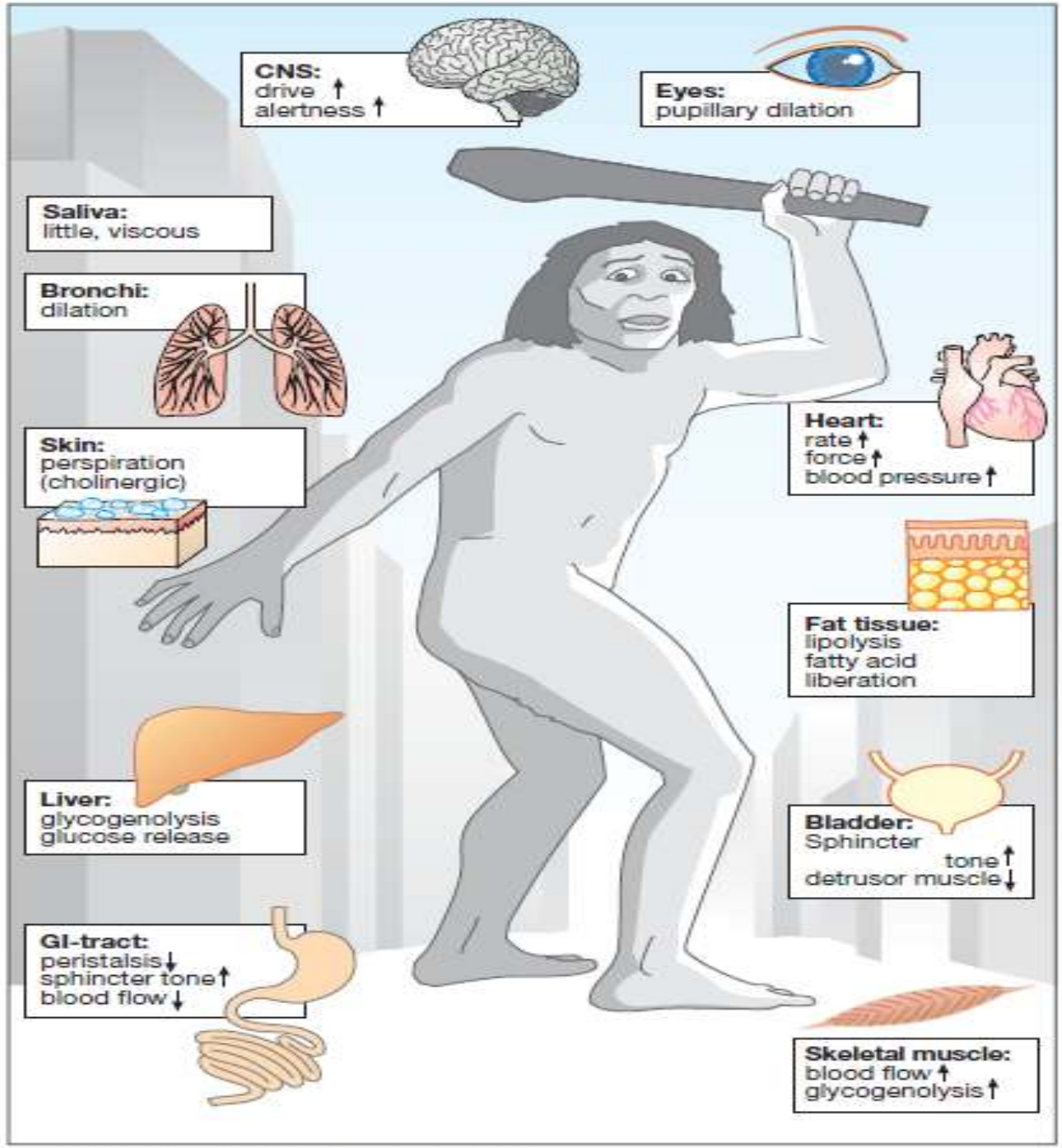
Specific body responses include:

1. **Increased arterial blood pressure and cardiac output**
2. **Increased blood flow to the brain, heart, and skeletal muscles; decreased blood flow to viscera, skin, and other organs.**
3. **Increased rate of cellular metabolism—increased oxygen consumption and carbon dioxide production**

4. Increased breakdown of muscle glycogen for energy
5. Increased blood sugar
6. Increased mental activity and ability to think clearly
7. Increased muscle strength
8. Increased rate of blood coagulation
9. Increased rate and depth of respiration
10. Pupil dilation to aid vision
11. Increased sweating.

Adrenergic Receptors

Type	Location	Effects of Stimulation
$Alpha_1$	Blood vessels	Vasoconstriction
	Kidney	Decreased renin secretion
	Intestinal smooth muscle	Relaxation
	Liver	Glycogenolysis, gluconeogenesis
	Genitourinary smooth muscle	Contraction
	Eye (radial muscle)	Blinking, mydriasis
	Pregnant uterus	Uterine contraction
	Male sexual organs	Ejaculation
$Alpha_2$	Nerve endings	Inhibit release of norepinephrine
	Vascular smooth muscle	Vasoconstriction
	Pancreatic beta cells	Inhibit insulin secretion
	Platelets	Aggregation
$Beta_1$	Heart	Increased heart rate, force of contraction, automaticity, and rate of atrioventricular conduction
	Kidney	Increased renin release
$Beta_2$	Bronchioles	Bronchodilation
	Blood vessels	Vasodilation
	Gastrointestinal tract	Decreased motility and tone
	Liver	Glycogenolysis, gluconeogenesis
	Urinary bladder	Relaxed detrusor muscle
	Pregnant uterus	Relaxation
Dopamine	Blood vessels of kidney, heart, other viscera	Vasodilation



A. Responses to sympathetic activation

Effects of adrenergic drugs:

1. Heart:

- increase Heart rate,
- increase force of contraction
- increase cardiac output .

Uses: cardiogenic shock, bradycardia, resuscitation, heart block.

2. Blood vessels:

- Systemic vasoconstriction of abdominal viscera, cerebrum & skin.

Uses: Hypotension, nasal decongestion, biliary colic, nose bleeds, migraine, headache, allergic reactions.

Effects of adrenergic drugs:

3. GIT and Genitourinary tracts:

- Decrease glandular secretions,
- constriction of sphincters, decrease muscle tone & motility of GIT & urinary bladder,
- increase muscle tone & motility of the ureters.

Uses: Enuresis, dysmenorrhea, biliary colic.

4. Lungs:

Relaxation of muscles of bronchial tree.

Uses: Bronchial asthma, emphysema, chronic bronchitis.

Effects of adrenergic drugs:

5. Eyes:

- Dilate iris,
- increase ocular pressure,
- relaxes ciliary muscle.

6. CNS:

- Excitory action,
- Respiratory stimulation,
- wakefulness.

7. Metabolism:

- increase in Glycogenolysis (sugar metabolism).
- Increase in lypolysis (release of fatty acids).

Pharmacotherapeutics

- Catecholamines that stimulate alpha-adrenergic receptors are used to treat low blood pressure (hypotension).
- blood loss (such as from hemorrhage).
- Catecholamines that stimulate beta1-adrenergic receptors are used to treat:
    - bradycardia
    - heart block
    - low cardiac output.

Pharmacotherapeutics

- beta1-adrenergic drugs are used to treat:
  - ventricular fibrillation (quivering of the ventricles resulting in no pulse)
  - asystole (no electrical activity in the heart)
  - cardiac arrest.
- Catecholamines that exert beta2-adrenergic activity are used to treat:
  - acute and chronic bronchial asthma
  - emphysema
  - bronchitis
  - acute hypersensitivity (allergic) reactions to drugs.

Adverse reactions

- restlessness
- anxiety
- dizziness
- headache
- palpitations
- cardiac arrhythmias
- hypertension and hypertensive crisis
- stroke
- angina
- increased glucose levels
- tissue necrosis and sloughing (if a catecholamine given IV leaks into the surrounding tissue).

1. Epinephrine:

Trade name: Adrenaline

Class. : Direct acting-adrenergic agent.

Action:

A natural hormone produced from adrenal medulla, induce marked stimulation of alpha, β1 + β2 receptors causing cardiac stimulation, bronchodilation & decongestion.

1. Epinephrine:

Uses:

1. Relief of respiratory distress due to bronchospasm.
2. Rapid relief of hypersensitivity reactions.
3. Cardiac arrest.
4. Open- angle glaucoma.
5. To prolong the action of anesthesia.
6. Topically to stop bleeding.

1. Epinephrine:

Contraindications:

- Narrow angle glaucoma.
- Shock
- Lactation.
- Tachycardia
- During labor (it may delay the 2nd stage of labor).

1. Epinephrine:

Side effects:

a) Cardiovascular

- Tachycardia
- Hypertension
- Chest pain
- Palpitations
- Ventricular fibrillation

c) Gastrointestinal

- Nausea/vomiting

b) Neurological

- Anxiety
- Dizziness
- Headache
- Tremors
- Seizures
- blurring of vision
- photophobia.

1. Epinephrine:

Dose:

Available in ampules of 1ml containing 1 mg adrenaline
Can be given by I.M injection., I.V. & S.C. 0.2 – 0.5 mg, IM or S.C. + Q 20 min – 4 hr as needed.

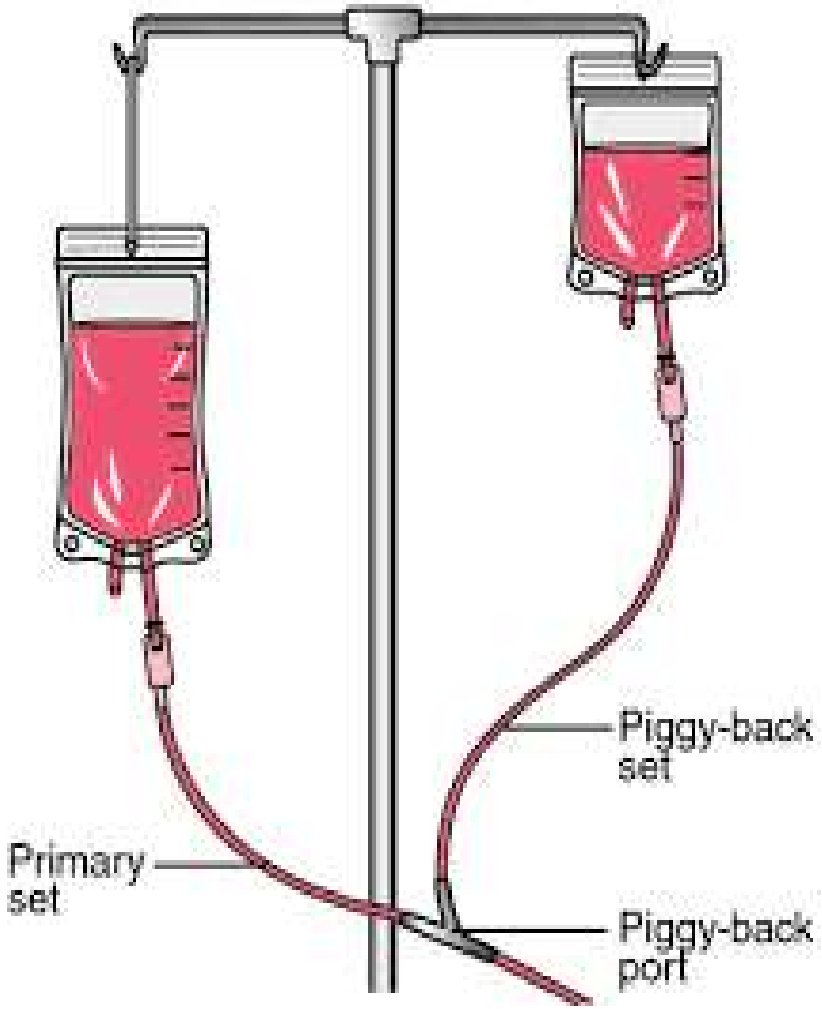
N.B. : For cardiac resuscitation 0.5 mg diluted to 10 ml with normal saline may be administered I.V. or intracrdiac to restore myocardial contractility.

1. Epinephrine:

Nursing considerations:

- **Never administer 1 : 100 solution IV., use 1 : 1000 mg sol. For I.V. use.**
- **Use tuberculine (1cc) syringe to measure adrenaline.**
- **Administer adrenaline using piggyback set to adjust the rate of infusion.**
- **Administer infusion by electronic infusion device for safety & accuracy.**

1. Epinephrine:



1. Epinephrine:

Nursing considerations:

- **Closely monitor patients receiving I.V. adrenaline infusion.**
- **Note the client for signs of shock “loss of consciousness, clammy, cold skin, cyanosis…. etc.).**
- **Briskly massage site of S.C. or I.M. injection to hasten the action of the drug.**

2. Albuterol: Salbutamol

Trade name: ventolin, ventocare, ventol, ventomin

Class.: sympathomimetic agent, bronchodilator.

Action: stimulate β 2 receptors of the bronchi

Uses:

a. Relaxes bronchial smooth muscle (Bronchial asthma).
b. Decreases airway resistance (Bronchospasm due to bronchitis or emphysema).
c. Parenteral for treatment of status asthmatics.

Side effects:

Neurological

- Tremors
- Headache/dizziness
- Sweating
- Anxiety

Gastrointestinal

- Nausea/vomiting

Cardiovascular

- Tachycardia
- Hypertension
- Dysrhythmias
- Palpitations

2. Albuterol: Salbutamol









2. Albuterol: Salbutamol

Dosage:

- **Aerosol for inhalation: 0.18 – 0.2 mg (2 inhalations) Every 4 – 8 hours.**
- **Solution for inhalation: 1.25 mg in 2 – 5 ml.**
- **Oral syrup, tablets: 2 – 6 mg tid – qid.**

2. Albuterol: Salbutamol

Nursing considerations:

- Don’t exceed the recommended dose.
- The contents of the container are under pressure, don’t store near heat or open flames.
- When given by neubilization, use facemask or mouth-piece.
- Compress O2 or air at 6 – 10 L\min for 5-15 minutes.
- Observe client for evidence of allergic response.
- NEVER give the solution prepared to be given as inhalation by the IV Route. It may cause severe tachycardia.

3. Dopamine Hydrochloride:

Trade names: Intropin, docard

Class.: Direct & indirect acting adrenergic agent.

N.B. : Available for hospital use only on 5 ml ampules containing 200 mg dopaminc hydrochloride.

3. Dopamine Hydrochloride:

Action:

- It is the immediate precursor of epinephrine in the body.
- It produces direct stimulation of β1 receptors resulting in increasing myocardial contraction, cardiac output as well as increase renal blood flow & sodium excretion.
- It exerts little effect on diastolic B.P. & induce fewer arrhythrrias that seen on other adrenergics.
- Alpha receptors, which are stimulated by higher doses of dopamine exerts vasoconstriction effects which can override the vasodilating effect.
- In higher doses it stimulates alpha receptors leading to peripheral vasoconstriction.

3. Dopamine Hydrochloride:

Indications:

- Symptomatic bradycardia persisting after prior therapies
- Cardiogenic shock with signs/symptoms of CHF.
- Hypotension due to poor cardiac output.
- Shock associated with septicemia, trauma, heart surgery, renal failure & C.H.F.
- Cardiomyopathy.
- In lower doses (1-5 Mcg/kg/min) used in renal failure.

3. Dopamine Hydrochloride:

Contraindications:

- **Hypovolemia**
- **Pheochromocytoma, uncorrected tachycardia, arrhythmias.**
- **Safety and efficacy is not established in children.**

3. Dopamine Hydrochloride:

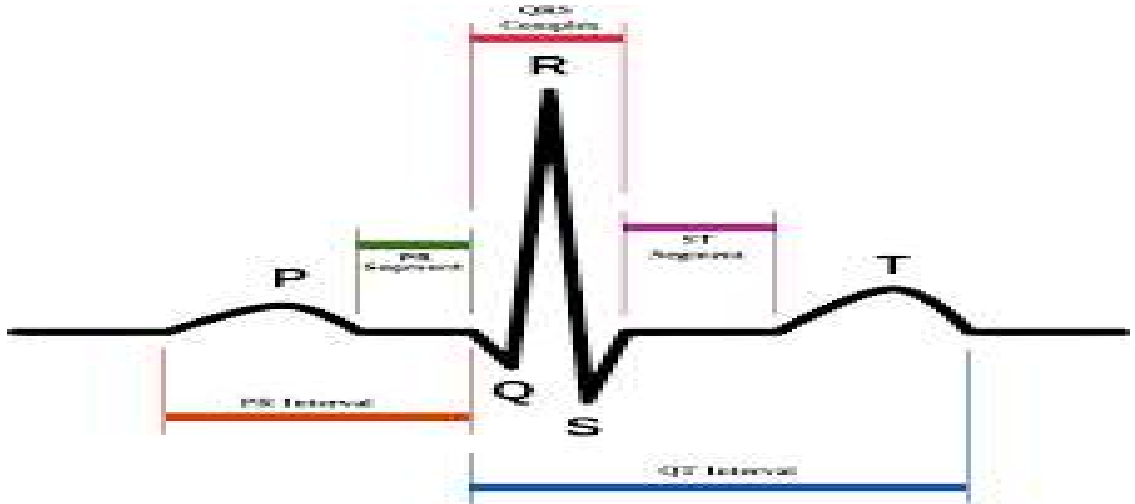
Side effects:

Cardiovascular

- Tachycardia
- Hypertension
- Increased O2 demand
- Ventricular irritability
- Chest pain (dyspnea)
- wide QRS complex.

Gastrointestinal

- Nausea/Vomiting
- Others:
- headache.



3. Dopamine Hydrochloride:

Dosage and administration:

it is a potent drug and must be diluted before administration to the patient.

Transfer contents of one ampule (5ml containing 200 mg of dopamine) by aseptic technique to either 250 or 500 ml bottle of sterile I.V. solution (saline, dextrose 5% or reinger lactate). These dilutions will yield a final concentration for administration as follows.

- **250 ml dilution contains 800 mcg\ml of dopamine.**
- **500 ml dilution contains 400 mcg\ml of dopamine.**

3. Dopamine Hydrochloride:

Dosage and administration:

1- Solution stable after dilution for 24 hours.

2- Don’t add dopamine to $NaHCO_3$ or other alkaline I.V. solutions since the drug is inactivated in alkaline solution.

3. Dopamine Hydrochloride:

Rate of administration.:

Through a suitable I.V. needle or a catheter & through an electronic infusion pump, rate is regulated according to required dose.

Dose:

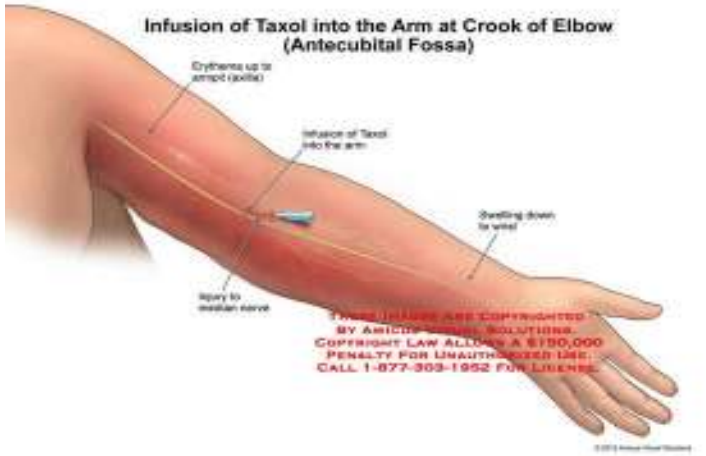
Renal dose: 1-5 Mcg\kg\minute.

Cardiac dose: start initially of 5 Mcg/kg/min then increase by increments

3. Dopamine Hydrochloride:

Nursing considerations:

- Administer only by IV INFUSION (Not IV bolus nor IM)
- Drug must be diluted before use.
- Administer drug through a central line or a big vein (vein in the anticupital fossa is preferred over those in the hand).
- Stop the drug by small increments.
- Solution is stable for 24 hrs, protect it from light.
- Monitor B.P. & ECG continuously during drug administration.
- Monitor patient for occurrence of side effects.
- Check I.V. site for signs of extravasation.
- Drug should be administered through electronic



4. Ephedrine sulfate:

Trade names: **Numacin, efedron nasal “nasal decongestant”.**

Class. : **Direct & indirect- acting adrenergic agent.**

4. Ephedrine sulfate:

Action:

Release norepinephrine from storage sites, stimulate alpha, β1 + β2 receptors, bronchodilation, nasal decongestant and increase strength of skeletal muscle so it may be used on myasthemia gravis .

4. Ephedrine sulfate:

Uses:

- Bronchial asthma
- Topically as a nasal decongestant.
- Myasthenia gravis.

4. Ephedrine sulfate:

Side effects:

Urinary retention, painful urination, dry mouth, drowsiness, blurring of vision.

Pregnancy Category C

Dose :

As a nasal decongestant 2-3 drops at each nostril every 4 hours.

N.B. : Not recommended for children under 6 years. Not to be used for 3-4 consecutive days.

Adrenergic blocking (sympatholytic) Agents

A.alpha adrenergic blocking agent

Classification:

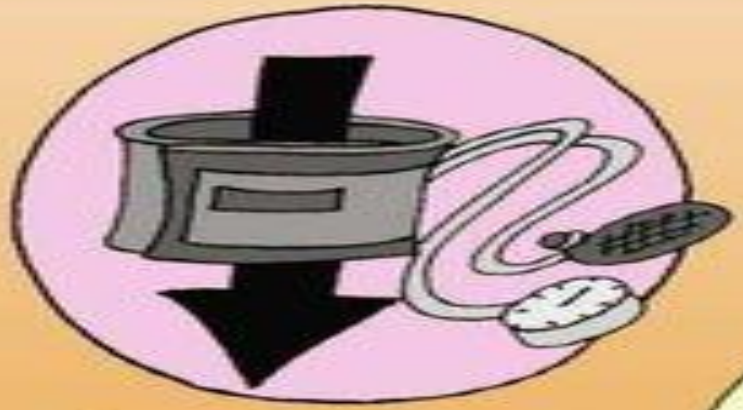
- α1-blockers or antagonists act at α1-adrenoceptors
- α2-blockers or antagonists act at α2-adrenoceptors

Uses:

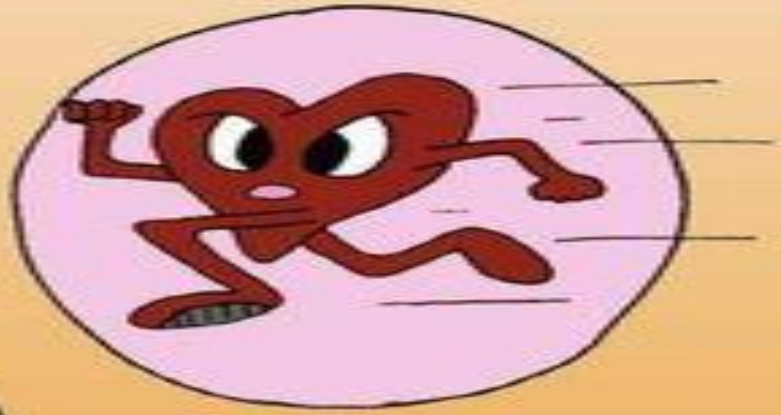
- Raynaud's disease, hypertension, and scleroderma
- Benign prostatic hyperplasia.



ALPHA-ADRENERGIC ANTAGONISTS (ALPHA-BLOCKERS) SIDE EFFECTS



Orthostatic Hypotension



Reflex Tachycardia

Examples:

doxazosin (Cardura)
prazosin (Minipress)



Dizziness



Sexual Dysfunction (inhibition of ejaculation)

Adrenergic blocking (sympatholytic) Agents

A.alpha adrenergic blocking agent

1. Phenoxybenzamine hydrochloride

Trade name: Dibenzyline

Action:

- potent, nonselective, irreversible α-adrenoceptor antagonist
- blocks catecholamine induced vasoconstriction decrease in total peripheral resistance and BP.
- Increase cardiac output due to some blockage of alpha 2 in cardiac sympathetic nerves.

Adrenergic blocking (sympatholytic) Agents

1. Phenoxybenzamine hydrochloride

Indications:

- **Hypertensive in phaeochromocytoma.**
- **Relieve vasospasm in Raynaud's phenomenon and peripheral vasospasm**
- **To relieve urinary obstruction in patient with prostatic hypertrophy.**

Adrenergic blocking (sympatholytic) Agents

1. Phenoxybenzamine hydrochloride

Adverse effects:

- **CNS: fatigue, sedation, convulsions following rapid intravenous infusion.**
- **Eye: miosis**
- **CVS: postural hypertension, tachycardia, idiosyncratic profound hypotension within few minutes of starting infusion,**
- **Respiratory tract: nasal stuffiness.**
- **GIT: nausea, vomiting (oral administration), dry mouth after intravenous infusion**
- **Reproduction: inhibition of ejaculation.**

Adrenergic Receptors

Type	Location	Effects of Stimulation
$Alpha_1$	Blood vessels	Vasoconstriction
	Kidney	Decreased renin secretion
	Intestinal smooth muscle	Relaxation
	Liver	Glycogenolysis, gluconeogenesis
	Genitourinary smooth muscle	Contraction
	Eye (radial muscle)	Blinking, mydriasis
	Pregnant uterus	Uterine contraction
	Male sexual organs	Ejaculation
$Alpha_2$	Nerve endings	Inhibit release of norepinephrine
	Vascular smooth muscle	Vasoconstriction
	Pancreatic beta cells	Inhibit insulin secretion
	Platelets	Aggregation
$Beta_1$	Heart	Increased heart rate, force of contraction, automaticity, and rate of atrioventricular conduction
	Kidney	Increased renin release
$Beta_2$	Bronchioles	Bronchodilation
	Blood vessels	Vasodilation
	Gastrointestinal tract	Decreased motility and tone
	Liver	Glycogenolysis, gluconeogenesis
	Urinary bladder	Relaxed detrusor muscle
	Pregnant uterus	Relaxation
Dopamine	Blood vessels of kidney, heart, other viscera	Vasodilation

Adrenergic blocking (sympatholytic) Agents

1. Phenoxybenzamine hydrochloride

Contra-indications:

- history of cerebrovascular accident; during recovery period after myocardial infarction (usually 3–4 weeks);
- avoid infusion in hypovolemia
- Renal impairment
- Breast-feeding may be present in milk

Adrenergic blocking (sympatholytic) Agents

1. Phenoxybenzamine hydrochloride

Dose:

Capsules, phenoxybenzamine hydrochloride 10 mg.

- **Oral dose, phaeochromocytoma, initially 10 mg daily increased by 10mg daily until hypertension controlled or treatment not tolerated; usual dose 1–2 mg/kg daily in 2 divided doses.**
- **Injection concentrate, phenoxybenzamine hydrochloride 50 mg/mL. To be diluted before use.**

Adrenergic blocking (sympatholytic) Agents

1. Phenoxybenzamine hydrochloride

Nursing considerations:

- Dose by intravenous infusion (preferably through large vein), adjunct in severe shock (but rarely used) and phaeochromocytoma,
- 1mg/kg daily over at least 2 hours; do not repeat within 24 hours.
- Caution Owing to risk of contact sensitization healthcare professionals should avoid contamination of hands.

Adrenergic blocking (sympatholytic) Agents

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

Trade name:

- **Prazosin (hypotens),**
- **Terazosin (Hytrin)**

Adrenergic blocking (sympatholytic) Agents

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

- **Prazosin, terazosin, doxazosin, and tamsulosin are selective competitive blockers of the alpha 1 receptor.**
- **The first three drugs are useful in the treatment of hypertension.**
- **Tamsulosin and alfuzosin are indicated for the treatment of benign prostatic hypertrophy (also known as benign prostatic hyperplasia or BPH).**

Adrenergic blocking (sympatholytic) Agents

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

Therapeutic uses:

- **Mild to moderate chronic hypertension (more effective when used in combination with a diuretic or propranolol.**
- **Acute congestive heart failure (prazosin).**
- **To relieve urineray obstruction (terazosin).**

Adrenergic blocking (sympatholytic) Agents

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

Adverse action:

- **CNS:** dizziness, headache.
- **CVS:** postural hypotension, first dose phenomenon (a precipitous drop in standing BP after the first dose that results in syncope and occur especially in patients who are salt and volume depleted.
- **Respiratory tract:** nasal congestion.
- **GIT:** GI hyper-motility.
- **Kidneys:** salt and fluid retention.

Adrenergic blocking (sympatholytic) Agents

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

Dose:

1. Prazosin
    a. First dose, .5 mg at bed time, orally
    b. Then 0.5 mg BD or TDS for 3-7 days.
    c. Followed by 1 mg BD or TDS, for 3-7 days.
    d. Thereafter increases gradually as required, upto max. of 20mg/day.
2. Terazosin
    a. Initially at bed time, orally.
    b. Titrate by approximately doubling dose at weekly intervals.
    c. Usual maintenance dose is 2-10mg OD.

B. Beta-adrenargic blocking agents:

Action: it combines with beta-adrenergic receptors to block the response to sympathetic nerve impulses, circulating catecholamines or adrenergic drugs.

- **β-adrenergic receptors have been classified as beta 1 (in the cardiac muscle) and beta 2 (in the bronchi and blood vessels).**
- **Blocking of β1 receptors decrease HR, myocardial contractility and cardiac output as a result it decrease blood pressure (BP).**

B. Beta-adrenargic blocking agents:

- **Blocking of β2 receptors lead to increase airway resistance (bronchospasm) , and vasoconstriction.**
- **These drugs could be selective (working on one receptor such as β1 selective drugs (Atenolol) or it could be nonselective (such as Propranolol).**

B. Beta-adrenargic blocking agents:

Uses:

1. hypertension,
2. Angina pectoris.
3. Cardiac arrhythmias.
4. Myocardial infarction.
5. Prophylaxis of migraine.

B. Beta-adrenargic blocking agents:

Contraindications:

- **Bradycardia, C.H.F., cardiogenic shock, diabetes, thyrotoxicosis, chronic bronchitis, asthma, bronchospasm, emphysema.**

Side effects:

- **Bradycardia, C.H.F., hypotension, cold extremities (due to peripheral vasoconstriction), edema, dyspnia, shortness of breath, nausea, vomiting, hepatomegaly and bronchospasm.**

B. Beta-adrenargic blocking agents:

Treating overdose:

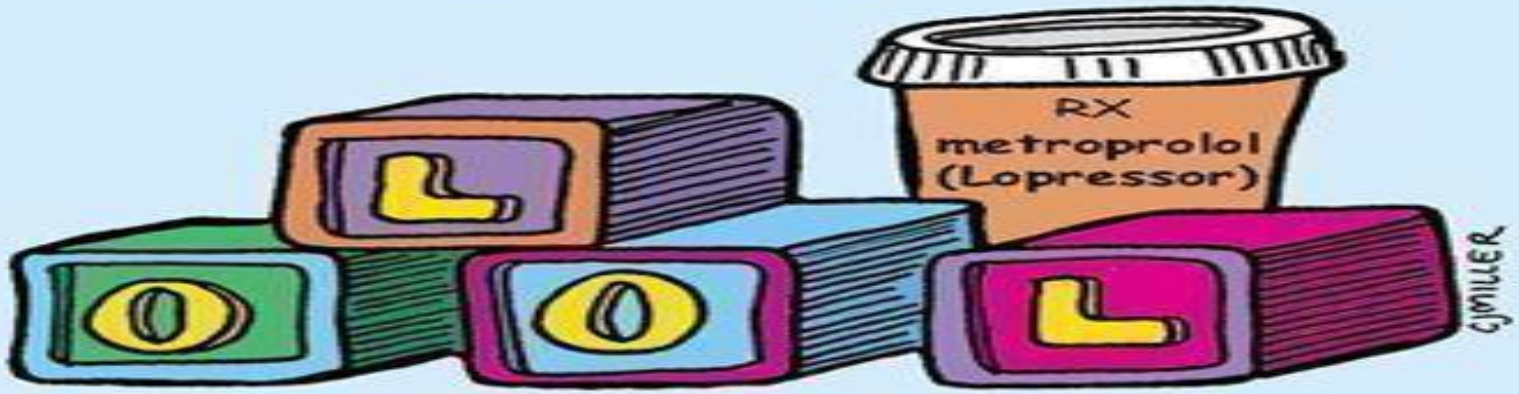
1. Inducing vomiting, gastric lavage.
2. Artificial respiration.
3. Give atropine sulfate 0.6 mg (up to 3 mg) and glycogan for the treatment of bradycardia.
4. Treat hypoglycemia and hypokalemia.
5. I.V fluids.
6. Adrenaline or dopamine to increase Blood pressure.

B. Beta-adrenargic blocking agents:

Nursing considerations:

1. Instruct patient/family to take blood pressure and pulse.
2. Provide written instructions as when to call physician (e.g. HR below 50 beat/min).
3. Consult the physician before interrupting the therapy.
4. Some drugs lead to blurring of vision, so that tell patients not to engage in activities need mental alertness.
5. Instruct patient to dress warmly during cold weather.
6. Diabetic patient should be very careful about symptoms of hypoglycemia.
7. Report any asthma-like symptoms.

BETA-BLOCKERS



(Ends in OLOL)



Action: Blocks beta-receptors in the heart causing:

- ↓ Heart rate
- ↓ Force of contraction
- ↓ Rate of atrioventricular (AV) conduction

Side Effects:

- Bradycardia
- Lethargy
- Gastrointestinal (GI) disturbance
- Congestive heart failure (CHF)
- ↓ Blood pressure (BP)
- Depression

B. Beta-adrenargic blocking agents:

1. Atenolol:

Trade name: Normatin.

Classification:

- **Beta1-selective adrenergic blocking agent, Antihypertensive**



B. Beta-adrenargic blocking agents:

1. Atenolol:

Uses:

- Treatment of angina pectoris due to coronary atherosclerosis
- Hypertension, as a step 1 agent, alone or with other drugs, especially diuretics
- Treatment of MI
- Unlabeled uses: Prevention of migraine headaches; alcohol withdrawal syndrome, treatment of ventricular and supraventricular arrhythmias

B. Beta-adrenargic blocking agents:

1. Atenolol:

Adverse effects:

- **Allergic reactions: Pharyngitis, respiratory distress**
- **CNS: Dizziness, vertigo, fatigue, emotional depression, paresthesias, sleep disturbances, hallucinations, disorientation, slurred speech.**
- **Dermatologic: Rash, pruritus, sweating, dry skin**
- **CV: Bradycardia, CHF, cardiac arrhythmias, sinoatrial or AV nodal block, tachycardia, claudication, pulmonary edema.**
- **Eyes : Eye irritation, dry eyes, conjunctivitis, blurred vision**
- **GI: Gastric pain, constipation, diarrhea, nausea, vomiting, anorexia.**
- **GU: Impotence, decreased libido, frequent urination.**
- **Musculoskeletal: Joint pain, arthralgia, muscle cramps**
- **Respiratory: Bronchospasm, dyspnea, cough, bronchial obstruction, nasal stuffiness, rhinitis, (less likely than with propranolol).**

B. Beta-adrenargic blocking agents:

Contraindications

Contraindicated with sinus bradycardia, second- or third-degree heart block, cardiogenic shock, CHF, pregnancy.

B. Beta-adrenargic blocking agents:

Dosage:

Tablets: 50 mg or 100 mg daily

- **Initial dose is 50 mg, if there is no response, increase dose to 100 mg daily.**
- **Angina pectoris: Initially, 50 mg PO daily. If optimal response is not achieved in 1 wk, increase to 100 mg daily; up to 200 mg/day may be needed.**

B. Beta-adrenargic blocking agents:

Dosage:

- **Acute MI: Initially, 5 mg IV given over 5 min as soon as possible after diagnosis; follow with IV injection of 5 mg 10 min later. Switch to 50 mg PO 10 min after the last IV dose; follow with 50 mg PO 12 hr later. Thereafter, administer 100 mg PO daily or 50 mg PO bid for 6–9 days or until discharge from the hospital.**

B. Beta-adrenargic blocking agents:

Nursing considerations:

- For IV use, the drug may be diluted in sodium chloride, dextrose, on dextrose saline.
- Do not discontinue drug abruptly after long-term therapy (hypersensitivity to catecholamines may have developed, causing exacerbation of angina, MI, and ventricular arrhythmias). Taper drug gradually over 2 wk with monitoring.
- Take drug with meals if GI upset occurs.
- Do not stop taking this drug unless told to do so by a health care provider.
- Avoid driving or dangerous activities if dizziness or weakness occur.
- You may experience these side effects: Dizziness, light-headedness, loss of appetite, nightmares, depression, sexual impotence.



B. Beta-adrenargic blocking agents:

2. Valolol:

Trade name: Corgard

Class: Beta-adrenergic blocking agent.

Action: manifests both beta 1 and beta 2 adrenergic blocking.

Uses: hypertension, angina pectoris.

Dose: 40-80 mg daily.

B. Beta-adrenargic blocking agents:

3. Propranolol hydrochloride

- **Trade name:** Inderal, Deralin

- **Classification:** beta-adrenergic blacking agent, antiarrhythmic.

- **Action:** manifests both beta1 and beta 2 adrenergic blocking activity.

B. Beta-adrenargic blocking agents:

Indication:

1. Angina pectoris.
2. Cardiac arrhytmias.
3. Prophylaxis of MI.
4. Hypertension.
5. Prophylaxis of migraine.
6. Pheochromocytoma

B. Beta-adrenargic blocking agents:

3. Propranolol hydrochloride

side effects: psoriasis-like eruptions.

Dosage:

Tablets: initial dose of 10, 40 mg bid, then 120-240 mg in 2-3 divided doses.

B. Beta-adrenargic blocking agents:

Propranolol hydrochloride

Contraindications:

- Contraindicated with allergy to beta-blocking agents, sinus bradycardia, second- or third-degree heart block, cardiogenic shock, CHF, bronchial asthma, bronchospasm, COPD, pregnancy (neonatal bradycardia, hypoglycemia, and apnea, and low birth weight with long-term use during pregnancy), lactation.

B. Beta-adrenargic blocking agents:

Dosage:

Tablets:

For hypertension: initial dose of 40 mg bid, then 120-240 mg in 2-3 divided doses.

B. Beta-adrenargic blocking agents:

4. Temolol maleate:

- **Trade name:** Timolin, Blocadren, Timpotic.
- **Classification:** Ophthalmic agent, beta-adrenergic agent.
- **Action:** both beta1 and beta 2 receptors blocking activity.
- **Uses:**
    - Tablets: for hypertension.
    - Ophthalmic solution: chronic open angle glaucoma



Beta Blocker Indications

MNEMONICS!

Beta blockers by indication include:

Cardiac arrhythmia
Esmolol
Sotalol
Acebutolol

Migraine
Propanolol
Timolol

Heart failure
Metoprolol
Bisoprolol
Carvedilol

Hypertension
Bisoprolol
Metoprolol
Nebivolol
Atenolol
Labetalol
Acetbutolol

Glaucoma (TBC)
Timolol
Betaxolol
Carteolol

Heart attack (CAB)
Carvedilol
Atenolol
Bisoprolol


Thank
You

Parasympathomimetic (cholinergic) Drugs

- The parasympathetic division maintains essential bodily functions, such as digestive processes and elimination of wastes, and is required for life.
- It usually acts to oppose or balance the actions of the sympathetic division and is generally dominant over the sympathetic system in rest and digest.

Parasympathomimetic (cholinergic) Drugs

The receptors of parasympathetic nervous system are classified into

a) muscarinic type

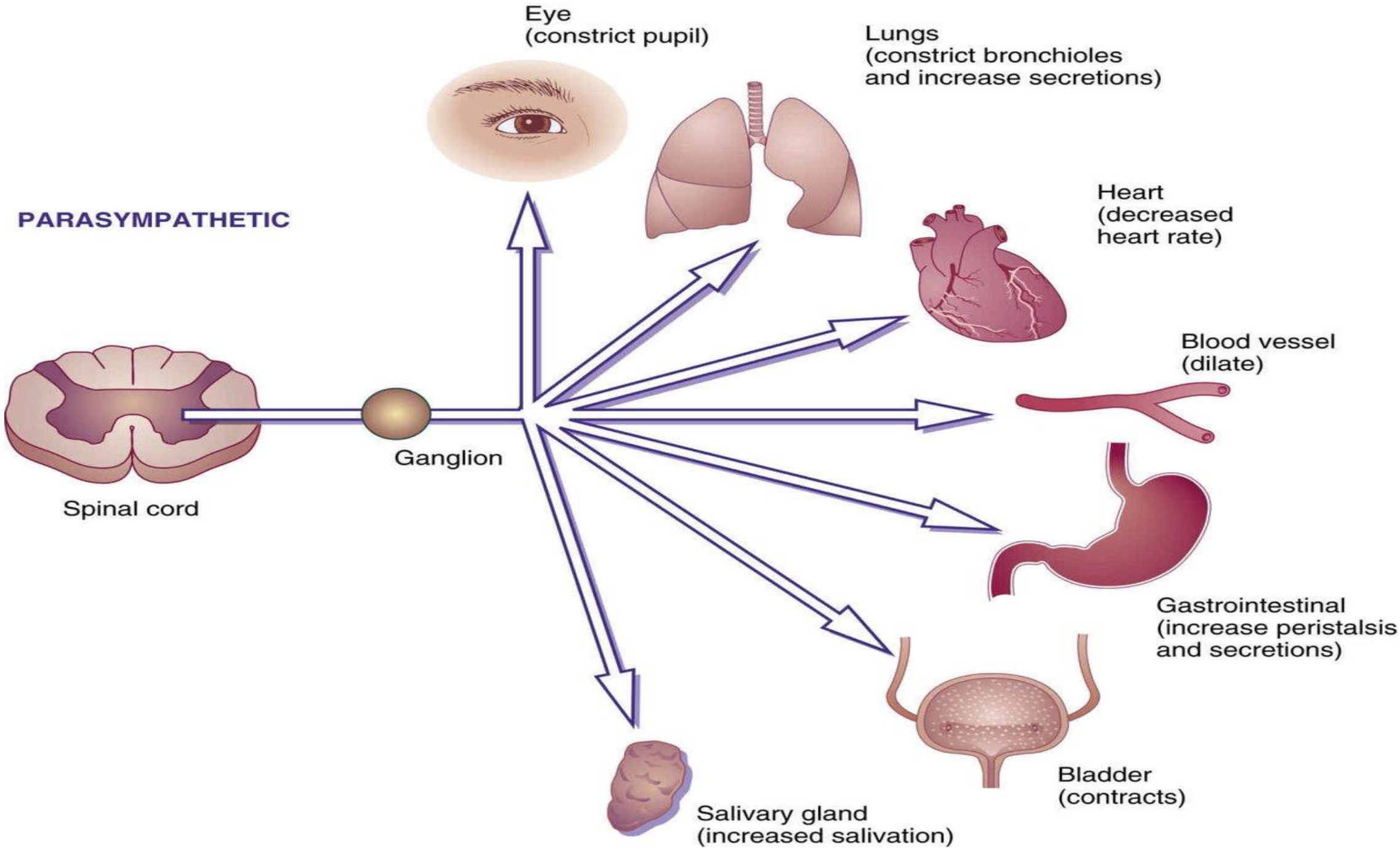
b) nicotinic type.

Parasympathomimetic (cholinergic) Drugs

Cholinergic drugs can be divided into 2 classes:

1- Directly acting drugs that mimic the action of acetylcholine e.g. muscarine, pilocarpine, carbachol.

2- Indirectly acting drugs that increase the concentration of acetylcholine, usually by inhibiting acetylcholinestrase, e.g neostigmine, physostigmine,.


Eye
(constrict pupil)
Lungs
(constrict bronchioles
and increase secretions)
PARASYMPATHETIC
Heart
(decreased
heart rate)
Blood vessel
(dilate)
Ganglion
Spinal cord
Gastrointestinal
(increase peristalsis
and secretions)
Bladder
(contracts)
Salivary gland
(increased salivation)

Physiological Responses to Cholinergic Stimulation:

Eye	**contraction - miosis** **contraction - near vision**
heart	decrease in HR decrease in contractility decrease in conduction ± block ± decrease in contractility
Blood vessels	Vasodilation
lung	constriction increased secretion
GIT	increased motility & tone sphincter relaxation increased secretion

Physiological Responses to Cholinergic Stimulation:

Bladder	**Detrusor contraction** **Trigone & internal sphincter relaxation**
Pancreas **Salivary** **Lacrimal** **Pharyngeal**	increased secretion
Sweat Glands	increased secretion
Sexual organs	erection (male)

Cholinergic agonists are used to:

- Treat atonic (weak) bladder conditions and postoperative and postpartum urine retention
- Treat GI disorders, such as postoperative abdominal distention and GI atony
- Reduce eye pressure in patients with glaucoma and during eye surgery
- Treat salivary gland hypofunction caused by radiation therapy

Adverse reactions

- Nausea and vomiting
- Cramps and diarrhea
- Blurred vision
- Decreased heart rate and low blood pressure
- Shortness of breath
- Urinary frequency
- Increased salivation and sweating.

The “eyes” (and the mouth, and the skin) have it!

Cholinergic agonists are usually administered:

- topically, with eye drops
- orally
- by subcutaneous (sub cut) injection.

Sub cut injections begin to work more rapidly than oral doses.

The cholinergic agonists rarely are administered by IM or IV injection because they're almost immediately broken down by cholinesterase in the interstitial spaces between tissues and inside the blood vessels.

Moreover, they begin to work rapidly and can cause a *cholinergic crisis* (a drug overdose resulting in extreme muscle weakness and possible paralysis of the muscles used in respiration).

1. Pyridostigmine Bromide:



Trade name: Mestinon.

Class. : Indirectly acting, cholinergic – acetylcholinestrase inhibitor.

Action: By inhibiting the enzyme cholinestrase, that lead to increase the concentration of acetylcholine at the myoneural junction which facilitate the transmission of nerve impulse across the myoneural junction, that lead to increase muscle strength in myasthenia gravis.

1. Pyridostigmine Bromide:

- ❖ It has a slower onset, longer duration of action & fewer side effects than neostigmine.

- ❖ Atropine may be given to control side effects.

1. Pyridostigmine Bromide:

- **Uses:** Myasthenia Gravis.

- **Contraindication:**

- Hypersensitivity
- Hypotension
- Hyperthyroidism
- peptic ulcer.
- Bradycardia
- Asthma
- GI obstruction

1. **Pyridostigmine Bromide:**

- **Side effects:**

- **Skin rash, thrombophlebitis after I.V. use, Nausea, vomiting, diarrhea.**
- **Bradycarda, Hypotension, headache, dizziness. Urinary frequency (Incontinence) , bronchospasm.**

- **Dose: Tab. 600 mg daily.**

2. Neostigmine Bromide:

- **Trade name: Prostigmin.**
- **Dose: Tab. 150 mg daily.**

- **All information as pyridostigmine**

3. Ophthalmic cholinergic (Miotic) Agents:

- Commonly used for the treatment of glaucoma.

Action:

- These agents inhibit the enzyme cholinestrase which lead to accumulation of acetylcholine & stimulate the ciliary muscles & increases contraction of the iris sphincter muscle.

- This opens the angle of the eye & results in increased the outflow of aqueous humor & consequently in decreased of intraocular pressure .

3. Ophthalmic cholinergic (Miotic) Agents:

Uses:

- **Glaucoma.**
- **Diagnosis & treatment of esotropia.**
- **Antidote against harmful effects of atropine-like drugs in patients suffering from glaucoma.**

3. Ophthalmic cholinergic (Miotic) Agents:

Contraindications:

inflammatory eye diseases (critis), Asthma, History of retinal detachment, Bradycardia, peptic ulcer.

3. Ophthalmic cholinergic (Miotic) Agents:

Side effects:

- **Pain in eye, blurring of vision, Headache.**
- **Failure to accommodate to darkness, retinal detachment.**
- **Diarrhea, hypotension, salivation, bronchial constriction**

3. Ophthalmic cholinergic (Miotic) Agents:

Nursing considerations:

a. Have adrenaline & atropine available for emergency treatment.
b. Stress the importance of taking eye drops exactly as prescribed
c. Minimize side effects by taking medication at bed-time.
d. Advice client not to drive a car for 1-2 hrs after administering eye drops.
e. Notify the physician, if side effects occurred.
f. Cold compresses for painful eye spasms.
g. Frequent eye examinations.

3. Ophthalmic cholinergic (Miotic) Agents:

1. Carbachol:

Trade name: Isopto carbachol.

Uses: glaucoma.

Dose: 1 gtt of the solution in the conjunctiva 1-2 times daily.

3. Ophthalmic cholinergic (Miotic) Agents:

2) Physostigmine salicylate:

Trade name: Eserine salicylate.

Uses: glaucoma.

Dose: 1 gtt of the 0.25 or 0.5 % solution bid-tid.

Ophthalmic cholinergic (Miotic) Agents:

3. Pilocarpine nitrate:

Uses: glaucoma.

Dose: 1 gtt of 1-4 % solution tid.

Cholinergic Blocking (Parasympatholytic) Drugs*

These agents prevent the neurotransmitter acetylcholine from combining with receptors on the muscarinic site & nicotonic site..

Cholinergic Blocking (Parasympatholytic) Drugs*

The main effects:

1. Reduce spasm of smooth muscle such as spasm of the urinary bladder or intestines.
2. To block vagal impulses to the heart which will increase heart rate & conductivity.
3. To suppress or decrease gastric secretions, perspiration, salivation and secretion of bronchial mucus.
4. To relax the sphincter muscles of the iris & cause pupillary dilation (mydriasis) & loss of accommodation for near vision.
5. Act on CNS producing such reactions as depression (scoplamine) or stimulation (toxic dose of atropine) to produce antiparkinsonism effect.

Cholinergic Blocking (Parasympatholytic) Drugs*

Contraindications:

- **Glaucoma, tachycardia, myocardial ischemia**

- **Prostate hypertrophy, myasthenia gravis, paralytic ileus, Mental impairment, lactation, hepatic disease.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Side effects:

Nausea, vomiting, dry mouth, constipation, heartburn, dizziness, drowsiness, headache, insomnia, blurring of vision, photophobia, euphoria, hallucination, flushing of the skin.

Cholinergic Blocking (Parasympatholytic) Drugs*

1. Atropine sulfate:

Class.: Cholinergic blocking agent.

Action:

It is a parasympatholytic agent which cause relaxation of smooth muscles & inhibition of secretary glands.

Cholinergic Blocking (Parasympatholytic) Drugs*

Uses:

- Adjunct in peptic ulcer treatment.
- Irritable bowel syndrome.
- Treatment of spastic disorders of biliary tract .
- During anesthesia to control salivation & bronchial secretions.
- Parkinsonism.
- Anti-arrhythmic (prophylaxis).
- Prophylaxis and treatment of toxicity due to cholinestrase inhibitor including organophosphate pesticides.
- Ophthalmologic treatment of uveitis.

Cholinergic Blocking (Parasympatholytic) Drugs*

- **Contraindications: See parasympatholytics**
- **Side effects: See parasympatholytics**
- **Dose:**
  - **Tablets: 0.3 – 1.2 mg Q 4-6 hr**
  - **Available in 1ml-ampoule containing 1 mg atropine.” IM, I.V. & S.C. 0.4 – 0.6 mg Q 4-6 hour for anticholinergic action**

Cholinergic Blocking (Parasympatholytic) Drugs*

- **For treatment of toxicity from cholinestrase inhibitors**
- **“ organophosphorus poisoning”, give 2-4 mg IV initially then 2 mg every 5-10 minutes until muscarinic symptoms disappear and signs of atropine toxicity begins to appear like dilation of pupils, flushing of face & tachycardia.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Nursing considerations:

- Check dosage & measure the drug exactly.
- Assess for history of asthma, glaucoma, ulcer .. etc.
- Determine the age of the client.
- Frequent mouth care.
- Assess client for change in pulse rate.
- In case of blurring of vision, assist on ambulating & give safety measures.

Cholinergic Blocking (Parasympatholytic) Drugs*

2) Scopalamine Hydrobromide :

Trade name: Hyoscine.

Class.: cholinergic blocking agent.

Action :

It is a parasympatholytic agent, depress the cerebral cortex, especially the motor area, act as a powerful hypnotic.

Cholinergic Blocking (Parasympatholytic) Drugs*

Uses:

- Motion sickness (prevention and control of nausea and vomiting).
- Preanesthetic.
- Antiarrhythemic.
- Mydriatic and cycloplegic.
- Adjunctive with other drugs to treat GIT ulcers.
- With other narcotics to treat biliary colic.

Cholinergic Blocking (Parasympatholytic) Drugs*

Contraindications:

- **Hypersensitivity.**
- **Glaucoma**
- **Cardiac arrhythmias**
- **Pregnancy**
- **Lactation.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Dose:

- **Oral: O.25 mg I hour before travel (for motion sickness).**
- **Parenteral: 0.32-0.6 mg SC or IM**

Cholinergic Blocking (Parasympatholytic) Drugs*

Side effects:

Pupil dilation, photophobia, blurred vesion, headache, drowziness. Dry mouth, constipation, nausea, vomiting. tachycardia, arrhythmia Suppression of lactation, flushing, nasal congestion.

Cholinergic Blocking (Parasympatholytic) Drugs*

Nursing considerations:

As atropine.

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Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Trade name: atrovent, irovent

Class.: cholinergic blocking agent, antimuscarinic drug.

Action: It is a parasympatholytic agent, chemically related to atropine antagonized the action of acetylcholine. It causes bronchodilation and inhibits secretion from nasal mucosa gland.

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Uses:

- **Bronchodilator for maintenance treatment of bronchospasm associated with COPD, chronic bronchitis, and emphysema (solution, aerosol).**
- **Nasal spray: symptomatic relief of rhinorrhea with perennial rhinitis, common cold.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Contraindications:

Hypersensitivity to atropine or its derivative, soy bean or peanut allergy (aerosol).

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Dosage:

- **Aerosol inhalation, ipratropium bromide 20 micrograms/ metered inhalation. two inhalations qid. Do not exceed 12 inhalations/ 24 hr.**
- **Nebulizer solution, ipratropium bromide 250 micrograms/ mL.**
  - **By inhalation of nebulized solution, reversible airways obstruction in chronic obstructive pulmonary disease, 250–500 micrograms 3–4 times daily.**
  - **Acute bronchospasm; 500 micrograms repeated as necessary; CHILD under 5 years 125–250 micrograms, max. 1mg daily; 6–12 years 250 micrograms, max. 1mg daily.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Adverse effects:

- **CNS: headache, dizziness, fatigue, insomnia, blurred vision.**
- **GI: nausea, dry mouth.**
- **Respiratory: bronchitis, dyspnea, cough.**

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:

Nursing considerations:

- Protect solution for inhalation from light and store unused vials in foil pouch.
- Use nebulizer mouthpiece instead of face mask to avoid blurred vision or aggravation of narrow angle glaucoma.
- Can mix albuterol in nebulizer for up to 1 hr.
- Ensure adequate hydration, control temperature to avoid hyperpyrexia.
- Teach patient proper use of inhalator.

Cholinergic Blocking (Parasympatholytic) Drugs*

Ipratropium bromide:





nebulizer mouthpiece

QUIZ

Anti – infectives

Antimicrobial agents:

These are agents that kills microorganisms or suppresses their multiplication or growth.

Classification of Antimicrobials:

1. **Antibacterial agents.**
2. **Antiviral agents.**
3. **Antiprotozoal agents.**
4. **Antifungal agents.**
5. **Antihelmintic agents.**

Anti – infectives

Antibiotics:

drugs produced by microorganisms or other live organisms to kill other microorganisms. Nowadays, many are produced synthetically.





Anti – infectives

a. **Bacteriostatic:** These drugs temporarily inhibit the growth of a microorganism. When the drug is removed, organism will resume growth and infection or disease may recur e.g. tetracycline.

b. **Bactericidal:** eradicate (kill) all living microorganisms e.g. penicillins.

Anti – infectives

Broad-spectrum antibiotics: anti-infectives that are stop the growth or eradicate many microorganism.

Narrow- spectrum antibiotics: anti-infectives that are affect only certain specific organisms.

Narrow-spectrum	Broad-spectrum
Penicillin G	Tetracyclines
Streptomycin	Chloramphenicol
Erythromycin	

Anti – infectives

Choice of antibiotic:

a. It follows automatically from clinical diagnosis.

b. It should be based wherever possible on bacteriological identification and sensitivity test.

Anti – infectives

Uses of anti-infectives:

A: Prophylactic

a) To protect persons exposed to a known specific organism.
b) To inhibit spread of infection from a clearly defined focus as after surgery or accidents.
c) To sterilize the bowel or other areas of the body in preparation for extensive surgery.

Anti – infectives

Uses of ant-infectives:

B. Acute infections:

Antibiotics are used to treat acute infections such as tonsillitis and acute upper respiratory tract infections.

Selecting an anti-infective drug

Selecting an appropriate anti-infective drug to treat a specific infection involves several important factors:

- First, the microorganism must be isolated and identified—generally through growing a culture.
- Then its susceptibility to various drugs must be determined. Because culture and sensitivity results take 48 hours, treatment typically starts at assessment and is then reevaluated when test results are obtained.
- The location of the infection must be considered. For therapy to be effective, an adequate concentration of the anti-infective drug must be delivered to the infection site.

Drug considerations:

The choice depends on the individualities of the patient, the infecting organism and the drug.

A. Patient factors

1. **Age**
   - May affect kinetics of many AMAs.
   - Conjugation and excretion of chloramphenicol is inefficient in the newborn: larger doses produce gray baby syndrome.
   - Sulfonamides displace bilirubin from protein binding sites—can cause kernicterus in the neonate because their blood brain barrier is more permeable.
   - Tetracyclines deposit in the developing teeth and bone—discolour and weaken them—are contraindicated below the age of 6 years.

A. Patient factors

2. Renal and hepatic function

- Cautious use and modification of the dose of an AMA (with low safety margin) becomes necessary when the organ of its disposal is defective

Antimicrobials needing dose reduction/ avoidance in renal failure

Reduce dose even in mild failure

Aminoglycosides	Amphotericin B
Cephalosporins	Ethambutol
Vancomycin	Flucytosine

Reduce dose only in moderate-severe failure

Metronidazole	Carbenicillin
Cotrimoxazole	Fluoroquinolones
Aztreonam	Clarithromycin
Meropenem	Imipenem

Drugs to be avoided

Cephalothin	Talampicillin
Nalidixic acid	Tetracyclines
Nitrofurantoin	(except doxycycline)

Antimicrobials in liver disease

Drugs to be avoided

Erythromycin estolate
Pyrazinamide
Talampicillin
Tetracyclines
Nalidixic acid
Pefloxacin

Dose reduction needed

Chloramphenicol
Metronidazole
Clindamycin
Isoniazid
Rifampin

3. Local factors

The conditions prevailing at the site of infection greatly affect the action of AMAs.

a. Presence of pus and secretions decrease the efficacy of most AMAs, especially sulfonamides and aminoglycosides.

b. Presence of necrotic material or foreign body makes eradication of infection practically impossible.

c. Lowering of pH at the site of infection reduces activity of macrolide and aminoglycoside antibiotics.

4. Impaired host defence

Integrity of host defence plays a crucial role in overcoming an infection (ADIS).

5. Pregnancy

- All AMAs should be avoided in the pregnant because of risk to the foetus.
- Penicillins, many cephalosporins and erythromycin are safe, while safety data on most others is not available. Therefore, manufacturers label ‘contraindicated during pregnancy’.
- Tetracyclines carry risk of acute yellow atrophy of liver, pancreatitis and kidney damage in the mother.

6. Genetic factors

Primaquine, nitrofurantoin, sulfonamides, chloramphenicol and fluoroquinolones are likely to produce haemolysis in G-6-PD deficient patient.

B. Drug factors

When any one of a number of AMAs could be used to treat an infection, choice among them is based upon specific properties of these AMAs

1. Spectrum of activity:

- For definitive therapy, a narrow-spectrum drug which selectively affects the concerned organism is preferred, because it is generally more effective than a broad-spectrum AMA, and is less likely to disturb the normal microbial flora. However, for empirical therapy, often a broad-spectrum drug has to be used to cover all likely pathogens.

2. Type of activity:

- Many infections in patients with normal host defense respond equally well to bacteriostatic and bactericidal AMAs. But several acute infections resolve faster with a cidal than a static drug, because the cidal drug directly reduces the number of bacteria at the site of infection, while the static drug only prevents increase in their number.

2. Type of activity:

- Many bactericidal drugs exert prolonged post antibiotic effect so that maintenance of drug level continuously above the MIC is not essential.
- With bacteriostatic AMAs the bacteria start multiplying quickly when drug level falls below the MIC, resulting in relapse of infection.

3. Sensitivity of the organism:

- assessed on the basis of MIC values (if available) and consideration of postantibiotic effect.

4. Relative toxicity:

- Obviously, a less toxic antibiotic is preferred, e.g. a β-lactam over an aminoglycoside or erythromycin over clindamycin.

5. Pharmacokinetic profile:

- For optimum action the antibiotic has to be present at the site of infection in sufficient concentration for an adequate length of time.
- Antimicrobial action depends on the length of time the concentration remains above MIC; division of daily dose has better effect. However, the doses should be so spaced that the surviving organisms again start multiplying and a cidal action is exerted.
- A drug which penetrates better and attains higher concentration at the site of infection is likely to be more effective.

6. Route of administration:

- Many AMAs can be given orally as well as parenterally, but aminoglycosides, penicillin G, carbenicillin, many cephalosporins, vancomycin, etc. have to be given by injection only. For less severe infections, an oral antibiotic is preferable; but for serious infections, e.g. meningitis, spreading cellulitis, septicaemias, a parenteral antibiotic may be chosen.

7. Cost:

- Less expensive drugs are to be preferred.

Drug considerations:

- Check expiration date on the container.
- Check for recommended method of storage and reconstitution.
- Clear mark the date & time of reconstitution, your initials and the strength of the solutions of all drugs.
- Note the length of time that the drug may be stored after dilution and store under appropriate condition.

Problems that arise with the use of antimicrobial agents (AMAs)

1. Toxicity

(a) Local irritancy:

- at the site of administration.
- Gastric irritation,
- pain and abscess formation at the site of I.M. injection, thrombophlebitis
- e.g. erythromycin, tetracyclines, certain cephalosporins and chloramphenicol are irritants.

Problems that arise with the use of antimicrobial agents (AMAs)

1. Toxicity

(b) Systemic toxicity:

- Almost all AMAs produce dose related and predictable organ toxicities.
- Characteristic toxicities are exhibited by different AMAs.
- Some have a high therapeutic index—doses up to 100-fold range may be given without apparent damage to host cells. These include penicillins, some cephalosporins and erythromycin. Others have a lower therapeutic index—doses have to be individualized and toxicity watched for.

Problems that arise with the use of antimicrobial agents (AMAs)

1. Toxicity

Examples:

(a) Aminoglycosides: 8th cranial nerve and kidney toxicity.
(b) Tetracyclines : liver and kidney damage.
(c) Chloramphenicol: bone marrow depression.

Problems that arise with the use of antimicrobial agents (AMAs)

Still others have a very low therapeutic index—use is highly restricted to conditions where no suitable alternative is available, e.g. :

a. **Polymyxin B: neurological and renal toxicity.**
b. **Vancomycin: hearing loss, kidney damage.**
c. **Amphotericin B: kidney, bone marrow and neurological toxicity.**

Problems that arise with the use of antimicrobial agents (AMAs)

2. Hypersensitivity reactions

- Practically all AMAs are capable of causing hypersensitivity reactions.
- These are unpredictable and unrelated to dose.
- Range of reactions from rashes to anaphylactic shock can be produced.
- The more commonly involved AMAs are—penicillins, cephalosporins, sulfonamides, fluoroquinolones.

Problems that arise with the use of antimicrobial agents (AMAs)

3.Drug resistance

- It refers to unresponsiveness of a microorganism to an AMA, and is similar to the phenomenon of tolerance seen in higher organisms

- (What the pathogen does to the drug).

Problems that arise with the use of antimicrobial agents (AMAs)

Natural resistance:

- They lack the metabolic process or the target site which is affected by the particular drug.
- This is generally a group or species characteristic, e.g. gram-negative bacilli are normally unaffected by penicillin G, or M. tuberculosis is insensitive to tetracyclines. This type of resistance does not pose a significant clinical problem.

Problems that arise with the use of antimicrobial agents (AMAs)

Acquired resistance:

- It is the development of resistance by an organism (which was sensitive before)
- due to the use of an AMA over a period of time.
- This can happen with any microbe and is a major clinical problem.
- However, development of resistance is dependent on the microorganism as well as the drug.

Problems that arise with the use of antimicrobial agents (AMAs)

Acquired resistance:

- Some bacteria are notorious for rapid acquisition of resistance, e.g. staphylococci, coliforms, tubercle bacilli. Others like Strep. pyogenes and spirochetes have not developed significant resistance to penicillin despite its widespread use for > 50 years.
- Gonococci quickly developed resistance to sulfonamides, but only slowly and low-grade resistance to penicillin. However, in the past 30 years, highly penicillin resistant gonococci producing penicillinase have appeared.

Problems that arise with the use of antimicrobial agents (AMAs)

Resistance may be developed by mutation or gene transfer.

- **Mutation: It is a stable and heritable genetic change that occurs spontaneously and randomly among microorganisms. It is not induced by the AMA.**

- **Gene transfer: infectious resistance from one organism to another.**

Problems that arise with the use of antimicrobial agents (AMAs)

Prevention of drug resistance:

It is of utmost clinical importance to control development of drug resistance.

Problems that arise with the use of antimicrobial agents (AMAs)

Measures are:

a. Selective and adequate or correctly use of AMAs should be made.
b. determined shorter courses of AMAs are being advocated now.
c. Prefer rapidly acting and selective (narrowspectrum) AMAs whenever possible;

Problems that arise with the use of antimicrobial agents (AMAs)

Measures are:

d. broad-spectrum drugs should be used only when a specific one cannot be determined or is not suitable.
e. Use combination of AMAs whenever prolonged therapy is undertaken, e.g. tuberculosis.
f. Infection by organisms that have high ability for developing resistance, e.g. Staph. aureus, E. coli, M. tuberculosis, Proteus, etc. must be treated intensively.

Problems that arise with the use of antimicrobial agents (AMAs)

4. Superinfection (Suprainfection)

- *This refers to the appearance of a new infection as a result of antimicrobial therapy.*
- **Use of most AMAs causes some alteration in the normal microbial flora of the body.**
- **The normal flora contributes to host defence by elaborating substances called bacteriocins which inhibit pathogenic organisms.**

Problems that arise with the use of antimicrobial agents (AMAs)

4. Superinfection (Suprainfection)

- complete the suppression of body flora, greater are the chances of developing superinfection.
- it is commonly associated with the use of broad/extended-spectrum antibiotics, such as tetracyclines, chloramphenicol, ampicillin, newer cephalosporins; especially when combinations of these are employed.

Problems that arise with the use of antimicrobial agents (AMAs)

4. Superinfection (Suprainfection)

- **Tetracyclines are more prone than chloramphenicol and ampicillin is more prone than amoxicillin to cause superinfection diarrhoeas because of incomplete absorption—higher amounts reach the lower bowel and cause greater suppression of colonic bacteria.**
- **Superinfections are more common when host defence is compromised.**

Problems that arise with the use of antimicrobial agents (AMAs)

4. Superinfection (Suprainfection)

- **Sites involved in superinfection are those that normally harbor commensals, i.e. oropharynx; intestinal, respiratory and genitourinary tracts; occasionally skin.**
- **Superinfections are generally more difficult to treat.**

Problems that arise with the use of antimicrobial agents (AMAs)

4. Superinfection (Suprainfection)

Conditions predisposing to superinfections

- Corticosteroid therapy
- Leukaemias and other malignancies, especially when treated with anticancer drugs (these are also immunosuppressants and decrease WBC count)
- Acquired immunodeficiency syndrome (AIDS)
- Agranulocytosis
- Diabetes, disseminated lupus erythematosus

Problems that arise with the use of antimicrobial agents (AMAs)

5. Nutritional deficiencies

- Some of the B complex group of vitamins and vit K synthesized by the intestinal flora is utilized by man.
- Prolonged use of antimicrobials which alter this flora may result in vitamin deficiencies.
- e.g, Neomycin causes morphological abnormalities in the intestinal mucosa—steatorrhoea and malabsorption syndrome can occur.

INHIBIT		CLASIFICATION		ANTIBIOTICS		
Cell Wall Synthesis	Beta Lactams	Penicillins	***Penicinillase – Sensible***			
			Natural Penicillins (narrow spectrum)	Penicillin G: Na, K, Procainic, Benzathine (IV, IM) Penicillin V: VO		
			Aminopenicillins (broad spectrum)	Ampicillin Amoxicillin		
			Penicinillase – Resistant *(very narrow spectrum)*			
			Nafcillin	Oxacillin	Dicloxacillin	
			Antipseudomonal *(extended spectrum)*			
			Carboxipenicillins	Ticarcillin Carbenicillin		
			Ureidopenicillins	Piperacillin Azlocillin Mezlocillin		
		Cephalosporins	1° Generation	Cefazolin	Cephalexine	Cephapirin
				Cefadroxil	Cephadrine	Cephalotin
			2° Generation	Cefuroxime	Cefamandole	Cefprozil
				Cefoxitin	Cefonicid	Cefmetazole
				Cefotetan	Cefaclor	
			3° Generation	Cefoperazone	Ceftriaxone	Ceftazidime
				Cefpodoxime	Ceftizoxime	Cefotaxime
				Cefdinir	Ceftibuten	Cefixime
				Cefditoren		
			4° Generation	Cefepime	Cefpirome *	
			5° Generation	Ceftaroline		
		Carbapenems	Meropenem	Ertapenem	Doripenem	Imipenem + Cylastatine
		Monobactams	Aztreonam			
		**** ***Beta-lactamase inhib.***	Sulbactam	Tazobactam	Clavulanic Acid	
	No lactam	***Glycopeptides***	Vancomycin	Bacitracin		
			Teicoplanin	Polymyxin B		

Cell Wall Synthesis

Beta Lactams
Penicillins
Cephalosporins
Carbapenems
Monobactams

Vancomycin
Bacitracin

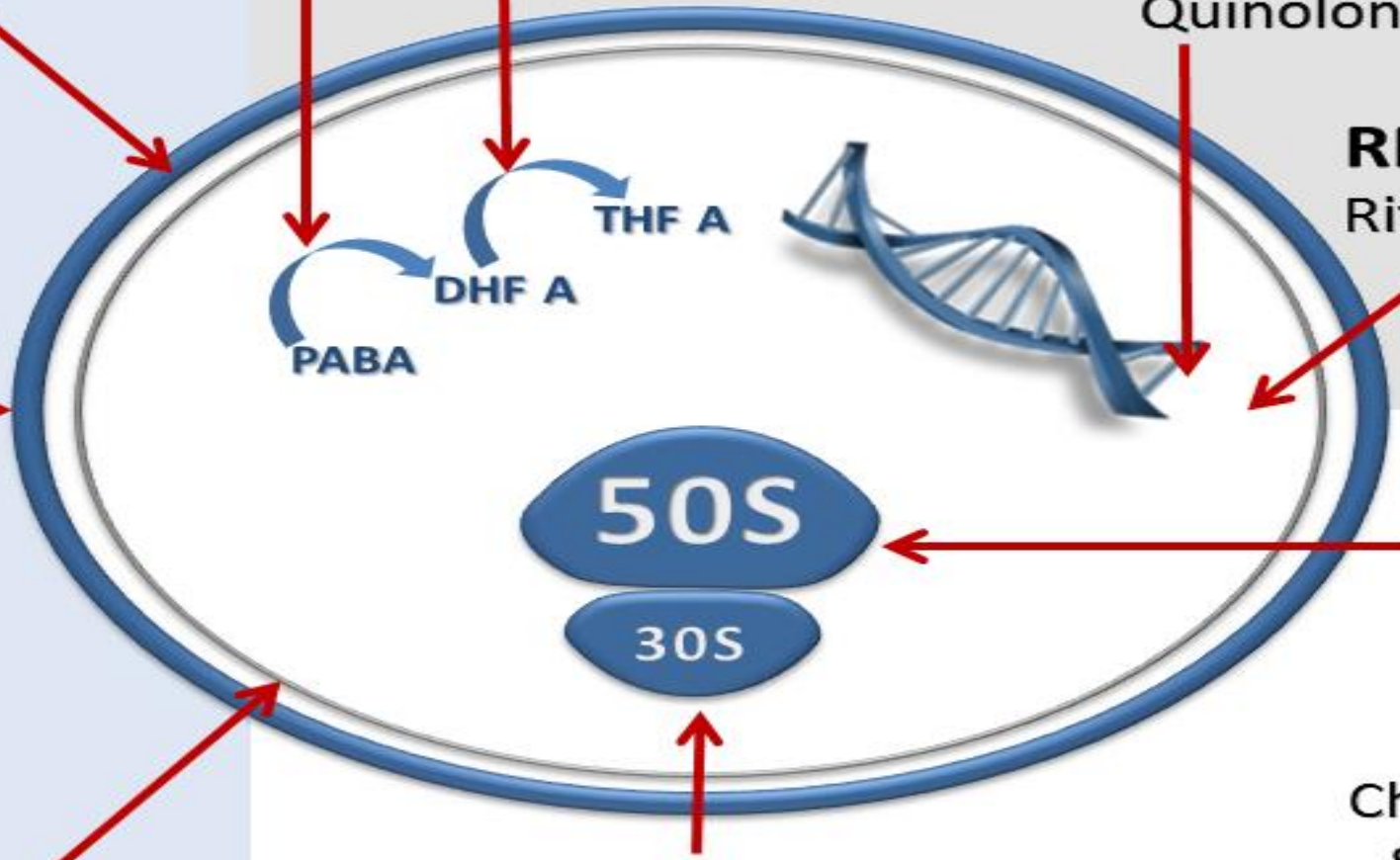
Cell Membrane
Polymyxins

Nucleic Acid Synthesis

Folate synthesis
Sulfonamides
Trimethoprim

DNA Gyrase
Quinolones

RNA Polymerase
Rifampin



50S subunit
Macrolides
Clindamycin
Linezolid
Chloramphenicol
Streptogramins

30S subunit
Tetracyclines
Aminoglycosides

Protein Synthesis

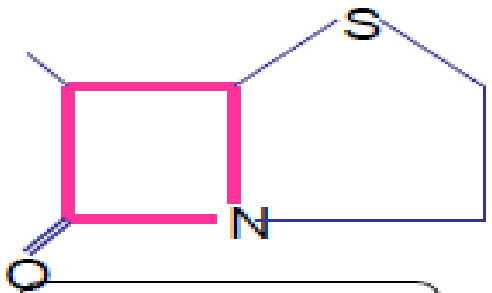
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Penicillins remain one of the most important and useful antibacterial drugs, despite the availability of numerous others. The penicillins can be divided into four groups:

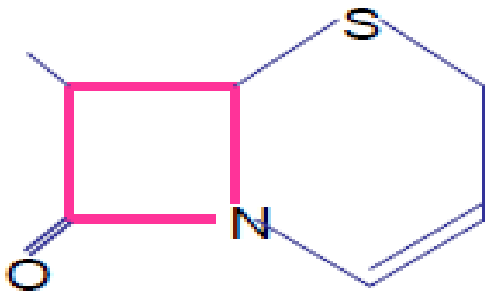
- **natural penicillins** (penicillin G benzathine, penicillin G potassium, penicillin G procaine, penicillin G sodium, penicillin V potassium)
- **penicillinase-resistant penicillins** (dicloxacillin sodium, cloxacillin sodium, nafcillin sodium)
- **aminopenicillins** (amoxicillin, ampicillin, amoxicillin–clavulanate potassium)
- **extended-spectrum penicillins** (carbenicillin indanyl sodium, ticarcillin disodium).

β-lactam Antibiotics

All of the drugs in this group contain a β -lactam ring in their structure



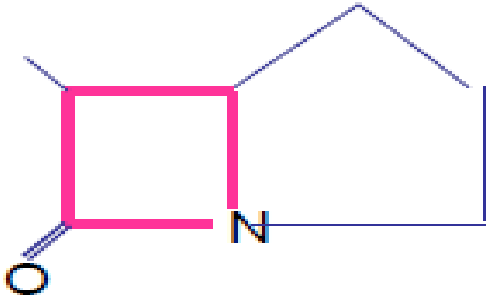
Penicillins



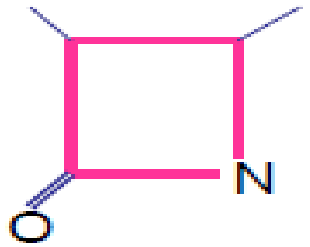
Cephalosporins

share similar

- features of chemistry,
- mechanism of action,
- pharmacologic and clinical effects.



Carbapenems



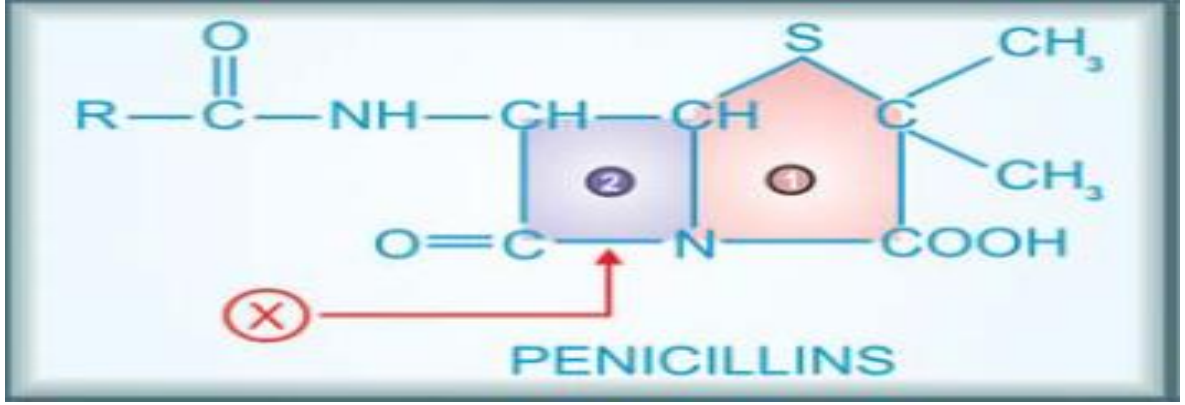
Monobactams

β-lactamAntibiotics

Mechanisms of Resistance

1. Production of β-lactamase enzymes
2. Trapping mechanism
3. Modification of target PBPs
4. Impaired penetration of drug to target PBPs

Penicillins



- Penicillin antibiotics were among the first medications to be effective against many bacterial infections caused by staphylococci and streptococci.

- Penicillins are still widely used today, though many types of bacteria have developed resistance following extensive use.
- All penicillins are β-lactam antibiotics.

Class: Anti-infective.

Pharmacokinetics

- After oral administration, penicillins' are absorbed mainly in the duodenum and the upper jejunum of the small intestine.
- Most penicillins should be given on an empty stomach (1 hour before or 2 hours after a meal) to enhance absorption.
- Penicillins that can be given without regard to meals include amoxicillin, penicillin V, and amoxicillin–clavulanate potassium.

Pharmacokinetics

- Penicillins are metabolized to a limited extent in the liver to inactive metabolites.
- Most penicillins are excreted 60% unchanged by the kidneys. Nafcillin and oxacillin, however, are excreted in bile.

Action: Inhibit cell wall synthesis; they inhibit cell division & growth of bacteria.

Pharmacokinetics

Uses:

- Gram +ve cocci “streptococci, meningococci, pneumococei ….”
- Subacute bacterial endocarditis caused by group A streptococci.
- Gonorrhea due to gonococci.
- Diphtheria, tetanus, anthrax, gas gangrene.
- Prophylaxis for rheumatic fever.

Contraindications:

- Allergy to penicillins & cephalosporins.

Side effects:

- Diarrhea, nausea, vomiting, abdominal cramps/pain.
- Hypersensitivity: nausea, rash, neurotoxicity, urticaria, and superinfection wheezing, fever etc.
- Psendomembranous colitis, thrombocytopnea, leukopnea
- Thrombophlebitis and electrolytes imbalance following I.V. use.
- Hepatotoxicity.

Dosage: Individualized.

General considerations for penicillin:

a) These antibiotics should be injected slowly to minimize local irritation.

b) I.M. injections are made deeply into the gluteal muscle.

c) Assess regularly for allergic reactions. If reaction occurs the drug must be discontinued immediately, Epinephrine, O2, antihistamines, & corticosteroids must be immediately available.

General considerations for penicillin:

d. **Used only for I.M**. use & do not massage after injection because rate of absorption should not be increased.

e. Take oral penicillin 1 hr before or 2-3 hr after meals.

f. Complete entire prescribed course of therapy.

NB: Probenecid increases the plasma concentration of penicillins.

Anti – infectives

Penicillin G: Procaine Penicillin, G Benzyl Penicillin

PnG is a narrow spectrum antibiotic; activity is limited primarily to gram-positive bacteria and few others.

Trade name: Bepen, Duplo-penicillin, procaine benzyl penicillin. .

Class.: Antibiotic, long acting penicillin.

Action: inhibit bacterial cell wall and causing cell death.

Anti – infectives

Uses: Streptococcal infections (without bacteremia) of:

- Upper respiratory tract such as Tonsillitis, pharyngitis.
- Otitis media
- Skin and soft tissue infections.
- Scarlet fever.
- Acute glomerulonephrites.
- Gonorrhea and syphilis.
- Rheumatic fever.
- Pneumococcal infections

Anti – infectives

Dosage form:

- Vials containing 1.2 MU, 2 MU.
- Penicillin G is inherently rapid in action while procaine penicillin G provides penicillin G in a prolonged action.



Contraindications:

- Allergy to penicillins & cephalosporins.

Side effects:

- nausea, vomiting , stomatitis, skin rash, anaphylaxis.

Anti – infectives

Benzathine Penicillin G, Parenteral.

Trade name: Bicillin L-A (long acting penicillin), Depocillin

Class. : Antibiotic, penicillin.

- **The parenteral product is long-acting form of penicillin in an aqueous vehicle.**
- **It is administered as a sterile suspension.**

Anti – infectives

Uses: Most gram +ve (streptococci , staphylococci, pneumoeocci) & some gram -ve (gonococci) organisms .

- Syphillis
- Prophylaxis of glomerulonephritis & rheumatic fever.
- Surgical infections.
- Secondary infections following tooth extraction + tonsillectomy.

Contraindications: Hypersensitivity to penicillin.

Anti – infectives

Dosage form: vial IM only, containing 600,000 unit – 1.2 mu unit.

Dosage: 200,000 – 600,000 unit q 4-6 hr.

- **Parenteral suspension: 1,200,000 units as single dose for adults and for pediatric 300,000 – 900,000 units as a single dose.**
- **Early syphilis: 2,400,000 units q 7 days for 3 weeks.**
- **Prophylaxis of Rheumatic fever: 1,200,000 units as a single dose every 4 weeks.**
- **Children: half the adult dose.**

Side effects and Nursing considerations:

- **See the previous drug.**

Anti – infectives

Benzyl Penicillin G

Trade name: Crystapen, crystaline penicillin .

Class: Antibiotic , penicillin.

- The low cost of penicillin G still makes it the first choice for treatment of many infections.
- Rapid onset makes it especially suitable for fulminating infections.
- Destroyed by acid & penicillinase.

Anti – infectives

Add side effects:

Rapid I.V. administration may cause hyperkalemia & cardiac arrhythmias.

Uses: Infections due to penicillin- sensitive organisms.

N.B.: Penicillin G potassium is indicated as on alternative to penicillin G sodium in those patients in whom intake of sodium must be restricted.

Contraindications: Hypersensitivity.

Anti – infectives

Dosage forms: Penicillin G potassium (injection) vial of 1 MU and Penicillin G sodium vials of 1,5,10 Mu.

Dosage: I.M. , continuous I.V. infusion 300,000 – 30 million unit depending on the use.

- Pediatric: 100,000 – 250,000 units \kg daily in divided dose.
- Pediatric: 25,000 – 90,000 units \kg daily in divided doses.

Anti – infectives

Ampicillin:

Trade name: Penbritin, Ampipharm.

Class: Antibiotic, penicillin.

Action: bactericidal, inhibit bacterial cell wall and causing cell death.

- Destroyed by penicillinase (lactamase enzymes)
- 30-60% absorbed from GIT after oral use.
- Broad – spectrum antibiotic.

Anti – infectives

Ampicillin:

Uses: treatment of infections caused by susceptible strains e.g. salmonella, E. coli, H. influenza, enterococci etc. particularly it is recommended in respiratory, urinary & GI tract infections & meningitis caused by Neisseria meningitis.

Anti – infectives

Ampicillin:

Dosage forms:

- Capsules 250 mg, suspension 125 mg\5ml, vials 250, 500, 1000 mg.
- Ampicillin forte capsules 500 mg, ampicillin forte suspension 250 mg\5ml.

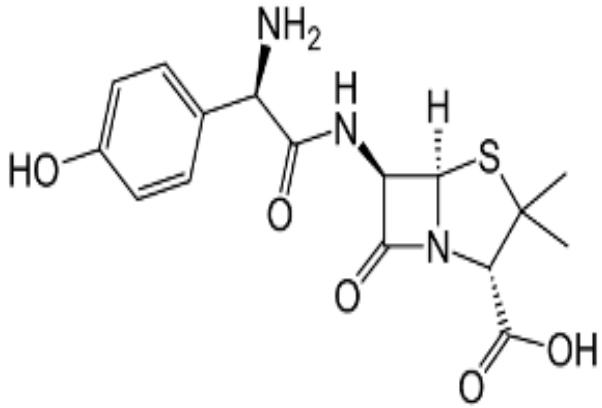
Dosage:

- P.O., I.V. or I.M. 250 mg – 500 mg q 6 hr (adult).
- Children 50 mg\kg 1 day in 4 divided doses.

For bacterial meningitis

- Adults: 12gm are given in divided doses (3 gm q 6 hr).
- Children: up to 400 mg\kg daily in divided doses q 4 hr.

Anti – infectives



Amoxycillin:

Trade name: **Moxypen**, Amoxitid, Amoxicare, Moxepharm.

Class: Antibiotic, penicillin.

Action:

- bactericidal, inhibit bacterial cell membrane and causing cell death.
- Inactivated by penicillinase, including those produced by staphylococcus aureus and gram negative baccili.
- Acid stable
- Broad-spectrum antibiotic.
- Absorbed well from the GIT.

Anti – infectives

Amoxycillin:

Dosage Forms:

- Caps. 250 mg, suspension 125 mg \5 ml
- Moyxpen forte caps. 500 mg, Moxypen forte suspension, 250 mg\5ml

Dosage: 250-500 mg q 8 hr.

Anti – infectives

Amoxycillin:

Side effects:

- **GI:** stomatitis, gastritis, sore mouth, nausea, vomiting, diarrhea, abdominal pain, bloody diarrhea etc.
- **Hematologic:** anemia, thrombocytopenia, leukopenia etc.
- **Hypersensitivity:** rash, fever, wheezing, anaphylaxis.
- **Other:** super-infection, vaginitis.

Anti – infectives

Amoxycillin:

Specific consideration:

- Reconstituted suspension is stable for 1 week at room temperature & for 2 weeks at 2- 8º Centigrade degrees.
- Teach patient about side effects specially on GI.

Anti – infectives

Penicillin V Potassium:

Trade name: Rafapen, Bepen.

Class.: Antibiotic, penicillin.

- Is acid stable and resist inactivation by gastric secretions.
- Well absorbed from the GIT & not affected by food.

Dosage forms:

- Tab.: 250 mg (400,000 units), 500 mg (800,000 units).
- Syrup:125 mg\5ml (200,000 units), 250 mg (400,000 units).

Dosage: according to causative agent, 125 mg-500mg q 6-8 hr for 10 days.

Anti – infectives

Amoxycillin & Potassium Clavulanate:

Trade name: Augmentin, , moclav, clamoxin, ogmin.

Class: Antibiotic, penicillin.

Note: Potassium clavulanate inactivates beta-lactamase enzymes which are responsible for resistance to penicillin.

Dose: tablets 500mg q8 hr. or suspension 125, 250 mg q 8 hr.

Anti – infectives

Cloxacillin:

Trade name: Orbenin

Class. : Antibiotic, penicillin. resistant to penicillinase.

Action: inhibit bacterial cell membrane and causing cell death.

Uses: Infections caused by penicillinase- producing staphylococci, streptococci, pneumococci that are resistant to Benzyl penicillin such as.

- Abscesses (skin and soft tissue.
- Osteomyelitis (bone and joints)
- infected wounds & burns
- Septic arthritis.
- pneumonia

Anti – infectives

Cloxacillin:

Dosage forms:

- **Vials containing 250 mg, 1g .**
- **Capsules 250 mg - 500 mg.**
- **Syrup 125 mg\ 5 ml, 250 mg\5ml.**

Dose:

- **Adult: 250 – 500 mg q 6 hr.**
- **Children: 50 mg\kg per day in 4 divided doses.**

Anti – infectives

Cloxacillin:

Considerations:

- Administer on an empty stomach.
- Refrigerate reconstituted solution & discard remaining amount after 14 days.
- N.B.: To prepare oral suspension, add amount of water stated on label and shake well.
- Shake the bottle well before each use.

Anti – infectives

Cephalosporins

- Cephalosporins share with penicillins a common mechanism, susceptibility to B-lactamases and similar side effects.
- Like penicillins, the original "generation" of cephalosporins was most effective against Gram-positive organisms. Chemical modifications increased the effectiveness of subsequent generations against Cram negatives and reduced their ability to fight Gram positive infections.
- Are semisynthetic antibiotics that resemble the penicillin both chemically and pharmacologically.

First generation cephalosporins

Parenteral	*Oral*
Cephalothin*	Cephalexin
Cefazolin	Cephradine
	Cefadroxil

Second generation cephalosporins

Parenteral	*Oral*
Cefuroxime	Cefaclor
Cefoxitin*	Cefuroxime axetil

Third generation cephalosporins

Parenteral	*Oral*
Cefotaxime	Cefixime
Ceftizoxime	Cefpodoxime proxetil
Ceftriaxone	Cefdinir
Ceftazidime	Ceftibuten
Cefoperazone	Ceftamet pivoxil

Fourth generation cephalosporins

Parenteral
Cefepime
Cefpirome

Anti – infectives

Cephalosporins

- Are absorbed rapidly from the GI tract and quickly reach effective concentrations in the urinary, GI and respiratory tracts except in patients with pernicious anemia or obstructive jaundice & eliminated rapidly in patients with normal renal function.
- They are broad- spectrum antibiotics that have been classified as first, second and third generation drugs. The difference among generations is based on antibacterial spectra.
- Third generation cephalosporins have more activity against gram negative organisms and resistant organisms & less activity against gram positive organisms than first generation drugs. They are also stable against beta-lactemase enzymes.
- The cost increases from 1st to 3rd generation cephalosporins.

Anti – infectives

Cephalosporins

Action:

- They interfere with the final step in the formation of the bacterial cell wall resulting in unstable cell membranes that undergo lysis, also cell division & growth are inhibited.
- 1st & 2nd generation drugs don't enter the CSF well, but 3rd generation drugs enter inflamed meninges readily.
- Cephabsporins are excreted rapidly by the kidneys.

Cephalosporins

Uses:

- Cephalosporins are indicated for the prophylaxis and treatment of infections caused by bacteria susceptible to this particular form of antibiotic.
- First-generation cephalosporins are active predominantly against Gram-positive bacteria, and successive generations have increased activity against Gram-negative bacteria
- They are effective against infections of biliary tract, GU tract, bones & joints, upper & lower respiratory tract, Skin, meningitis, osteomyelitis, peritonitis, otitis media, gonorrhea and prophylaxis prior to surgery

Anti – infectives

Cephalosporins

Contraindications:

Hypersensitivity to cephalosporins or Penicillin, renal failure, Pregnancy, Lactation.

Side effects:

Nausea, vomiting, diarrhea, anorexia, abdominal pain, flatulence, skin rashes, super-infection, heartburn, sore mouth, bone marrow depression: (Decrease WBC, decreased platelets, decreased Hct), Nephrotoxicity, (pain, abscess at injection site, phlebitis and inflammation at IV site).

Anti – infectives

Cephalosporins

Drug considerations:

- Infuse over 30 minutes unless otherwise indicated.
- Therapy should be continued for at least 2-3 days after symptoms of infection have disappeared.
- Assess client with a history of hypersensitivity reaction. “for penicillin or cephalosporins.”
- Assess client financial status. These drugs are usually expensive.
- If GI upset occurs administer drugs with meals. “Should be administered on empty stomach”.
- Obtain liver & renal studies.

Anti – infectives

Cephalosporins

Cephalexin monohydrate

Trade names Keflex , Jeflex.

Class.: antibiotic, cephalosporin (first generation).

Use : infections of respiratory tract, skin, bones, GU and otitis media.

Additional Side-effects: Nephrotoxicity, jaundice.

Dosage:

- Caps. 250-500 mg q 6 hr.
- Suspension 125-250mg q 6 hr.

Anti – infectives

Cephalosporins

Cephalothin sodium:

Trade name: Keflin.

Class: Broad-spectrum antibiotic, cephalosporin (first generation).

N.B.: poorly absorbed from GI tract & must be given parenterally.

Dosage: Deep I.M. or I.V. 500-1000 mg a 4-6 hr.

Uses: see Keflex.

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Anti – infectives

Cephalosporins

Cefaclor:

Trade name: Ceclor

Classification: antibiotic, cephalosporin " second generation"

Uses: infections of upper & lower respiratory tract. UTI & dermatological infections and otitis media.

Dosage: tablet (500,750 mg), oral suspension. 250 mg every 8 hr & may be doubled in case of severe infection (for adults). Children 20 mg/kg/day in 3 divided doses. Maximum 1 gm daily.

.

Anti – infectives

Cephalosporins

Cefaclor:

Specific drug considerations:

- The suspension should be refrigerated after reconstitution and discarded after 14 days.
- Give drug with meals or food to decrease GI disturbance.
- Discontinue drug if hypersensitivity reaction occurs.

.

Anti – infectives

Cephalosporins

Cefuroxime Sodium:

Trade names: Zinacef, Zinnat.

Class: Antibiotic, cephalosporin (second generation).

Uses:

- P.O. :pharyngitis, tonsillitis, UTI, bronchitis.
- I.M. or I.V.: pneumonia, UTI, osteomyelitis, meningitis, septicemia, pre-op. Prophylaxis.

Add. Side effects: decrease in Hb & HCT (hematocrit).

.

Anti – infectives

Cephalosporins

Cefuroxime Sodium:

Dosage:

- Tablets 250-500 mg q 12 hr (adult).
- 125- 250 mg bid (children).
- I.M. –I.V. 750 mg q 8 hr
- Severe infection, 1.5 g q 6-8 hr.

.

Anti – infectives

Cephalosporins

Cefixime, oral:

Trade name: suprax, supran .

Classification: antibiotic, cephalosporin (third generation).

- The third generation drug can be given orally.
- Stable in the presence of beta-lactamase enzymes.

Uses: Uncomplicated UTI, pharyngitis, tonsillitis & acute bronchitis.

Additional side effects: Flatulence, alkaline phosphatase levels.

.

Anti – infectives

Cephalosporins

Cefixime, oral:

Dosage: oral suspension, tablets. Adults either 400 mg once daily or 200 mg every12 hr, children 8 mg/kg / day.

Specific drug consideration:

- Once reconstituted, the suspension should be kept at a room temp, where it maintains potency for 14 days.
- Therapy should be taken once a day at the same time each day.

Anti – infectives

Cephalosporins

Cefotaxime Sodium:

Trade name: claforan.

Classification: Antibiotic, cephalosporin (third generation).

- **N.B.: Treatment should be continued for a minimum of 10 days for group A beta-hemolytic streptococcal infections to minimize the risk of glomarulonephritis or rheumatic fever.**
- **The I.V. route is preferable for patients with severe or life – threatening infections.**

Anti – infectives

Cephalosporins

Cefotaxime Sodium:

Uses:

- **Infections in respiratory tract (Pneumonia).**
- **GU tract infections**
- **Meningitis.**
- **Peritonitis.**
- **Septicemia.**
- **Pelvic cellulitis.**
- **Endometritis.**

N.B. used with aminoglycosides when causative agent has not been identified.

Anti – infectives

Cephalosporins

Cefotaxime Sodium:

Dosage:

- I.V., I.M. 1 gm q 12 hr (adult).
- Moderate to severe infection. 1-2 g every 6-8 hr.
- Pre-op. Prophylaxis, 1 gm 30-90 min prior to surgery.

Anti – infectives

Cephalosporins

Cefotaxime Sodium:

Drug considerations:

- **Should not be mixed with aminoglycosides. (each should be given separately)**
- **For I.V. use, should be mixed with 10 ml sterile water & administer over 3-5 minutes.**
- **Observe for signs and symptoms of anaphylaxis during first dose; CBC with differential specially in long course treatment.**

Anti – infectives

Cephalosporins

Ceftriaxone Soduim:

Trade name: Rocephin, cefixon.

Class: Cephalosporin (third generation), antibiotic.

Uses: lower respiratory tract (pneumonia), UTI, infections of skin, bone & abdomen. Meningitis, bacterial septicemia, pre-op. prophylaxis.

Anti – infectives

Cephalosporins

Ceftriaxone Soduim:

Dosage: I.V., I.M. adult 1-2g daily in single or divided doses every 12 hr.

- Pediatric: 50-75 mg\kg \day (other than meningitis) meningitis 100 mg\kg 1 day.

Drug considerations:-

- I.M. injection should be deep into the body of large muscle.
- I.V. injection should be diluted.
- For stability of solution the package insert should be checked carefully.
- Dosage should be maintained for at least 2 days after symptoms of infection have disappeared (usual course is 4-14 days).

Anti – infectives

Cefpodoxime proxetil

Trade name: Cefpo

Class:

- **prodrug of 3rd generation cephalosporin**
- **highly active against Enterobacteriaceae and streptococci, it inhibits Staph. aureus.**

Uses: It is used mainly for respiratory, urinary, skin and soft tissue infections.

Dose:

- **Tablet 200 mg BID (max 800 mg/day)**
- **Susp. 50 mg/5 ml.**

Protein synthesis inhibitor

- Bacterial ribosomes differ structurally from mammalian cytoplasmic ribosomes
- It composed of 30S and 50S subunits (mammalian ribosomes have 40S and 60S subunits).
- In general, selectivity for bacterial ribosomes minimizes potential adverse.
- high concentrations of drugs such as chloramphenicol or the tetracyclines may cause toxic effects as a result of interaction with mitochondrial mammalian ribosomes, since the structure of mitochondrial ribosomes more closely resembles bacterial ribosomes.

Tetracycline

Action:

- The drugs bind reversibly to the 30S subunit of the bacterial ribosome. This action prevents binding of tRNA to the mRNA–ribosome complex, thereby inhibiting bacteri protein synthesis.
- Bacteriostatic antibiotics effective against a wide variety of organisms, including gram-positive and gram-negative bacteria, protozoa, spirochetes and mycobacteria.
- Well absorbed from the stomach & upper GIT.
- it can pass through the placental barrier.
- They are deposited in the fetal skeleton and calcifying teeth.

Tetracyclines

Elimination:

- **Tetracycline is not hepatically metabolized.**
- **Tetracycline is primarily eliminated unchanged in the urine, whereas minocycline undergoes hepatic metabolism and is eliminated to a lesser extent via the kidney.**
- **In renally compromised patients, doxycycline is preferred, as it is primarily eliminated via the bile into the feces.**

Tetracyclines

Uses:

- They are the drugs of choice for rickettsial infections as rocky mountain spotted fever and endemic typhus.
- Urethritis due to mycoplasma.
- Prophylaxis of in cases of plague after exposure.
- Treatment of rectal infections caused by C. trachomatis and acute intestinal amebiasis.
- They are the drug of choice for gram –ve bacteria causing cholera and brucellosis.
- Alternative treatment to penicillin for uncomplicated gonorrhea.
- Vaginitis.
- Severe acne.
- Topical uses include skin granuloma.

Tetracyclines

Contraindications:

- Hypersensitivity.
- During tooth development stage:
  - Last trimester of pregnancy
  - During breast-feeding.
  - Neonatal period and during childhood up to 8 years.

Tetracyclines

Side effects:

- GI: Nausea, vomiting, thirst, diarrhea flatulence, sore throat & anorexia, black tongue and esophageal ulcer .
- Dermatological: Skin rashes and photosensitivity.
- Dental: Discoloration of nails, infants' & children's teeth.
- Bone: Delayed bone growth.
- Blood: I.V. use may cause thrombophlebitis.

Tetracyclines

Superinfection

- Tetracyclines are the most common antibiotics responsible for superinfections, because they cause marked suppression of the resident flora. Though mouth, skin or vagina may be involved.
- intestinal superinfection by Candida albicans is most prominent.

Tetracyclines

Fanconi like syndrome:

- It results from ingestion of outdated and degraded tetracycline
- It characterized by nausea, vomiting, acidosis, proteinuria, hypokalemia, polyuria & polydipsia.

Tetracyclines

Considerations:

- Don’t use outdated or deteriorated drugs.
- Discard unused medications.
- Administer on an empty stomach (1hr before or 2 hr after meals).
- Avoid direct sunlight, which can cause severe sunburn-like reaction.

Examples:

1. **Tetracycline Hcl: Brimocyclin**
   - **Dose: cap. 250, 500mg 1-2 gm every 6 hr**
   - **NB: all doses given on empty stomach.**
   - **Recycline 1% as eye oint.**

2. **Chlortetracycline HCl (ophthalmic & topical 3% on affected area 1-5 times daily).**

Examples:

3- Doxycycline :

- **Trade names:** Doxylin , Doxypal , Doxypharm, doxacylin.
- **Class:** Antibiotic, tetracycline.
- **Dosage form:** Caps., film-coated tab. 100 mg
- **Dose:** The usual dose for adults.
  - First-day 100 mg q 12hr followed by 1 tab. daily.
  - In severe cases: 1 tablets daily in divided doses q 12 hr.

Examples:

3- Doxycycline :

- **Side effects:** nausea, vomiting, diarrhea, headache, hypertension, colitis, hemolytic anemia.
- **Consideration:**
  - Do not give with antacid, milk, or any product that contains Calcium, zinc, aluminum, magnesium, and ferrous salts, because these products decrease the absorption of the drug.

Examples:

4- Oxytetracycline: oxycin

Trade name: Terramycin

Class: antibiotic , tetracycline.

Dose: ointment 5mg , vaginal tablet with polymyxine B sulphate

Examples:

5- Minocycline: Minocycline, Minocin

Indications: add. meningococcal carrier state; acne vulgaris

Contra-indications: see notes above

Side-effects:

- see notes above; also dizziness and vertigo (more common in women); rarely anorexia, tinnitus, impaired hearing, acute renal failure, pigmentation (sometimes irreversible), alopecia, discoloration of conjunctiva, tears, and sweat

Dosage form: tablet, capsules 100mg

Dose: 50, 100 mg twice daily (initial 200mg followed by 100mg every 12 hr.

Examples:

6- Tigecycline: Tygacil

- A derivative of minocycline, is the first available of the glycylcycline antimicrobial class

- **Action:** It exhibits bacteriostatic action by reversibly binding to the 30S ribosomal subunit and inhibiting protein synthesis.

6- Tigecycline: Tygacil

Indications:

- It exhibits broad-spectrum activity.
- it is active against Gram-positive and Gram-negative bacteria, including tetracycline-resistant organisms, and some anaerobes. It is also active against meticillin-resistant Staphylococcus aureus and vancomycin-resistant enterococci
- Tigecycline was developed to overcome the recent emergence of tetracycline class–resistant organisms.

6- Tigecycline: Tygacil

Indications:

- **It should be reserved for the treatment of complicated skin and soft-tissue infections and complicated abdominal infections caused by multiple-antibacterial resistant organisms when other antibacterials cannot be used; it is not recommended for the treatment of foot infections in patients with diabetes.**

6- Tigecycline: Tygacil

Adverse effects

- Significant nausea and vomiting.
- Acute pancreatitis, including fatality, has been reported with therapy.
- Photosensitivity, pseudotumor cerebri, discoloration of permanent teeth when used during tooth development, and fetal harm when administered in pregnancy.

6- Tigecycline: Tygacil

Drug interaction:

- It may decrease the clearance of warfarin and increase prothrombin time. Therefore, the international normalized ratio should be monitored closely when it is coadministered with warfarin.

6- Tigecycline: Tygacil

Dosage form: Following IV infusion,

- The primary route of elimination is biliary/fecal. No dosage adjustments are necessary for patients with renal impairment. However, a dose reduction is recommended in severe hepatic dysfunction.

Dose: ADULT over 18 years, initially 100 mg, then 50mg every 12 hours for 5–14 days

Chloramphenicol:

Trade names: chloromycetin , chloram, synthomycin

Action:

- It inhibits protein synthesis in bacteria by binding to ribosomes (50S).
- Chloramphenicol is primarily bacteriostatic, though high concentrations have been shown to exert cidal effect on some bacteria, e.g. H. influenzae.
- It is a broad-spectrum antibiotic, active against nearly the same range of organisms (gram-positive and negative bacteria, rickettsiae, mycoplasma) as tetracyclines.

Chloramphenicol: Pharmacokinetics

- Chloramphenicol is administered intravenously and is widely distributed throughout the body.
- It reaches therapeutic concentrations in the CSF.
- Chloramphenicol primarily undergoes hepatic metabolism to an inactive glucuronide, which is secreted by the renal tubule and eliminated in the urine.
- Dose reductions are necessary in patients with liver dysfunction or cirrhosis.
- It is also secreted into breast milk and should be avoided in breastfeeding mothers.

Notable differences between chloramphenicol and tetracycline are:

a) Chloramphenicol was highly active against Salmonella including S. typhi, but resistant strains are now rampant.

a) It is more active than tetracyclines against H. influenzae, B. pertussis, Klebsiella, N. meningitides and anaerobes including Bact. fragilis.

a) It is less active against gram-positive cocci, spirochetes, certain Enterobacteriaceae and Chlamydia. Entamoeba and Plasmodia are not inhibited. Like tetracyclines, it is ineffective against Mycobacteria, Pseudomonas, many Proteus, viruses and fungi.

Chloramphenicol

Uses:

- Treatment of choice for typhoid fever (not for carrier state).
- CVS infections (Meningitis).
- Topically for skin infections.
- Brain abscesses.
- Eye and ear infections.

Chloramphenicol

Contraindications:

- Hypersensitivity to chloramphenicol.
- Pregnancy
- Renal and hepatic failure.

Side effects:

- **A plastic anemia, pancytopenia, bone marrow depression, neutropenia, nausea, vomiting, abdominal distention, Gray syndrome in infants, super infections, fever.**
- **Gray baby syndrome: Neonates have a low capacity to glucuronidate the antibiotic, and they have underdeveloped renal function. Therefore, neonates have a decreased ability to excrete the drug, which accumulates to levels that interfere with the function of mitochondrial ribosomes. This leads to poor feeding, depressed breathing, cardiovascular collapse, cyanosis (hence the term "gray baby"), and death.**
- **Adults who have received very high doses of the drug can also exhibit this toxicity.**

Drug interactions:-

- Acetaminophen (Acamol) increases the effect of chlpramphenicol.
- Chloramphenicol inhibits metabolism of tolbutamide, chlorpropamide, warfarin, cyclophosphamide and phenytoin. Toxicity can occur if dose adjustments are not done.
- Phenobarbitone, phenytoin, rifampin enhance chloramphenicol metabolism → reduce its concentration → failure of therapy may occur.
- Being bacteriostatic, chloramphenicol can antagonize the cidal action of β-lactams/ aminoglycosides on certain bacteria.

Formulation:

- Cap. 250, 500 mg, vial 1g.
- Eye drops, chloramphenicol 0.5%.
- Eye ointment, chloramphenicol 1%.
- Skin ointment chloramphenicol 3%.

Dose:

- **50 mg\kg daily in 4 divided doses q 6hr (500 mg to 1 gm q 6 hr).**

Drug considerations:

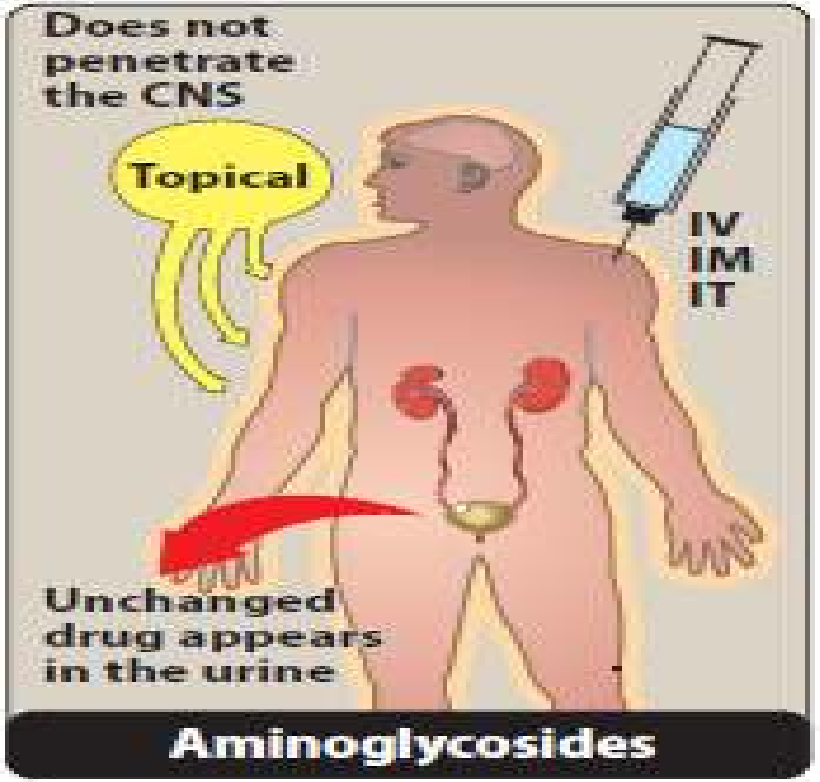
- Administer I.V. as 10% solution over at least 1 min.
- Neonates should be observed closely (greater hazards of toxicity).
- Arrange for further hematologic studies to be conducted every 2 days to detect early signs of bone marrow depression.
- The drug should be taken regularly.
- The drug should be taken on empty stomach or 2 hr after meals (if GI upset occurs it can taken e food).

Aminoglycosides

- Are broad–spectrum antibiotics that inhibit bacterial ribosomal function and primarily used for the treatment of serious gram negative infection caused by Pseudomonas , E.coli, proteus and klebsiella.
- They are more toxic than penicillins and cephalospoines and must be maintain within a certain therapeutic range.
- They are distributed in extracellular fluids, cross the placenta barriers but not the BBB because it isn't lipid-soluble.
- They are poorly absorbed from the GI tract, therefore are usually administered parenterally, not orally.

Aminoglycosides

- Are excreted largely in the urine, so it is suitable for urinary tract infections.
- Are powerful antibiotics that can induce serious side effects (nephrotoxicity, ototoxicity).



Aminoglycosides

Systemic aminoglycosides

Streptomycin	Amikacin
Gentamicin	Sisomicin
Kanamycin	Netilmicin
Tobramycin	

Topical aminoglycosides

Neomycin	Framycetin

They are inhibit protein synthesis by binding irreversibly to ribosome which leads to production of nonfunctional protein.

Aminoglycosides

Uses:

- Gram negative bacteria causing: bone and joint infections, respiratory tract infections, septicemia (including neonatal sepsis), urinary tract infection (UIT), post operative infections, intra–abdominal infections (as peritonitis), skin infections (including burns).
- It should be used for gram +ve bacteria only when other less toxic drugs either ineffective or contraindicated.

Aminoglycosides

Contraindications:-

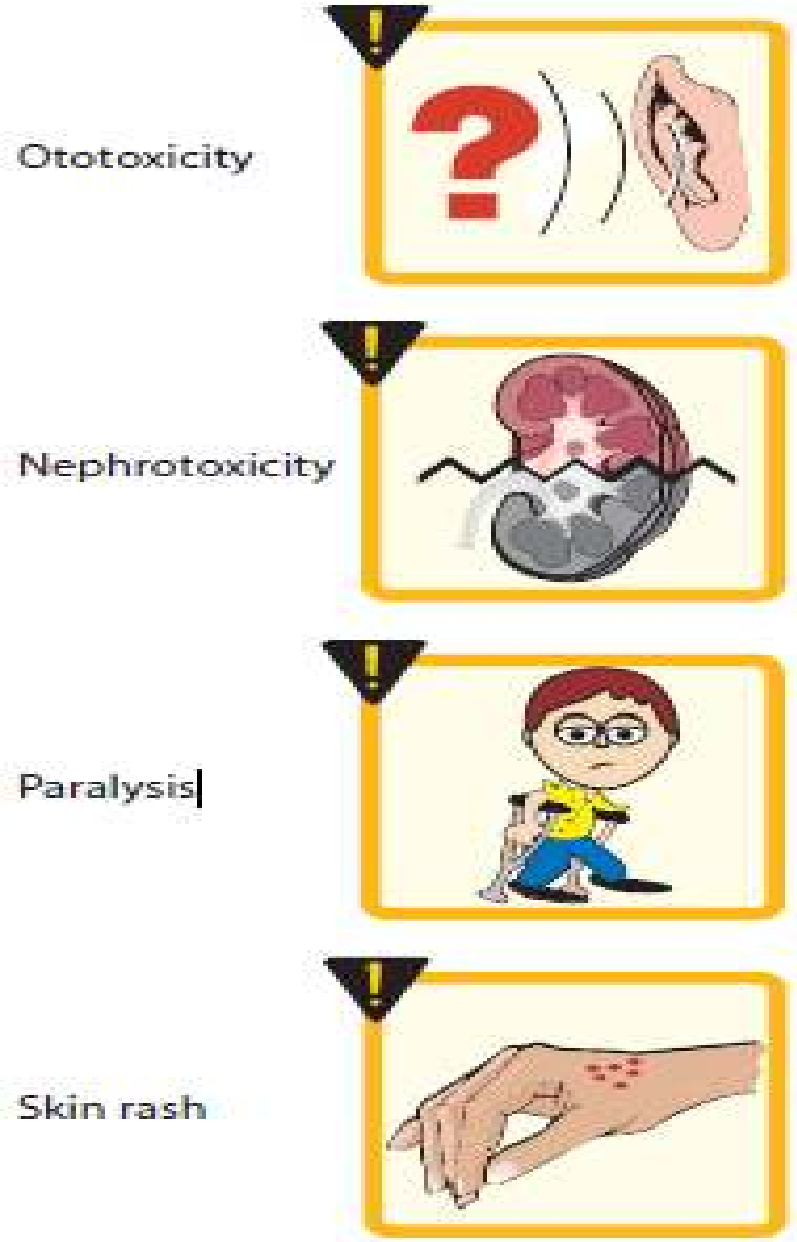
- **Hypersensitivity to aminoglycosides.**
- **Long–term therapy “except streptomycin for tuberculosis”.**
- **For patients with impaired renal function or pre-existing hearing impairment.**

Aminoglycosides

Side effects

- **Ototoxicity**: tinnitis , hearing impairment, ataxia & dizziness.
- Renal impairment (**Nephrotoxicity**) hematuria, proteinuria
- **Neurotoxicity**: headache, tremor, lethargy, numbness, burning of face.
- **Others**: nausea, vomiting , skin rash & superinfection.
- **Allergic reactions**: Contact dermatitis is a common reaction to topically applied neomycin.

Aminoglycosides



Some adverse effects of aminoglycosides.

Aminoglycosides

Precautions and interactions

1. Avoid aminoglycosides during pregnancy: risk of foetal ototoxicity.
2. Avoid concurrent use of other ototoxic drugs, e.g. high ceiling diuretics, minocycline.
3. Avoid concurrent use of other nephrotoxic drugs, e.g. amphotericin B, vancomycin, cyclosporine and cisplatin.
4. Cautious use in patients past middle age and in those with kidney damage.

Aminoglycosides

Drug considerations:

- IM administration inject deep into muscle to minimize pain.
- Administer for only 7-10 days.
- Assess history of hypersensitivity.
- Obtain laboratory studies for renal function.
- Continue to monitor for ototoxicity.
- The importance of administered medications at the appropriate prescribed time intervals.

Amikacin Sulfate:

Trade name: Amikin

Action:

- Inhibits protein synthesis in susceptible bacteria by binding to 30S ribosome subunits.
- It has the widest spectrum of activity, including many organisms resistant to other aminoglycosides. However, relatively higher doses are needed for Pseudomonas, Proteus and Staph. infections.
- It is recommended as a reserve drug for hospital acquired gram-negative bacillary infections where gentamicin/tobramycin resistance is high.

Amikacin Sulfate:

Classification: Antibiotic, aminoglycoside .

Uses:

- Serious infections caused by pseudomonas, proteus, serratia and gram positive bacilli including bone infections, respiratory tract infections, endocarditis, and septicemia.

Dosage:

- IM (preferred) I.V. adults, children 15 mg/ kg / day in 2-3 equally divided doses every 8-12 hr for 7-10 days. Newborns loading dose, 10 mg/ kg followed by 7.5 mg\ kg every 12 hr.

Gentamycin Sulfate:

Trade name: Garamyein.

Class: antibiotic, aminoglycosides.

Action:

- Inhibits protein synthesis in susceptible bacteria by binding to 30S ribosome subunits.
- The most commonly used aminoglycoside for acute infections.
- It is the drug of choice for hospital- acquired gram negative sepsis including neonatal sepsis and serious staphylococcal infections.

Gentamycin Sulfate:

Side effects:

- Alopecia
- CNS: ototoxicity, neurotoxicity, tinnitus, dizziness, ringing in the ears, vertigo.
- GI: nausea, vomiting, anorexia, weight loss, increased salivation.
- C.V.: palpitation, hypotension or hypertension.
- Hematologic: Decrease number of blood cells.
- GU: nephrotoxicity
- Local: Pain and irritation at IM injection site.

Gentamycin Sulfate:

Dosage form:

- Ampules 20, 40, 80 mg
- Vial 240 mg per vial inj.
- 0.3% eye/ear drops,
- 0.1% skin cream.

Gentamycin Sulfate:

Dose:

- IM (usual) . I.V. adults 3 mg\kg q 8 hr up to 5 mg\kg daily.
- Children 2-2.5 mg\kg every 8 hr.
- Newborns 2.5 mg\ kg every 12 hr.
- Ophthalmic solution 0.3% , 1-2 drops every 15-30 minutes.
- Topical ointment 0.1% , 1-5 times daily to the affected area.
- N.B.: should not be mixed with other drugs for parenteral use.

Gentamycin Sulfate:

Drug Consideration:

- When given IM, give it slowly and deep in a large muscle.
- Dilute dose when given IV.
- Monitor for kidney function tests (BUN & Creatinine), complete blood count when used for more than 7 days. Consult with Dr. accordingly.
- The daily dose of gentamicin (and other aminoglycosides) should be reduced in patients with impaired renal function according to measured creatinine clearance.

Neomycin Sulfate:

- It is a wide-spectrum aminoglycoside, active against most gram negative bacilli and some gram-positive cocci. However, Pseudomonas and Strep. pyogenes are not sensitive.
- It is highly toxic to the internal ear (mainly auditory) and to kidney. It is, therefore, not used systemically.
- Absorption from the GIT is minimal.
- Oral and topical administration does not ordinarily cause systemic toxicity.

Neomycin Sulfate:

Class.: antibiotic, aminoglycoside.

Action:

- Inhibits protein synthesis in susceptible bacteria by binding to 30S ribosome subunits.

Neomycin Sulfate:

Indications (uses):

a) P.O

- Diarrhea due to enteropathogenic E.coli.
- Suppression of intestinal bacteria for short-term therapy prior to intestinal surgery.
- Suppression of the intestinal flora for long-term therapy in hepatic coma. Neomycin attacks ammonia- forming bacteria in the intestinal tract causing reduction in blood ammonia and resulting in neurological improvement.

b) Topical:

- Inflamed skin conditions, burns.

Neomycin Sulfate:

Additional Contraindication:

- intestinal obstruction, pregnancy.

Additional Side effects:

- steatorrhea & electrolytes imbalance (mal-absorption).

Formulation:

Neomycin Sulphate

- 350, 500 mg tab,
- 0.3% skin oint,
- 0.5% skin cream, eye oint.

Dose: 15mg/kg/day IM in divided doses.

Oral for hepatic coma: 4-12 g/d in divided doses for 5-6 days.

Streptomycin Sulfate:

- It is the oldest aminoglycoside antibiotic.
- It used extensively in the past, but now practically restricted to treatment of tuberculosis.
- **It is less potent (MICs are higher) than other aminoglycosides. The antimicrobial spectrum of streptomycin is relatively narrow: active primarily against aerobic gram-negative bacilli, but potency is low.**

Streptomycin Sulfate:

- Sensitive organisms are, Brucella, granulomatis, M. tuberculosis. Only few strains of E. coli, H. influenzae, V. cholerae, Shigella, Klebsiella, enterococci and some gram-positive cocci are now inhibited, that too at higher concentrations. All other organisms including Pseudomonas are unaffected.
- Streptomycin is highly ionized. It is neither absorbed nor destroyed in the GIT However, absorption from injection site in muscles is rapid.

Streptomycin Sulfate:

- Streptomycin is not metabolized—excreted unchanged in urine.
- Half-life is prolonged and accumulation occurs in patients with renal insufficiency, in the elderly and neonates who have low GFR Reduction in dose or increase in dose-interval is essential in these situations.

Streptomycin Sulfate:

Class: antibiotic, aminoglycosides.

Additional uses:

- Tuberculosis (T.B.), plague, cholera, bacterial endocarditis caused by hemophillus influenza.

Additional Contraindications:

- contact dermatitis, myasthenia gravis.

Dosage:

- 0.75, 1 g dry powder per vial for inj.
- Acute infections: 1 g (0.75 g in those above 50 yr age) i.m. BD for 7–10 days.
- Tuberculosis: 1 g or 0.75 g i.m. twice weekly for 30–60 days.

Macrolides and Ketolides

- Erythromycin was the first of these drugs to find clinical application,
- Used as a drug of first choice and as an alternative to penicillin in individuals with an allergy to β-lactam antibiotics.
- Clarithromycin (a methylated form of erythromycin) and azithromycin (having a larger lactone ring) have some features in common with, and others that improve upon, erythromycin.
- Telithromycin, a semisynthetic derivative of erythromycin, is the first “ketolide” antimicrobial agent.

Macrolides and Ketolides

- Ketolides and macrolides have similar antimicrobial coverage. However, the ketolides are active against many macrolide-resistant gram-positive strains.
- The macrolides bind irreversibly to a site on the 50S subunit of the bacterial ribosome, thus inhibiting translocation steps of protein.
- Erythromycin and azithromycin are primarily concentrated and excreted in the bile as active drugs. In contrast, clarithromycin and its metabolites are eliminated by the kidney as well as the liver. The dosage of this drug should be adjusted in patients with renal impairment.

Macrolides and Ketolides

Erythromycin:

Trade names: Erthrocare, Erthrolet, Erthrotab, Erthroteva

Action:

- Erythromycin acts by inhibiting bacterial protein synthesis. It combines with 50S ribosome subunits and interferes with ‘translocation.
- Bacteriostatic and bactericidal in high doses.
- Absorbed well from the upper GIT.
- Enteric or coated forms to prevent destruction by gastric acid.
- Diffuse poorly to C.S.F. & primarily excreted in bile.

Erythromycin:

Uses:

- The drug of choice to treat respiratory tract infections.
- Intestinal amebiasis.
- Acne.
- An alternative drug to treat the following conditions in patients who are allergic to penicillin.
  - Infections due to group A- beta hemolytic streptococci.
  - To prevent bacterial endocarditis prior to dental procedures.
  - Gonorrhea or syphilis.
  - Conjunctivitis of the newborn.
  - Long term prophylaxes in rheumatic fever.

Erythromycin:

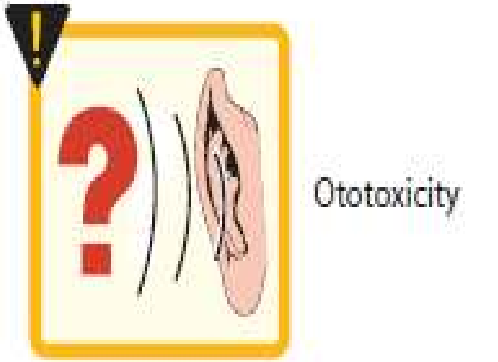
Contraindication: Hypersensitivity.

Side effects:

- GI: Nausea, vomiting, diarrhea, skin rashes, pseudomembranous colitis.
- Hepatotoxicity, confusion.
- CVS: reversible hearing loss in patients receiving high doses or with renal function impairment.







Some adverse effects of macrolide antibiotics.

Erythromycin:

Formulation:

- Tablets 250, 500 mg , susp, 125 mg, 200 mg, 400mg.

Dosage:

- The usual adult dose is 250 mg q 6 hr (can be doubled in severe infections).
- Children, 30-50 mg\kg in divided doses.

Erythromycin:

Drug Consideration:

- Should be taken on an empty stomach.
- In the treatment of sterptococcal infections, should be administered for at least 10 days.
- Don’t administer with fruit juice or other acidic drinks because it may decrease the activity of the drug.
- Erythromycin inhibits hepatic oxidation of many drugs. The clinically significant interactions are—rise in plasma levels of theophylline, carbamazepine, valproate, ergotamine and warfarin.

Clarithromycin

Trade name: Klaricare, klacid

- The antimicrobial spectrum of clarithromycin is similar to erythromycin.
- It is more active against sensitive strains of gram-positive cocci, Moraxella, Legionella, Mycoplasma pneumonia and Helicobacter pylori.

Clarithromycin

- **Bacteria that have developed resistance to erythromycin are resistant to clarithromycin also.**
- **Clarithromycin is more acid-stable than erythromycin, and is rapidly absorbed; oral bioavailability is ~50% due to first pass metabolism; food delays but does not decrease absorption.**
- **It has slightly greater tissue distribution than erythromycin and is metabolized by saturation kinetics—t½ is prolonged from 3–6 hours at lower doses to 6–9 hours at higher doses.**
- **An active metabolite is produced. About 1/3 of oral dose is excreted unchanged in urine, but no dose modification is needed in liver disease or in mild to- moderate kidney failure.**

Clarithromycin

Uses:

- indicated in upper and lower respiratory tract infections,
- sinusitis,
- otitis media,
- whooping cough,
- atypical pneumonia,
- skin and skin structure infections due to Strep.
- pyogenes and some Staph. aureus.
- Used as a eradicates H. pylori in 1–2 weeks.

Clarithromycin

Side effects:

- Similar to erythromycin, but gastric tolerance is better.
- High doses can cause reversible hearing loss.
- Few cases of pseudomembranous enterocolitis, hepatic dysfunction.
- Its safety in pregnancy and lactation is not known.
- The drug interaction potential is also similar to erythromycin.

Dose:

- 250, 500 mg BID for 7 days; severe cases 500 mg BD up to 14 days.
- 125, 250 mg/5 ml dry syr;

Azithromycin

Trade name: Azicare

- This new azalide congener of erythromycin has an expanded spectrum,
- Improved pharmacokinetics, better tolerability and drug interaction profiles.
- It is more active than other macrolides against H. influenzae, but less active against gram-positive cocci.
- High activity is exerted on respiratory pathogens—Mycoplasma, Chlamydia pneumoniae, Legionella, Moraxella and on others like Campylobacter, granulomatis, N. gonorrhoeae.
- it is not active against erythromycin resistant bacteria. Penicillinase producing Staph. aureus are inhibited but not MRSA.

Azithromycin

- Because of higher efficacy, better gastric tolerance and convenient once a day dosing, azithromycin is now preferred over erythromycin as first choice drug for infections such as: Legionnaires' pneumonia, Chlamydia trachomatis, urethritis
- The other indications of azithromycin are pharyngitis, tonsillitis, sinusitis, otitis media, pneumonias, acute exacerbations of chronic bronchitis, streptococcal and some staphylococcal skin and soft tissue infections.
- Other potential uses are in typhoid, toxoplasmosis and malaria.

Dose:

- 250, 500 mg cap and 250 mg per 5 ml dry once daily 1 hour before or 2 hours after food (food decreases bioavailability);

LINCOSAMIDE ANTIBIOTICS

Clindamycin:

Trade name: Dalacin C,

- **This potent lincosamide antibiotic is similar in mechanism of action (inhibits protein synthesis by binding to 50S ribosome) and spectrum of activity to erythromycin.**
- **It inhibits most gram-positive cocci (including penicillinase producing Staph., but not MRSA), C. diphtheriae, Nocardia, Actinomyces, Toxoplasma, but the distinctive feature is its high activity against a variety of anaerobes, especially Bact. fragilis. Aerobic gram-negative bacilli, spirochetes, Chlamydia, Mycoplasma and Rickettsia are not affected.**

Clindamycin:

- Oral absorption of clindamycin is good. It penetrates into most skeletal and soft tissues, but not in brain and CSF.
- It is largely metabolized and metabolites are excreted in urine and bile.
- Side effects are rashes, urticaria, abdominal pain, but the major problem is diarrhoea and pseudomembranous enterocolitis due to Clostridium difficile superinfection which is potentially fatal.
- The drug should be promptly stopped and metronidazole (alternatively vancomycin) given to treat it.

Clindamycin:

- Because of potential toxicity, use of clindamycin is restricted to anaerobic and mixed infections, especially by Bact. fragilis causing abdominal, pelvic and lung abscesses.
- It is generally combined with an aminoglycoside or cephalosporin.
- Metronidazole and chloramphenicol are the alternatives to clindamycin for covering the anaerobes
- It has also been employed for prophylaxis of endocarditis in penicillin allergic patients with valvular defects who undergo dental surgery, as well as to prevent surgical site infection in colorectal/pelvic surgery.
- Topically it can be used for infected acne vulgaris.

Clindamycin:

Dose:

- 150–300 mg cap. QID oral;
- 200–600 mg i.v. 8 hourly;
- Vaginal cream 2%, 1 application intravaginal at bed time for 7 consecutive days.
- Topical solution: 10mg /ml twice daily on affected area.

Quinolones, Folic Acid Antagonists, and Urinary Tract Antiseptics

Fluoroquinolones

- **These are synthetic antimicrobials having a quinolone structure that are active primarily against gram-negative bacteria, though newer fluorinated compounds also inhibit gram-positive ones.**

Quinolones, Folic Acid Antagonists, and Urinary Tract Antiseptics

- The first member Nalidixic acid had usefulness limited to urinary and GI tract infections because of low potency, modest blood and tissue levels, limited spectrum and high frequency of bacterial resistance.

- A quinolone derivatives called fluoroquinolones had high potency, expanded spectrum, and slow development of resistance, better tissue penetration and good tolerability.

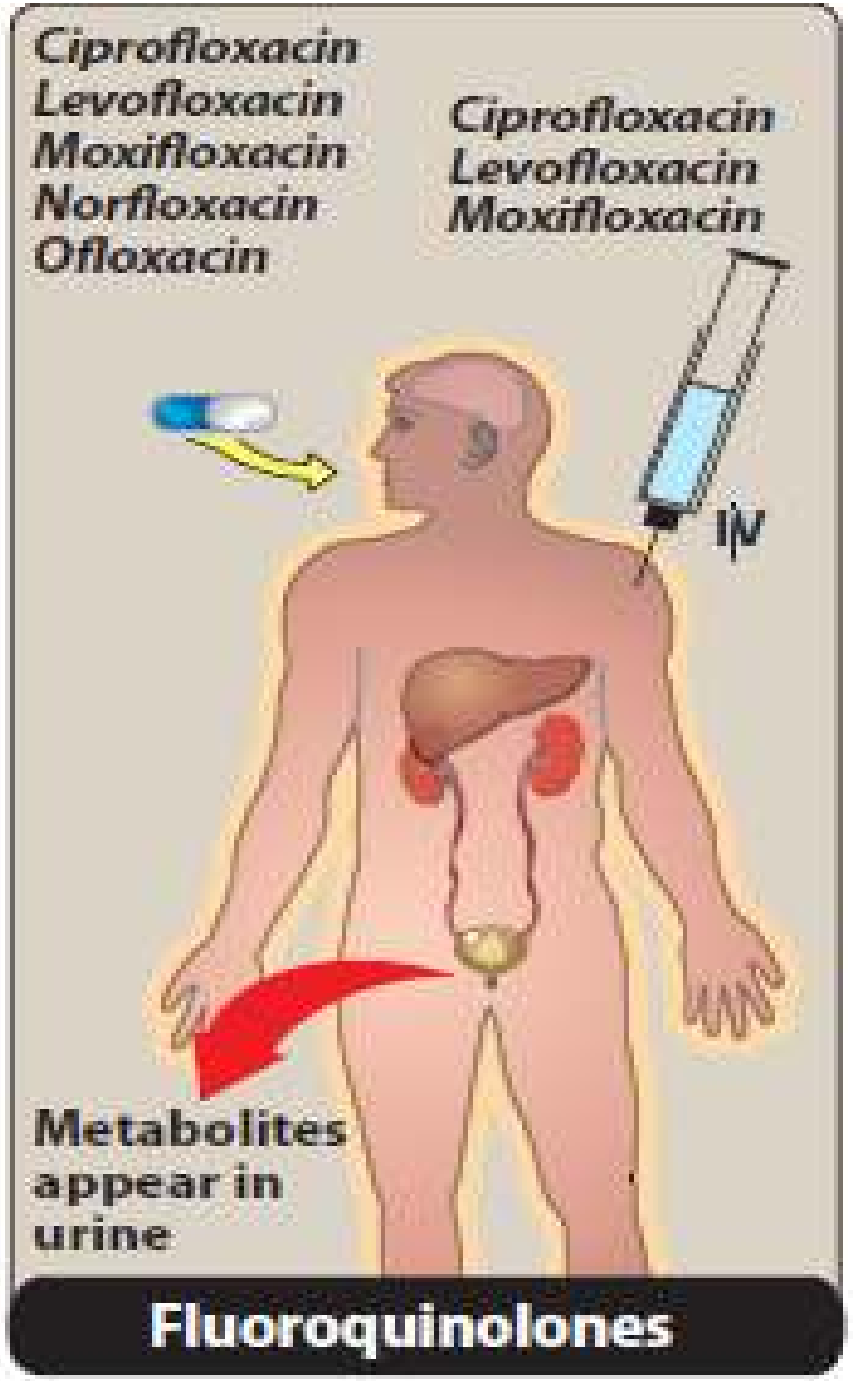
Classification of fluoroquinolones

First generation fluoroquinolones

- Norfloxacin
- Ofloxacin
- Ciprofloxacin

Second generation fluoroquinolones

- Levofloxacin
- Gatifloxacin
- Moxifloxacin


Ciprofloxacin
Levofloxacin
Moxifloxacin
Norfloxacin
Ofloxacin
Ciprofloxacin
Levofloxacin
Moxifloxacin
I/V
Metabolites appear in urine
Fluoroquinolones

Nalidixic acid

Trade name: Negram, Ugram

- It is active against gram-negative bacteria, especially coliforms: E. coli, Proteus, Klebsiella, Enterobacter, Shigella but not Pseudomonas.

Action:

- It acts by inhibiting bacterial DNA gyrase, urinary germicide.
- It is bactericidal.
- Resistance to nalidixic acid develops rather rapidly.

Nalidixic acid

Pharmacokinetics:

- Nalidixic acid is absorbed orally,
- highly plasma protein bound and
- partly metabolized in liver: one of the metabolites is active.
- It is excreted in urine.

NB. Concentration of the free drug in plasma and most tissues attained with the usual doses is nontherapeutic for systemic infections. However, high concentration attained in urine (20–50 times that in plasma) is lethal to the common urinary pathogens.

Nalidixic acid

Uses:

- Nalidixic acid is primarily used as a urinary antiseptic, generally as a second line drug in recurrent cases or on the basis of sensitivity reports.
- Nitrofurantoin should not be given concurrently—antagonism occurs.
- It has also been employed in diarrhoea caused by Proteus, E. coli, Shigella or Salmonella, and has a special place in ampicillin resistant Shigella enteritis.

Nalidixic acid

Contraindications:

- To be administered with caution in patients with: Liver disease, impaired kidney function, epilepsy, pregnancy, lactation, children, and cerebral arteriosclesois.

Side effect: Nausea, vomiting, diarrhea, seizure, headache dizziness, leukopnea, thrombocytopnea, hemolytic anemia due to G-6-PD deficiency.

Dose: 0.5–1 g TDS or QID 0.5 g tab, 0.3 g/5 ml syrup for 1-2 weeks.

Ciprofloxacin Hydrochloride:

It is the most potent first generation FQ active against a broad range of bacteria, the most susceptible ones are the aerobic gram-negative bacilli, especially the Enterobacteriaceae and Neisseria.

Ciprofloxacin Hydrochloride:

Pharmacokinetics:

- Ciprofloxacin is rapidly absorbed orally, but food delays absorption, and first pass metabolism occurs.
- It is high tissue penetrability: concentration in lung, sputum, muscle, bone, prostate and phagocytes exceeds that in plasma, but CSF and aqueous levels are lower.
- It is excreted primarily in urine, both by glomerular filtration and tubular secretion. Urinary and biliary concentrations are 10–50 fold higher than plasma.

Ciprofloxacin Hydrochloride:

Trade name: cipro, ciproxin, ciprocare.

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Uses:

- Acute otitis media
- UIT, infectious diarrhea
- Infection of lower respiratory tract, bone, joints & skin.

Ciprofloxacin Hydrochloride:

Contraindications:

- Hypersensitivity
- Children
- Lactation.
- Pregnancy

Side effects:

- Headache, dizziness, insomnia, fatigue, blurred vision.
- Nausea vomiting, dysphagia, crystalluria, hematuria. Because gut anaerobes are not affected diarrhea is infrequent
- Rashes, bad taste, GI bleeding, Headache, insomnia.

Dosage: 250-500 mg q 12 hr for 7-14 days.

Ciprofloxacin Hydrochloride:

Drug considerations:

- Arrange for culture and sensitivity test before beginning therapy.
- Give medication 2 hr after meals.
- Stress importance of drinking increased amounts of fluids to keep urine acidic & to minimize the risk of crystalluria.
- Plasma concentration of theophylline, caffeine and warfarin are increased by ciprofloxacin (also by norfloxacin and pefloxacin) due to inhibition of metabolism: toxicity of these drugs can occur.
- NSAIDs may enhance the CNS toxicity of FQs; seizures are reported.

Norfloxacin

- **It is less potent than ciprofloxacin: MIC values for most gram-negative bacteria are 2–4 times higher.**

- **Many Pseudomonas and gram positive organisms are not inhibited at clinically attained concentrations. Moreover, it attains lower concentration in tissues. It is metabolized as well as excreted unchanged in urine.**

Norfloxacin

Trade name: Apirol

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Norfloxacin

Indications:

- Norfloxacin is primarily used for urinary and genital tract infections.
- It is also good for bacterial diarrhoeas, because high concentrations are present in the gut and anaerobic flora is not disturbed.
- Norfloxacin is not recommended for respiratory and other systemic infections, particularly where gram-positive cocci are involved.

Dose: 200, 400, 800 mg tab, 3 mg/ ml eye drops, 100 mg/5 ml syr.

Ofloxacin

- This FQ is intermediate between ciprofloxacin and norfloxacin in activity against gram-negative bacteria, but it is comparable to or more potent than ciprofloxacin for gram-positive organisms and certain anaerobes.
- Good activity against Chlamydia and Mycoplasma has been noted:
- It is an alternative drug for nonspecific urethritis, cervicitis and atypical pneumonia.
- It also inhibits M. tuberculosis; can be used in place of ciprofloxacin.
- It is highly active against M. leprae: is being used in alternative multidrug therapy regimens.

Ofloxacin

Trade name: Taracin, oflox

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Pharmacokinetics:

- Ofloxacin is relatively lipid soluble; oral bioavailability is high: attains higher plasma concentrations.
- Food does not interfere with its absorption.
- It is excreted largely unchanged in urine; dose needs to be reduced in renal failure.

Ofloxacin

Indications:

- Ofloxacin is comparable to ciprofloxacin in the therapy of systemic and mixed infections.
- It is particularly suitable for chronic bronchitis and other respiratory or ENT infections.
- Inhibition of theophylline metabolism is less marked.
- Gonorrhoea has been treated with a single 200 mg dose.
- It is also useful in nongonococcal urethritis.

Dose:

- 100, 200, 400 mg tab; 200 mg/100 ml i.v. infusion, 50 mg/5 ml susp. 0.3% eye drops.

Levofloxacin

Trade name: levox

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Levofloxacin

Indications:

- The primary indication of levofloxacin is community acquired pneumonia and exacerbations of chronic bronchitis in which 87–96% cure rate has been obtained.
- High cure rates have been noted in sinusitis, enteric fevers, and pyelonephritis and skin/soft tissue infections as well.
- It is the levoisomer of ofloxacin having improved activity against Strep. Pneumonia and some other gram-positive and gram negative bacteria.

Levofloxacin

Indications:

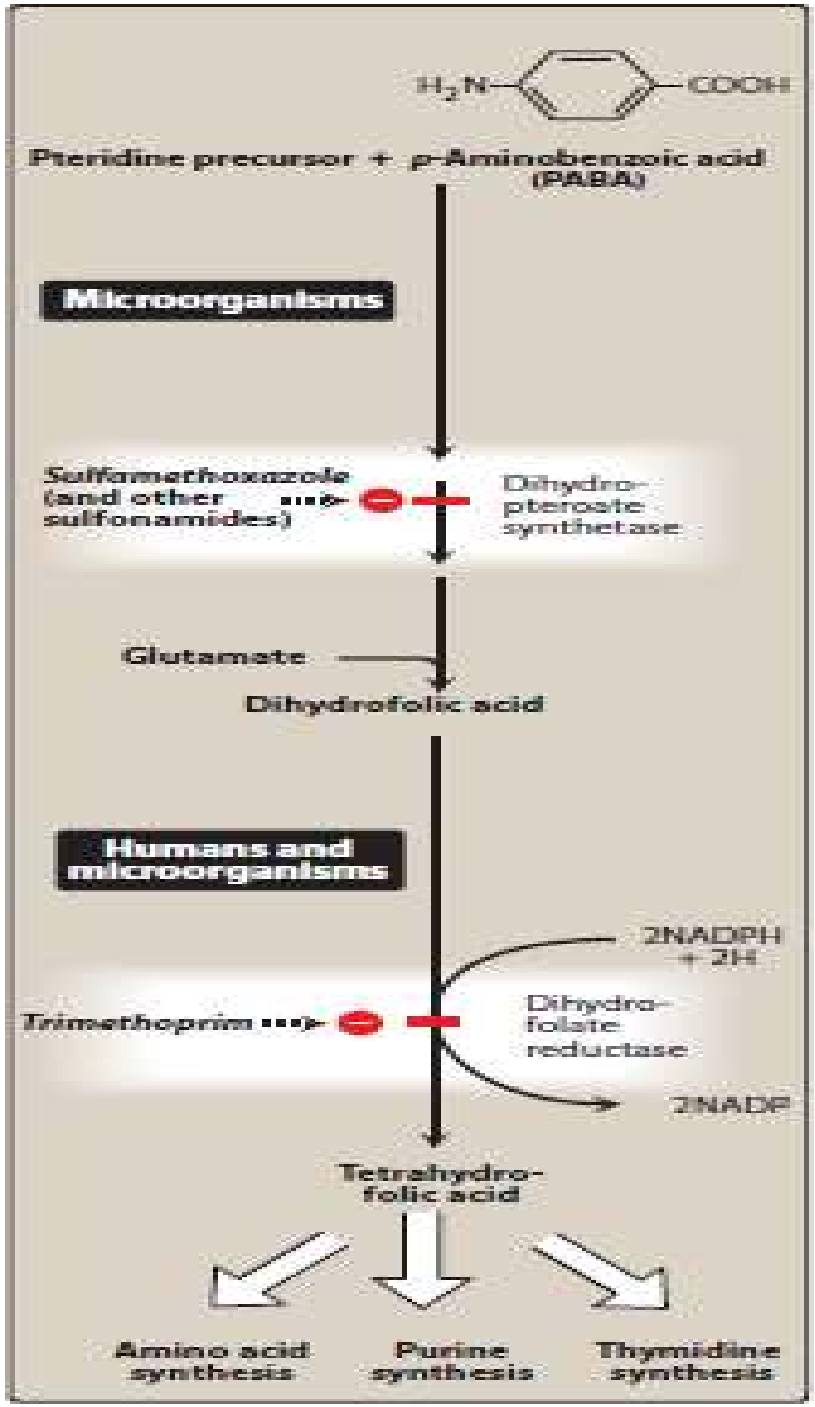
- **Anaerobes are moderately susceptible.**
- **Oral bioavailability of levofloxacin is nearly 100%; oral and i.v. doses are similar.**
- **It is mainly excreted unchanged and a single daily dose is sufficient because of slower elimination.**
- **Theophylline, warfarin, cyclosporine and zidovudine pharmacokinetics has been found to remain unchanged during levofloxacin treatment.**

Folic Acid Antagonists
Sulfonamides

Action:

- **sulfonamides are structurally related to para-aminobenzoic acid, the bacteria take it instead of these acids. Therefore, tetrahydrofolic acid is not formed which is necessary for synthesis of DNA; therefore, there will be no cell multiplication.**

Folic Acid Antagonists
Sulfonamides



Sulfonamides

- Sulfonamides were the first antimicrobial agents (AMAs) effective against pyogenic bacterial infections.
- because of rapid emergence of bacterial resistance and the availability of many safer and more effective antibiotics, their current utility is limited, except in combination with trimethoprim (as cotrimoxazole)
- Sulfonamides are primarily bacteriostatic against many gram-positive and gram-negative bacteria. However, bactericidal concentrations may be attained in urine.
- Relatively inexpensive drugs.
- Excreted mainly via kidneys.

Sulfonamides

Sulfonamides that are still of clinical interest are:

- **Short acting (4–8 hr): Sulfadiazine**
- **Intermediate acting (8–12 hr): Sulfamethoxazole**
- **Long acting (~7 days): Sulfadoxine, Sulfamethopyrazine**
- **Special purpose sulfonamides: Sulfacetamide sod., Mafenide, Silver sulfadiazine, Sulfasalazine**

Sulfonamides

Uses:

- Less useful because of bacterial resistance & development of other antibiotics.
- UTI, e.g. E. coli, proteus & staph. Aureus. “cephalosporines are given now”.
- To eliminate meningococci from nasopharynx (carriers).
- As alternative to penicillin for prophylaxis from rheumatic fever.
- As alternative to tetracycline in treatment of chlamydia, trachoma & conjunctivitis.
- In combination with penicillin to treat otitis media.

Sulfonamides

Contraindications:

- Newborns and infants less than 2 months.
- Pregnancy except for treatment of toxoplasmosis.
- Used with caution in patients with impaired liver & kidney function & blood disorders.

Sulfonamides

Side effects

- Nausea, vomiting, stomatitis, abdominal pain, headache, dizziness, ataxia, photosenitivity, rash.

- Allergic reactions: Steven –Johnson syndrome (rhinitis, conjunctivitis, fever, stomatites, & rash).

- Renal damage: crystaluria, hematuria, proteinuria. Adequate hydration and alkalinization of urine can prevent the problem by reducing the concentration of drug and promoting its ionization.

Sulfonamides

Side effects

- **Hemolytic anemia is encountered in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency. Super-infection.**

- **Kernicterus: This disorder may occur in newborns, because sulfa drugs displace bilirubin from binding sites on serum albumin.**

- **The bilirubin is then free to pass into the CNS, because the blood–brain barrier is not fully developed.**

Sulfonamides

Resistance to sulfonamides:

- Most bacteria are capable of developing resistance to sulfonamides.
- Prominent among these are gonococci, pneumococci, Staph. aureus, meningococci, E. coli, Shigella and some Strep. pyogenes, Strep. viridans and anaerobes.

The resistant mutants either:

a) produce increased amounts of PABA, or
b) their folate synthase enzyme has low affinity for sulfonamides, or
c) adopt an alternative pathway in folate metabolism.

Sulfonamides nursing consideration:

- Obtain a complete nursing and drug history.
- If the client is pregnant, the physician should be informed so that, another type of medication not harmful to a developing fetus may be used.

Reactions that may require withdrawal of the drugs:

- Skin rashes, abdominal pain & anorexia.
- Jaundice (hepatic involvement).
- Renal colic, oliguria, anuria, proteinuria (renal involvement).
- Jaundice, pallor, weakness (blood dyscrasias).
- Rhinitis, stomatitis, fever, headache, conjunctivitis (Steven- Johnson syndrome)
- Hemorrhage, due to vitamin K deficiency and it should be administered to patient take long-term sulfonamide since, it kills normal flora.

Sulfonamides

Silver sulfadiazine:

Trade name: Silvadene, Silverol cream.

Uses:

- Topically 1% cream for prevention and treatment of sepsis in second and third degree burns.
- Minor bacterial skin infections and dermal ulcers.
- it is not good for treating established infection

Dosage:

- Once or twice daily topically by sterile gloves over a clean or debrided burn.

Contraindications:

- Pregnancy, lactation, infant below 2 months.

Mafenide Acetate:

Trade name: Sulfamylon

Class: Topical sulfonamide.

Uses:

- for prevention of infection in 2nd and 3rd degree burns.
- limitation is that mafenide produces burning sensation and severe pain when applied to raw surface.

Contraindications:

- Not to be used for already established infection
- Used with caution in pregnancy & lactation.

Dosage: As silver sulfadiazine.

Sulfacetamide sodium :

Trade name: optizole eye drops, Ramasulf.

Class: sulfonamide.

Uses:

- Topically for treatment of ophthalmic infections including trachoma.

TRIMETHOPRIM

- A potent inhibitor of bacterial dihydrofolate reductase,
- Exhibits an antibacterial spectrum similar to that of the sulfonamides.
- Trimethoprim is most often compounded with sulfamethoxazole, producing the combination called **COTRIMOXAZOLE.**

COTRIMOXAZOLE

Sulfamethoxazole and trimethoprim:

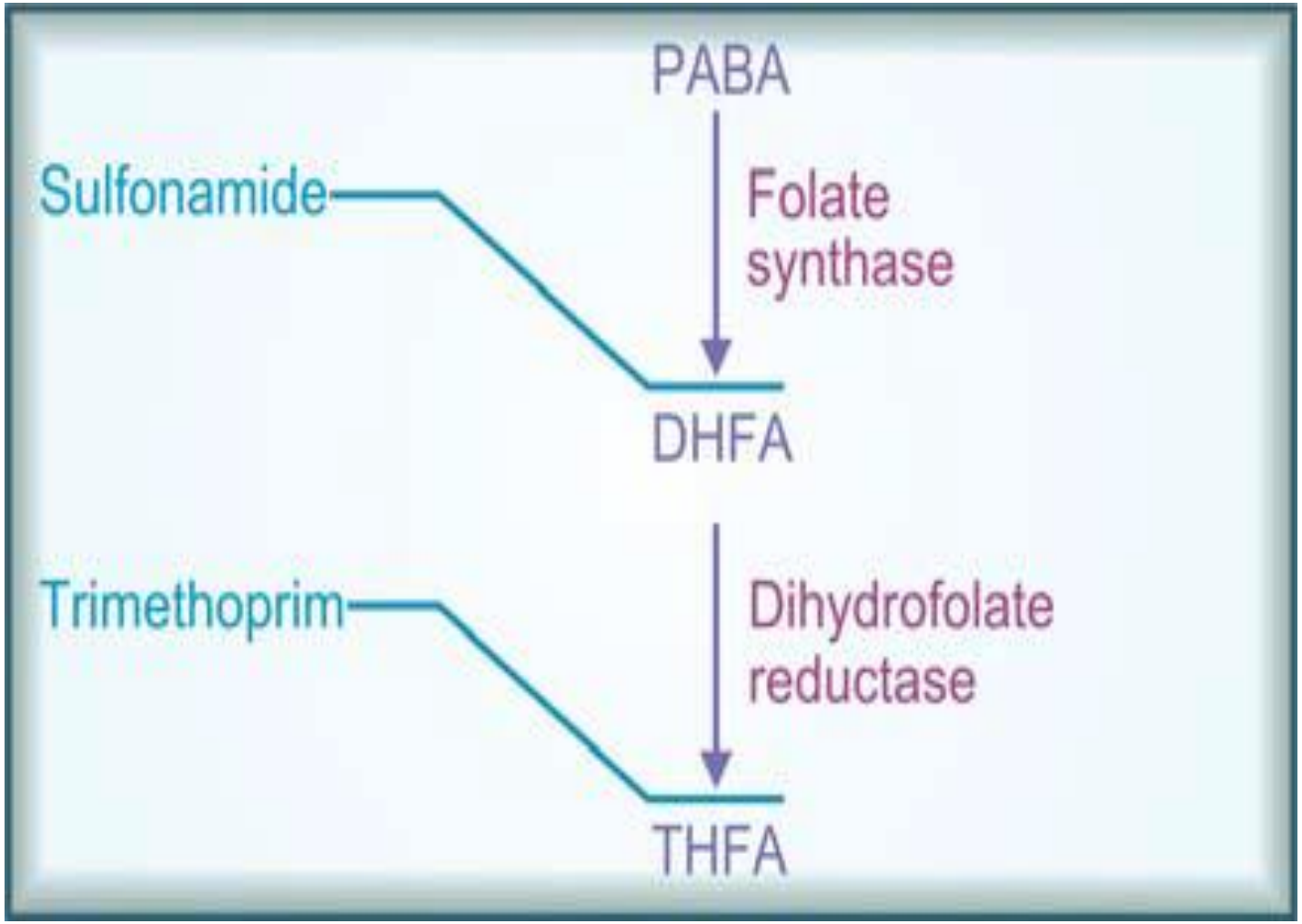
- The combination of trimethoprim with sulfamethoxazole, called cotrimoxazole , shows greater antimicrobial activity than equivalent quantities of either drug used alone.

Mechanism of action

- The synergistic antimicrobial activity of cotrimoxazole results from its inhibition of two sequential steps in the synthesis of tetrahydrofolic acid. Sulfamethoxazole inhibits the incorporation of PABA into dihydrofolicacid precursors, and trimethoprim prevents reduction of dihydrofolate to tetrahydrofolate

COTRIMOXAZOLE

Sulfamethoxazole and trimethoprim:



COTRIMOXAZOLE

Sulfamethoxazole and trimethoprim:

Trade name: Resprim, pathoprim, Bactil, septin, ultrasept

Dosage forms:

- Oral suspension: Sulfamethexazole 200 mg + trimethoprin 40 mg \5ml.
- Tablets: Sulfamethexazole 400 mg & trimethoprin 80/tab.
- Double strength tab. (forte): Sulfamethexazole 800 mg + trimethoprin 160 mg\tab.

Uses: UTI, otitis media , enterites (shigella), bronchitis (adult)

COTRIMOXAZOLE

Sulfamethoxazole and trimethoprim:

Additional Contraindications:

- Megaloblastic anemia due to folate deficiency.
- Infants less than one month of age.

Dose:

- **one or 2 tablets q 12 hours/adults, or 4 teaspoonful 12 hrs for 10-14 days.**

Miscellaneous Antibiotics

Vancomycin Hydrochloride:

Trade name: vancocin.

Class: Antibiotic, miscellaneous.

Action:

- It inhibits bacterial cell wall synthesis
- It is bactericidal for most organisms and bactereostatic for enterococci.
- It is poorly absorbed from GIT so it should be administered parenteral only for life threatening infections.

Miscellaneous Antibiotics

Vancomycin Hydrochloride:

Uses:

- Drug should be used for treatment of life threatening infections when other treatment have been ineffective.
- Patients with severe staphylococcal infections resistant or allergic to penicillins or cephalosporins such as: Endocarditits, Osteomylitis, Pneumonia, Septicemia.

Contraindications:

- Hypersensitivity

Miscellaneous Antibiotics

Vancomycin Hydrochloride:

Side effects:

- Ototoxicity (deafness).
- Nephrotoxicity (uremia).
- Red-Neck syndrome: “chills, erythema of neck and back fever”.
- Skin rashes, Drug fever
- Hypotension (due to rapid administration).
- Thrombophlepitis at site of injection.

Drug interactions:

- **Never give with other ototoxic or nephrotoxic agents especially aminoglycosides.**

Miscellaneous Antibiotics

Vancomycin Hydrochloride:

Dosage: I.V. 0.5 g q 6 hr or 1g q 12 hr.

Drug considerations:

- Mix as indicated on package insert.
- Intermittent infusion is the preferred route.
- Avoid rapid I.V. administration.
- Avoid extravastion during injections.

Urinary Germicides/Analgesics

1- Methanamine Hippurate : Hiprex.

Methanamine Mandelate : Mandelamin.

Class: urinary tract anti-infective.

Action:

- **The drug is converted in acid media into amonia and formaldehyde (the active principle) which denaturate protein. Effective when pH of urine is 5.5 or less (maintained) by hippurate or mandealte.**

Uses: acute, chronic & recurrent UTI specially gram negative organisms.

Urinary Germicides/Analgesics

1- Methanamine Hippurate : Hiprex.

Contraindications:

- renal insufficiency, liver damage, severe dehydration

Special concern: Pregnancy & gout.

Side effects:

- Nausea, vomiting, diarrhea, anorexia, proteinuria , crystaluria, urinary frequency & urgency.

Note:

- Na Hco3 decrease drug effect since it increases alkalinity of urine.

Urinary Germicides/Analgesics

2- Nalidixic Acid: discussed before

3- Nitrofurantoin: Macrofuran, Furadantin

Class: urinary germicide

Action:

- Interfere with bacterial metabolism.
- It also interferes with bacterial cell wall synthesis.
- It is Bacteriostatic at low concentration, bactericidal at high concentration.

Uses: severe UTI (pyelonephritis, cystitis).

Contraindications:

- Anuria, ologuria, impaired renal function. pregnancy, lactation, infant less than 1 month of age.

Urinary Germicides/Analgesics

3- Nitrofurantoin: Macrofuran, Furadantin

Special concerns:

- **Used with caution in patients with anemia, DM, electrolyte imbalance, avitaminosis, debilitating disease.**

Side effect:

- **Nausea, vomiting, pancreatitis, abdominal pain, leukopnea, megaloblastic anemia, headache, vertigo, hemolytic anemia in patients with G-6-PD deficiency.**

Dose: 100 mg QID with food for one week

Urinary Germicides/Analgesics

4- Norfloxacin: Apirol: discussed before

5- Phenazopyridine: sedural, utised

Class: Urinary analgesic.

Action: Azo dye has local anesthetic effect on UTI.

Uses: pain relief in chronic UTI, irritation, trauma instrumentation. (Not For Treatment of UTI).

Contraindications: Renal insufficiency.

Urinary Germicides/Analgesics

5- Phenazopyridine: sedural, utised

Side effects:

- **hemolytic anemia (G6 PD), nephrotoxicity, hepatotoxicity, nausea. yellow-brownish discoloration of urine.**

Note: *Yellow discoloration of skin and sclera indicates accumulation as a result of renal insufficiency.*

Dose: 200 mg t.i.d with or after meal.

Treatment of overdose: methylene blue IV or ascorbic acid P.O.

Urinary Germicides/Analgesics

5- Phenazopyridine: sedural, utised

Consideration:

- The drug may turn urine color into deep yellow or orange color.
- Renal function test is necessary to avoid toxicity.
- Encourage fluid intake.

Antiviral Drugs

- Viruses are obligate intracellular parasites.
- They lack both a cell wall and a cell membrane, and they do not carry out metabolic processes.
- Viruses use much of the host's metabolic machinery, and few drugs are selective enough to prevent viral replication without injury to the infected host cells.

Antiviral Drugs

- **Viral diseases is further complicated by the fact that the clinical symptoms appear late in the course of the disease, at a time when most of the virus particles have replicated. At this stage of viral infection, administration of drugs that block viral replication has limited effectiveness. However, some antiviral agents are useful as prophylactic agents.**

Antiviral Drugs

- Antiviral drugs are a class of medication used specifically for treating viral infections, Vaccines have been widely used to prevent certain viral infections e.g measles, small pox, polio etc.
- Most of the antiviral drugs now available are designed to help deal with HIV, herpes viruses, the hepatitis B and C viruses, and influenza A and B viruses.

Antiviral Drugs

1. Anti-Herpes virus: Idoxuridine, Acyclovir, Valacyclovir, Famciclovir
2. Anti-Retrovirus
    a. Nucleoside reverse transcriptase inhibitors (NRTIs): Zidovudine (AZT), Didanosine, Zalcitabine, Stavudine, Lamivudine, Abacavir
    b. Nonnucleoside reverse transcriptase inhibitors (NNRTIs): Nevirapine, Efavirenz,
    c. Protease inhibitors: Ritonavir, Indinavir, Nelfinavir, Saquinavir, Amprenavir, Lopinavir
3. Anti-Influenza virus: Amantadine, Rimantadine
4. Nonselective antiviral drugs: Ribavirin, Lamivudine, Adefovir dipivoxil, Interferon α

Antiviral Drugs

1. **Acyclovir : Zovirax, Virax**

Class. : Antiviral , anti-infective

Action:

- **Drug is converted to acyclovir triphosphate which interferes with herpes simplex virus DNA polymerase and therefore, inhibit DNA replication.**

1. Acyclovir

Uses:

- **P.O.: for initial and recurrent herpes infection in immunocompromised & nonimmunocompromised patients.**
- **Parenteral: initial therapy of severe genital herpes, varicella zoster infection in immunocompromised patients.**
- **Herpes simplex encephalitis.**
- **Topical: it decreases duration of healing in limited nonlife threatening infection.**

1. Acyclovir

Contraindications: Hypersensitivity.

Side effects:

- P.O: nausea, vomiting, anorexia, sore throat.
- Parenteral: phlebitis, hypotension , skin rash.
- Topical: Burning pain.

Dose:

- 200, 400, 800 mg tab.
- Caps. & tabs: Initial 200 mg /4 hrs for a total of 5 caps\day for 10 days.
- Susp. 200mg/5ml
- 3% eye oint, 5% skin cream;

2. Amantadine: Symmetrel

Class: Antiviral, antiparkinsonian, anti-influenza virus drug

Action:

- Prevent penetration of the virus into cell, may be by preventing uncoating of RNA.

Uses:

- Influenza A viral infection.
- Symptomatic treatment of idiopathic Parkinson.
- Parkinsonism syndrome postencephalitis.

2. Amantadine: Symmetrel

Contraindications:

- Hypersensitivity, epilepsy and other CNS disease, gastric ulcer, pregnancy.

Side effects:

- Nausea, vomiting, constipation, anorexia, CHF, leukopnea, dizziness, nightmares, lack of mental concentration

Dose: 100 mg BD, elderly—half dose, children 5 mg/kg/ day.

3. Idoxuridine:

Trade name: virusan oint. , iduridine eye drops.

Class: ophthalmic antiviral.

Uses: Herpes simplex keratitis.

Contraindications:

- Hypersensitivity, deep ulceration, and lactation.

Side effects:

- localized to eye (visual haze , irritation, pain , edema of eye lids, photosensitivity) .

4. Interferon α: intron A

- Interferons are low molecular weight glycoprotein cytokines produced by host cells in response to viral infections and some other inducers.
- They have nonspecific antiviral as well as other complex effects on immunity and cell proliferation.
- Interferons bind to specific cell surface receptors and affect viral replication at multiple steps: viral penetration, synthesis of viral mRNA, assembly of viral particles and their release, but the most widespread effect is direct or indirect suppression of viral protein synthesis.

4. Interferon α: intron A

- Three types of human interferons (α, β and γ) have been produced by recombinant DNA technology.

Uses:

- Chronic hepatitis B and C
- AIDS-related Kaposi's sarcoma (but not to treat HIV as such).
- H. simplex, H. zoster in immunocompromised patients
- Rhinoviral cold: intranasal interferon is prophylactic, but not beneficial in those already having cold.
- Interferon is not effective orally. Clinical utility of s.c. or i.m. injected interferon is limited by substantial adverse effects.

4. Interferon α: intron A

Adverse effects

- Flu-like symptoms—fatigue, aches and pains, malaise, fever, dizziness, anorexia, taste and visual disturbances: develop a few hours after each injection, but become milder later.
- Neurotoxicity—numbness, neuropathy, tremor, sleepiness, rarely convulsions.
- neutropenia, thrombocytopenia.
- Thyroid dysfunction (hypo as well as hyper).
- Hypotension, transient arrhythmias, alopecia and liver dysfunction.

Dose: Interferon α2B 3MU, 5MU vials for inj.

Antiviral drug	Mechanism of action	Viruses or diseases affected
Acyclovir	Metabolized to acyclovir triphosphate, which inhibits viral DNA polymerase	Herpes simplex, varicella-zoster, cytomegalovirus
Amantadine	Blockage of the M2 protein ion channel and its ability to modulate intracellular pH	Influenza A
Cidofovir	Inhibition of viral DNA polymerase	Cytomegalovirus; indicated only for virus-induced retinitis
Famciclovir	Same as *penciclovir*	Herpes simplex, varicella-zoster
Foscarnet	Inhibition of viral DNA polymerase and reverse transcriptase at the pyrophosphate-binding site	Cytomegalovirus, *acyclovir*-resistant herpes simplex, *acyclovir*-resistant varicella-zoster
Ganciclovir	Inhibits viral DNA polymerase	Cytomegalovirus
Interferon-α	Induction of cellular enzymes that interfere with viral protein synthesis	Hepatitis B and C, human herpesvirus 8, papilloma virus, Kaposi sarcoma, hairy cell leukemia, chronic myelogenous leukemia
Lamivudine	Inhibition of viral DNA polymerase and reverse transcriptase	Hepatitis B (chronic cases), human immunodeficiency virus type 1
Oseltamivir	Inhibition of viral neuraminidase	Influenza A
Penciclovir	Metabolized to penciclovir triphosphate, which inhibits viral DNA polymerase	Herpes simplex
Ribavirin	Interference with viral messenger RNA	Lassa fever, hantavirus (hemorrhagic fever renal syndrome), hepatitis C (in chronic cases in combination with *interferon-α* and in combination both with *interferon-α* and HCV protease inhibitor for HCV genotype I), RSV in children and infants
Rimantadine	Blockage of the M2 protein ion channel and its ability to modulate intracellular pH	Influenza A
Valacyclovir	Same as *acyclovir*	Herpes simplex, varicella-zoster, cytomegalovirus
Zanamivir	Inhibition of viral neuraminidase	Influenza A

Anti-tubercular Agents

Tuberculosis:

- it is acommunicable chronic granuloumatous disease caused by Mycobacterium tuberculosis.
- It usually involved the lung, but may affect any organ or tissue in the body.
- Take more than one drug.
- Every 3-4 months drugs are changed to prevent resistance.
- The long duration of drug therapy increases the risk for complications.

Anti-tubercular Agents

Tuberculosis:

- Drugs of T.B. are categorized as 1st line and 2nd line depending on their efficacy, activity & incidence of side effects.

  - 1st line drugs: Ethambutol, Isoniazide, Rifampin, Streptomycin.
  - 2nd line drugs: Capreomycin, Kanamyein, PAS.

Anti-tubercular Agents

Considerations:

- Conduct laboratory tests for nephrotoxicity, ototoxixity and hepatotoxicity caused by most of antilubercular agents.
- Incorporate safety precautions to protect the client manifesting vestibular difficulties during ambulation to prevent falls and injury.

Anti-tubercular Agents

Ethambutol Hcl:

Trade name: Myambutol.

Class: 1st line antitubercular agent.

Action:

- Arrest multiplication of rapidly dividing tubercle bacilli
- Absorbed from GIT & widely spread in body tissues except CSF.
- 65% of metabolized & unchanged drug execrated in urine & accumulate in patients with renal insufficiency.

Anti-tubercular Agents

Ethambutol Hcl:

Uses: Pulmonary T.B. in combination therapy.

Contraindications:

- Hypersensitivity, less than 13 years of age & preexisting optic neuritis.

Special concerns: pregnancy, impaired renal function, gout.

Side effects:

- Optic neuritis, loss of color (green) discrimination.
- Loss of vision (temporary).
- Peripheral neuritis (tingling), Hyperuricemia.

Note: side effects appear in early months, and then disappear.

Isoniazide: INH

- "Is the most effective tuberculostatic agent"

Class: 1st line antilubercular agent.

Action:

- It interfere with lipid & nucleic acid metabolism of growing bacteria, resulting in alteration of bacterial wall.
- Absorbed after oral or IM administration & is widely spread in body tissues.

.

Isoniazide: INH

Uses: T.B. caused by human, BCG., & Prophylaxis of T.B.

Contraindications: Hypersensitivity.

Side effects:

- peripheral neuritis, ataxia, jaundice optic neuritis and thrombocytopnea.

Note:

- Vitamin B-complex or pyridoxine 10-15mg \day may be given to CNS side effects.
- Liver function & ophthalmic test are recommended periodically.

Streptomycin sulfate:

Discussed before

Rifampin: **Rifadin, Rimactane**

Class: **1st line antitubercular agent.**

Action: **suppress RNA synthesis.**

- **Both bacteriostatic + bactericidal**
- **Well absorbed from GIT & widely distributed in body tissues.**
- **Metabolized in liver & execrated in feces.**

Rifampin: Rifadin, Rimactane

Uses:

- Pulmonary T.B. in combination therapy.
- In treatment of meningococcal carrier in Niesseria meningitis infection.

Contraindication: Hypersensitivity.

Special concerns:

- Lactation, pregnancy, children less than 5 years old, hepatic dysfunction.

.

Rifampin: Rifadin, Rimactane

Side effects:

- Nausea, vomiting, diarrhea, cramps, heartburn ataxia, jaundice, increase SGOT & SGPT., Hemolytic anemia.

Dose:

- T.B. 600 mg single dose daily.
- Meningitis 600 daily for 4 days

.

Aminosalicylate: PAS " para-aminosalicylic acid".

Class: 2nd line antitubercular.

Action:

- **Interfere with folic acid synthesis of susceptible tubercle.**
- **Bacteriostatic**
- **absorbed from GIT, 50% metabolized + execrated in urine.**

Uses: with Isoniazide or streptomycin in treatment of T.B.

Aminosalicylate: PAS " para-aminosalicylic acid".

Contraindication: Hypersensitivity.

Special concerns:

- Pregnancy, impaired liver & kidney function, peptic ulcer, congestive heart failure.

Side effect:

- nausea, vomiting diarrhea, thrombocytopnea, leukopnea, hemolytic anemia.

Antifungal Agents

- An antifungal agent is a drug that used to
  - Treat and prevent superficial mycoses such as athlete's foot, ringworm, candidiasis (thrush),
  - Serious systemic infections such as cryptococcal meningitis, and others.
- An accurate diagnosis of the infection before therapy is most important for the choice of the therapeutic agents.
- It is important that drug therapy be continued until the infectious agent has been completely eradicated to avoid the emergence of resistant strains.

1. Amphotericin B:

Trade name: Fungizone, Amphotec

Class: antifungal.

Action:

- It binds to fungal cell membrane in presence of ergosterol and alter cell membrane and causing loss of cellular macromolecules and ions producing irreversible damage.
- Fungistatic or fungicidal "depending on the dose". It is poorly absorbed from the GIT, It can be administered I.V., intrathecally or topically.

1. Amphotericin B:

Uses:

- The drug is toxic and should be used only for patients under close medical supervision with a relatively certain diagnosis of deep mycotic infections.
- I.V. administration is usually reserved for life threatening diseases because it is toxic.
- Systemic fungal infections, e.g. histoplasmosis, disseminated candidiasis & monilial over growth resulting from oral antibiotic therapy.
- Topical coetaneous & mucocutaneous infections of candida "monilia" infections.

1. Amphotericin B:

Contraindications: Hypersensitivity, lactation

Side effects:

- Topical use: pruritis, dry skin, and irritation, rash.
- IV use: nausea, vomiting, diarrhea, melena, abdominal cramps.
- Fever, headache, thrombophlebitis, bone marrow depression, pancytopnea. anaphylaxis

2. Nystatin:

Trade name: Mycostatin.

Class: antifungal.

Action: similar to Amphotricin B.

Uses:

- candida infections of the skin, mucous membranes, GI tract, vagina & mouth (thrush).

Side effects:

- nausea, vomiting, diarrhea with oral adminesration.

2. Nystatin:

Dosage forms:

- Suspension: contains 100,000 units\ ml “12 ml bottle with dropper”.
- Ointment: contains 100,000 units\g “15 g tube”.
- Vaginal tab.: Contains 100,000 units.

Dosage:

- Suspension: For adult, 2 ml, 4 times daily.
- Pediatric: 1 ml 4 times daily.
- Vaginal cream or tab., 1 tab. inserted into vagina once or twice each day for 2 weeks.

2. Nystatin:

Considerations:

- Don’t mix oral suspension in foods since the mediation will be inactivated.
- Instruct client to keep medication in mouth as long as possible before swallowing.
- Insert vaginal tab. high in vagina with an applicator.
- Treatment should be continued for at least 48 hr after clinical cure has been achieved to prevent relapse.
- Discontinue topical or vaginal administration if rash or sensitivity occurs.

3. Tolnaftate:

Trade name: pitrex.

Class: Topical antifungal.

Action:

- The exact mechanism is not known although the drug is thought to stunt mycelial growth causing antifugal effect.

Uses:

- Tinea pedis , tinea cruris, tinea corpuris & tinea versicolor.
- Fungal infections of moist skin areas.

3. Tolnaftate:

Contraindications:

- Scalp & nail infections “ avoid getting into eyes”

Side effects:

- mild skin irritation.

Dosage form:

- cream tubes of 10 g. Solution Bottles of 10 ml.

Dosage:

- Apply bid. For 2-3 weeks.

4. Miconazole:

Trade name: Daktarin, Dactazol.

Class: antifungal agent.

Actions:

- it alter fungal cell membrane permeability by blocking biosynthesis of fungal lipids especially ergosterol in cell membrane

4. Miconazole:

Uses:

- Systemically: fungal infection e.g. mucocutaneous candiasis.
- Topically: Skin, nail, hair infections, tinea pedis (athlete's foot) and tinea corpuris.
- Oral thrush (oral gel).

Contraindication: **Hypersensitivity.**

4. Miconazole:

Side effects:

- Topical: skin rashes, headache, burning, and irritation.
- Systemic: nausea, vomiting, diarrhea, anorexia fever, thrombocytopnea.

Dosage forms:

- Cream: tubes of 15g or 30g.
- Vaginal Ovule 50, 150mg
- Oral gel: tubes of 40 g.

Dosage:

- Topical: apply to cover affected areas in morning & evening.
- Vaginal cream\supp. One supp. daily at bedtime for 7 days.

5. Ketoconazole:

Trade name: Nizoral, ketoral.

Class.: Broad- spectrum antifungal.

Action: similar to miconazole.

Uses: candidiasis, oral thrush.

Contraindications: Hypersensitivity, Lactation, Ophthalmic use.

5. Ketoconazole:

Side effects:

- nausea, vomiting, diarrhea,
- fever, headache, photophobia,
- Hepatotoxicity,
- Gynecomastia, impotence,
- thrombocytopnea.

Dosage:

- tab. oral suspension:
- 200 – 400 mg. As a single dose\day.

6. Clotrimazole

Trade name: Canesten, Mycoten.

Class.: Broad-spectrum fungistatic agent.

- Topical drug for vaginal and oral candidiasis.

Action: similar to miconazole.

6. Clotrimazole

Uses:

- Local treatment of oropharyngeal candidiasis
- topically for fungal infection in otitis externa
- Vaginal candidiasis.
- Skin infections: for athlete's foot and candidal intertrigo

Contraindications: Hypersensitivity

Side effects:

- nausea, vomiting, diarrhea,
- fever, headache, photophobia,
- Hepatotoxicity

Anthelmintic

- The term anthelmintic is restricted to drugs acting locally to expel parasites from gastro intestinal tract.
- There are several types of worms which penetrate other tissues.
- All family members should be examined since infestations are transmitted by sharing bathrooms (hygiene is very important).

A. Anthelmintic

A. Worms divided into 3 groups:
   A. Cestodes:
   B. Tape worm e.g. taenia saginata (beef), taenia solium (pork).

   C. Nematodes:
      A. 1. Filaria: filariasis
      B. 2. Hook worm: ancylostoma.
      C. 3. Pinworm: oxyuriasis (common in school age).
      D. 4. Round worm: ascaris (can cause GI & respiratory obstruction).
      E. 5. Threadworm: strongoloid.
      F. 6. Whipworm.

   D. Trematodes: Bilharziasis

A. Anthelmintic

Considerations:

Instruct responsible family member how to prevent infestation with pinworm by :

- **Washing hands after toiling and before meals.**
- **Keep nails short.**
- **Applying antipruritic ointment to anal area to reduce scratching which transfers pinworms.**
- **Alert all family members to be examined for pinworms.**
- **Emphasize the need for follow-up examinations to check the results of treatment.**

1. Mebendazole:

Trade name: Vermox, Vermacare
Class: anthelmintic.
Action: By blocking the glucose uptake of the organism which leads to worm death.
Uses: whipworm, pinworm, roundworm.
Contraindications: Hypersensitivity.
Special concerns: Pregnancy, children less than 2 years.

1. Mebendazole:

Side effects:

- nausea, vomiting, transient abdominal pain, diarrhea.

Dosage: for pinworm, whipworm, roundworm, hookworm 1 tablet (100 mg) in the morning and evening on 3 consecutive days.

Note: All treatments can be repeated after 2-3 weeks.

2. Niclosamide:

Trade name: Yomesan

Class: anthelmintic.

Action:

- Inhibit oxidative phosphorylation in the mitochondria of the worm.
- The drug is not absorbed from GIT and execrated in feces.

2. Niclosamide: Yomesan

Uses: For tape worn only.

Special concerns:

- pregnancy, lactation, children under 2 years.

Side effects:

- nausea, vomiting, anorexia, bad month odor, shin rash, alopecia, dizziness, fever, sweating.

Dose: 2g as a single dose (4 tabs X 0.5 g).

3. Piperazine:

Trade name: ascarol, vermizine

Class: anthelmintic

Action:

- Paralyze the muscle of the parasite, which leads to dislodge & excretion of parasite.
- Absorbed via GIT, metabolize in liver & excreted in urine.

Uses: Pinworm (oxyuriasis), round worm (ascariasis).

- Recommended for pediatric use.

3. Piperazine:

Contraindications:

- impaired liver & kidney function, seizures, and hypersensitivity.

Side effects:

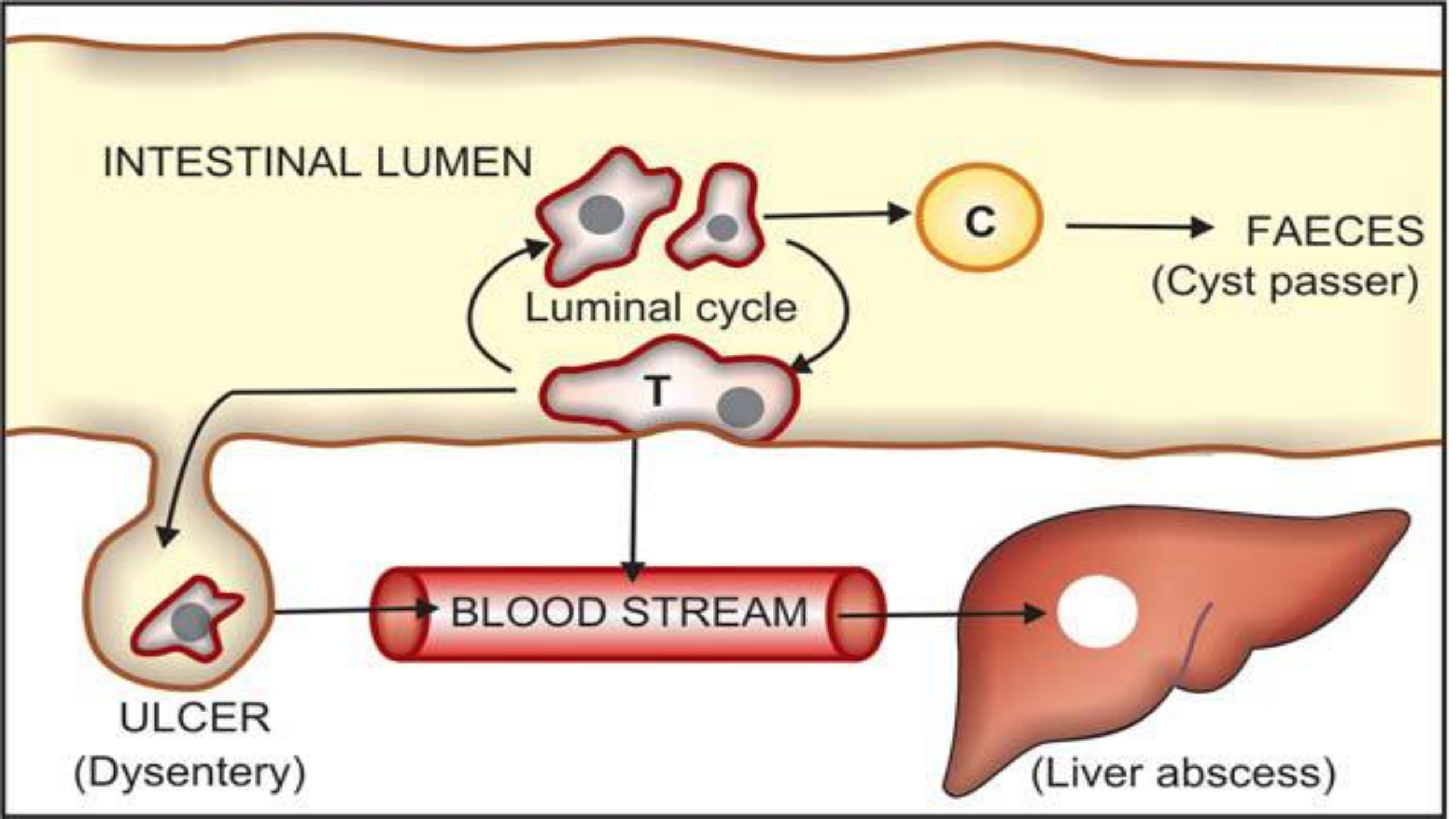
- nausea, vomiting, diarrhea, tremors, ataxia, blurring of vision, cataract, bronchospasm, rash.

Dosage: 75mg/kg orally for two successive days.

Disease/worm	Drugs used
'Thread' worm infection (*Enterobius vermicularis* in UK, *Strongyloides stercoralis* in US)	Mebendazole, Albendazole, Piperazine For US 'threadworm': **Ivermectin**
Common round worm (*Ascaris lumbricoides*)	Mebendazole, Levamisole, Piperazine
Whip worm (*Trichuris trichiura*)	Mebendazole
Tape worm infection: *Taenia saginata* (beef tapeworm), *Taenia solium* (pork tape worm)	Praziquantel
Cysticercosis (infection with larval *Taenia solium*)	Albendazole, Praziquantel
Hydatid disease (*Echinococcus granulosus*)	Albendazole
Bilharziasis caused by blood flukes /schistosomes (*Schistosoma haematobium, S. mansoni, S. japonicum*)	Praziquantel
Hookworm infection (*Ankylostoma duodenale, Necator americanus*)	Mebendazole
Cutaneous larva migrans (*Ankylostoma caninum*)	Albendazole, **Ivermectin**
Visceral larva migrans (*Toxacara canis*)	Albendazole

Amebicides drugs

- **Antiamoebic drugs:** these are drugs useful in infection caused by the protozoa entamoeba histolytica.
- Amoebiasis has a worldwide distribution (over 40 million people are infected), but it is endemic in other developing countries.
- Poor environmental sanitation and low socio-economic status are important factors in the spread of the disease, which occurs by faecal contamination of food and water.


INTESTINAL LUMEN
C
FAECES
(Cyst passer)
Luminal cycle
T
BLOOD STREAM
ULCER
(Dysentery)
(Liver abscess)

Amebicides drugs classification

1. Tissue amoebicides
    a) For both intestinal and extraintestinal amoebiasis:
        - Nitroimidazoles: Metronidazole, Tinidazole, Secnidazole, Ornidazole, Satranidazole
        - Alkaloids: Emetine, Dehydroemetine
    b) For extraintestinal amoebiasis only: Chloroquine

2. Luminal amoebicides
    a) Amide : Diloxanide furoate, Nitazoxanide
    b) 8-Hydroxyquinolines: Quiniodochlor (Iodochlorohydroxyquin, Clioquinol), Diiodohydroxyquin (Iodoquinol)
    c) Antibiotics: Tetracyclines

1- Metronidazole: Flagyl

Class: systemic trichomonocide, amebicide.

- Effective against anaerobic bacteria & protozoa.
- Well absorbed from GIT & widely distributed in tissues.
- Eliminated in urine (primarily), 20% unchanged, resulting in red, brown discoloration in urine following P.O. or I.V. use.

Action:

- Inhibit growth of trichomona & amebae by binding to DNA & inhibit nucleic acid synthesis; resulting in cell death.

1- Metronidazole: Flagyl

Uses:

1- Amebiasis, trichomoniasis.
2- Amebic dysentery.
3- Amebic liver abscess.
4- Septicemia
5- Endocarditis
6- Giardiasis
7- To control anaerobic infections of the abdomen following colorectal surgery, emergency appendectomy.

1- Metronidazole: Flagyl

Contraindications:

- Active organic disease of CNS.
- Blood disorders
- Lactation.
- 1st trimester of pregnancy.
- Topical Hypersensitivity.

1- Metronidazole: Flagyl

Side effects:

- Dry mouth, metallic taste, diarrhea, dizziness abdominal discomfort, furry tongue,
- Ataxia, vertigo & leukopenia.



Adverse effects of *metronidazole*.

1- Metronidazole: Flagyl

Dose: 500-750 mg, 3 times daily for 5-10 days.

Considerations:

- If used IV, drug should not be given by IV bolus.
- The drug may turn urine brown.
- Explain for the male partner, the necessity to have therapy

2- Tinidazole:

Trade name: Tinogyn

It is an equally efficacious congener of metronidazole, similar to it in every way except:

- **Metabolism is slower; t½ is ~12 hr;**
- **Duration of action is longer; dosage schedules are simpler. Thus, it is more suited for single dose or once daily therapy.**
- **Some comparative trials in amoebiasis have reported higher cure rates.**

2- Tinidazole:

- It appears to be better tolerated; the incidence of side effects is lower: metallic taste (2%), nausea (1%), rash (0.2%).
- Dosage form: 300, 500, 1000 mg tabs; 800 mg/400 ml i.v. infusion
- Anaerobic infections: prophylactic—2 g single dose before colorectal surgery;
- H. pylori: 500 mg BD for 2 weeks in triple combination.

3- Diloxanide furoate:

Class: amebicide

Action: probably directly amebicidal but exact mechanism is unknown.

Uses:

- It is a highly effective luminal amoebicide: directly kills trophozoites responsible for production of cysts.
- Diloxanide furoate exerts no antibacterial action.

3- Diloxanide furoate:

- It is less effective in invasive amoebic dysentery, because of poor tissue amoebicidal action. However, a single course produces high (80–90%) cure rate in mild intestinal amoebiasis and in asymptomatic cyst passers.

- It is primarily metabolized by glucuronidation and is excreted in urine.

3- Diloxanide furoate:

Side effects:

- **GIT: flatulence, nausea, vomiting, diarrhea, dryness of mouth, abdominal cramps.**
- **Renal: proteinuria.**
- **Skin: urticarial, tingling sensation.**

Contraindications: **hypersensitivity, pregnancy and children under two years.**

Dose: **500mg 3 time daily with meals for 10 days.**

4- Halogenated Hydroxyquinolines:

Iodoquinol

Mechanism of Action:

- Unknown
- They are effective against organisms in the GIT only not intestinal walls or liver.

Uses:

- Luminal amebiasis
- Eradication of infection given along with tissue amebicide (metronidazole)

4- Halogenated Hydroxyquinolines:

Pharmacokinetics:

- Absorption is poor (90%); they are excreted in feces.
- 10% enters the circulation, and they are excreted as glucouronides in urine.
- They have a half-life of 11-14 hours.

4- Halogenated Hydroxyquinolines:

Adverse Effects:

- Peripheral neuropathy including optic neuritis
- Nausea, vomiting, diarrhea
- Enlargement of the thyroid gland
- Agranulocytosis
- Iodine sensitivity
- They interfere with thyroid function tests (increase protein-bound serum iodine, decrease in measured 131I uptake).

4- Halogenated Hydroxyquinolines:

Contraindications:

- Optic neuropathy
- Thyroid disease
- Sensitivity to iodine
- Severe liver disease

Dosage:

- Tablet 200mg, 3.5% suspension
- mixed with other drugs for treatment of bacterial diarrhea, giardiasis, amebiasis, trichomoiasis

5- Emetine & Dehydroemetine:

- **Emetine hydrochloride is an alkaloid plant derived from ipeca.**
- **Dehydroemetine is a synthetic analogue.**

5- Emetine & Dehydroemetine:

Pharmacokinetics:

- Erratic oral absorption
- They’re preferably given subcutaneously, but it could be given intramuscularly.
- THEY COULD NEVER EVER BE GIVEN INTRAVENOUSLY.
- Emetine is concentrated in the liver, lungs, spleen, kidneys, cardiac muscles, and intestinal walls.
- Emetine is metabolized and excreted slowly via the kidneys; it has a cumulative effect.
- Trace amounts could be detected in urine 1-2 months after the last dose.
- Emetine should not be used for more than 10 days (usually 3-5 days).

5- Emetine & Dehydroemetine:

Pharmacological Actions:

- Emetine acts on trophozoites causing irreversible block of protein synthesis.

Clinical Uses:

- Amebic liver abscess
- Intestinal wall infections
- Severe forms of acute amebic dysentery (dehydroemetine and tetracyclines for a short period followed by metronidazole)

5- Emetine & Dehydroemetine:

Adverse Effects:

- Dehydroemetine is less toxic than emetine.
- Pain at the site of injection, abscesses
- Nausea, vomiting, and diarrhea
- Neuromuscular weakness
- Serious toxicities (Cardiotoxicity)
  - Arrhythmias
  - Hypotension
  - Congestive heart failure

5- *Emetine & Dehydroemetine:*

Contraindications:

- **Heart disease**
- **Renal disease**
- **Pregnancy**
- **Children**

Clinical Syndrome	Drug
Asymptomatic Cyst Carrier	Iodoquinol
	Paromycin
	Diloxanide furoate
Diarrhea/Dysentery Extraintestinal	Metronidazole + Iodoquinol
	Paromycin
	Diloxanide furoate
Amebic Liver Abscess	Chloroquine + Metronidazole
	Emetine

Miscellaneous drugs:

Gentian Violet

Class: Topical & vaginal anti-infective.

Action:

- **Rosaniline dye is effective against some gram positive bacteria, many fungi & many strains of candida.**
- **Treatment should continue until symptoms subside.**

Gentian Violet

Uses:

- Topically for treatment of cautoneous and mucocautaneous condida albicans as thrush.
- Vulvovaginal candidiasis.
- Candidiasis in general.

Contraindications:

- Hypersensitivity.
- Ulcerative lesions of face and vagina.

Gentian Violet

Side effects:

- Irritation, ulceration of mucus membranes, laryngitis, esophagitis, laryngeal obstruction, & vaginal irritation.

Dose:

- Vaginal; one tampon (5 mg), inserted for 3 hours once or twice daily for 12 consecutive days.
- Topical solution: 1% - 2% solution to affected area 2-3 times for 3 days.

Povidone Iodine: Betadine

Class: Antiseptic, germicide.

Action:

- Is a non-stinging, non-staining iodine complex with all of the antiseptic properties of iodine but without skin & mucus membrane irritation.

- Bactericidal for gram positive & negative bacteria, antibiotic resistant organisms, fungi, viruses, protozoa & yeast .

Povidone Iodine: Betadine

Uses:

- Topical dressing
- Antiseptic for wounds & burns.
- Degerming of skin
- Preoperatively.
- Treatment of dandruff.

Contraindications: skin hypersensitivity.

Dose: on full strength (solution or ointment) only once.

Cardiovascular system drugs

- Components of the cardiovascular system include the heart, arteries, capillaries, veins, and lymphatics.

- These structures transport life-supporting oxygen and nutrients to cells, remove metabolic waste products, and carry hormones from one part of the body to another.

- Because this system performs such vital functions, any problem with the heart or blood vessels can seriously affect a person's health.

Types of drugs used to improve cardiovascular function include:

- inotropic drugs
- antiarrhythmic drugs
- antianginal drugs
- antihypertensive drugs
- diuretics
- antilipemic drugs.

Diuretics

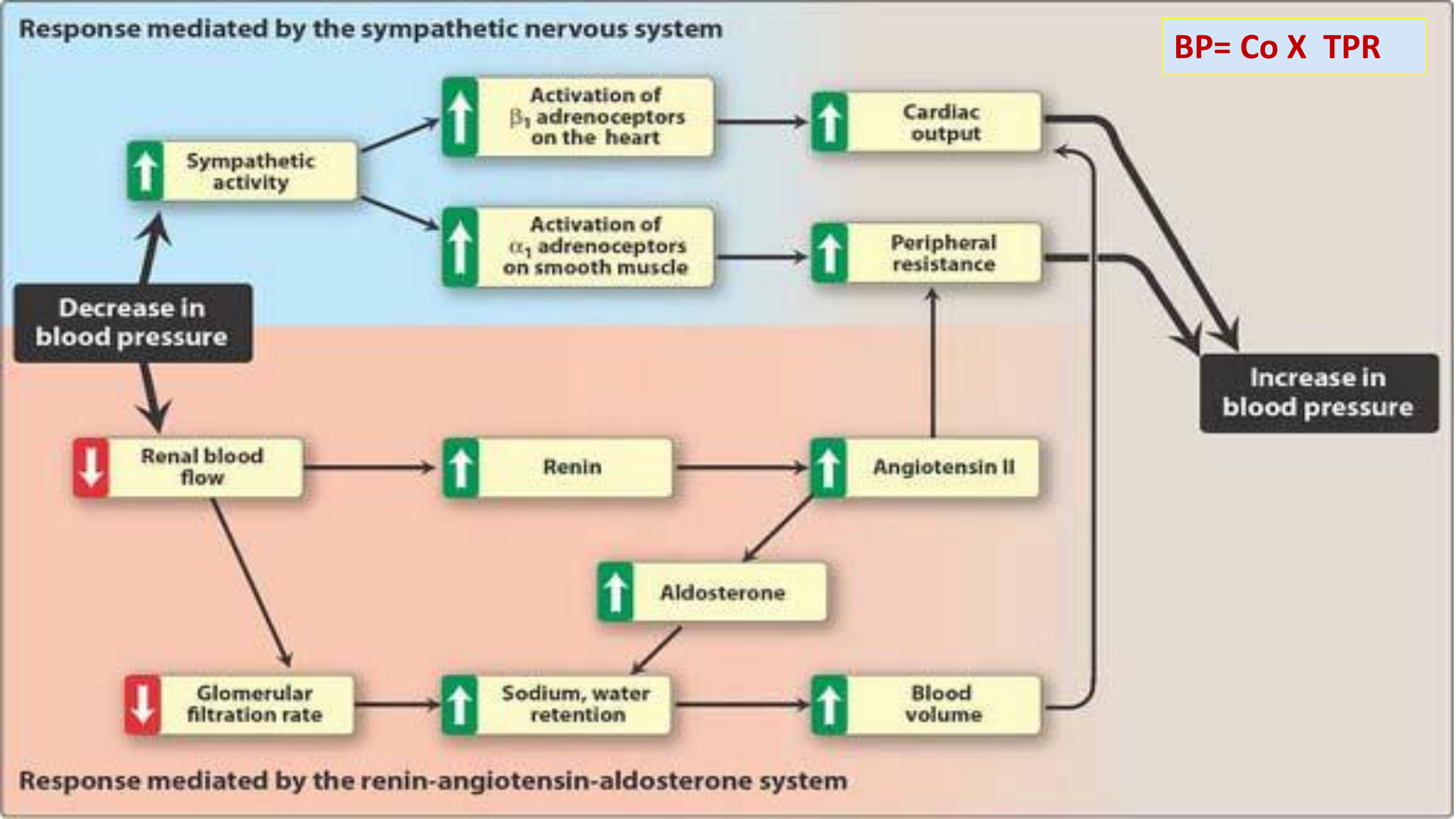
- A **diuretic** **is any drug that elevates the rate of urination and thus provides a means of forced diuresis.**
- Diuretic drugs include several classes of diuretics that reduce blood pressure by increasing urine output and decreasing edema, circulating blood volume, and cardiac output.

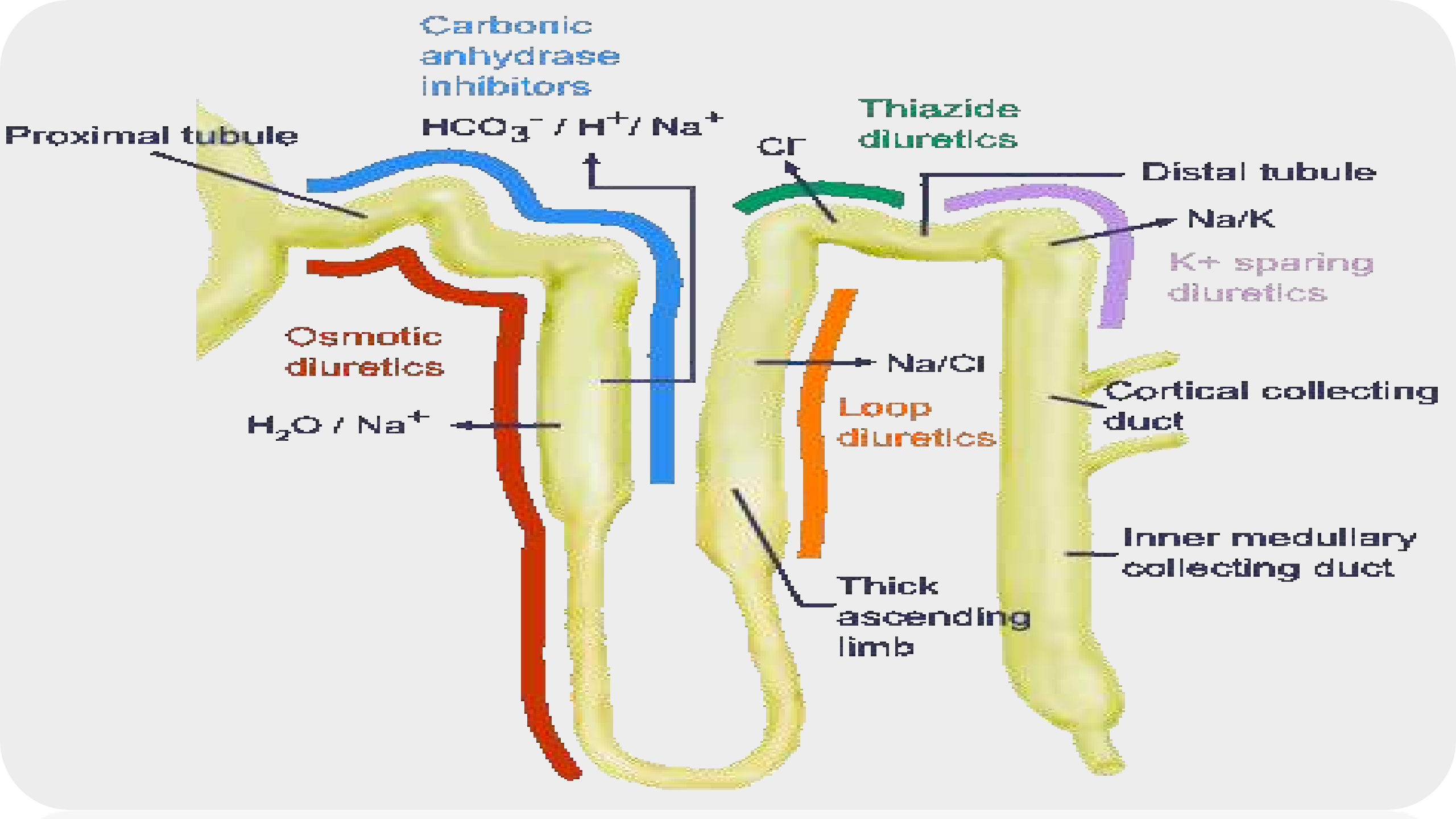
Action of diuretics:

It increases the urinary output of water and sodium "prevention or treatment of edema" through one of the following mechanisms:

1. Increasing the glomerular filtration rate.
2. Decreasing the rate at which sodium is reabsorbed from the glomerular filtrate by the renal tubules, therefore water is excreted along with sodium.
3. Promoting the excretion of sodium & therefore water by the kidney.

- **Uses:** Congestive heart failure, hypertension, edema.


Response mediated by the sympathetic nervous system
BP= Co X TPR
Activation of β_1 adrenoceptors on the heart
Cardiac output
Sympathetic activity
Activation of α_1 adrenoceptors on smooth muscle
Peripheral resistance
Decrease in blood pressure
Increase in blood pressure
Renal blood flow
Renin
Angiotensin II
Aldosterone
Glomerular filtration rate
Sodium, water retention
Blood volume
Response mediated by the renin-angiotensin-aldosterone system


Carbonic anhydrase inhibitors
$HCO_3^- / H^+ / Na^+$
Proximal tubule
Cl⁻
Thiazide diuretics
Distal tubule
Na/K
K+ sparing diuretics
Osmotic diuretics
H_2O / Na^+
Na/Cl
Loop diuretics
Cortical collecting duct
Inner medullary collecting duct
Thick ascending limb

1- Carbonic Anhydrase Inhibitor Diuretics:

Generic name: Acetazolamide

Trade name: Diamox, uramox.

Class. : anticonvulsant, diuretic.

Therapeutic actions:

- **In the kidneys, carbonic anhydrase inhibitors decrease the availability of hydrogen ions, which blocks the sodium-hydrogen exchange mechanisms. As a result, urinary excretion of sodium, potassium, bicarbonate, and water increases.**
- **Inhibition of carbonic anhydrase in the eye results in decreased secretion of aqueous humor results in lowering of intraocular pressure**
- **CNS inhibition of carbonic anhydrase and resultant diuresis may decrease abnormal neuronal firing**

1- Carbonic Anhydrase Inhibitor Diuretics:

Indications:

- **Lowering of intraocular pressure in the treatment of glaucoma**
- **Management of acute altitude sickness**
- **Diuretic**
- **Adjunct to the treatment of refractory seizures**
- **Prevention of renal calculi composed of uric acid or cystine**

1- Carbonic Anhydrase Inhibitor Diuretics:

Contraindications:

- ❑ **Low serum level of sodium & potassium.**
- ❑ **Renal & hepatic dysfunction.**
- ❑ **Adrenal insufficiency.**
- ❑ **Hypersensitivity to thiazide diuretics.**

Side effects: hypokalemia, anorexia, polyuria, drowsiness, confusion, & acidosis.

Dosage forms: Tablets, 125 mg, 250 mg, ER capsules: 500 mg
Injection: 500 mg/vial

Dose: Tab. 4-30 mg\kg\day in divided doses.

2- Loop Diuretics:

Generic name: Furosemide

Trade name: Fused, Lasix

Class: Loop diuretic.

Action:

- **Inhibits sodium and chloride reabsorption in the ascending loop of Henle, thus increasing renal excretion of sodium, chloride, and water**
- **Increases excretion of potassium (like thiazide diuretics)**
- **Produces greater maximum diuresis and electrolyte loss than a thiazide diuretic**

2- Loop Diuretics:

Indications

- Oral, IV: Edema associated with CHF, cirrhosis, renal disease
- IV: Acute pulmonary edema
- Oral: Hypertension.

Contraindications

- Allergy to furosemide, sulfonamides.
- Anuria
- Severe renal failure
- Hepatic coma
- Pregnancy and Lactation.

2- Loop Diuretics:

Side effects:

- **Dehydration, hypovolemia.**
- **Hypokalemia ,hyperglycemia, Hyponatremia**
- **Nausea, vomiting, diarrhea, anorexia.**
- **Tinnitus, ototoxicity, blurring of vision, headache, orthostatic hypotension, rashes & photosensitivity.**
- **Thrombophlebitis & cardiac arrest after I.V use.**
- **Pain at injection site after I.M use.**
- **Metabolic alkalosis**

2- Loop Diuretics:

Forms: tablets 40 mg, Ampules 20 mg /2ml.

Dose:

- Oral: 20-80 mg as a single dose.
- I.V: 20-40 mg as a single dose.
- For hypertensive crisis:100-200 mg.

2- Loop Diuretics:

Nursing considerations:

1. **Observe client for signs of dehydration & circulatory collapse.**
2. **Monitor pulse & blood pressure.**
3. **Advise the client to take medication in the morning to avoid interruption of sleep.**
4. **Monitor for adverse effects, such as pancreatitis, hematologic disorders, and electrolyte imbalances (especially hypokalemia).**
5. **Signs of hypokalemia may include leg cramps and muscle aches and discuss the need for a diet high in potassium.**

3- Osmotic Diuretics

Generic name: Mannitol

Trade name: Osmitrol

Class: Osmotic diuretic.

Action:

- Increase the osmolality of the glomerular filtrate, which decrease the reabsorption of water while increasing the excretion of sodium and chloride.
- It also increases the osmolality of the plasma which cause increase flow of water from tissues to the interstitial fluid & plasma, thus cerebral edema, increased intracranial pressure & CSF volume & pressure are reduced.

3- Osmotic Diuretics

Uses:

- Acute renal failure
- Cerebral edema
- To decrease intracranial pressure
- Glaucoma

Contraindications:

- Anemia
- Dehydration
- pulmonary edema

3- Osmotic Diuretics

Side effects:

Hypernatremia , electrolyte imbalance, acidosis, Dehydration, dry mouth , thirst, edema, hypotension , blurring of vision , headache , dizziness.

Dose:

50 ml of 25%, 75 ml of 20 % or 100 ml of 15% solution infused over 3-5 minutes.

3- Osmotic Diuretics

Nursing considerations:

1. Mannitol should not be added to other I.V. solutions nor should it be mixed with other medications .
2. If blood is to be administered at the sometime , add 20 mEq of sodium chloride to each liter of mannitol to prevent pseudoagglutination.
3. Monitor & record vital signs .
4. Observe for signs of electrolyte imbalance or dehydration.
5. Observe for signs, & symptoms of pulmonary edema (dyspnea,cyanosis, frothy sputum). “Slow the rate & notify the physician”.
6. Assess for signs of orthostatic blood pressure changes, which could lead to falls and injuries.

4- Potassium – Sparing Diuretics:

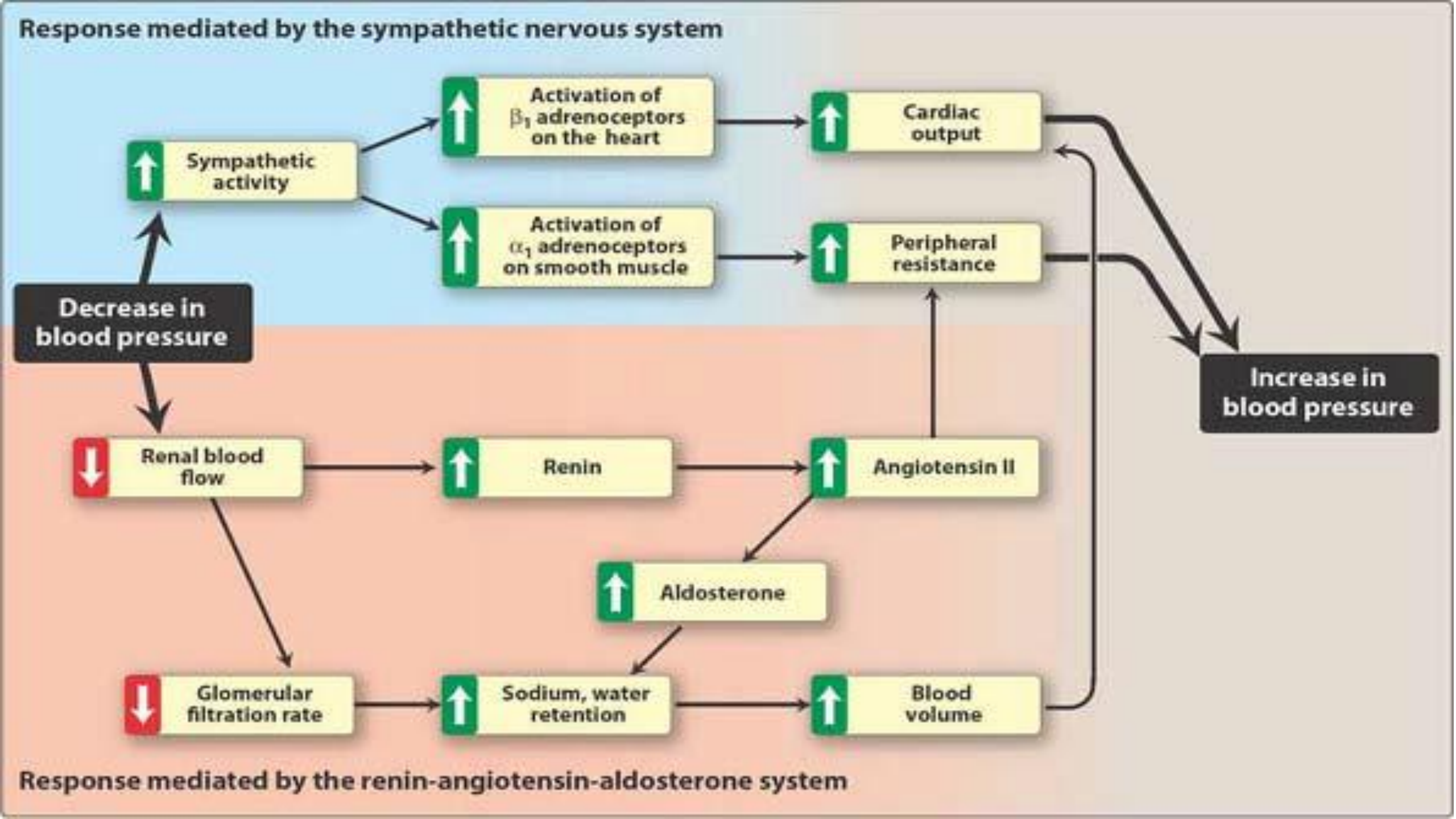
Generic name: Spironolactone

Trade name: Aldactone.

Class: Diuretic – potassium sparing.

Action:

- **Is a mild diuretic that acts on the distal tubule to inhibit sodium exchange for potassium which results in increased secretion of sodium and water & conservation of potassium**
- **It is also aldosterone antagonist.**
- **It has slight antihypertensive effect.**
- **It also interferes with syntheses of testosterone & may increase formation of estradiol from testosterone thus leading to endocrine abnormalities.**


Response mediated by the sympathetic nervous system
Activation of β_1 adrenoceptors on the heart
Cardiac output
Sympathetic activity
Activation of α_1 adrenoceptors on smooth muscle
Peripheral resistance
Decrease in blood pressure
Increase in blood pressure
Renal blood flow
Renin
Angiotensin II
Aldosterone
Glomerular filtration rate
Sodium, water retention
Blood volume
Response mediated by the renin-angiotensin-aldosterone system

4- Potassium – Sparing Diuretics:

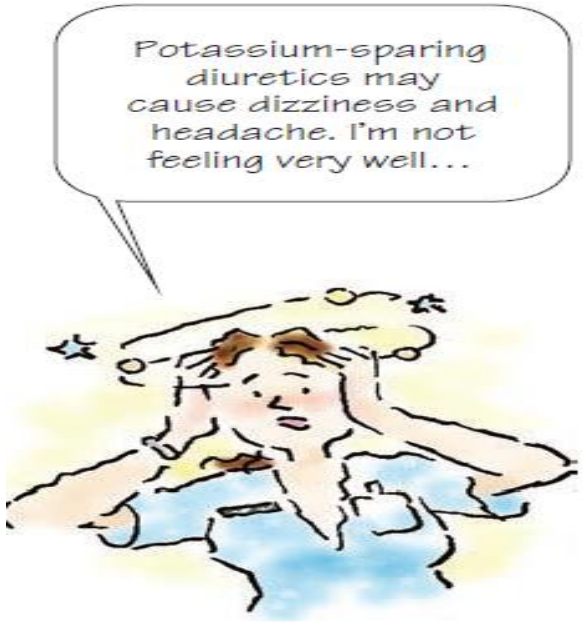
Uses:

- Edema due to congestive heart failure
- Liver cirrhosis.
- Nephrotic syndrome .
- Essential hypertension.
- Primary hyperaldosteronism.
- Hypokalemia (as in CHF).

4- Potassium – Sparing Diuretics:

Contraindications:

- Acute renal insufficiency.
- Progressive renal failure.
- Patients receiving potassium supplement.
- Hyperkalemia.



Side effects:

- Hyperkalemia, hyponatremia (dry mouth, lethargy, thirst & easy fatigability). Metabolic acidosis
- Vomiting, diarrhea, cramps.
- Menstrual irregularities, gynecomastia, hirsutism & deeping of voice, impotence.
- Skin rashes & breast carcinoma.

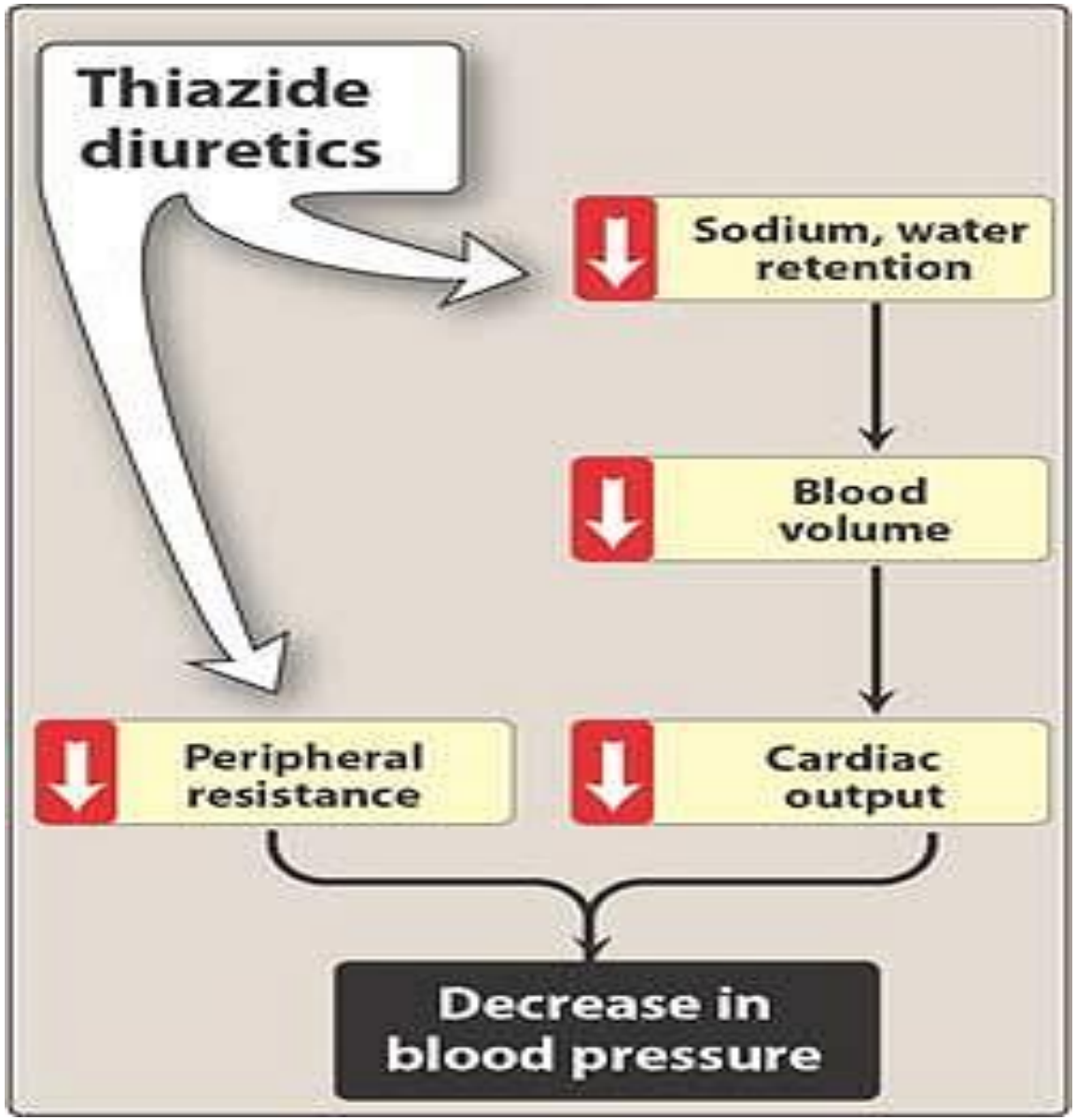
4- Potassium – Sparing Diuretics:

Dose: 100 mg (1 tab) per day.

Nursing consideration:

- Protect drug from light.
- Food may increase absorption of aldactone.
- Obtain serum electrolyte levels prior to starting therapy.
- Record vital signs, intake & output & body weight.
- Advise the client to avoid food high in potassium.

5- Thiazides & related diuretics:



5- Thiazides & related diuretics:

Generic name: Hydrochlorothiazide

Trade name: Esidrex.

Action:

- Increases renal excretion of sodium, chloride, water, potassium, and calcium
- Increases bicarbonate, magnesium, phosphate, bromide, and iodide excretion
- Decreases excretion of ammonia, causing increased serum ammonia levels

5- Thiazides & related diuretics:

Uses:

Edema due to congestive heart failure , nephrosis , liver cirrhosis & renal failure.

Contraindications:

- hypersensitivity
- impaired renal function
- Edema due to toxemia of pregnancy (adverse of effect on the newborn) .
- Hypokalemia.

5- Thiazides & related diuretics:

Side effects:

- Hypokalemia (cardiac arrhythmias).
- Hyponatremia (nausea, vomiting, lethargy, epigastric distress).
- Dry mouth, diarrhea, and easy fatigability.
- Skin rashes, muscle cramps.
- Hyperglycemia.
- Metabolic alkalosis
- Hyperglycemia

Nursing considerations: See Furosemide (lasix)

Side-Effects of Diuretics

Thiazide Diuretics	Loop Diuretics	K-Sparing Diuretics	CarbonicAhydrase
Hypo-K	Hypo-K	Hyper-K	Hypo-K
Hypo-Na	Hypo-Mg	Metabolic Acidosis	Metabolic Acidosis
Hypovolemia	Hypovolemia	Gynecomastia	
Hypotension	Hypotension		
Hyperuricemia	Hyperuricemia		
Hyperglycemia	Ototoxicity		
Azotemia	Metabolic Alkalosis		
Metabolic Alkalosis			

Potassium Chloride

Trade name: KCL for IV preparation, Slow K for oral preparations.

Classification: Electrolyte, mineral.

Uses:

- Patients receiving high doses of potent diuretics.
- Patients having secondary hyperaldosteronism.
- When there is excessive loss of potassium in feces.
- When potassium intake is not adequate (especially in elderly).

Potassium Chloride

N.B. maintenance of adequate potassium level is necessary for patients receiving digoxin to avoid digitalis toxicity.

Dose:

- **Available in slow release tablets of 600mg/tablets and in ampules of 20 mmol/10 ML.**
- **Vary according to patients' needs**
- **Maintenance dose for patients who receive normal diet is 2-4 gm daily in divided doses.**
- **IV: 20 mmol is added to 500 ML of IV solution (D5W or NS) and to infuse slowly over 2-3 hours.**

Potassium Chloride

Contraindications:

- Severe renal failure.
- If serum potassium level is over 5 mmol/liter.

Side effects:

- Hyperkalemia
- Phlebitis at injection site.
- Heart toxicity if infused rapidly.
- With oral tablet: Dyspepsia, nausea and vomiting, esophageal or bowel ulceration.

Potassium Chloride

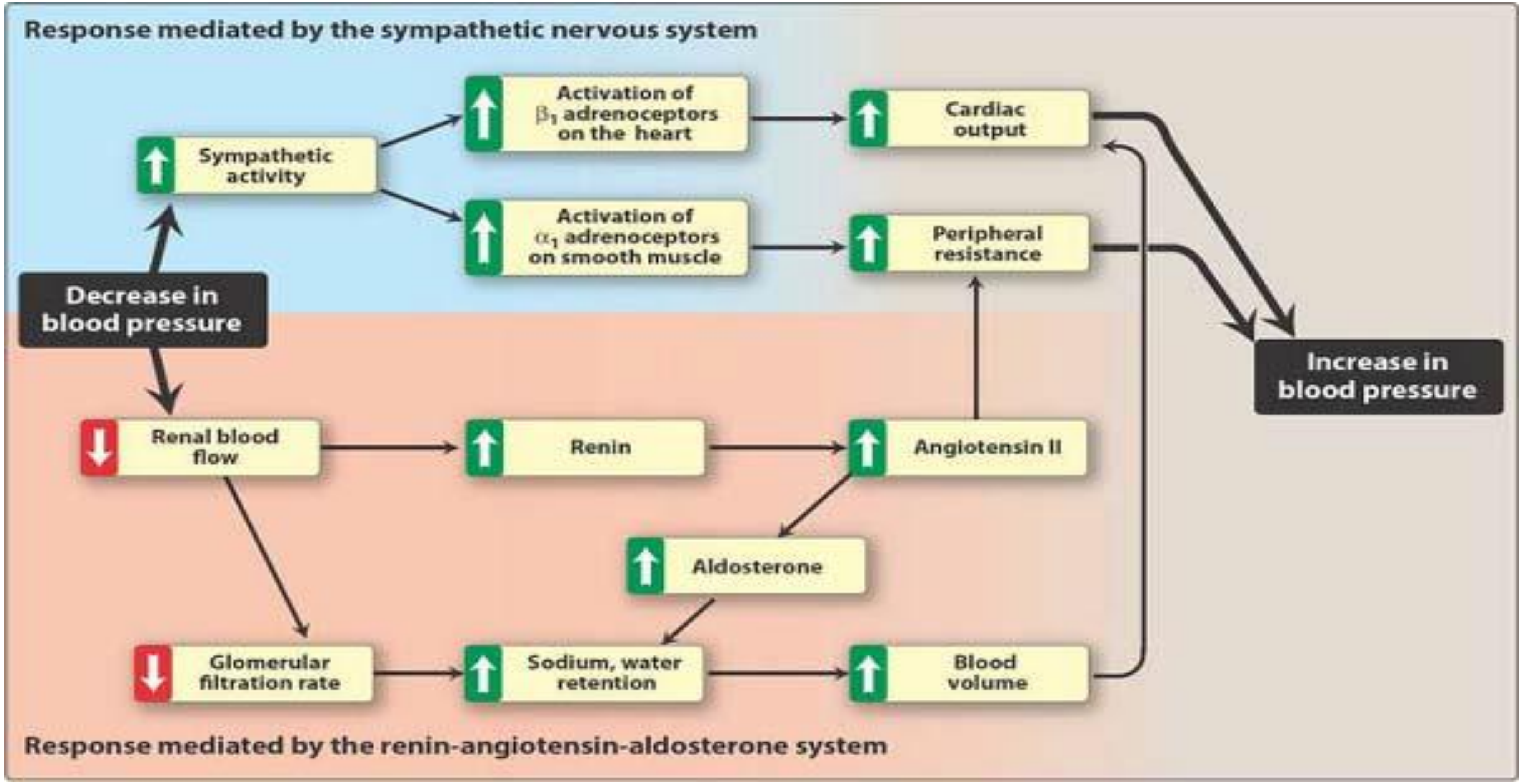
Nursing considerations:

- Mix solution well.
- Administer solution slowly (500 ML of solution over 2-3 hours)
  Monitor serum potassium level.
- Assess insertion site for signs of phlebitis.
- Give oral preparation on full stomach.

B: drugs that alter sympathetic nervous system.

1. **Beta-adrenergic blocking agents: discussed before**
2. **Centrally acting agents:**
   Used for treatment of chronic hypertension, since it affects CNS.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



Antihypertensive

Methyldopa:

Trade name: (Aldomin)

Classification: Antihypertensive, centrally acting antiadrenergic agent.

Action:

- **Converted to alpha methyl norepinephrine**, which is stored in adrenergic nerve granules where it replaces norepinephrine and is released by nerve stimulation to interact with presynaptic central alph2 adrenoceptors and decrease sympathetic outflow and decrease arterial pressure.
- **Inhibit dopa decarboxylase** lead to decrease stores of norepinephrine in the sympathetic nervous system and decrease BP.

Antihypertensive

Uses: Hypertension & hypertension crises (parenteral).

Side effects:

- Headache, dizziness, general weakness, depression and sedation.
- Bradycardia, orthostatic hypotension.
- Dry mouth, nausea, vomiting, sore (black) tongue.
- Jaundice, liver disorders.
- Hemolytic anemia & leukopenia.
- Male impotence.

Antihypertensive

Contraindications:-

- Hypersensitivity,
- Active hepatic diseases.
- Pheochromocytoma.

Dosage: Tabs. : initially 250 mg bid or tid.

Antihypertensive

Nursing considerations:-

1. Dose increases should be made with the evening dose to minimize drowsiness
2. Avoid activities that need mental awareness such as driving.
3. Note any evidence of jaundice and do liver function test on intervals.
4. Advise pt. to rise from the bed slowly.
5. Instruct pt. about reportable Signs & Symptoms.
6. Explain to pt. that urine rarely may be turn into dark / red-black color.
7. Advise patient to consult health care professional before taking any Rx, OTC, or Herbal products, especially cough, cold, or allergy remedies.

Antihypertensive

C: Direct acting vasodilators.

- **Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large arteries, smaller arterioles and large veins.**

- **The process is essentially the opposite of vasoconstriction, or the narrowing of blood vessels. When vessels dilate, the flow of blood is increased due to a decrease in vascular resistance. Therefore, dilation of arterial blood vessels (mainly arterioles) leads to a decrease in blood pressure.**

Antihypertensive

C: Direct acting vasodilators.

Hydralazine hydrochloride:

Trade name: Apresoline

Classification: Antihypertensive, direct action on vascular smooth muscles.

Action: directly affect smooth muscles results in vasodilation and increase a cardiac output and blood flow to the brain and kidneys.

Antihypertensive

C: Direct acting vasodilators.

Uses:

- Management of moderate to severe hypertension.
- Management of congestive heart failure.
- Management of hypertension secondary to pre-eclampsia/eclampsia.
- Given parenterally in hypertension emergencies.

Antihypertensive

C: Direct acting vasodilators.

Contraindications:

- Angina pectoris.
- Rheumatic heart disease.
- Chronic glomerulonephritis.
- Systemic lupupus erthrmatosis (S.L.E.)

Side effects:

- Orthostatic hypotension, tachycardia, nausea, vomiting.
- Headache, dizziness, constipation and male impotence.

Antihypertensive

C: Direct acting vasodilators.

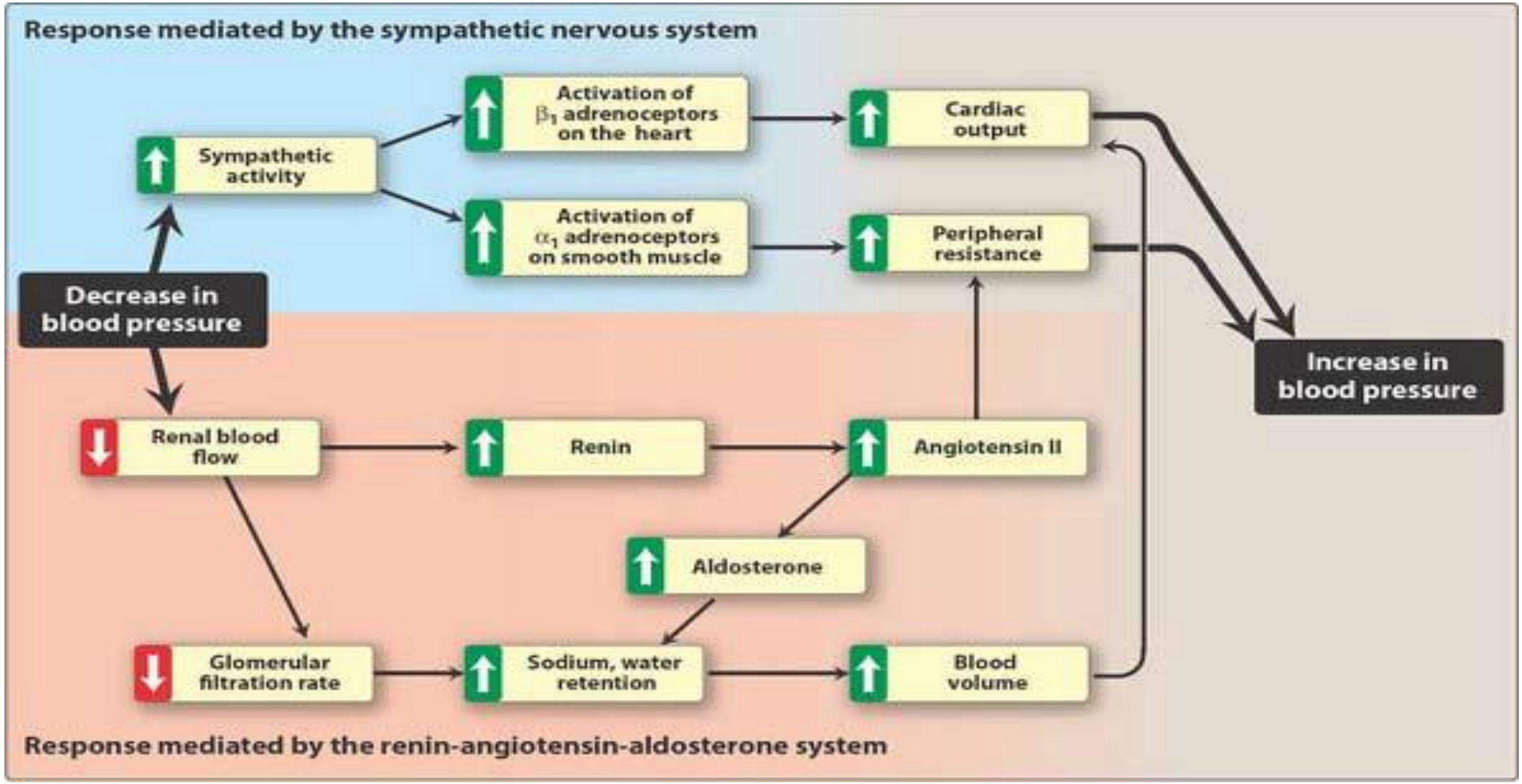
Available forms: Injection, solution, as hydrochloride: 20 mg/mL (1 mL), Tablet as hydrochloride: 10 mg, 25 mg, 50 mg, 100 mg

Dose:

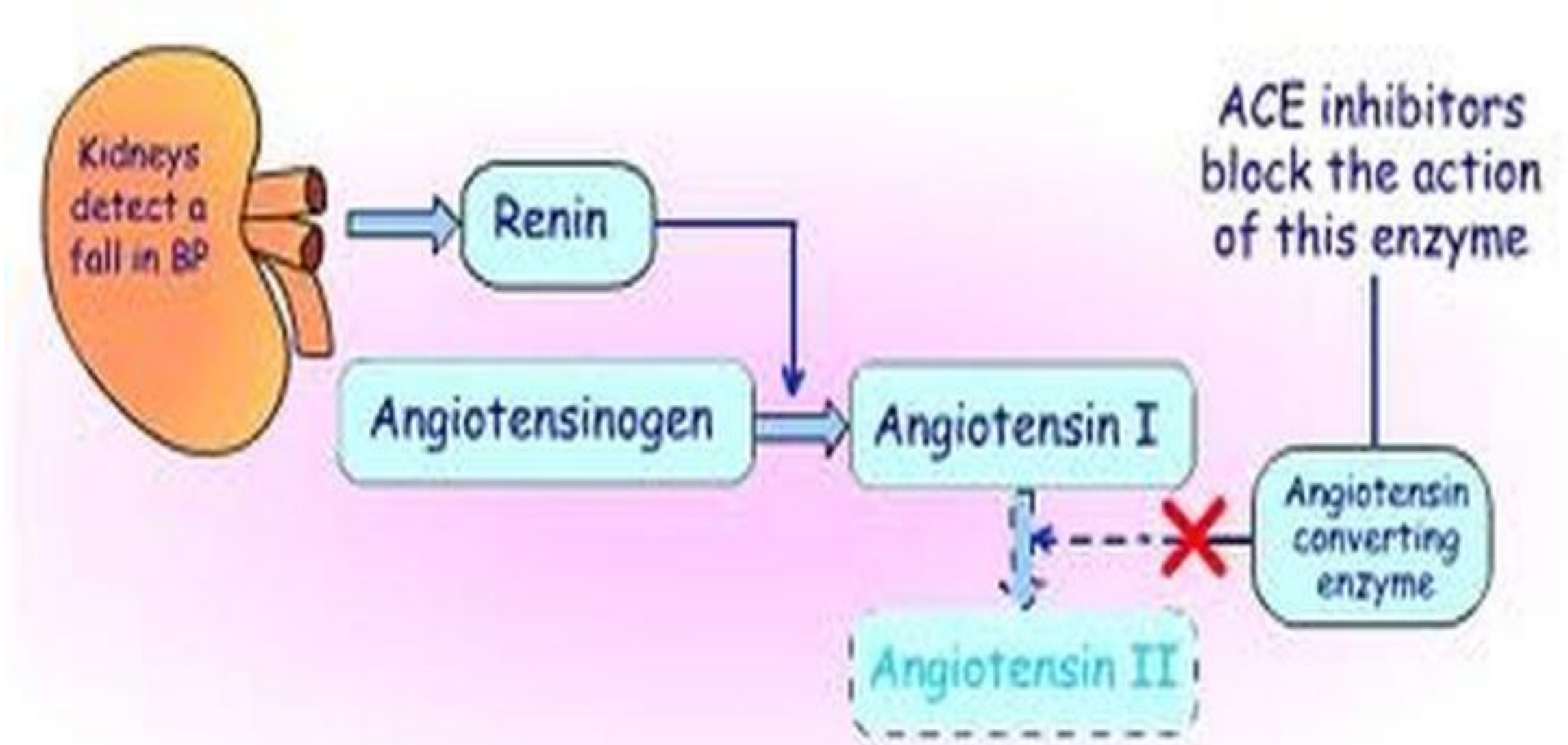
- **Tabs. : Initially 10 mg qid for 2-4 days, then 25 mg bid**
- **I.V, I.M: 50 mg (IV slowly) repeated as necessary. (May decrease Bp in 5 minutes.)**

Other drugs: diazoxide, Minoxidil

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



ACE INHIBITORS (Angiotensin-Converting Enzyme)



Actions: ↓ Peripheral vascular resistance without:

- ∅ ↑ Cardiac output
- ∅ ↑ Cardiac rate
- ∅ ↑ Cardiac contractility

Side Effects: Dizziness
Orthostatic hypotension
GI distress
● Cough
Headache

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Captopril:

Trade name: Capotin, inhabace.

Class: antihypertensive, inhibitor of angiotensin synthesis.

Action:

- Captopril is a highly specific competitive inhibitor of angiotensin I converting enzyme. The enzyme is responsible for the conversion of angiotensin I to angiotensin II which decrease BP.
- Reduce peripheral arterial resistance.
- Decrease aldosterone secretion which works to increase level of serum potassium.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Indications:

- Hypertension.
- In combination with diuretics and digitalis in the treatment of CHF.

Contraindication: Hypersensitivity, renovascular disease, pregnancy and lactation.

Side effects:

- Skin rash, loss of taste, neutropnea, nausea, vomiting, hypotension, proteinuria, renal failure and hyperkalemia.

ACE inhibitor side effects (CAPTOPRIL)

C	Cough
A	Angioneurotic oedema
P	Proteinuria
T	Taste disturbance/ Teratogenic in 1st trimester
O	Other (fatigue, headache)
P	Potassium increased
R	Renal impairment
I	Itch
L	Low BP (1st dose)

MORE INFORMATION:

An angiotensin-converting-enzyme inhibitor (ACE inhibitor) is a pharmaceutical drug used primarily for the treatment of hypertension (elevated blood pressure) and congestive heart failure. Examples of ACE inhibitors include: Benazepril (Lotensin), Captopril, Enalapril (Vasotec), Fosinopril, Lisinopril (Zestril), Moexipril (Univasc), Perindopril (Aceon), Quinapril (Accupril), Ramipril (Altace), Trandolapril (Mavik)



D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Dosage:

- **Tablets: 12.5 mg 2-3 time per day. If there is no response, after 1-2 weeks, increase dose to 25 mg 2-3 time per day.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Nursing considerations:

1. In case of overdose, give normal saline to restore BP.
2. Should not be discontinued without medical order.
3. Obtain baseline hematological studies, liver & renal functions tests prior to beginning the treatment.
4. Determine client's understanding of the therapy and if he/she takes other medications.
5. Observe client closely for hypotension 3 hours after the initial dose.
6. Withhold potassium sparing diuretics and consult with physician (hyperkalemia may occur).
7. Take captopril 1 hour before meal or on an empty stomach.
8. Report skin rash, heartburn, and chest pain to physician.
9. Explain to client that he may develop loss of taste for 2-3 months, if it persist, notify the physician.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Trade name: convertin, anapril

Class: ACE-inhibitor

Uses: Hypertension

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Side effects:

- CNS: headache, dizziness, fatigue, vertigo, syncope.
- CV: hypotension, chest pain, angina pectoris.
- GI: diarrhea, nausea, abdominal pain, vomiting.
- GU: decreased renal function (in patients with bilateral renal artery stenosis or heart failure).
- Hematologic: bone marrow depression.
- Respiratory: dyspnea, dry, persistent, nonproductive cough.
- Skin: rash.
- Other: angioedema.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Interactions

- **Diuretics**: May excessively reduce blood pressure. Use together cautiously.
- **Insulin**, oral antidiabetics: **May cause hypoglycemia**, especially at start of enalapril therapy. Monitor patient closely.
- **Lithium**: **May cause lithium toxicity**. Monitor lithium level.
- **NSAIDs**: May reduce antihypertensive effect. Monitor blood pressure.
- **Potassium-sparing** diuretics, potassium supplements: May cause hyperkalemia. Avoid using together unless hypokalemia is confirmed.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Contraindications & cautions

- **Contraindicated in patients hypersensitive to drug and in those with a history of angioedema related to previous treatment with an ACE inhibitor.**

- **Use cautiously in patients with renal impairment or those with aortic stenosis or hypertrophic cardiomyopathy.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Dose:

- **Initially, 5 mg P.O. once daily; then adjusted based on response. Usual dosage range is 10 to 40 mg daily as a single dose or two divided doses. Or, 1.25 mg I.V. infusion over 5 minutes q 6 hours.**

- **Adjust-a-dose: If patient is taking diuretics or creatinine clearance is 30 ml/minute or less, initially, 2.5 mg P.O. once daily.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Nursing considerations

1. Closely monitor blood pressure response to drug.
2. Monitor CBC with differential counts before and during therapy.
3. Diabetic patients, those with impaired renal function or heart failure, and those receiving drugs that can increase potassium level may develop hyperkalemia. .
4. Instruct patient to report breathing difficulty or swelling of face, eyes, lips, or tongue. Swelling of the face and throat (including swelling of the larynx) may occur, especially after first dose.
5. Advice patient to report signs of infection, such as fever and sore throat.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Nursing considerations

6. Inform patient that light-headedness can occur, especially during first few days of therapy. Tell him to rise slowly to minimize this effect and to notify doctor if symptoms develop. If he faints, he should stop taking drug and call prescriber immediately.
7. Tell patient to use caution in hot weather and during exercise. Inadequate fluid intake, vomiting, diarrhea, and excessive perspiration can lead to light-headedness and fainting.
8. Tell woman of childbearing age to notify doctor if pregnancy occurs. Drug will need to be stopped.

Calcium channel blocking agents:

Action:

- for contraction of cardiac and smooth muscle to occur, extracellular calcium must move into the cell through openings called calcium channels.
- These agents inhibit the influx of calcium through the cell membrane resulting in a depression of automatically and conduction velocity in both smooth and cardiac muscles leading to:

1. Decrease myocardial contractility (negative inotropic effect).
2. Dilatation of coronary arteries.
3. Decrease peripheral vasodilatation and peripheral resistance.
4. Decrease S. A. node automatically and conduction and decrease heart rate (negative chronotropic effect).

E: Calcium channel blocking agents:

Diltazem Hcl:

Trade name: Cardiazem, dilatam.
Class: calcium channel blocking agent (anti-angina, antihypertensive).
Uses: vasospastic angina, essential hypertension.
Contraindications: hypotension, pulmonary congestion, and MI.
Side effects: AV block, bradycardia, CHF. Hypotension.
Dosage: 30 mg qid before meals and at bedtime.

E: Calcium channel blocking agents:

Nifedipine:

Trade name: Adalat
Class: calcium channel blocking agent (anti-angina, antihypertensive).
Uses: vasospastic angina, essential hypertension.
Contraindications: hypersensitivity, lactation.
Side effects: pulmonary and peripheral edema, MI, hypotension, dizziness, light- headedness, palpitation, headache, muscle cramps.
Dosage: 10- 30 mg tid.

- In hypertensive emergencies: 10-20 mg given orally or sublingually by puncturing the capsule and squeezing contents under the tongue.

E: Calcium channel blocking agents:

Amlodipine besylate

Trade name: Amicore

Class: calcium channel blocking agent (anti-angina, antihypertensive).

Use: Angina and hypertension.

Adverse reactions

- CNS: headache, fatigue, dizziness, light-headedness, paresthesia.
- CV: edema, flushing, palpitations.
- GI: nausea, abdominal pain.
- Musculoskeletal: muscle pain.
- Respiratory: dyspnea.
- Skin: rash, pruritus.

E: Calcium channel blocking agents:

Amlodipine besylate

Contraindications & cautions

- Contraindicated in patients hypersensitive to drug.
- Use cautiously in patients receiving other peripheral vasodilators, especially those with severe aortic stenosis, and in those with heart failure.
- Because drug is metabolized by the liver, use cautiously and in reduced dosage in patients

Dose: Initially, 2.5 to 5 mg P.O. daily. Dosage adjusted according to patient response and tolerance. Maximum daily dose is 10 mg.

E: Calcium channel blocking agents:

Amlodipine besylate

Nursing considerations

- Monitor blood pressure frequently during initiation of therapy.
- Notify doctor if signs of heart failure occur, such as swelling of hands and feet or shortness of breath.
- Teach patient to continue taking drug, even when feeling better.
- Grapefruit juice may increase drug level and adverse reactions. Discourage use together.

E: Calcium channel blocking agents:

Verapamil:

Trade name: Ikacor

Class: Uses: calcium channel blocking agent (anti-angina, antihypertensive, antiarrhythmia).

- P.O:
  - Angina pectoris.
  - Arrhythmia (atrial fibrillation, and flutter).
  - Essential hypertension.
- IV: supraventricular tachycardia.

E: Calcium channel blocking agents:

Verapamil:

Contraindications: hypotension, cardiac shock, and MI.

Side effects: AV block, bradycardia, headache, dizziness, abdominal cramps, blurring of vision, and edema.

Dosage: Initial 80-120 mg tid then 240-480 mg /day.

E: Calcium channel blocking agents:

Verapamil:

Nursing considerations for calcium channel blocking agents:

1. Discuss with the patient/family the goals of therapy.
2. Teach them how to take pulse and blood pressure. Hold the medication in case of hypotension or bradycardia and consult the treating physician.
3. Instruct the client to report any untoward sings as dizziness.
4. In case of postural hypotension, advise the client to change position slowly.
5. Advise client to sit down immediately if fainting occurs.
6. Calcium antagonists should be taken with meals to decrease GI irritation.

Cardiac Glycoside

- Cardiac glycosides are the most effective drugs for treatment of C.H.F.
- Digitoxins are plant alkaloids.
- They increase myocardial contractions, which will increase blood supply to all organs including the kidneys therefore causing diuresis, which will decrease the edema.
- They are used to treat cardiac arrhythmia because they decrease heart rate.

Cardiac Glycoside

Action:

- They increase the force of myocardial contractions (positive inotropic).
- It increases the contractility of the heart muscle by minimizing the movement of Na+ and K+ ions and increasing the release of Ca++ ions in the myocardial cells.
- It decreases the heart rate due to increase in parasympathetic nervous system and decrease in the sympathetic tone.
- They are primarily excreted through the kidneys.
- The initial dose is the larger dose (the loading or digitalizing dose), the subsequent doses are referred to as (Maintenance doses).

Cardiac Glycoside

Results:

- Decrease in venous pressure.
- Coronary dilatation.
- Reduce heart size.
- Marked diuresis and decreasing edema.

Cardiac Glycoside

Indications:

- Congestive heart failure (C.H.F).
- Cardiac arrhythmia (atrial fibrillation, atrial flutter and sinus tachycardia).

Contraindication:

- Hypersensitivity.
- Angina pectoris in absence of CHF.
- Given with caution for elderly and people who have kidney failure.

Cardiac Glycoside

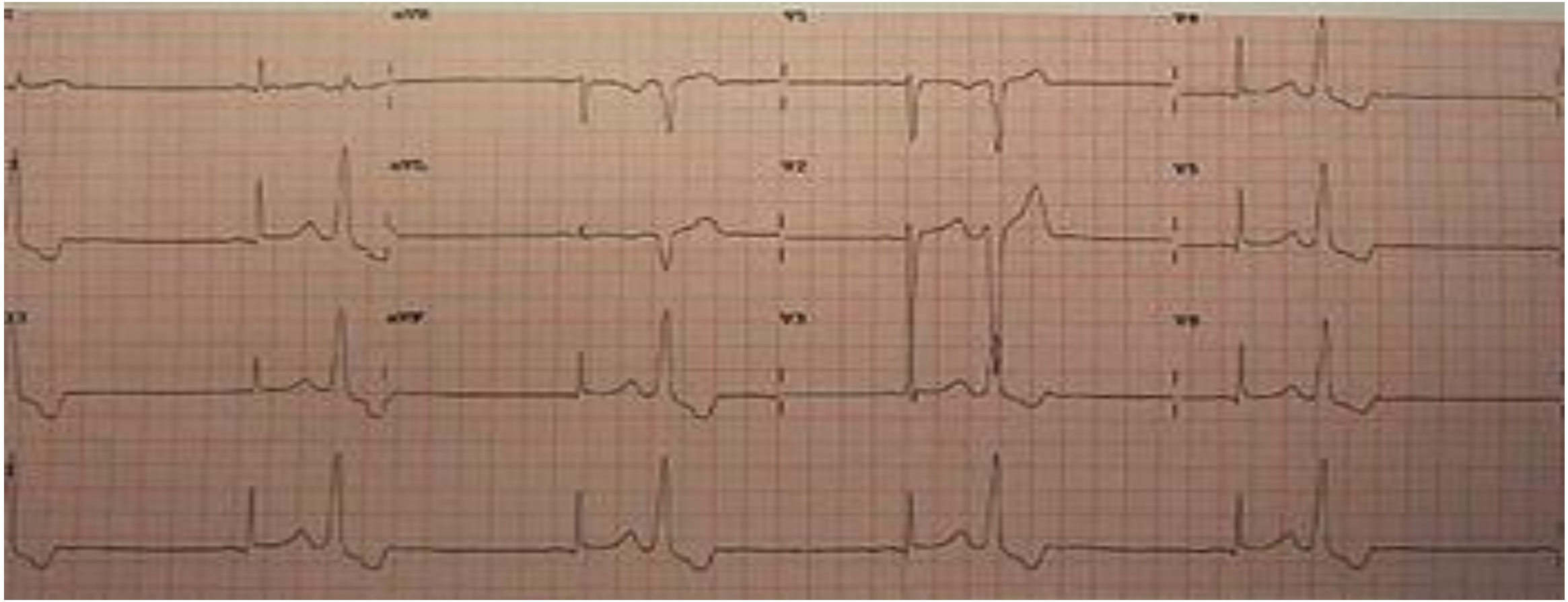
Side effects:

1. They are extremely toxic and may cause death.
2. There is a narrow margin of safety between the therapeutic dose and the toxic dose.
3. Could cause overdose by cumulative effects of the drug, so frequent assessment of the serum level is essential.
4. May cause cardiac arrhythmia such as bradycardia (below than 60 beat/ minute), ventricular fibrillation (which may lead to cardiac arrest and death), and Bigeminal rhythm.
5. Nausea, vomiting, and diarrhea.
6. Headache, malaise and muscle weakness.
7. Skin rashes, blurring of vision, diplopia and white halos.

Cardiac Glycoside

Side effects:

Bigeminal rhythm.



Cardiac Glycoside

Note:

1. **Digitalization: If there is no urgent need for a desired effect, an oral digitalizing dose is first administered and then the maintenance dose is adjusted on the bases of clinical and laboratory assessment.**
2. **Patients suffering from digitalis intoxication should be admitted to the ICU for continuous monitoring of ECG. Administration of digitalis should be stopped.**
3. **If serum potassium is below normal, administer K+ salts and give antiarrhythmic drugs as Lidocain as ordered by physician.**

Cardiac Glycoside

Drug interactions:

- Antacid (they decrease the effect of digitalis).
- Furosemide (Lasix): it increase K+ loss and increase the chance for digitalis toxicity.

Predisposing factors for digitalis toxicity:

1. K+ loss (hypokalemia) which results from: diuretics, NPO, gastric suction, and poor K+ intake.
2. Pathological conditions;
    a. Liver disease: they decrease metabolism and therefore increase digitalis level.
    b. Kidney disease: they decrease the excretion of drug and therefore increase digitalis levels.

Cardiac Glycoside

Nursing considerations:

1. Check doctor's order, medication record and bottle label accurately.
2. Observe & monitor for evidence of bradycardia or arrhythmia.
3. Measure intake and output accurately.
4. Weigh the patient in daily basis.
5. Pulse should be checked by 2 nurses.
6. Provide the client with food high in potassium as banana, orange.

Cardiac Glycoside

Nursing considerations:

7. Monitor serum digoxin level.
8. Elderly people should be assessed for early signs of toxicity.
9. Teach patients that bradyacadia, nausea, vomiting, diarrhea, appetite loss, and visual disturbances could be early signs of toxicity.
10. Teach client if heart rate is less than 60/minute to hold the medication and see the doctor.
11. Have digoxin antidote available (digoxin immune FAB).

Cardiac Glycoside

Drugs:

1. Digitoxin:

Trade name: crystodigin
Class: cardiac glycoside
Uses: drug of choice for maintenance in CHF.
Dose:

- Digitalizing dose is 0.6 mg in 4-6 hours.
- Maintenance: 0.05 – 0.3 mg/day.

Cardiac Glycoside



2. Digoxin:

Trade name: Lanoxin, DIGOXIN 0.25 MG TAB

Class: cardiac glycoside.

It is the drug of choice for CHF because of:

1. It has rapid onset.
2. It has short duration.
3. It can be administered P.O. or IV.

Dose:

Digitalization dose = 0.4 – 0.6 mg followed by 0.05 – 0.35 mg once or twice daily.

Cardiac Glycoside

3. Digoxin Immune FAB:

Trade name: (Ovine)

Class: digoxin antidote.

Action: antibodies bind to digoxin and excreted through the kidneys.
Uses: life threatening digitalis toxicity or overdose.
Note: cardiac arrest can be expected if an adult ingests 10 mg or if a child ingests 4 mg.

Coronary vasodilators

Antianginal drugs

Angina pectoris: is a clinical syndrome characterized by paroxysm of pain in the anterior chest caused by insufficient coronary blood flow and/or inadequate oxygen supply to the myocardial muscle.

Causes:

1. Atherosclerosis.
2. Vasospasim.

Coronary vasodilators

Antianginal drugs

There are three groups of drugs used for treatment of angina:

1. **Nitrates/nitrites.**
2. **Beta-adrenergic blocking agents.**
3. **Calcium channel blocking agents.**

Coronary vasodilators

Antianginal drugs

Nitrates/nitrites:

Action:

- Direct relaxation of blood vessels and smooth muscles vasodilatation lead to decrease O2 requirements.
- Relaxation of smooth muscles of coronary arteries lead to coronary vasodilatation and increase blood supply to the myocardium.
- Relaxation of arteries and veins lead to decrease BP and decrease workload in the heart.

Coronary vasodilators

Antianginal drugs

Objectives of treatment:

1. Treatment of anginal attack and thus relief pain.
2. Prophylactic treatment to prevent or delay the occurrence of MI.
3. Prolongs intervals between attacks.

Indications:

1. Prophylaxis and treatment of acute angina pectoris.
2. Treatment of chronic angina pectoris.
3. Treatment of hypertension associated with MI or CHF.
4. Nitroglycerin ointment for treatment of Raynaud's disease.

Coronary vasodilators

Antianginal drugs

Contraindications:

1. Sensitivity to nitrates (Hypotension).
2. Severe anemia.
3. Hypotension.
4. Head trauma.
5. Cerebral hemorrhage.

Side effects:

1. Headache, syncope, dizziness.
2. Postural hypotension, transient flushing, and palpitation.
3. Topical application may lead to dermatitis.

Coronary vasodilators

Antianginal drugs

Drug interaction: Antihypertensive agents, Beta-adrenergic blocking agents, and calcium-channel blocking agent (they may lead to additive hypotension).

Dosage: there are several forms available:

1. Sublingual: Cordil 5 mg PRN.
2. PO: Isotard 20 – 40 mg twice a day.
3. Topical: available as patches or ointment.
4. Parental (IV infusion).

Coronary vasodilators

Antianginal drugs

Nursing considerations:

1. Medications should be taken on an empty stomach.
2. Carry sublingual tablets in a glass bottle, tightly capped.
3. If angina pain is not relieved in 5 minutes by first sublingual tablet, to take up to 2 more tablets at 5 minutes interval. If pain has not subsided 5 minutes after the 3rd tablet, client should be taken to the emergency room as this case could be an infarction (MI) and not angina.
4. Take sublingual tablets 5-15 minutes prior to any situation likely to cause anginal pain such as climbing stairs.
5. Take sublingual tablets while sitting to avoid postural hypotension.

Coronary vasodilators

Antianginal drugs

Isosorbide dinitrate:

Present in the forms of capsules chewable, sublingual, tablets.

Trade names: Isoral, Isotard.

Class: coronary vasodilator.

Dosage forms: caps 20-40 mg, tabs 20-40 mg.

Coronary vasodilators

Antianginal drugs

Uses:

- Tabs for only prophylaxis of anginal pain.
- Chewable, sublingual to terminate acute attack and relieve acute pain.
- Esophageal spasm.

Side effects: Headache, hypotension.

Dosage:

- Sublingual: acute attack 5 mg Q 2-3 hrs.
- Oral caps/tabs: 5-20 mg Q 6 hrs.
- Extended release tabs: 20 –80 mg Q 8-12 hrs.

Note: Isosorbide mononitrate given for patients with liver impairments.

Peripheral vasodilators:

Isoxsuprine:

Trade name: Vasodin.

Class: peripheral vasodilator.

Action: Direct relaxation of vascular smooth muscles increasing peripheral blood flow, the drug also has a cardiac stimulation and uterine relaxation effect “Alpha receptor blocking and Beta receptor stimulation.

Peripheral vasodilators:

Isoxsuprine:

Uses:

1. **Symptomatic treatment of cardiovascular insufficiency.**
2. **Buerger's disease.**
3. **Raynaud's disease.**

Contraindications: **postpartum period, arterial bleeding.**

Side effects: **tachycardia, hypotension, chest pain, nausea, vomiting, rash, dizziness.**

Dosage: **10-20 mg 3-4 time daily.**

Peripheral vasodilators:

Papverine:

Class: peripheral vasodilator.

Action: direct spasmolytic effect on smooth muscle and vascular system, bronchial muscle, GI & urinary tract.

Uses:

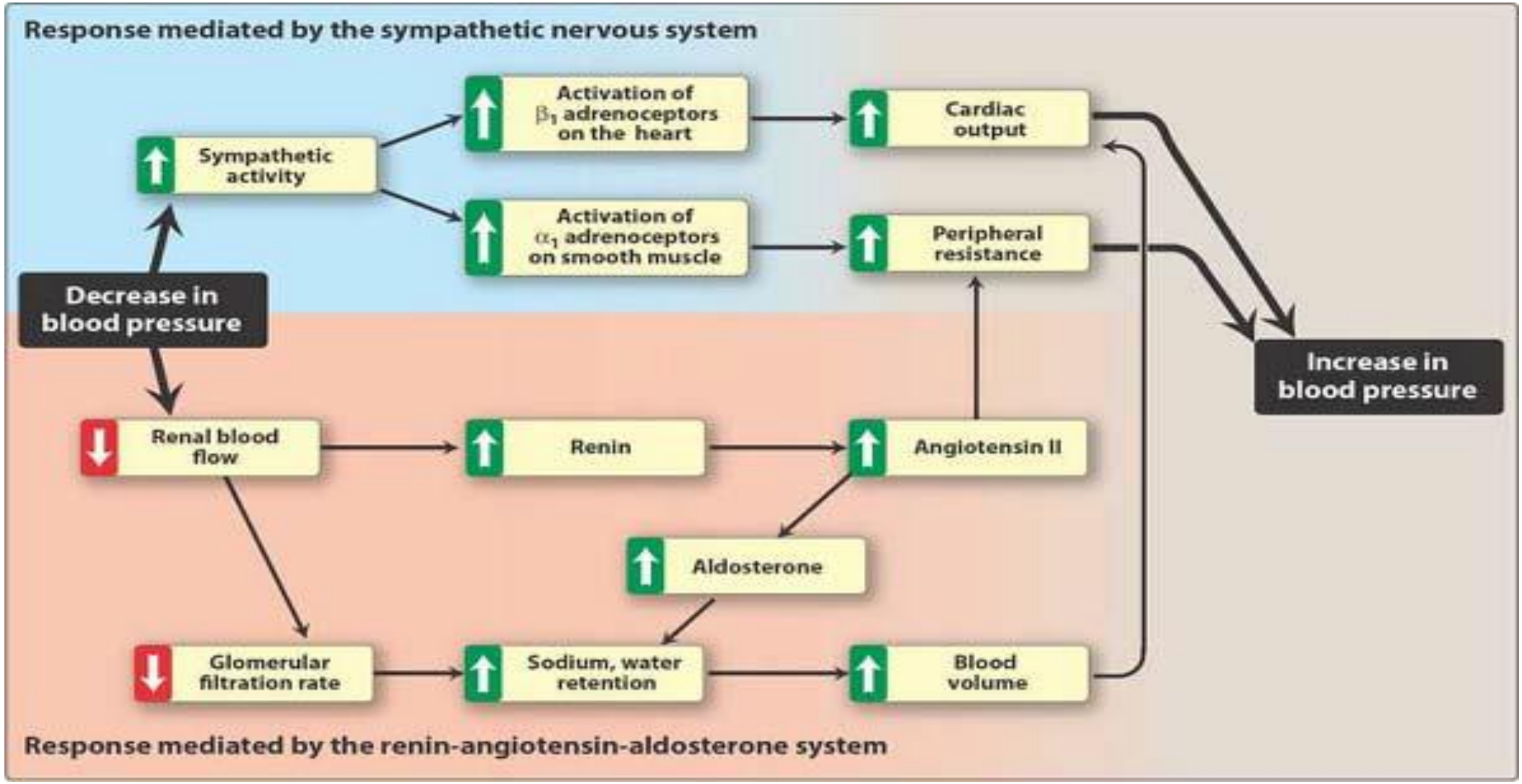
1. Cerebral and peripheral ischemia.
2. Smooth muscle relaxant/
3. Paraenteral use for: Acute MI and angina pectoris, Pulmonary embolism, and ureteral, biliary, & GI colic.

Side effects: Flushing of face, hypertension, tachycardia, constipation, dry mouth and throat, headache.

B: drugs that alter sympathetic nervous system.

1. **Beta-adrenergic blocking agents: discussed before**
2. **Centrally acting agents:**
   Used for treatment of chronic hypertension, since it affects CNS.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



Antihypertensive

Methyldopa:

Trade name: (Aldomin)

Classification: Antihypertensive, centrally acting antiadrenergic agent.

Action:

- **Converted to alpha methyl norepinephrine**, which is stored in adrenergic nerve granules where it replaces norepinephrine and is released by nerve stimulation to interact with presynaptic central alph2 adrenoceptors and decrease sympathetic outflow and decrease arterial pressure.
- **Inhibit dopa decarboxylase** lead to decrease stores of norepinephrine in the sympathetic nervous system and decrease BP.

Antihypertensive

Uses: Hypertension & hypertension crises (parenteral).

Side effects:

- Headache, dizziness, general weakness, depression and sedation.
- Bradycardia, orthostatic hypotension.
- Dry mouth, nausea, vomiting, sore (black) tongue.
- Jaundice, liver disorders.
- Hemolytic anemia & leukopenia.
- Male impotence.

Antihypertensive

Contraindications:-

- Hypersensitivity,
- Active hepatic diseases.
- Pheochromocytoma.

Dosage: Tabs. : initially 250 mg bid or tid.

Antihypertensive

Nursing considerations:-

1. Dose increases should be made with the evening dose to minimize drowsiness
2. Avoid activities that need mental awareness such as driving.
3. Note any evidence of jaundice and do liver function test on intervals.
4. Advise pt. to rise from the bed slowly.
5. Instruct pt. about reportable Signs & Symptoms.
6. Explain to pt. that urine rarely may be turn into dark / red-black color.
7. Advise patient to consult health care professional before taking any Rx, OTC, or Herbal products, especially cough, cold, or allergy remedies.

Antihypertensive

C: Direct acting vasodilators.

- **Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large arteries, smaller arterioles and large veins.**

- **The process is essentially the opposite of vasoconstriction, or the narrowing of blood vessels. When vessels dilate, the flow of blood is increased due to a decrease in vascular resistance. Therefore, dilation of arterial blood vessels (mainly arterioles) leads to a decrease in blood pressure.**

Antihypertensive

C: Direct acting vasodilators.

Hydralazine hydrochloride:

Trade name: Apresoline

Classification: Antihypertensive, direct action on vascular smooth muscles.

Action: directly affect smooth muscles results in vasodilation and increase a cardiac output and blood flow to the brain and kidneys.

Antihypertensive

C: Direct acting vasodilators.

Uses:

- Management of moderate to severe hypertension.
- Management of congestive heart failure.
- Management of hypertension secondary to pre-eclampsia/eclampsia.
- Given parenterally in hypertension emergencies.

Antihypertensive

C: Direct acting vasodilators.

Contraindications:

- Angina pectoris.
- Rheumatic heart disease.
- Chronic glomerulonephritis.
- Systemic lupupus erthrmatosis (S.L.E.)

Side effects:

- Orthostatic hypotension, tachycardia, nausea, vomiting.
- Headache, dizziness, constipation and male impotence.

Antihypertensive

C: Direct acting vasodilators.

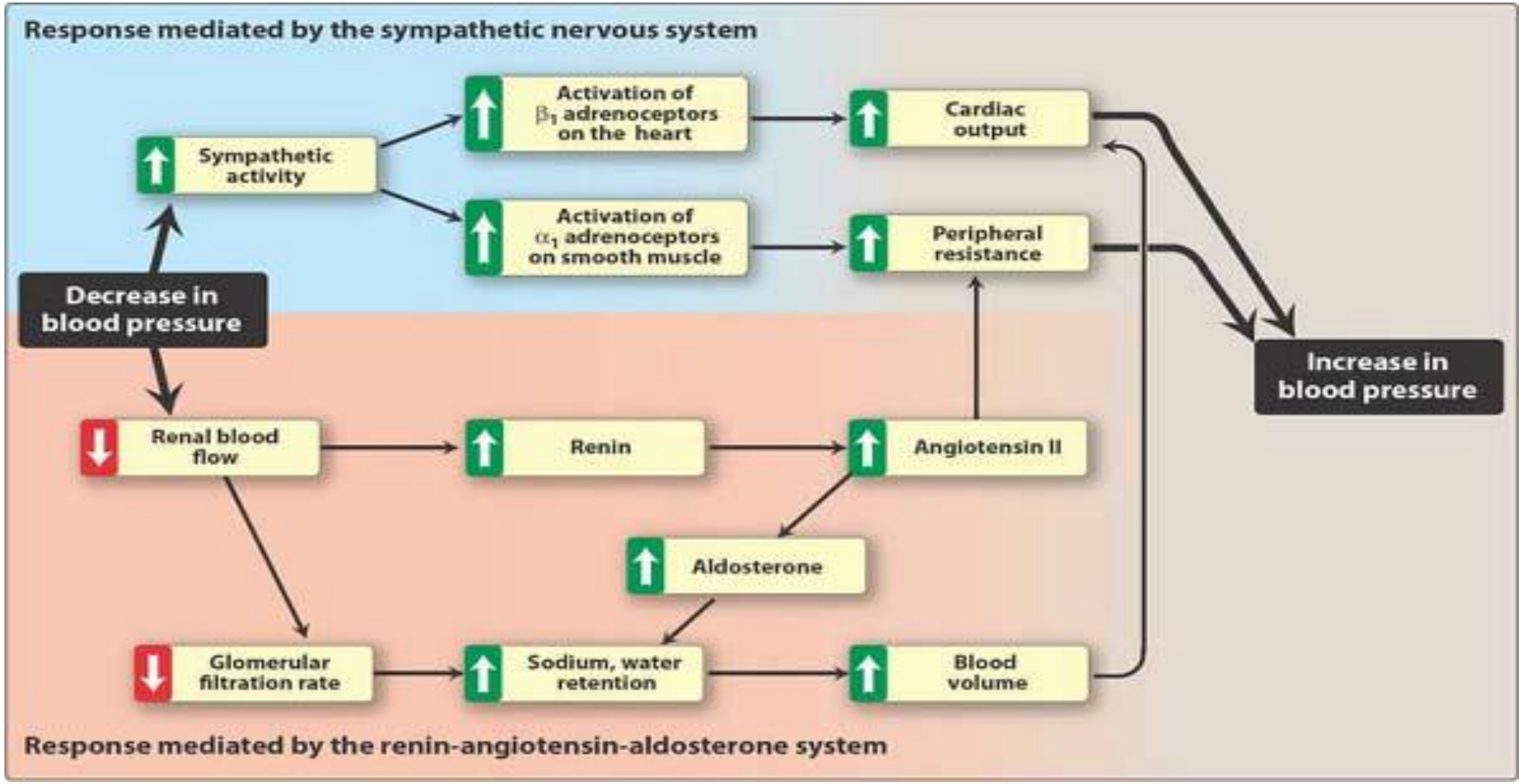
Available forms: Injection, solution, as hydrochloride: 20 mg/mL (1 mL), Tablet as hydrochloride: 10 mg, 25 mg, 50 mg, 100 mg

Dose:

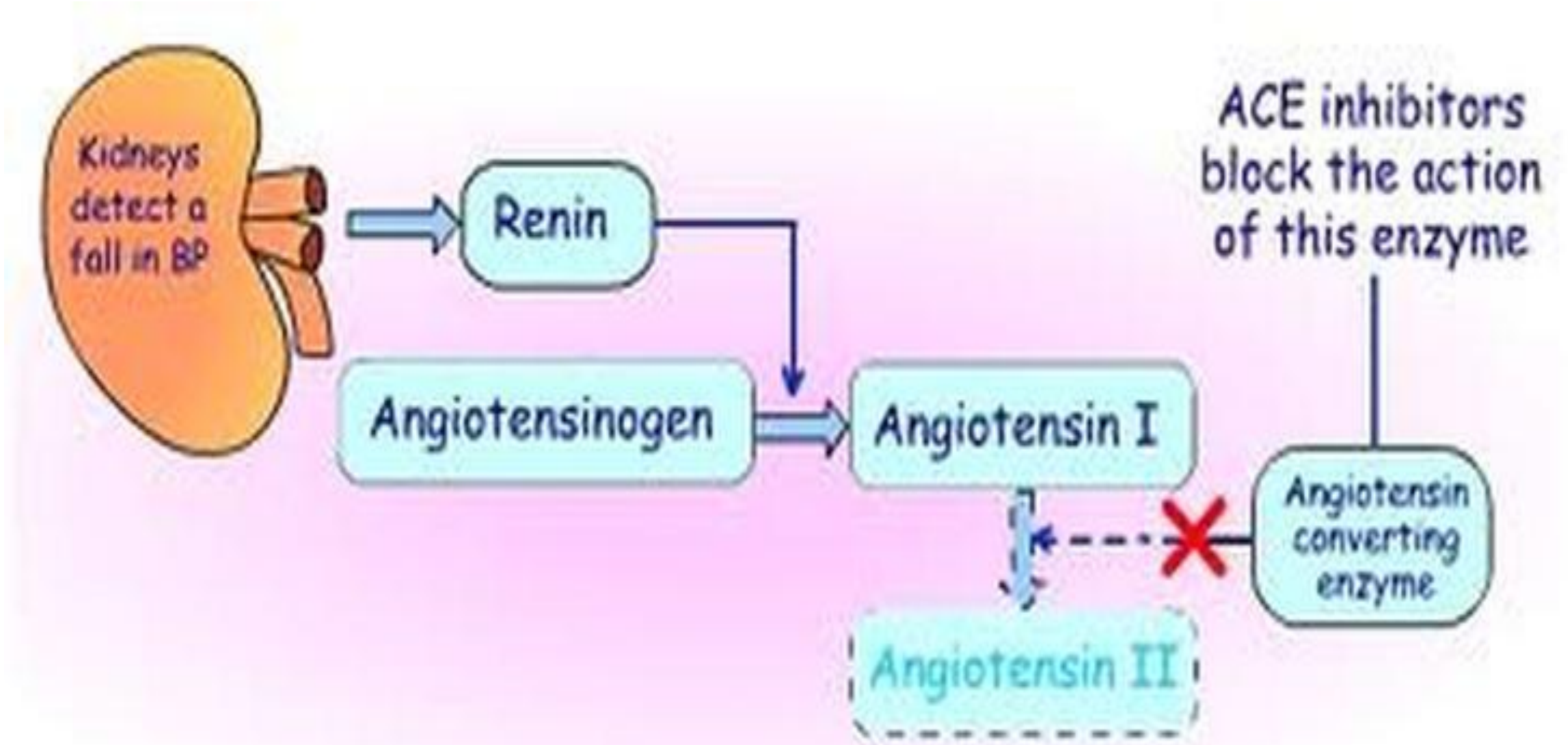
- **Tabs. : Initially 10 mg qid for 2-4 days, then 25 mg bid**
- **I.V, I.M: 50 mg (IV slowly) repeated as necessary. (May decrease Bp in 5 minutes.)**

Other drugs: diazoxide, Minoxidil

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):



ACE INHIBITORS (Angiotensin-Converting Enzyme)



Actions: ↓ Peripheral vascular resistance without:

- ∅ ↑ Cardiac output
- ∅ ↑ Cardiac rate
- ∅ ↑ Cardiac contractility

Side Effects: Dizziness
Orthostatic hypotension
GI distress
● Cough
Headache

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Captopril:

Trade name: Capotin, inhabace.

Class: antihypertensive, inhibitor of angiotensin synthesis.

Action:

- Captopril is a highly specific competitive inhibitor of angiotensin I converting enzyme. The enzyme is responsible for the conversion of angiotensin I to angiotensin II which decrease BP.
- Reduce peripheral arterial resistance.
- Decrease aldosterone secretion which works to increase level of serum potassium.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Indications:

- Hypertension.
- In combination with diuretics and digitalis in the treatment of CHF.

Contraindication: Hypersensitivity, renovascular disease, pregnancy and lactation.

Side effects:

- Skin rash, loss of taste, neutropnea, nausea, vomiting, hypotension, proteinuria, renal failure and hyperkalemia.

ACE inhibitor side effects (CAPTOPRIL)

C	Cough
A	Angioneurotic oedema
P	Proteinuria
T	Taste disturbance/ Teratogenic in 1st trimester
O	Other (fatigue, headache)
P	Potassium increased
R	Renal impairment
I	Itch
L	Low BP (1st dose)

MORE INFORMATION:

An angiotensin-converting-enzyme inhibitor (ACE inhibitor) is a pharmaceutical drug used primarily for the treatment of hypertension (elevated blood pressure) and congestive heart failure. Examples of ACE inhibitors include: Benazepril (Lotensin), Captopril, Enalapril (Vasotec), Fosinopril, Lisinopril (Zestril), Moexipril (Univasc), Perindopril (Aceon), Quinapril (Accupril), Ramipril (Altace), Trandolapril (Mavik)



D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Dosage:

- **Tablets: 12.5 mg 2-3 time per day. If there is no response, after 1-2 weeks, increase dose to 25 mg 2-3 time per day.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Nursing considerations:

1. In case of overdose, give normal saline to restore BP.
2. Should not be discontinued without medical order.
3. Obtain baseline hematological studies, liver & renal functions tests prior to beginning the treatment.
4. Determine client's understanding of the therapy and if he/she takes other medications.
5. Observe client closely for hypotension 3 hours after the initial dose.
6. Withhold potassium sparing diuretics and consult with physician (hyperkalemia may occur).
7. Take captopril 1 hour before meal or on an empty stomach.
8. Report skin rash, heartburn, and chest pain to physician.
9. Explain to client that he may develop loss of taste for 2-3 months, if it persist, notify the physician.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Trade name: convertin, anapril

Class: ACE-inhibitor

Uses: Hypertension

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Side effects:

- CNS: headache, dizziness, fatigue, vertigo, syncope.
- CV: hypotension, chest pain, angina pectoris.
- GI: diarrhea, nausea, abdominal pain, vomiting.
- GU: decreased renal function (in patients with bilateral renal artery stenosis or heart failure).
- Hematologic: bone marrow depression.
- Respiratory: dyspnea, dry, persistent, nonproductive cough.
- Skin: rash.
- Other: angioedema.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Interactions

- **Diuretics**: May excessively reduce blood pressure. Use together cautiously.
- **Insulin**, oral antidiabetics: **May cause hypoglycemia**, especially at start of enalapril therapy. Monitor patient closely.
- **Lithium**: **May cause lithium toxicity**. Monitor lithium level.
- **NSAIDs**: May reduce antihypertensive effect. Monitor blood pressure.
- **Potassium-sparing** diuretics, potassium supplements: May cause hyperkalemia. Avoid using together unless hypokalemia is confirmed.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Contraindications & cautions

- **Contraindicated in patients hypersensitive to drug and in those with a history of angioedema related to previous treatment with an ACE inhibitor.**

- **Use cautiously in patients with renal impairment or those with aortic stenosis or hypertrophic cardiomyopathy.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Dose:

- **Initially, 5 mg P.O. once daily; then adjusted based on response. Usual dosage range is 10 to 40 mg daily as a single dose or two divided doses. Or, 1.25 mg I.V. infusion over 5 minutes q 6 hours.**

- **Adjust-a-dose: If patient is taking diuretics or creatinine clearance is 30 ml/minute or less, initially, 2.5 mg P.O. once daily.**

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Nursing considerations

1. Closely monitor blood pressure response to drug.
2. Monitor CBC with differential counts before and during therapy.
3. Diabetic patients, those with impaired renal function or heart failure, and those receiving drugs that can increase potassium level may develop hyperkalemia. .
4. Instruct patient to report breathing difficulty or swelling of face, eyes, lips, or tongue. Swelling of the face and throat (including swelling of the larynx) may occur, especially after first dose.
5. Advice patient to report signs of infection, such as fever and sore throat.

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Enalapril maleate

Nursing considerations

6. Inform patient that light-headedness can occur, especially during first few days of therapy. Tell him to rise slowly to minimize this effect and to notify doctor if symptoms develop. If he faints, he should stop taking drug and call prescriber immediately.
7. Tell patient to use caution in hot weather and during exercise. Inadequate fluid intake, vomiting, diarrhea, and excessive perspiration can lead to light-headedness and fainting.
8. Tell woman of childbearing age to notify doctor if pregnancy occurs. Drug will need to be stopped.

Calcium channel blocking agents:

Action:

- for contraction of cardiac and smooth muscle to occur, extracellular calcium must move into the cell through openings called calcium channels.
- These agents inhibit the influx of calcium through the cell membrane resulting in a depression of automatically and conduction velocity in both smooth and cardiac muscles leading to:

1. Decrease myocardial contractility (negative inotropic effect).
2. Dilatation of coronary arteries.
3. Decrease peripheral vasodilatation and peripheral resistance.
4. Decrease S. A. node automatically and conduction and decrease heart rate (negative chronotropic effect).

E: Calcium channel blocking agents:

Diltazem Hcl:

Trade name: Cardiazem, dilatam.
Class: calcium channel blocking agent (anti-angina, antihypertensive).
Uses: vasospastic angina, essential hypertension.
Contraindications: hypotension, pulmonary congestion, and MI.
Side effects: AV block, bradycardia, CHF. Hypotension.
Dosage: 30 mg qid before meals and at bedtime.

E: Calcium channel blocking agents:

Nifedipine:

Trade name: Adalat
Class: calcium channel blocking agent (anti-angina, antihypertensive).
Uses: vasospastic angina, essential hypertension.
Contraindications: hypersensitivity, lactation.
Side effects: pulmonary and peripheral edema, MI, hypotension, dizziness, light- headedness, palpitation, headache, muscle cramps.
Dosage: 10- 30 mg tid.

- In hypertensive emergencies: 10-20 mg given orally or sublingually by puncturing the capsule and squeezing contents under the tongue.

E: Calcium channel blocking agents:

Amlodipine besylate

Trade name: Amicore

Class: calcium channel blocking agent (anti-angina, antihypertensive).

Use: Angina and hypertension.

Adverse reactions

- CNS: headache, fatigue, dizziness, light-headedness, paresthesia.
- CV: edema, flushing, palpitations.
- GI: nausea, abdominal pain.
- Musculoskeletal: muscle pain.
- Respiratory: dyspnea.
- Skin: rash, pruritus.

E: Calcium channel blocking agents:

Amlodipine besylate

Contraindications & cautions

- Contraindicated in patients hypersensitive to drug.
- Use cautiously in patients receiving other peripheral vasodilators, especially those with severe aortic stenosis, and in those with heart failure.
- Because drug is metabolized by the liver, use cautiously and in reduced dosage in patients

Dose: Initially, 2.5 to 5 mg P.O. daily. Dosage adjusted according to patient response and tolerance. Maximum daily dose is 10 mg.

E: Calcium channel blocking agents:

Amlodipine besylate

Nursing considerations

- Monitor blood pressure frequently during initiation of therapy.
- Notify doctor if signs of heart failure occur, such as swelling of hands and feet or shortness of breath.
- Teach patient to continue taking drug, even when feeling better.
- Grapefruit juice may increase drug level and adverse reactions. Discourage use together.

E: Calcium channel blocking agents:

Verapamil:

Trade name: Ikacor

Class: Uses: calcium channel blocking agent (anti-angina, antihypertensive, antiarrhythmia).

- P.O:
  - Angina pectoris.
  - Arrhythmia (atrial fibrillation, and flutter).
  - Essential hypertension.
- IV: supraventricular tachycardia.

E: Calcium channel blocking agents:

Verapamil:

Contraindications: hypotension, cardiac shock, and MI.

Side effects: AV block, bradycardia, headache, dizziness, abdominal cramps, blurring of vision, and edema.

Dosage: Initial 80-120 mg tid then 240-480 mg /day.

E: Calcium channel blocking agents:

Verapamil:

Nursing considerations for calcium channel blocking agents:

1. Discuss with the patient/family the goals of therapy.
2. Teach them how to take pulse and blood pressure. Hold the medication in case of hypotension or bradycardia and consult the treating physician.
3. Instruct the client to report any untoward sings as dizziness.
4. In case of postural hypotension, advise the client to change position slowly.
5. Advise client to sit down immediately if fainting occurs.
6. Calcium antagonists should be taken with meals to decrease GI irritation.

Cardiac Glycoside

- Cardiac glycosides are the most effective drugs for treatment of C.H.F.
- Digitoxins are plant alkaloids.
- They increase myocardial contractions, which will increase blood supply to all organs including the kidneys therefore causing diuresis, which will decrease the edema.
- They are used to treat cardiac arrhythmia because they decrease heart rate.

Cardiac Glycoside

Action:

- They increase the force of myocardial contractions (positive inotropic).
- It increases the contractility of the heart muscle by minimizing the movement of Na+ and K+ ions and increasing the release of Ca++ ions in the myocardial cells.
- It decreases the heart rate due to increase in parasympathetic nervous system and decrease in the sympathetic tone.
- They are primarily excreted through the kidneys.
- The initial dose is the larger dose (the loading or digitalizing dose), the subsequent doses are referred to as (Maintenance doses).

Cardiac Glycoside

Results:

- Decrease in venous pressure.
- Coronary dilatation.
- Reduce heart size.
- Marked diuresis and decreasing edema.

Cardiac Glycoside

Indications:

- Congestive heart failure (C.H.F).
- Cardiac arrhythmia (atrial fibrillation, atrial flutter and sinus tachycardia).

Contraindication:

- Hypersensitivity.
- Angina pectoris in absence of CHF.
- Given with caution for elderly and people who have kidney failure.

Cardiac Glycoside

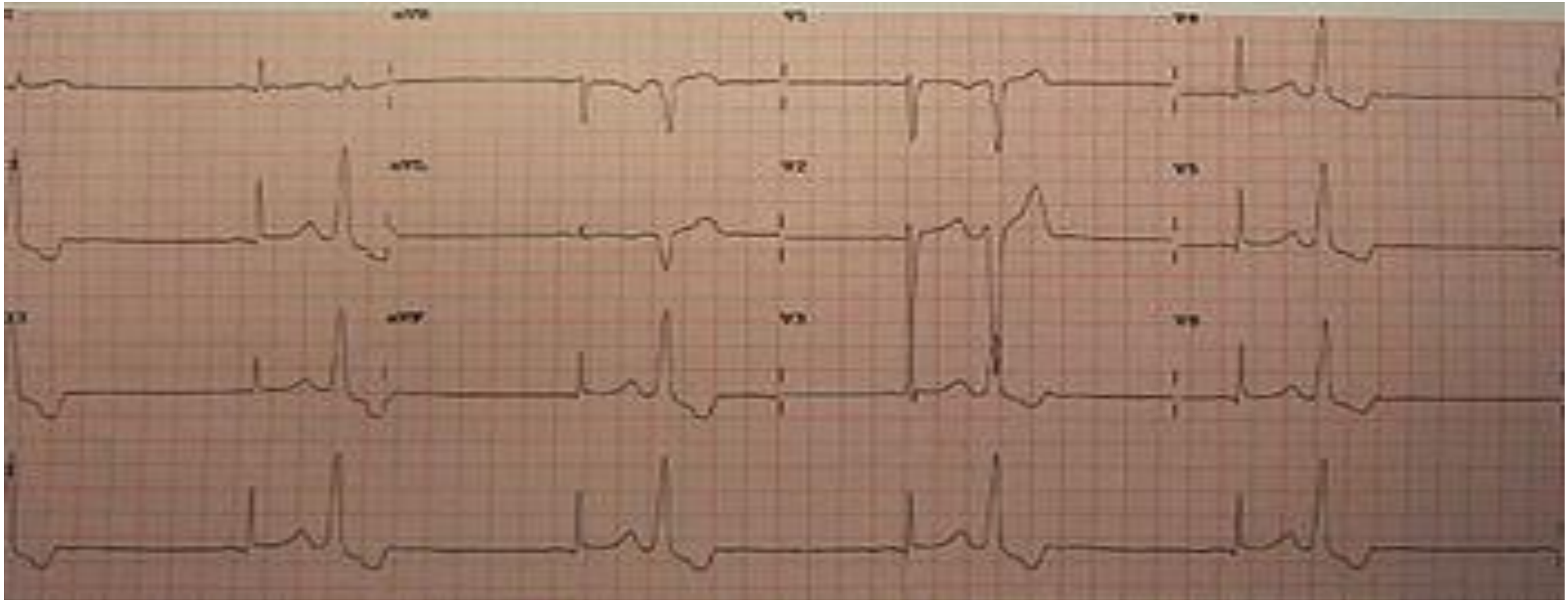
Side effects:

1. They are extremely toxic and may cause death.
2. There is a narrow margin of safety between the therapeutic dose and the toxic dose.
3. Could cause overdose by cumulative effects of the drug, so frequent assessment of the serum level is essential.
4. May cause cardiac arrhythmia such as bradycardia (below than 60 beat/ minute), ventricular fibrillation (which may lead to cardiac arrest and death), and Bigeminal rhythm.
5. Nausea, vomiting, and diarrhea.
6. Headache, malaise and muscle weakness.
7. Skin rashes, blurring of vision, diplopia and white halos.

Cardiac Glycoside

Side effects:

Bigeminal rhythm.



Cardiac Glycoside

Note:

1. **Digitalization: If there is no urgent need for a desired effect, an oral digitalizing dose is first administered and then the maintenance dose is adjusted on the bases of clinical and laboratory assessment.**
2. **Patients suffering from digitalis intoxication should be admitted to the ICU for continuous monitoring of ECG. Administration of digitalis should be stopped.**
3. **If serum potassium is below normal, administer K+ salts and give antiarrhythmic drugs as Lidocain as ordered by physician.**

Cardiac Glycoside

Drug interactions:

- Antacid (they decrease the effect of digitalis).
- Furosemide (Lasix): it increase K+ loss and increase the chance for digitalis toxicity.

Predisposing factors for digitalis toxicity:

1. K+ loss (hypokalemia) which results from: diuretics, NPO, gastric suction, and poor K+ intake.
2. Pathological conditions;
    a. Liver disease: they decrease metabolism and therefore increase digitalis level.
    b. Kidney disease: they decrease the excretion of drug and therefore increase digitalis levels.

Cardiac Glycoside

Nursing considerations:

1. Check doctor's order, medication record and bottle label accurately.
2. Observe & monitor for evidence of bradycardia or arrhythmia.
3. Measure intake and output accurately.
4. Weigh the patient in daily basis.
5. Pulse should be checked by 2 nurses.
6. Provide the client with food high in potassium as banana, orange.

Cardiac Glycoside

Nursing considerations:

7. Monitor serum digoxin level.
8. Elderly people should be assessed for early signs of toxicity.
9. Teach patients that bradyacadia, nausea, vomiting, diarrhea, appetite loss, and visual disturbances could be early signs of toxicity.
10. Teach client if heart rate is less than 60/minute to hold the medication and see the doctor.
11. Have digoxin antidote available (digoxin immune FAB).

Cardiac Glycoside

Drugs:

1. Digitoxin:

Trade name: crystodigin
Class: cardiac glycoside
Uses: drug of choice for maintenance in CHF.
Dose:

- Digitalizing dose is 0.6 mg in 4-6 hours.
- Maintenance: 0.05 – 0.3 mg/day.

Cardiac Glycoside



2. Digoxin:

Trade name: Lanoxin, DIGOXIN 0.25 MG TAB

Class: cardiac glycoside.

It is the drug of choice for CHF because of:

1. It has rapid onset.
2. It has short duration.
3. It can be administered P.O. or IV.

Dose:

Digitalization dose = 0.4 – 0.6 mg followed by 0.05 – 0.35 mg once or twice daily.

Cardiac Glycoside

3. Digoxin Immune FAB:

Trade name: (Ovine)

Class: digoxin antidote.

Action: antibodies bind to digoxin and excreted through the kidneys.
Uses: life threatening digitalis toxicity or overdose.
Note: cardiac arrest can be expected if an adult ingests 10 mg or if a child ingests 4 mg.

Coronary vasodilators

Antianginal drugs

Angina pectoris: is a clinical syndrome characterized by paroxysm of pain in the anterior chest caused by insufficient coronary blood flow and/or inadequate oxygen supply to the myocardial muscle.

Causes:

1. Atherosclerosis.
2. Vasospasim.

Coronary vasodilators

Antianginal drugs

There are three groups of drugs used for treatment of angina:

1. **Nitrates/nitrites.**
2. **Beta-adrenergic blocking agents.**
3. **Calcium channel blocking agents.**

Coronary vasodilators

Antianginal drugs

Nitrates/nitrites:

Action:

- Direct relaxation of blood vessels and smooth muscles vasodilatation lead to decrease O2 requirements.
- Relaxation of smooth muscles of coronary arteries lead to coronary vasodilatation and increase blood supply to the myocardium.
- Relaxation of arteries and veins lead to decrease BP and decrease workload in the heart.

Coronary vasodilators

Antianginal drugs

Objectives of treatment:

1. Treatment of anginal attack and thus relief pain.
2. Prophylactic treatment to prevent or delay the occurrence of MI.
3. Prolongs intervals between attacks.

Indications:

1. Prophylaxis and treatment of acute angina pectoris.
2. Treatment of chronic angina pectoris.
3. Treatment of hypertension associated with MI or CHF.
4. Nitroglycerin ointment for treatment of Raynaud's disease.

Coronary vasodilators

Antianginal drugs

Contraindications:

1. Sensitivity to nitrates (Hypotension).
2. Severe anemia.
3. Hypotension.
4. Head trauma.
5. Cerebral hemorrhage.

Side effects:

1. Headache, syncope, dizziness.
2. Postural hypotension, transient flushing, and palpitation.
3. Topical application may lead to dermatitis.

Coronary vasodilators

Antianginal drugs

Drug interaction: Antihypertensive agents, Beta-adrenergic blocking agents, and calcium-channel blocking agent (they may lead to additive hypotension).

Dosage: there are several forms available:

1. Sublingual: Cordil 5 mg PRN.
2. PO: Isotard 20 – 40 mg twice a day.
3. Topical: available as patches or ointment.
4. Parental (IV infusion).

Coronary vasodilators

Antianginal drugs

Nursing considerations:

1. Medications should be taken on an empty stomach.
2. Carry sublingual tablets in a glass bottle, tightly capped.
3. If angina pain is not relieved in 5 minutes by first sublingual tablet, to take up to 2 more tablets at 5 minutes interval. If pain has not subsided 5 minutes after the 3rd tablet, client should be taken to the emergency room as this case could be an infarction (MI) and not angina.
4. Take sublingual tablets 5-15 minutes prior to any situation likely to cause anginal pain such as climbing stairs.
5. Take sublingual tablets while sitting to avoid postural hypotension.

Coronary vasodilators

Antianginal drugs

Isosorbide dinitrate:

Present in the forms of capsules chewable, sublingual, tablets.

Trade names: Isoral, Isotard.

Class: coronary vasodilator.

Dosage forms: caps 20-40 mg, tabs 20-40 mg.

Coronary vasodilators

Antianginal drugs

Uses:

- Tabs for only prophylaxis of anginal pain.
- Chewable, sublingual to terminate acute attack and relieve acute pain.
- Esophageal spasm.

Side effects: Headache, hypotension.

Dosage:

- Sublingual: acute attack 5 mg Q 2-3 hrs.
- Oral caps/tabs: 5-20 mg Q 6 hrs.
- Extended release tabs: 20 –80 mg Q 8-12 hrs.

Note: Isosorbide mononitrate given for patients with liver impairments.

Peripheral vasodilators:

Isoxsuprine:

Trade name: Vasodin.

Class: peripheral vasodilator.

Action: Direct relaxation of vascular smooth muscles increasing peripheral blood flow, the drug also has a cardiac stimulation and uterine relaxation effect “Alpha receptor blocking and Beta receptor stimulation.

Peripheral vasodilators:

Isoxsuprine:

Uses:

1. **Symptomatic treatment of cardiovascular insufficiency.**
2. **Buerger's disease.**
3. **Raynaud's disease.**

Contraindications: **postpartum period, arterial bleeding.**

Side effects: **tachycardia, hypotension, chest pain, nausea, vomiting, rash, dizziness.**

Dosage: **10-20 mg 3-4 time daily.**

Peripheral vasodilators:

Papverine:

Class: peripheral vasodilator.

Action: direct spasmolytic effect on smooth muscle and vascular system, bronchial muscle, GI & urinary tract.

Uses:

1. Cerebral and peripheral ischemia.
2. Smooth muscle relaxant/
3. Paraenteral use for: Acute MI and angina pectoris, Pulmonary embolism, and ureteral, biliary, & GI colic.

Side effects: Flushing of face, hypertension, tachycardia, constipation, dry mouth and throat, headache.

Antiarrhythmic agents:

Antiarrhythmic agents

are a group of pharmaceuticals that are used to suppress fast rhythms of the heart (cardiac arrhythmias), such as atrial fibrillation, atrial flutter, ventricular tachycardia, and ventricular fibrillation.

Antiarrhythmic agents:

The five main classes in the Vaughan Williams classification of antiarrhythmic agents are:

- **Class** I agents interfere with the sodium (Na^+) channel.
- **Class** II agents are anti-sympathetic nervous system agents. Most agents in this class are beta blockers.
- **Class** III agents affect potassium (K^+) efflux.
- **Class** IV agents affect calcium channels and the AV node.
- **Class** V agents work by other or unknown mechanisms.

Antiarrhythmic agents:

Class	Known as	Examples	Mechanism	Medical uses [4]
Ia	Fast-channel blockers	• Quinidine • Ajmaline • Procainamide • Disopyramide	(Na^+) channel block (intermediate association/dissociation) and K+ channel blocking effect; affects QRS complex class 1a prolong the action potential and has intermediate effect on the 0 phase of depolarization	• Ventricular arrhythmias • Prevention of paroxysmal recurrent atrial fibrillation (triggered by vagal overactivity) • Procainamide in Wolff-Parkinson-White syndrome • Increases QT interval
Ib		• Lidocaine • Phenytoin • Mexiletine • Tocainide	Na^+ channel block (fast association/dissociation); can prolong QRS complex in overdose class 1b shorten the action potential of myocardial cell and has weak effect on intiation of phase 0 of depolarization	• Treatment and prevention during and immediately after myocardial infarction, though this practice is now discouraged given the increased risk of asystole • Ventricular tachycardia
Ic		• Encainide • Flecainide • Propafenone • Moricizine	Na^+ channel block (slow association/dissociation) has no effect on action potential and has the strongest effect on the initiation phase 0 of depolarization	• Prevents paroxysmal atrial fibrillation • Treats recurrent tachyarrhythmias of abnormal conduction system • Contraindicated immediately after myocardial infarction

Antiarrhythmic agents:

II	Beta-blockers	• Carvedilol • Propranolol • Esmolol • Timolol • Metoprolol • Atenolol • Bisoprolol • Nebivolol	Beta blocking Propranolol also shows some class I action	• Decrease myocardial infarction mortality • Prevent recurrence of tachyarrhythmias • Propranolol has sodium channel-blocking effects
III		• Amiodarone • Sotalol • Ibutilide • Dofetilide • Dronedarone • E-4031 • Vernakalant	K^+ channel blocker Sotalol is also a beta blocker[5] Amiodarone has Class III mostly, but also I, II, & IV activity[6]	• In Wolff-Parkinson-White syndrome • (Sotalol:) ventricular tachycardias and atrial fibrillation • (Ibutilide:) atrial flutter and atrial fibrillation • (Amiodarone): prevention of paroxysmal atrial fibrillation,[7] and haemodynamically stable ventricular tachycardia[8]
IV	Calcium Channel Blockers	• Verapamil • Diltiazem	Ca^{2+} channel blocker	• Prevent recurrence of paroxysmal supraventricular tachycardia • Reduce ventricular rate in patients with atrial fibrillation
V		• Adenosine • Digoxin • Magnesium Sulfate	Work by other or unknown mechanisms (direct nodal inhibition)	Used in supraventricular arrhythmias, especially in heart failure with atrial fibrillation, contraindicated in ventricular arrhythmias. Or in the case of magnesium sulfate, used in *torsades de pointes*.

Antiarrhythmic agents:

1- Amiodarone hydrochloride:

Trade name: (Procor)
Classification:- Antiarrhythmic
Action: it blocking potassium rectifier currents that are responsible for the repolarization of the heart during phase 3 of the cardiac action potential.
Indications:

- Should be reserved for life threatening ventricular arrhythmias which don't respond to other therapy.

Antiarrhythmic agents:

1- Amiodarone hydrochloride:

Side effects:

- CVS: Bradycardia, CHF, Fatigue, tremors.
- CNS: Visual disturbances, photophobia, dry eyes.
- Blood: Hemolytic or aplastic anemia.
- Liver: Hepatotoxicity.

Contraindications:

- Sensitivity.
- Sinus bradycardia
- AV block
- Thyroid dysfunction

Antiarrhythmic agents:

1- Amiodarone hydrochloride:

Dose:

- **Tabs 200 mg.**
  - **Maintenance dose: 200-400 mg daily.**
  - **IV infusion: 5 mg/kg over 20-120 minutes.**

Antiarrhythmic agents:

2- Lidocaine Hydrochloride:

Trade name: xylocaine – Esracain

Class: antiarrhythmic and local anesthetic.

Action: decreases the depolarization, automaticity, and excitability in the ventricles during the diastolic phase by a direct action on the tissues, especially the Purkinje network, without involvement of the autonomic system. Neither contractility, systolic arterial blood pressure, atrioventricular (AV) conduction velocity, nor absolute refractory period is altered by usual therapeutic doses.

Indications: acute as which follow MI or cardiac surgeries and digitalis toxicity.

Antiarrhythmic agents:

2- Lidocaine Hydrochloride:

Contraindications:

- **Hypersensitivity**
- **Heart block.**

Side effects: **hypotension, bradycardia, dyspnea, dizziness.**

Dosage: **available in ampules of a concentration of 1% (100mg), 2% (200mg)**

Dose:

- **IV. Bolus 50-100 mg at rate of 25-50 mg\min**
- **Infusion: 20-50 mg\kg at a rate of 1-4 mg\m.**

Antiarrhythmic agents:

2- Lidocaine Hydrochloride:

Nursing considerations:

1. **Do not add** lidocaine to blood transfusion assembly.
2. Make certain that vials state “for cardiac arrhythmias”.
3. Use 5% dextrose solution to prepare drug (stable for 24 hours).
4. Assess for history of hypersensitivity.
5. Use electronic infusion device to regulate the infusion of the drug.

Antiarrhythmic agents:

2- Lidocaine Hydrochloride:

Nursing considerations:

6. Obtain B.P., pulse, respiratory rate to use as baseline data to evaluate response to treatment.
7. drug should be given in a monitored environment.
8. Assess B.P. frequently during administration.
9. Assess for respiratory depression.
10. If adverse reactions occur, discontinue infusion & prepare for emergency management.

Antiarrhythmic agents:

3- Phenytoin

Trade name: Epanutin, Dilantin

Class: anticonvulsant

Action:

- Effects are very similar to lidocaine
- Increase conduction through damaged purkinje fibers.
- Acts in the motor cortex of the brain to

in

this area. In addition, activity of centers in the brain stem responsible for the tonic phase of grand mal seizures.

Antiarrhythmic agents:

3- Phenytoin

Uses:

- Chronic epilepsy.
- Premature ventricular contractions. Tachycardia.

Contraindications:

- Hypersensitivity.

Side effects:

- Drowsiness, ataxia, dizziness, measles-like rash, gingival hyperplasia, Hirsutism (excessive hair growth) , hypoglycemia.

Antiarrhythmic agents:

3- Phenytoin

Note: rapid I.V. administration may result in hypotension & arrhythmia.

Dose for arrhythmias:

- Tabs 200-400 mg daily.
- I.V. 100 mg q 5 minutes up to a maximum of 1g.

Antiarrhythmic agents:

3- Phenytoin

Nursing Considerations:

1. I.V. phenytoin may forms precipitate, so flush tubing by saline (not dextrose) before & after administration.
2. Assess for hypersensitivity.
3. If a pregnant woman takes this drug, tell her not to breastfeed her baby.
4. Obtain liver & kidney function studies.
5. Monitor serum drug levels on a routine basis.
6. During I.V. therapy, monitor B.P. for signs of hypotension.
7. Take with food to minimize GI upset.
8. If the patient is diabetic, monitor for signs of hypoglycemia.
9. Oral hygiene to minimize bleeding from the gum.

Antiarrhythmic agents:

4- Procainamide:

Trade name: procan

Class: antiarrhythmic

Action: produce a direct cardiac effect to prolong the refractory period of the heart & depress the conduction of the cardiac impulse.

Uses:

- Ventricular tachycardia.
- Atrial fibrillation.
- Digitalis intoxication.

Antiarrhythmic agents:

4- Procainamide:

Contraindication: Hypersensitivity, complete AV heart block.

Side effects: Hypotension, abdominal pain, bitter taste.

5- Propranolol, bisoprolol

6- Verapamil

Nonopioid analgesics, antipyretics, and NSAIDS

- **Non-opioid analgesics, antipyretics, and NSAIDs are a broad group of pain medications.**
- **In addition to pain control, they produce antipyretic (fever control) and anti-inflammatory effects.**
- **They can be used alone or as adjuvant medications.**
- **These drugs have a ceiling effect, and no physical dependence is associated with them.**
- **They are not steroids and exhibited anti-inflammatory actions, they have analgesic, antipyretic, and platelet-inhibitory actions.**

Nonopioid analgesics, antipyretics, and NSAIDS

The drug classes included in this group are:

- Salicylates (especially aspirin), which are widely used
- Acetaminophen, a para-aminophenol derivative
- NSAIDs (nonselective and selective)
- Phenazopyridine hydrochloride, a urinary tract analgesic.

Non-steroidal Anti-inflammatory Drugs - Mechanism of Action

Arachidonic acid (component of the phospholipid of the cell membrane)

Mechanism

5- Lipoxygenase → Leukotrienes

Cyclooxygenases ← Inhibition ← **NSAIDs**

COX-1 → Prostaglandins

COX-2 → Prostaglandins



inhibit gastric acid secretion
stimulate synthesis of GI mucus

→ Epigastric distress
Ulceration
Hemorrhage



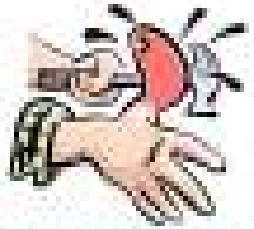
maintain renal blood flow

→ Na & water retention
→ Edema & Hyperkalemia
→ Interstitial nephritis



enhances platelet aggregation

→ Anti-coagulant



inflammation
(pain, fever, & edema)

→ Anti-inflammatory
Analgesic
Antipyretic

Topical irritation of the gastric epithelium

→ *decreases by the use of enteic-coated tablets, parenteral or rectal preparation*



- **Anti-inflammatory refers to the property of a substance or treatment that reduces inflammation or swelling.**

- **Anti-inflammatory drugs make up about half of analgesics, remedying pain by reducing inflammation as opposed to opioids, which affect the central nervous system.**

- They act by blocking the synthesis of prostaglandins by inhibiting cyclooxygenase, which converts arachidonic acid to cyclic endoperoxides, precursors of prostaglandins.

- Inhibition of prostaglandin synthesis accounts for their analgesic, antipyretic, and platelet-inhibitory actions

Salicylates:

Generic name: **Acetylsalicylic Acid:**

Trade name: **Aspirin**



Classification:

- **Non-narcotic analgesic, antipyretic, anti-inflammatory, antirhumatic, antiplatelet, NSAID.**

Salicylates:

Action:

- The antipyretic effect is due to an action on the hypothalamus that results in heat loss by vasodilation of peripheral blood vessels & promoting sweating.
- The anti-inflammatory effects probably by decreasing prostaglandin synthesis & other mediators of the pain response.
- The analgesic action is not fully known but may be due to improvement of the inflammatory condition .

N.B. : Aspirin also produces inhibition of platelet aggregation.

Salicylates:

Action:

- One salicylate, aspirin, inhibits platelet aggregation (the clumping of platelets to form a clot) by interfering with the production of thromboxane A2, a substance necessary for platelet aggregation. Unlike aspirin, NSAIDs' effects on platelet aggregation are temporary.

- As a result, aspirin can be used to enhance blood flow during myocardial infarction (MI) and to prevent an event such as recurrent MI.

Salicylates:

Indications

- Mild to moderate pain
- Fever
- Inflammatory conditions—rheumatic fever, rheumatoid arthritis, osteoarthritis
- Reduction of risk of recurrent Transient Ischemic Attack (TIAs) or stroke in males with history of TIA due to fibrin platelet emboli
- Reduction of risk of death or nonfatal MI in patients with history of infarction or unstable angina pectoris
- MI prophylaxis

Side effects

The most common adverse reactions to salicylates include gastric distress, nausea, vomiting, and bleeding tendencies.

Other adverse reactions include:

- **hearing loss (when taken for prolonged periods)**
- **diarrhea**
- **thirst**
- **tinnitus**
- **confusion**
- **dizziness**
- **impaired vision**
- **hyperventilation (rapid breathing)**
- **Reye's syndrome (when given to children with chickenpox or flulike symptoms).**

Salicylates:

Contraindications:

- Allergy to salicylates , NSAID and tartrazine
- Hemophilia
- bleeding ulcers
- Hemorrhagic states
- Blood coagulation defects
- Hypoprothrombinemia
- Vitamin K deficiency (increased risk of bleeding)
- Pregnancy, lactation, .
- Before and after surgery (tooth extraction)
- asthma

Dose:

- In minor conditions: 325-600 mg Q 4 hours.
- May reach up to 6 grams /day in divided doses in arthritis and rheumatic conditions.

Salicylate toxicity

- Acute aspirin poisoning: Respiratory alkalosis, hyperpnea, tachpnea, hemorrhage, confusion, pulmonary edema, convulsion, tetany, metabolic acidosis.

Salicylates:
Drug interactions:

- Risk for bleeding increase if taken with other anticoagulants.
- Risk of GI bleeding increase if taken with steroids, alcohol, or other NSAIDs.
- Increased risk for salicylate toxicity if taken with furosemide (Lasix)
- Hypotension may occur if taken with nitroglycerns.

Salicylates:

considerations:

1. Take drug with or after food or with milk to decrease GI irritation.
2. Assess for history of asthma and history of hypersensitivity.
3. Do not use with other anticoagulants.
4. Note any history of peptic ulcer.
5. Report signs of side effect e.g. gastric irritation if occurs.
6. Aspirin is not given 1 week before & after surgery to prevent bleeding.

Salicylates: considerations:

7. If patient is diabetic, discuss the possibility of hypoglycemia occurring to patients, and should monitor their blood glucose level frequently.
8. Teaches patient about the toxic symptoms (ringing in the ears dizziness, mental confusion-etc) and ask him/her to report it to physician.
9. Institute emergency procedures if overdose occurs: Gastric lavage, induction of emesis, activated charcoal, supportive therapy.

Salicylate warning

Before administering salicylates to a patient, be aware of its risks to special populations:

- **Children and teenagers: Avoid the use of aspirin and salicylates to treat flulike symptoms or chickenpox because Reye's syndrome may develop.**
- **Surgical patients: If possible, discontinue aspirin 1 week before surgery because of the**
- **risk of postoperative bleeding.**
- **Asthmatics: Be aware that these patients are more likely to develop bronchospasm, urticaria,**
- **angioedema, or shock when salicylates are administered.**

Acetaminophen:

Acetaminophen is an OTC drug that produces analgesic and antipyretic effects. It appears in many products designed to relieve pain and symptoms associated with colds and influenza.

Generic name: “paracetamol”

Trade names: acamol, panadol





Class. : non-narcotic analgesic, para-aminophenol type.

Action:

- Acetaminophen decrease fever by an effect on hypothalamus leading to sweating & vasodilation.
- It also inhibits the effect of pyrogens on the heat-regulating center on the hypothalamus.
- It may cause analgesia by inhibiting CNS prostaglandin syntheses (it has no anti-inflammatory effect).
- It doesn't manifest any anticoagulant effect or any ulceration of GIT.

Uses:

- **Pain due to Headache, dysmenorrhea, arthralgia, myalgia, muscluoskletal pain, immunization, teething, tonsillectomy.**
- **To reduce fever due to bacterial & viral infection .**
- **As a substitute for aspirin when contraindicated .**

Contraindications: renal insufficiency, anemia.

Side effects:

- Most patients tolerate acetaminophen well. Unlike the salicylates, acetaminophen rarely causes gastric irritation or bleeding tendencies. However, it may cause liver toxicity, and the total daily dose should be monitored.

Other adverse reactions include:

- skin rash
- hypoglycemia (with overdose)
- neutropenia.

Symptoms of over dosage:

- Hepatic toxicity (general malaise, delirium, depression seizures, coma & death, nausea, vomiting, fever, and vascular collapse).

Treatment of overdose:

1. Induction of emesis.
2. Gastric lavage.
3. Activated charcoal.
4. Oral N-acetyleystine (mucomyst) is said to reduce or prevent hepatic damage by inactivating acetaminophen metabolites which cause liver effects.

Dose: Tab. 500 mg Q 4 hrs or up to 1g Q 6 hrs.

Considerations:

1. Suppositories should be stored below 27c .
2. Liver function studies for long term therapy.
3. Note signs of met-hemoglobinemia: bluish discoloration of gum & finger nails.
4. Have mucomyst available for signs of toxicity.
5. Teach patient signs of toxicity to be reported immediately.

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

As in aspirin, the therapeutic actions of these substances are believed to result from the inhibition of the enzyme cyclo-oxygenase which results in decreased prostaglandin synthesis so it is effective in:

- **Reducing joint swelling, pain & morning stiffness.**
- **Increasing the mobility in arthritic patients.**
- **Antipyretic action due to decreased production of prostaglandin from the hypothalamus.**
- **Having irritating effect on the GIT.**

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

- As their name suggests, nonsteroidal anti-inflammatory drugs, or NSAIDs, are typically used to combat inflammation.
- Their anti-inflammatory action equals that of aspirin.
- They also have analgesic and antipyretic effects.

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

- **Nonselective NSAIDs inhibit prostaglandin synthesis by blocking two enzymes known as cyclooxygenase-1 (COX-1) and cyclooxygenase- 2 (COX-2).**
- **These drugs (called COX-1 and COX-2 inhibitors) include indomethacin, ibuprofen, diclofenac, etodolac, fenoprofen, flurbiprofen, ketoprofen, ketorolac, mefenamic acid, meloxicam, nabumetone, naproxen, oxaprozin, and sulindac.**

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

- **Selective NSAIDs selectively block COX-2 enzymes, thereby inhibiting prostaglandin synthesis.**
- **This selective inhibition of COX-2 produces the analgesic and anti-inflammatory effects without causing the adverse GI effects associated with COX-1 inhibition by nonselective NSAIDs.**

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

- As in aspirin, the therapeutic actions of these substances are believed to result from the inhibition of the enzyme cyclo-oxygenase which results in decreased prostaglandin synthesis so it is effective in:
  - Reducing joint swelling, pain & morning stiffness.
  - Increasing the mobility in arthritic patients.
  - Antipyretic action due to decreased production of prostaglandin from the hypothalamus.
  - Having irritating effect on the GIT.

Uses:

- Rheumatoid arthritis
- Osteorthritis.
- Gout
- Other musculoskletal diseases.
- Dental pain
- Strains & sprains.

Contraindications:

1. Children less than14 years of age.
2. Lactation.
3. Hypersensitivity (asthma, rashes, rhinitis).
4. Uses with caution in patients with a history of GI disease & reduced renal functions.

Side effects:

- Peptic, duodenal ulcer, GI bleeding nausea, vomiting, dyspepsia.
- Dizziness, drowsiness. Hypo + hyperglycemia.
- Bronchospasm, rhinitis. Blurring of vision.
- Tinnitus, loss of hearing.
- Bone marrow depression, C.H.F.

Considerations:

- Note any history of allergic responses to aspirin or nonsteroidal anti-inflammatory agents.
- Note the age of the client.
- Determine if patient is taking oral hypoglycemic or insulin and document it.
- Take these agents with milk or meal or antacids as prescribed.
- Encourage patient to take drug regularly.
- Report signs of GI irritation.
- Instruct client to report signs of bleeding, blurring of vision, tinnitus rashes – etc.
- If the client has Diabetes Mellitus, explain the possible in increasing hypoglycemic effect of the drugs, to test urine & blood for glucose. To adjust dose of these agents.

1. Diclofenac Sodium:



Trade name: Voltaren , Rufenal

Class. : Non steroidal anti-inflammatory analgesic

Dose: Suppositories, tabs or injection of 150-200 mg daily in 2-4 divided doses.

considerations:

1. Give on full stomach to avoid GIT irritation.
2. When given IM, Give it deep into a large muscle because drug is very irritant.



2. Indomethacin:

Trade name : Indocid.

Class. : Anti-inflammatory, analgesic, antipyretic.

Dose:

- suppositories & caps.
- 25mg – 50 mg bid-tid.

3. Naproxen:

Trade name: Naprex.

Class.: Non- steroidal anti-inflammatory analgesic.

Dose: 500 mg bid.

4. Ibuprofen:

Trade names: Brufen, artofen.
Class.: nonsteroidal anti-inflammatory analgesie.
Dose: 400 mg bid.

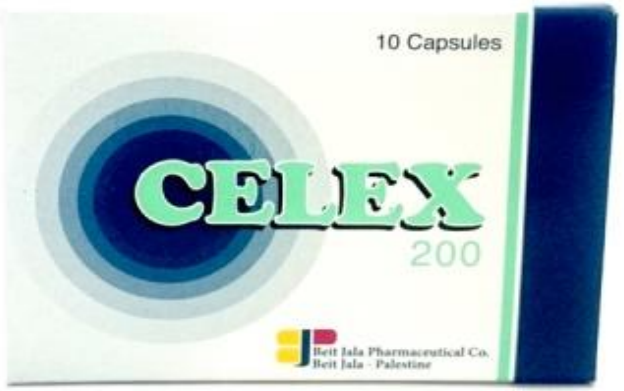




5. **Generic name :** Celecoxib

Trade name : Celex , Celebrex

Class: NSAID, Analgesic (non-opioid), Specific COX-2 enzyme blocker



Therapeutic actions :

- Analgesic and anti-inflammatory activities related to inhibition of the enzyme, which is activated in inflammation to cause the signs and symptoms associated with inflammation; does not affect the COX-1 enzyme, which protects the lining of the GI tract and has blood clotting and renal functions.

5. Celecoxib

Indications

- Acute and long-term treatment of signs and symptoms of
  and
- Reduction of the number of colorectal polyps in familial adenomatous polyposis (FAP)

- Treatment of
- Relief of signs and symptoms of anklylosing spondylitis

5. Celecoxib

Contraindications and cautions

- Contraindicated with , or aspirin; significant renal impairment; pregnancy (third trimester); lactation.
- Use cautiously with impaired hearing, hepatic, and CV conditions.

Dosage forms: Capsules 100, 200 mg

Dose:

Initially, 100 mg PO bid; may increase to 200 mg/day PO bid as needed.

5. Celecoxib

considerations

- Administer drug with food or after meals if GI upset occurs.
- Establish safety measures if CNS, visual disturbances occur.
- Arrange for periodic ophthalmologic examination during long-term therapy.
- If overdose occurs, institute emergency procedures—gastric lavage, induction of emesis, supportive therapy.
- Provide further comfort measures to reduce pain (eg, positioning, environmental control), and to reduce inflammation (eg, warmth, positioning, rest).

Assignment

Write mechanism of action, classification, side effects, contraindications, dose, dosage form and uses of Etoricoxib

Narcotic Analgesics & Antagonists

- Opioids produce the majority of their therapeutic and adverse effects by acting as agonists at mu and/or kappa opioid receptors.
- It include opium such as morphine, codeine & opium derivatives such as Meperidine.
- These substances have similar pharmacological properties.
- Following prolonged use, patients develop tolerance to most opioid effects. Constipation and miosis are notable exceptions,
- The relative activity of all narcotic analgesics in measured against morphine.

Narcotic Analgesics & Antagonists

Dependence & Tolerance:

- Drugs of this group may lead to addiction.
- Psychological & physical dependence & tolerance develop even when using clinical doses.
- Tolerance usually develops because the patient requires shorter periods between doses or larger doses for relief of pain.

Narcotic Analgesics & Antagonists

Effects of narcotic analgesics:

Central nervous system effects:

Analgesia: Related to their ability to change both pain perception and effectively raise the threshold for pain.

Euphoria: A pleasant floating sensation and freedom from anxiety and distress.

Sedation: Drowsiness and clouding of mentation are frequent concomitants action.

Narcotic Analgesics & Antagonists

Effects of narcotic analgesics:

Central nervous system effects:

Respiratory depression: By inhibiting brain, stem respiratory mechanisms.

Cough suppression: Suppression of the cough reflex is a well-recognized action

Miosis: Constriction of the pupils is seen with virtually all of the opioid agonists.

Nausea and vomiting: it can activate the brain stem chemoreceptor trigger zone to produce nausea and vomiting.

Narcotic Analgesics & Antagonists

Peripheral Effects:

Cardiovascular system: cerebral vascular dilation and concomitant decrease in cerebral vascular resistance, increase in cerebral blood flow, and increase in intracranial pressure.

Gastrointestinal tract: Constipation, decrease motility, decreased gastric. In the large intestine delays passage of the fecal mass and allows increased absorption of water, which leads to constipation (management of diarrhea).

Billiary tract: The opioids constrict biliary smooth muscle, which may result in biliary colic.

Narcotic Analgesics & Antagonists

Peripheral Effects:

Genitourinary tract: Renal function is depressed by opioids. It is believed that in humans this is chiefly due to decreased renal plasma flow.

Uterus: The opioid analgesics may prolong labor.

Miscellaneous: Flushing and warming of the skin accompanied sometimes by sweating and itching; central effects and histamine release may be responsible for these reactions.

Narcotic Analgesics & Antagonists

Acute toxicity:

Characterized by respiratory depression, deep sleep , stupor, coma, pinpoint pupil, , cyanosis, hypotension, decreased urinary output, decreased temperature, clammy skin, and finally Death (due to Respiratory failure).

Treatment of acute overdose:

1. **Induce vomiting or gastric lavage.**
2. **Artificial respiration.**
3. **Give narcotic antagonist (Narcan).**

Narcotic Analgesics & Antagonists

Signs & symptoms:

- Constricted pupil, constipation, skin infections, needle scare abscesses & itching on the anterior surface of the body.

- Withdrawal signs appear when drugs is withheld for 4-12 hrs. & characterized by intense craving for the drug, insomnia, yawning, sneezing, vomiting, diarrhea, tremors, sweating, mental depression, muscular aches, pain, chills & anxiety. (They are rarely life- threatening).

Narcotic Analgesics & Antagonists

Signs & symptoms:

Action of narcotic analgesics:

- **Narcotic analgesics attach to specific receptor in the CNS resulting in analgesia- action**
- **Action exactly is unknown but may be by decreasing cell membrane permeability to sodium transmission of pain impulses.**

Narcotic Analgesics & Antagonists

Uses:

- Sever pain
- Hepatic & renal colic.
- Pre-anesthetic medication
- Post-surgical pain.
- Diarrhea & dysentery
- Pain from MI, carcinoma.
- Postpartum pain & burns.
- Antitussive.

Narcotic Analgesics & Antagonists

Contraindications:

- Asthmatic conditions
- Emphysema
- Sever obesity
- Convulsions
- Diabetic acidosis
- Myxedema
- Addisson's disease
- Hepatic cirrhosis
- Children less than age of 6 months.

Narcotic Analgesics & Antagonists

Side effects:

Respiratory depression, apnea, dizziness, euphoria headache, mental clouding, insomnia, nausea, vomiting, constipation, dry mouth, skin rashes, laryngospasm, urinary retention, and decreased libido.

Narcotic Analgesics & Antagonists

Nursing considerations:

- Use supportive nursing measures as relaxation techniques to relieve pain before using narcotics.
- Explore the source of pain, use non-narcotic analgesia if possible.
- Administer the medication when needed, prolonging the medication administration will decrease the effect of the medication.
- Monitor vital signs & mental status.
- Monitor respiratory rate (drug may lead to respiratory depression).
- Monitor blood pressure (hypotension may occur)
- Monitor pulse rare (if 60\m withhold the drug).

Narcotic Analgesics & Antagonists

Nursing considerations:

- Watch for constricted pupils. Document it and notify the physician.
- Monitor bowel function, since drug may cause constipation .
- Encourage client to empty bladder every 3-4 hrs (since drug may cause urinary retention).
- If client is bed ridden, use side rails.
- Inform the client\family that the drug may become habit forming and leading to addiction.
- Document any history of asthma or other contraindications.
- Have emergency equipment and narcotic antagonist available .

Narcotic Analgesics & Antagonists

1. Codeine sulfate:

Trade name: Codivis, Rekod , (Tussosedan)
Class. : Narcotic analgesic, morphine type.

Action:

- Resembles morphine pharmacologically but produce less effect on respiratory system, less nausea & less vomiting.
- In high doses (more than 60 mg), it will irritate the cough center, but in lower doses, it is a potent antitussive and is an ingredient in many cough syrups.

Narcotic Analgesics & Antagonists

Uses:

- **Relief of mild to moderate pain.**
- **Antitussive.**

Dose:

- **Analgesic: 15-60 mg 4-6 hrs.**
- **Antitussive: 10-20 mg 9 4-6 hrs.**

Narcotic Analgesics & Antagonists

2. Meperidine Hydrochloride "Pethedine Hydrochloride":

Trade name: Demerol

Class. : Synthetic opioid agonist analgesic.

Action: Similar to opiates.

- It has no antitussive effect.
- The duration of action is less than that of opium.

Narcotic Analgesics & Antagonists

2. Meperidine Hydrochloride "Pethedine Hydrochloride":

Uses:

- Sever pain.
- Renal & hepatic colic.
- Obstetric pre-anasthetic medication.
- In minor surgeries.
- Spasm of GI tract, uterus.
- Prior some diagnostic procedures e.g. cystoscopy.
- Post operative pain.

Narcotic Analgesics & Antagonists

2. Meperidine Hydrochloride "Pethedine Hydrochloride":

Additional Contraindications:

Hypersensitivity, convulsive states, children less than 6 months, head injuries, diabetic acidosis.

Additional Side effects: Transient hallucinations, hypotension.

Dose:

- **Drug can is available in the form of tablets, syrup, I.M, S.C. Dose is 50-100 mg Q 4-6 hr.**
- **It can be given as I.V. continuous infusion on a concentration of 1 mg\ml.**
- **It also can be given IV slowly, and should be diluted in a concentration of 10mg/ml.**

Narcotic Analgesics & Antagonists

3. Methadone Hydrochloride:

Class.: Narcotic analgesic, morphine type.

Action:

- Produce only mild euphoria and used as a heroin withdrawal substitute.
- It produces physical dependence but the abstinence syndrome develops more slowly upon termination of the therapy.
- Withdrawal symptoms are less intense but more prolonged than those associated with morphine.
- It is not effective for preoperative or obstetric anesthesia.

Narcotic Analgesics & Antagonists

3. Methadone Hydrochloride:

Uses:

- Sever pain.
- Drug withdrawal.

Additional Contraindications:

- Pregnancy since it depresses respiration of neonate.
- I.V. use.
- Liver disease.

Additional side effects:

- Constipation and pulmonary edema.

Dose:

- Can be given oral, I.M., S.C. at a dose of 2.5 – 10 mg Q 3-4 hrs.

Narcotic Analgesics & Antagonists

4. morphine Sulfate:

Class.: Narcotic analgesic, morphine type.

Action: See narcotic analgesic.

Uses:

- Intrathecally, orally or I.V. infusion for acute or chronic pain.
- Preoperative medication.
- To facilitate induction of anesthesia or to decrease the dose of anesthesia.

Narcotic Analgesics & Antagonists

4. morphine Sulfate:

N.B.:

- It is given in lower doses for continuous pain & in higher doses in sharp intermittent & all kinds of pain.

Additional contraindications:

- It is not given intrathecal, if infection is present at injection site.
- In patients on anticoagulant therapy.
- Bleeding disorders.
- If patients have received parenteral corticosteroids within the past 2 weeks.

Narcotic Analgesics & Antagonists

4. morphine Sulfate:

Dose:

- **Oral: 10-30 mg Q 4 hr.**
- **I.M.: 5-20 mg\70 kg Q 4 hr as needed.**
- **I.V.: bolus of 2.5-15 mg for a person of average weight of 70 kg over 4-5 minutes (slowly).**
- **Continuous infusion: 0.1-1 mg\ ml in 5% dextrose in water by a controlled infusion pump.**

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Trade name: Ultram, Tramal

Class. : Narcotic analgesic, synthetic.

Action: Unknown. A centrally acting synthetic analgesic compound not chemically related to opiates. Thought to bind to opioid receptors and inhibit reuptake of norepinephrine and serotonin.

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Indications & dosages

- Moderate to moderately severe pain
- Adults: Initially, 25 mg P.O. Adjust by 25 mg q 3 days to 100 mg/day (on divided doses).

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Adverse reactions

- CNS: dizziness, vertigo, headache, CNS stimulation, anxiety, confusion, euphoria, nervousness, sleep disorder, seizures, malaise, visual disturbances.
- CV: vasodilation.
- GI: nausea, vomiting, constipation, dyspepsia, dry mouth, diarrhea, abdominal pain, anorexia, flatulence.
- GU: urine retention, urinary frequency, menopausal symptoms, proteinuria.
- Respiratory: respiratory depression.
- Skin: pruritus, diaphoresis, rash.

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Contraindications & cautions

- hypersensitive to drug or other opioids,
- in breast-feeding women
- acute intoxication from alcohol, hypnotics, centrally acting analgesics, opioids, or psychotropic drugs.
- Use cautiously in patients at risk for seizures or respiratory depression; in patients with increased intracranial pressure or head injury, acute abdominal conditions, or renal or hepatic impairment; or in patients with physical dependence on opioids.

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Nursing considerations

- Reassess patient's level of pain at least 30 minutes after administration.
- Monitor CV and respiratory status. Withhold dose and notify doctor if respirations decrease or rate is below 12 breaths/minute.
- Monitor bowel and bladder function. Anticipate need for laxative.
- For better analgesic effect, give drug before onset of intense pain.
- Monitor patients at risk for seizures. Drug may reduce seizure threshold. In the case of an overdose, naloxone may also increase risk of seizures.

Narcotic Analgesics & Antagonists

5. Tramadol hydrochloride

Nursing considerations

- Monitor patient for drug dependence. Drug can produce dependence similar to that of codeine and thus has potential for abuse.
- Withdrawal symptoms may occur if drug is stopped abruptly. Reduce dosage gradually.
- Caution ambulatory patient to be careful when rising and walking. Warn outpatient to avoid driving and other potentially hazardous activities that require mental alertness until drug's CNS effects are known.

Narcotic Antagonists:

The narcotic antagonists are able to prevent or reverse many of the pharmacological actions of morphine-type analgesics & meperedine as respiratory depression induced by these drugs within minutes.

Narcotic Antagonists:

Naloxone Hydrecholride:

Trade name: Narcan.

Class.: Narcotic antagonist

Action:

- Block the action of narcotic analgesic by displacing previously given narcotics from their receptor sites or preventing them from attaching to opiate receptors.
- The duration of action of naloxone is shorter than that of the narcotic analgesic so the respiratory depression may return when the narcotic antagonist has washed off the body.

Narcotic Antagonists:

Naloxone Hydrecholride:

Uses:

- **Respiratory depression induced by narcotics.**
- **Drug of choice when the depressant drug is unknown.**
- **Diagnosis of acute opiate overdose.**

Note: Naloxone is not effective when respiratory depression is induced by hypnotic, sedative or other narcotic drugs.

Contraindications:

- **Sensitivity to drug.**
- **Narcotic addicts since it will cause severe withdrawal symptoms.**
- **Neonates.**

Narcotic Antagonists:

Naloxone Hydrecholride:

Side effects:

- Nausea, vomiting, sweating, hypertension, tremors.
- If used postoperatively: tachycardia, pulmonary edema, hypo or hypertension.

Dose: 0.4-2 mg I.V , S.C. or I.M.

considerations:

1. Determine the etiology of respiratory depression.
2. Assess & obtain baseline vital signs.
3. Monitor respiration closely after the duration of action.
4. Have emergency drugs & equipment available.
5. If the patient is comatosed, turn him to his side to avoid aspiration.
6. Maintain safe environment (side rails & soft support).

Antihistamine “H 1 Blockers”

- **Antihistamines primarily act to block histamine effects that occur in an immediate hypersensitivity reaction, commonly called an allergic reaction.**

- **They’re available alone or in combination products and may be obtained by prescription or over the counter.**

Antihistamine “H 1 Blockers”

- **The term antihistamine refers to drugs that act as histamine-1 (H1)-receptor antagonists; that is, they compete with histamine for H1-receptor sites throughout the body. However, they don’t displace histamines already bound to receptors.**

Antihistamine “H 1 Blockers”

Appropriate stimuli including: tissue injury, antigen- antibody (allergic) reactions, and extreme cold trigger the release of histamine from its storage sites into the vascular system

Antihistamine “H 1 Blockers”

it induces the following responses:

1. Dilation & increased permeability of the small arterioles & capillaries results in increasing permeability to fluid leading to hypotension & edema nasal congestion & laryngeal edema “associated with allergies”.
2. Contraction of some smooth muscles such as those of bronchioles leading to bronchoconstriction “the role of histamine plays in bronchial asthma”. & uterine contraction.
3. Dilation of cerebral vessels (Headache).
4. Pain & itching because it stimulates the sensory nerve endings.

Action:

- The effect of histamines may be reversed either by drugs that block histamine receptors (antihistamine) or by drugs that have effects opposite to those of histamine e.g. epinephrine.
- Antihistamines used for the treatment of allergic conditions are referred to as H1-receptor blockers while those used for treatment of GI disorders as peptic ulcer are referred as H2-receptor blockers.
- They don't prevent the release of histamine
- They prevent or reduce increased permeability that results in decreasing edema & itching, & bronchospasm.
- H1-blockers manifest varying degrees of CNS depression, anti-cholinergic & antiemetic effect.

Antihistamine “H 1 Blockers”

Uses:

- Treatment of seasonal allergic rhinitis, allergic conjunctivitis.
- Treatment of urticarial transfusion reactions.
- Treatment of topic dermatitis.
- Treatment of insect bites.
- Sneezing & rhinorrhea due to common cold.
- Prophylaxis & treatment of motion sickness “nausea & vomiting”.
- Night – time sleep aid.



Antihistamine “H 1 Blockers”

Contraindications:

- Pregnancy.
- Glaucoma
- Prostatic hypertrophy
- CNS depression (phenothiazine type).
- Bone marrow depression
- Comatose patients.

Antihistamine “H 1 Blockers”

Side effects:

- **Sedation, deep sleep, Dizziness, Headache, muscle weakness, disturbed coordination, epigastric distress, dry mouth, nausea, vomiting, urinary frequency, anemia (pancytopnea).**
- **Paradoxical excitation (especially in children & elderly) including restlessness, irritability, insomnia, hysteria, tremors euphoria, nervousness, hallucinations, disorientation & convulsion. Usually caused by overdose (acute toxicity).**

Antihistamine “H 1 Blockers”

Treatment of overdose:

- Symptomatic & supportive.
- Vomiting is induced with syrup of ipecac.
- Gastric lavage.
- Vasopressors (to treat hypotension) –e.g. Dopamine , adrenaline.
- Phenytoin for treatment of convulsion.

N.B.: Do not use CNS depressants including diazepam.

Antihistamine “H 1 Blockers”

Nursing Considerations:

- Inject I.M. preparations deep into muscles.
- Oral preparations may cause gastric irritation, so give drug with meals.
- Note if the client has any medical history of ulcer, glaucoma & if the client is pregnant.
- Obtain a baseline B.P., Pulse & respiration.
- Note signs of CNS depression (signs of overdose so induce vomiting).
- If in hospital, use side rails (safety measures).
- Advice client to report signs of side effects immediately.
- Instruct client to avoid undue exposure to sun.
- If the drug is being used for motion sickness, it should be taken 30 minutes before transporting. Caution the client not to drive a car or operate other machinery.

Antihistamine “H 1 Blockers”

1- Astemizole:

Trade name: Hismanal
Class.: antihistamine
Action: It has no sedative, antiemetic or anticholinergic effects.

2- Chlorpheniramine maleate:

Trade name: Anaphyl, Ahistone
Class: Antihistamine
Action: sedation is less pronounced.
Dose: syrup each 5ml contains 5mg, 2 teaspoonful 3-4 times daily.

Antihistamine “H 1 Blockers”

3- Promethazine Hcl:

Trade name: phenergan, prothiazine.

Class:

- It is aphenothiazine derivative. It is a potent antihistamine with prolonged action. It may cause severe drowsiness. It also provides antiemetic effect (it chemoreceptor trigger zone).

- It also has a sedative action, effective in vertigo and decrease vestibular apparatus

Antihistamine “H 1 Blockers”

3- Promethazine Hcl:

Uses:

- Motion sickness.
- Nausea & vomiting due to anesthesia.

Forms:

- Syrup: 5ml contains 5 mg , 25 mg.
- Ampule: 50 mg \2ml.

Dose:

- Antihistamine: 125 mg 4 times daily.
- Sedative: 25 mg – 50 mg.
- Antivertigo: 25 mg 2 times daily.

Antihistamine “H 1 Blockers”

4- Loratadine

Trade name: lorastine
Class: Antihistamine
Action: Non-sedating antihistamines
Dose:

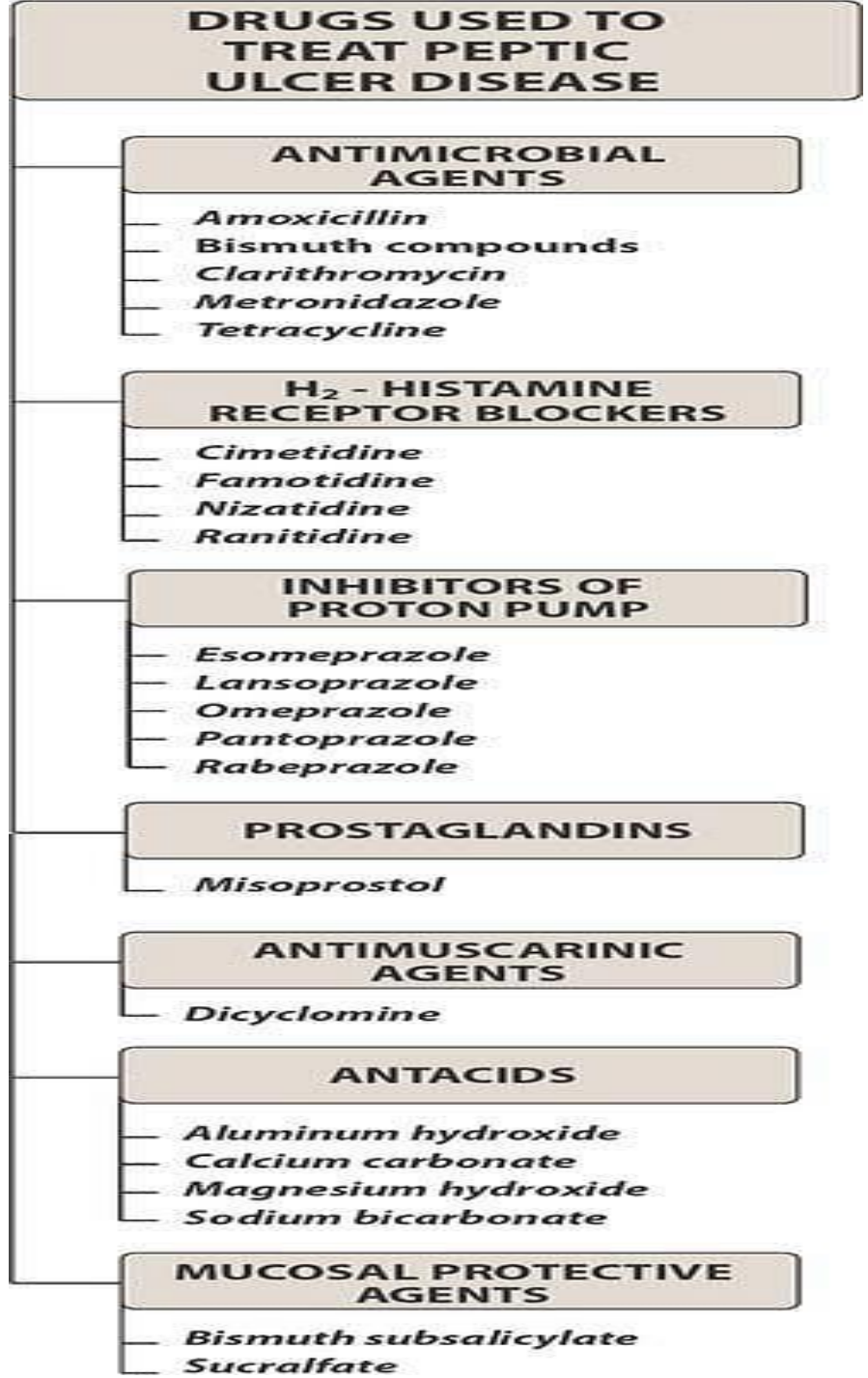
- Tablet 10 mg
- Syrup each 5ml contains 5mg, 1 teaspoonful 1-2 times daily.

5.cyproheptadine
6. Telfast (Fexofenadine)120, 180

Anti-peptic ulcer drugs

Drugs classification

a. Antacids
b. Gastric anti secretory drugs
    1. H2 receptor antagonists (cimetidine, famotidine, nizatidine, ranitidine).
    2. Anti muscarinics
    3. Proton pump inhibitors (omeprazole, lansoprazole)
c. Mucosal protective agents (sucralfate, colloidal bismuth compound, prostaglandine analogues e.g. misoprostol)


DRUGS USED TO TREAT PEPTIC ULCER DISEASE
ANTIMICROBIAL AGENTS
Amoxicillin
Bismuth compounds
Clarithromycin
Metronidazole
Tetracycline
H2 - HISTAMINE RECEPTOR BLOCKERS
Cimetidine
Famotidine
Nizatidine
Ranitidine
INHIBITORS OF PROTON PUMP
Esomeprazole
Lansoprazole
Omeprazole
Pantoprazole
Rabeprazole
PROSTAGLANDINS
Misoprostol
ANTIMUSCARINIC AGENTS
Dicyclomine
ANTACIDS
Aluminum hydroxide
Calcium carbonate
Magnesium hydroxide
Sodium bicarbonate
MUCOSAL PROTECTIVE AGENTS
Bismuth subsalicylate
Sucralfate

Anti-peptic ulcer drugs

A. Antacids

Action:

- They are weak bases that neutralizes gastric acidity by reacting with gastric HCL to form salt and water.
- **Properties of ideal antacids**
  - Immediate and prolonged neutralization of acid without increasing PH above 4.
  - Action should be confined to GIT.
  - No constipating or laxative effects
  - should not be absorbed systemically
  - Palatable to patients

Anti-peptic ulcer drugs

A. Antacids

Uses:

- Treatment of hyperacidity. (Heart- burns).
- Peptic ulcer
- Duodenal ulcer.
- Gastro-esophageal reflux.

Contraindications:

- Sodium containing products are contraindicated in C.H.F., hypertension, and other conditions requiring low sodium diet.
- Pregnancy
- Children less than 6 years of age.
- N.B.: chronic use of aluminum containing antacids may contribute to development of Alzheimer's disease.

Anti-peptic ulcer drugs

A. Antacids

Nursing considerations:

- **It is recommended that most antacids be taken at 3 hours after meals & at bed – time.**
- **Tablets should be thoroughly chewed before swallowing & followed by a glass of milk or water.**
- **Shake liquid suspensions thoroughly before pouring the medication.**
- **Clients' taking aluminum or calcium containing antacids should take 2500-3000 cc of fluids to prevent constipation.**
- **Advise clients to report persistent diarrhea or constipation physician.**

Anti-peptic ulcer drugs

A. Antacids

1. Aluminum Hydroxide :

Class.: Antacid

Action: it reacts with intestinal secretions to produce insoluble salts esp. phosphates that are eliminated in feces prevent phosphates urinary stones.

Additional uses: prevention of urinary stones, hypophosphatemia..

Side-effects: constipation, intestinal obstruction, bone pain, muscle weakness.

Drug interaction: reduce bioavailability of some drugs e.g. tetracycline, atropine.

Dose: 500-1800 mg 3-6 times daily after meals, between meals & at bed time.

Anti-peptic ulcer drugs

A. Antacids

2. Calcium carbonate:

Class: antacid.

Action: Non-systemic, since calcium carbonate is constipating it is often alternated or mixed with Mg+2 salts.

Additional uses: calcuim deficiency .

Side effects: constipation, flatulence, eructation, intestinal obstruction, metabolic alkalosis, hypercalcemia , rebound hyperacidity.

Milk alkali syndrome: occur with concomitant heavy milk drinking and characterized by: Headache, weakness, anorexia, nausea, vomiting, abdominal pain, constipation, thirst, polyuria.

Dose: 0.5 – 1 g as necessary.

Anti-peptic ulcer drugs

A. Antacids

3. Maalox:

Class. : Antacid.

Additional uses: hiatus hernia

Dosage:

- Tablet: antacids Al(OH)3 200 mg + Mg (OH)2 200 mg.
- Suspension: antacids 225 mg of Al(OH)3 /5ml + 200mg of Mg(OH)2.

Dose:

- 2-4 tabs - 20-60 m after meals & at bedtime.
- Suspension: 10- 20 ml aid- 20-60 m after meals & at bedtime.

Anti-peptic ulcer drugs

A. Antacids

4. Magnesium Hydroxide (magnesia) :

Class: antacid , laxative

Action:

- **Acts as an antacid by neutralizing Hcl, it does not produce alkalosis & has a demulcent effect.**
- **As a laxative, it increases the bulk of the stools by attracting & holding large amounts of fluids. The increased bulk results in the mechanical stimulation of peristalsis.**

Uses:

- **antacid**
- **laxative to empty the bowel prior to diagnostic or surgical procedures .**

Contraindications: - poor renal function.

Side effects: Diarrhea, abdominal pain, nausea, vomiting, hypermagnesemia .

Anti-peptic ulcer drugs

A. Antacids

5. Sodium Bicarbonate: NaHco3

Class: Alkalinizing agent, antacid, electrolyte.

Action: Is due to neutralization of Hcl by forming sodium chloride & CO2.

NB:- Rarely used as antacid because of:

1- It's high sodium content.
2- Short duration of action.
3- Ability to cause alkalosis.

Uses:

- Hyperacidity
- severe diarrhea (increased loss of Hco3 -)
- Metabolic acidoses (shock, dehydration, renal diseases…..)

Anti-peptic ulcer drugs

A. Antacids

Contraindications:

- C.H.F.
- Renal impairment
- Edema
- Cirrhosis of the liver
- Metabolic or respiratory alkalosis.
- Children less than 6 years of age.

Side effects:

- Metabolic alkalosis (nausea, vomiting, cramps, dizziness, decreased breathing).
- Extravasation following I.V. use may manifest: ulceration, sloughing, cellulitis or tissue necrosis

Nursing considerations:

- I.V. Dose should be determined by Arterial Blood Gases analysis.
- Should be administered. slowly.
- Periodically assess patient serum pH during the therapy.
- Observer for signs of edema.

Gastric anti secretory drugs

H2 receptor antagonists:

1- Cimetidine:

Trade name: Tagamet
Class: Histamine H2-receptor blocking agent.
Action: decreases the acidity of the stomach by blocking the action of histamine, which involved in triggering gastric acid secretion. It blocks the action of histamine by competitively occupying the histamine H2-receptors in the gastric mucosa leading to decrease secretion of HCl.

Gastric anti secretory drugs

H2 receptor antagonists:

1- Cimetidine:

Uses:

- Short-term (up to 8 wks) & maintenance treatment of duodenal ulcer & treatment of benign gastric ulcer.
- Management of hypersecretion of gastric acid.
- Reflux esophagitis.

Contraindications:

- Children under 16 years.
- Lactation
- Impaired renal & hepatic function.

Gastric anti secretory drugs

H2 receptor antagonists:

1- Cimetidine:

Side effects:

- **Diarrhea, hepatic fibrosis, Hepatitis, Pancreatitis ,**
- **Hallucinations, Dizziness, Headache, confusion, ataxia, double vision.**
- **Hypotension, Arrhythmias following I.V. administration. .**
- **Aplastic anemia, thrombocytopenea.**

Dose: 300 mg, 4 times daily with meals & at bedtime.

Gastric anti secretory drugs

H2 receptor antagonists:

1- Cimetidine:

Nursing considerations:

- Administer oral medication with meals, if I.V. dilute as prescribed.
- Note the general condition of the patient (If take chemotherapy or radiation)
- Note signs of infection.
- For diarrhea, maintain adequate hydration.
- Monitor renal function.
- Be alert for mold swings that may occur.

Gastric anti secretory drugs

H2 receptor antagonists:

2- Ranitidine Hcl:

Trade name: Zantac, Randine
Class.: H2-receptor antagonists.
Action: It competitively inhibits gastric acid secretion by blocking the effect of histamine on histamine H2-receptors.
Uses: see cimetidine.
Contraindications: Liver cirrhosis, impaired renal & hepatic function.

Gastric anti secretory drugs

H2 receptor antagonists:

Side effects:

- **Constipation , nausea , vomiting, diarrhea,**
- **headache Dizziness , malaise , vertigo ,**
- **bradycardia or tachycardia**
- **Pancytopnea , rashes, bronchospasm , alopecia.**

Dosage form:“ampule 50 mg \2m.” Tab. 150, 300 mg.

Dose: 150 mg 2 times daily. Maintenance 150 mg at bedtime.

Gastric anti secretory drugs

H2 receptor antagonists:

Nursing considerations:

- Dilute for I.V. use (50 mg in 20 ml of 0.9% Nacl).
- Note any evidence of renal or liver disease.
- Obtain baseline liver & kidney function.
- Note for signs of infection.
- Adequate hydration for problem of diarrhea.

Gastric anti secretory drugs

2. Proton pump inhibitors

c) Omeprazole

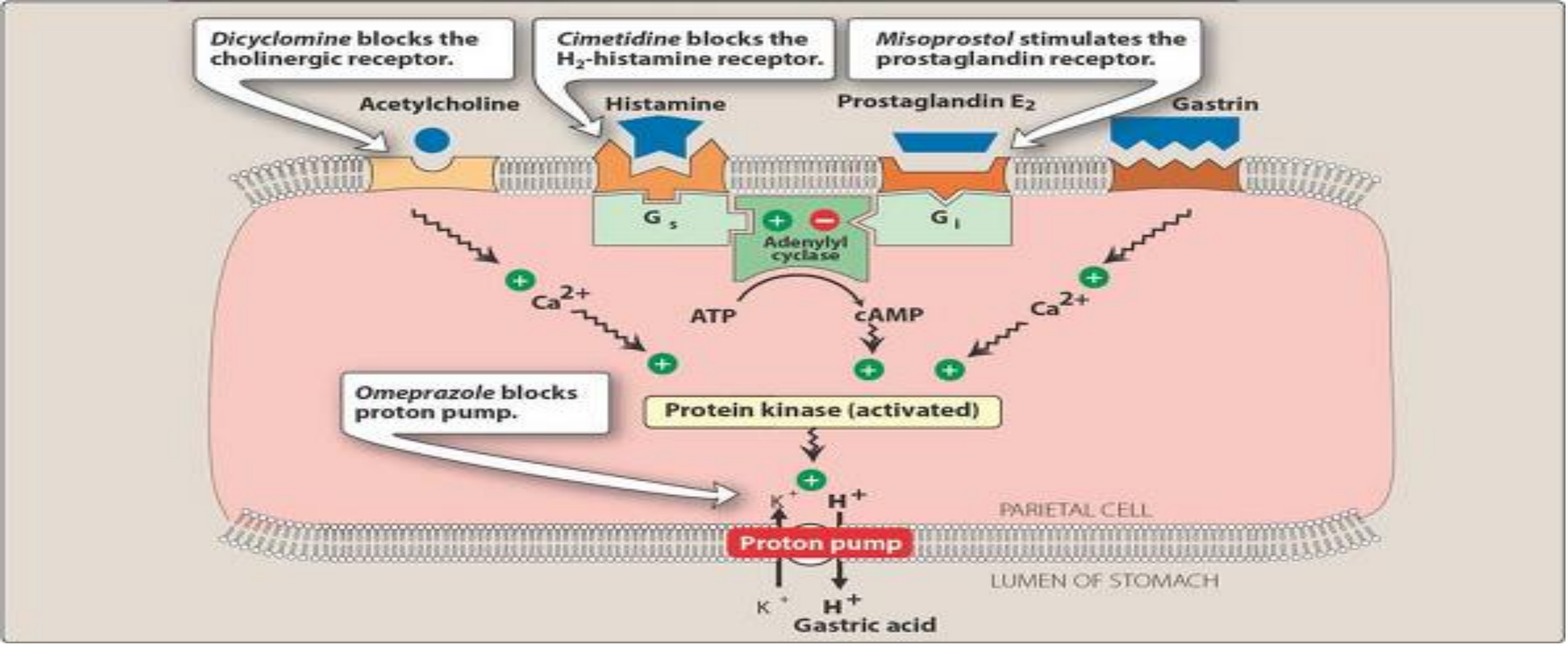
Trade name: Losec, Pepticum, Mepral

Uses:

- Gastroesophageal reflux disease, esophagitis, Duodenal ulcer (short-term treatment)
- to eradicate H. pylori, Short-term treatment of active benign gastric ulcer

Action

- Inhibits activity of acid (proton) pump and binds to hdrogen-potassium adenosine triphosphatase at secretory surface of gastric parietal cells to block formation of gastric acid.



It binds to the H+/K+-ATPase enzyme system (proton pump) of the parietal cell, thereby suppressing secretion of hydrogen ions into the gastric lumen. The membrane-bound proton pump is the final step in the secretion of gastric acid.

Gastric anti secretory drugs

2. Proton pump inhibitors

c) Omeprazole

Dose: 20-40 mg daily for 4-8 weeks

Side effects:

- **CNS: headache, dizziness.**
- **GI: diarrhea, abdominal pain, nausea, vomiting, constipation, flatulence. Musculoskeletal: back pain.**
- **Respiratory: Cough upper respiratory tract infection.**
- **Skin: rash.**

Contraindications & cautions

- **Contraindicated in patients hypersensitive to drug or its components. Use cautiously in patients with hypokalemia and respiratory alkalosis.**

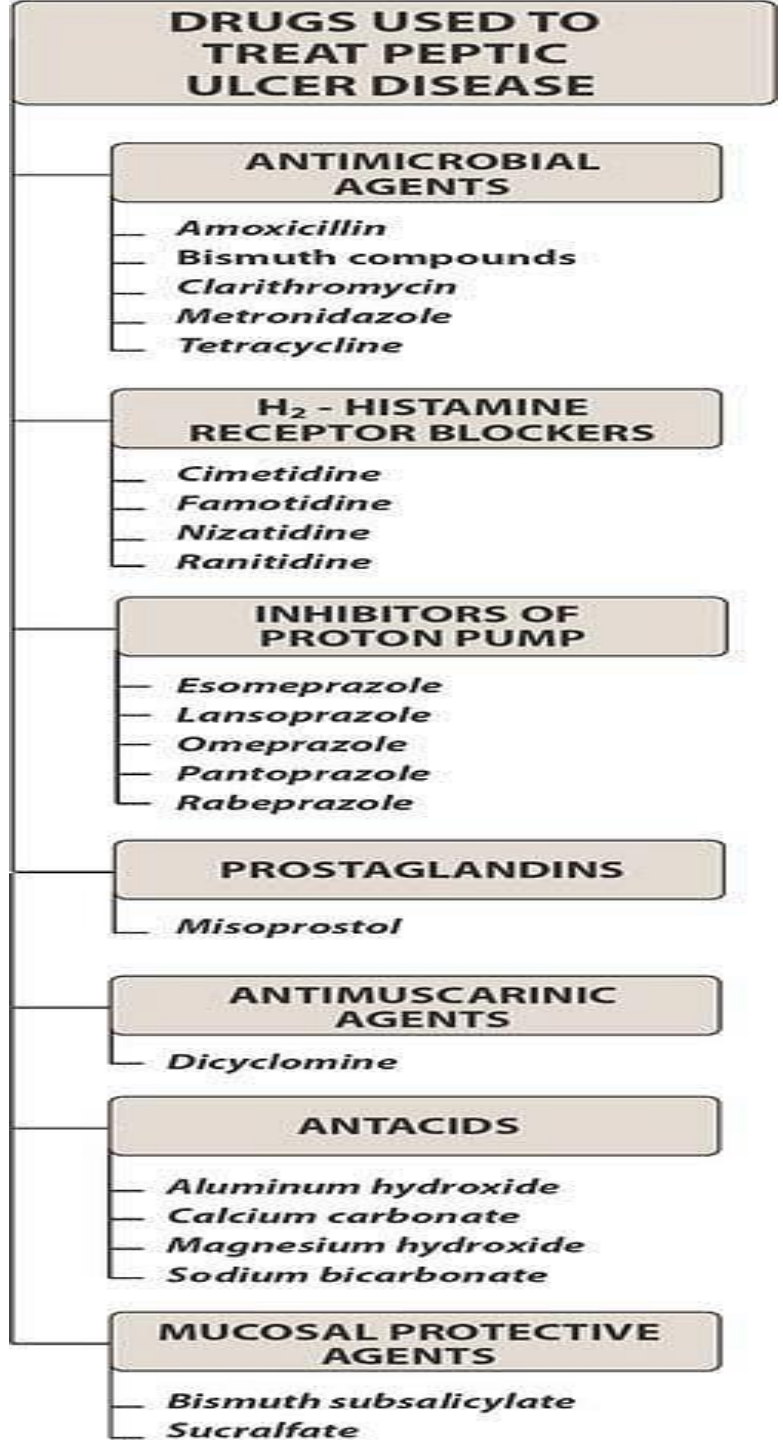
Gastric anti secretory drugs

2. Proton pump inhibitors

c) Omeprazole

Nursing considerations

1. Dosage adjustments may be necessary in patients with hepatic impairment.
2. Tell patient to swallow tablets or capsules whole and not to open, crush, or chew them.
3. Instruct patient to take drug 30 minutes before meals.
4. Caution patient to avoid hazardous activities if he gets dizzy.


DRUGS USED TO TREAT PEPTIC ULCER DISEASE
ANTIMICROBIAL AGENTS
Amoxicillin
Bismuth compounds
Clarithromycin
Metronidazole
Tetracycline
H2 - HISTAMINE RECEPTOR BLOCKERS
Cimetidine
Famotidine
Nizatidine
Ranitidine
INHIBITORS OF PROTON PUMP
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Omeprazole
Pantoprazole
Rabeprazole
PROSTAGLANDINS
Misoprostol
ANTIMUSCARINIC AGENTS
Dicyclomine
ANTACIDS
Aluminum hydroxide
Calcium carbonate
Magnesium hydroxide
Sodium bicarbonate
MUCOSAL PROTECTIVE AGENTS
Bismuth subsalicylate
Sucralfate

Anti-emetics

- Nausea & vomiting can be caused by a variety of conditions such as infections, drugs, motion, organic disease or psychological factors.
- The underlying cause of the symptoms must be elicited before emesis is corrected.
- The act of vomiting is complex. The vomiting center in the medulla responds to stimulation from many peripheral areas as well as stimuli from CNS itself, the CTZ in the medulla, the vestibular apparatus of the ear & the cerebral cortex.
- The selection of antiemetic depends on the cause of the symptom as well as on the manner in which the vomiting is triggered.

Anti-emetics

- Many drugs used for other conditions such as antihistamine, phenothiazines & barbiturates have antiemetic properties & can be so used.

Drug interaction:

- Because of their antiemetic and antinauseant action, the antiemetics may mask overdose caused by other drugs.

Anti-emetics

Nursing considerations:

1. Take a complete history, if it is unusual occurrence or if it is a recurring phenomenon.
2. Assess for other untoward symptoms as increased intracranial pressure or intestinal obstruction (antiemetic may mask signs of underlying pathology)
3. Caution the client that drug tends to cause drowsiness & dizziness, advise him\her to avoid hazardous tasks.

Anti-emetics

Metoclopromide Hcl:

Trade name: Pramin

Class: Antiemetic

Action:

- It is dopamine receptor antagonist acts both centrally & peripherally, centrally due to the effect in the CTZ (inhibition), peripherally it stimulate the motility of the upper GIT without affecting gastric & biliary or pancreatic secretions.
- It relaxes the pyloric sphincter & increases the peristalsis of the duodenum resulting in accelerated gastric emptying & intestinal transit.

Anti-emetics

Metoclopromide Hcl:

Indications:

1. **Digestive disorders leading to relief GIT pain, dyspepsia & regurgitation in peptic ulcer, reflux esophagitis & post anesthetic vomiting.**
2. **Nausea & vomiting as in chemotherapy.**
3. **Facilitate diagnostic procedure e.g. barium meal.**

Anti-emetics

Metoclopromide Hcl:

Side effects:

GI disturbances, transient hypertension, supraventricular tachycardia, dizziness & extrapyramidal effect "convulsion".

Forms:

- **Ampule 50 mg\10ml**
- **Tablet 10mg**
- **Syrup 5mg\5ml**
- **Suppository. 20mg (adult) , 5mg (children).**

Anti-emetics

Metoclopromide Hcl:

Dose:

- **10 mg, 30 minutes before meal & at bed time .**
- **For chemotherapy 2 mg /kg, 30 minutes before chemotherapy.**

Contraindications: ***Seizure (epilepsy) , Pheochromocytoma ,intestinal obstruction.***

Anti-emetics

Metoclopromide Hcl:

Nursing considerations:

1. Don’t give pramin to patients with epilepsy, pheochromocytomes or patients with intestinal obstruction.
2. Administer oral medication 30 minutes before meal & at bed time .
3. Administer I.V. injection slowly over 1-2 minutes.
4. Be aware of the extrapyramidal symptoms specially in children

Other antiemetics:

- Diphenidol Hcl “ ventrol” .
- Trimetho benzamide Hcl “ Tigan” .